

International Studies in Entrepreneurship

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David B. Audretsch *Editors*

# Entrepreneurial Ecosystems

Place-Based Transformations and  
Transitions



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# Entrepreneurial Ecosystems

Place-Based Transformations and Transitions

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# Preface

Interest in entrepreneurial systems has recently exploded. Policy-makers want to create one. Entrepreneurs want to be in one. This emergence of entrepreneurial ecosystems is as new as it is rapid. Earlier generations had thought of entrepreneurship as a rather lonely, or at least individualistic, journey. Both the decision to become an entrepreneur and the subsequent outcome or performance of the entrepreneurial venture seemed to be all about the individual, or team of individuals, and the entrepreneurial firm, but little else. Both policy-makers and scholars remained fixated on those individual characteristics and traits that might influence the decision to become an entrepreneur, as well as the eventual entrepreneurial performance, such as the preference for risk, need for autonomy, and desire for actualization.

Perhaps it was the large, imposing elephant in the entrepreneurship room, Silicon Valley, that persuaded thought leaders in business and policy that entrepreneurship was more than simply an individual calculation based on *homo economicus*. Rather, place matters, or at least what is at that place. One thing that no one seems to have doubted for well over two decades is that whatever it is that constitutes Silicon Valley is conducive to entrepreneurship. And not only Silicon Valley. Just as entrepreneurship has been found to thrive across a broad spectrum of cultural and national contexts, ranging from Bangalore to Berlin, a common denominator is the rich and supportive local firms, policies, and institutions—the ecosystem.

Perhaps the corollary of Hillary Rodham Clinton’s insight that “It takes a village to raise a child” is “It takes an ecosystem to foster an entrepreneur.” If “No man is an island,” as the English poet John Donne mused, then an entrepreneur who is unconnected and isolated is surely lost.

The emergence of entrepreneurial ecosystems as an important construct for policy-makers and scholars comes not just from the recognition that place matters for entrepreneurs. The reverse holds as well—entrepreneurs matter for the place. Those places, albeit a city, state, or region, that exhibit more entrepreneurship enjoy an enhanced economic performance. The performance premium, in terms of jobs, a higher standard of living and prosperity, and greater economic growth, accruing

from entrepreneurship has not gone unnoticed. Rather, constituents have responded with a mandate for making their place entrepreneurial. Rather than a top-down process of decision making, the mandate for entrepreneurship policy bubbles up from multiple directions, not just from the entrepreneurs themselves, but also from a myriad of constituents with a stake in the economic performance and vitality of a particular place.

The emerging literature on entrepreneurial ecosystems could be considered to constitute the intersection of entrepreneurial studies with spatial studies—the confluence between place and entrepreneurship. This important new book contributes to and makes considerable headway in shedding light on what exactly constitutes an entrepreneurial ecosystem, how and why entrepreneurial ecosystems matter in shaping the economic performance for both the entrepreneurs and the particular place, and what exactly needs to be done, in terms of policies and institutions, to generate and sustain an entrepreneurial ecosystem.

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# Entrepreneurial Ecosystems: The Foundations of Place-based Renewal

Allan O'Connor , Erik Stam, Fiona Sussan and David B. Audretsch

**Abstract** Regardless of how the concept of a ‘place’ is geographically defined, be it cities, regions, nations or otherwise, the impact of new technologies will influence much of our business, social, and economic landscapes. Evidently, there is an increasing pressure on ‘places’ to embrace new opportunities for strategic development and confront complacency that retards change. The solution may very well be in creating and sustaining entrepreneurial ecosystems where entrepreneurial action thrives and innovation drives the new economy. However, defining the entrepreneurial ecosystem remains difficult and the methods used to analyse them are inconsistent. This chapter deals with the theoretical foundations of an entrepreneurial ecosystem when it is specifically considered as a place-based change management instrument. As we introduce the variety of submitted works to this volume it becomes apparent that while capital, labour, resources and infrastructure are all important, equally, how these elements are mobilized through leadership, governance, and institutions are at least but perhaps even more important. While technology figures heavily, it is overshadowed to some extent by an emphasis on individual action and agency. Defining place-based transitions and transformations is dependent upon anchoring the point of departure. Entrepreneurship therefore has a key role to play in innovating the renewal of place and the value creation of entrepreneurs takes precedence. Technological advances offer great value creating opportunities in some places but in all, the value lies in the socioeconomic stimulus that entrepreneurs create through new opportunities for developing cohesive communities.

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**Keywords** Entrepreneurial ecosystems • Place-based • Transition Transformation • Regional development • Governance

## 1 Introduction

Entrepreneurial ecosystems have become a prominent concept in recent discussions about economic development in both the policy and business worlds. The concept of entrepreneurial ecosystems has gained popularity due to mainstream business books such as Feld's (2012) *Startup Communities*, work by Isenberg (2010) in the *Harvard Business Review*, insights related to policy making (OECD 2013; Stangler and Bell-Masterson 2015), and practitioner projects like Compass (Herrmann et al. 2015). These works have popularized the idea amongst entrepreneurial leaders and policymakers that the local context can have a significant impact on the entrepreneurship process. In spite of its popularity, there is not yet a widely shared definition nor an approach to researching entrepreneurial ecosystems that facilitates development.

Furthermore, how these entrepreneurial ecosystems relate to, influence, or inspire change in the economic fortunes of a place, fostering social and industrial transformations and transitions, is an even more puzzling question. These transformations involve creative destruction, in which entrepreneurs are key agents (Schumpeter 1934). However, creative destruction does not happen overnight, it often involves a long and tedious transition from the old regime to the new regime (Raven et al. 2012). Despite the fact that some places, Silicon Valley, Route 128 or Boulder Colorado for instance, are persistently named as places with strong entrepreneurial ecosystems, a broad based study of entrepreneurial ecosystems is yet to be undertaken. Governance of entrepreneurial ecosystems is difficult without well-developed measures of what comprises an entrepreneurial ecosystem and indicators that show how well it is functioning. Ultimately we are left with only a narrow base of evidence to show how healthy entrepreneurial ecosystems are critical to regional development.

In this book we deliberately problematize the idea of an entrepreneurial ecosystem by emphasizing 'place' as its location. In other words we assume first that every place has an entrepreneurial ecosystem but the performance, practices, strengths, weaknesses, opportunities, threats, objectives and actors each vary. Picking up the points made by Acs et al. (2017) that the lineages of the entrepreneurial ecosystems are rooted in the strategy and regional development literatures, we seek to better understand the strategic approaches, challenges and pathways that various places face and how the entrepreneurial ecosystems respond across various places, either through coordination or organically, to account for regional development. Transformations are an emerging property of entrepreneurial ecosystems, but they do not emerge automatically. For these transformations to emerge, many actors need to be aligned, to initiate and set in motion a large scale transition. We expect places will recognize the need for transformations of their regional economies (e.g. from traditional manufacturing to advanced knowledge-based industries) and work on the transition from

the old to the new regime. Through this work we seek to extend knowledge on how entrepreneurial ecosystems are strategically beneficial or otherwise to regional development.

Our current endeavours attempt to answer three questions: (1) What is an entrepreneurial ecosystem, (2) how are places affected by entrepreneurial ecosystems and affecting their entrepreneurial ecosystems, (3) how can we empirically grasp the relationship between entrepreneurial ecosystems, entrepreneurship and the transformation of places to ground research practice in this domain? In this chapter we will provide a framework that answers the first question. This volume provides further research that contributes to answering the second question to understand the relationship between place and the entrepreneurial ecosystem. Our third question—how can we empirically grasp the relationship between entrepreneurial ecosystems, entrepreneurship and the transformation of places to ground research practice in this domain?—will be addressed through synthesis of the discussion raised through this chapter.

## 2 Entrepreneurial Ecosystem Defined

What is an entrepreneurial ecosystem? In abstract terms, central to the definition of entrepreneurial ecosystems are (entrepreneurial) agency and (human made) context (i.e. the ecosystem), especially the humanly devised constraints that structure human interaction (i.e. rules of the game, institutions), that shape the presence and form of important entrepreneurial ecosystem elements such as capital, labour and knowledge (Acs et al. 2014; Stam 2015).

The ecosystem concept finds its roots in ecology (eco-logical system), in which the interaction of living organisms with their physical environment is at the center. Adopting an ecological lens to consider the entrepreneurial ecosystem also invites the consideration of two alternate ontologies (McKenzie and Sud 2009). The first ontological perspective considers the ecosystem as a steady-state natural regulated system maintaining balance among the organisms and elements. This balanced system is then open to disruption from external influence, and might differ in its ability to absorb external shocks (resilience). The second ontology assumes that ecosystems are in a constant change of state that shifts between levels of complexity. This latter ontology, in biological terms, suggests that survival, actor cooperation and external factors each need to be accounted for in understanding ecosystem development and change. These ontological views resonate with the competing economic debates with respect to entrepreneurship and contrasts between economic equilibrium and continuous change. Our perspective is firmly grounded in the second ecology, emphasizing transformations and transitions.

We do not stick to the strict natural systems version of the ecological approach, as entrepreneurial ecosystems largely involve artificial, social systems. The entrepreneurial ecosystem approach emphasizes that entrepreneurship takes place in a community of interactive (Audretsch and Belitski 2016) and interdependent actors (cf. Freeman and Audia 2006) set among social, political, economic, and cultural elements (Spigel 2017). In particular the literature on entrepreneurial ecosystems

focuses on the role of the (social) context in allowing or restricting entrepreneurship and in that sense is closely connected to other recent ‘systems of entrepreneurship’ or systemic entrepreneurship research approaches (Acs et al. 2014; Neck et al. 2004; Sternberg 2007; Ylinenpää 2009), which often aim to bridge the innovation system approach and entrepreneurship studies.

Whether to ponder casually or to apply systematic research rigour in the analysis of an entrepreneurial ecosystem, it requires the adoption of systems thinking. Systems thinking involves identifying the contextual patterns of organization of the elements and the relationships among the elements found in the whole of the system in focus. Rather than dealing with specific content, “systems thinking balances the focus between the whole and its parts, and takes multiple perspectives into account” (Cabrera et al. 2008, p. 301). Applying systems thinking in our case requires an examination of what comprises the entrepreneurial ecosystem and what does not. An understanding is needed about what function the elements of the system are intended to perform, what outcomes are desired from the ecosystem and how well these elements contribute to these expected outcomes. The elements of the system also do not sit idly but rather the actors’ form relationships among the elements that influence the overall system’s performance. The actors themselves also hold perspectives of the system and a full analysis of the system needs to account for these various perspectives. In this work we seek to bring together various perspectives of the entrepreneurial ecosystem to attempt to synthesize a more general view.

Entrepreneurial activity, is the first observable output of the entrepreneurial ecosystem and is considered to be the process by which individuals identify and pursue opportunities for innovation. Theoretically this innovation will eventually lead to new value in society and this is therefore the desirable and ultimate objective outcome of an entrepreneurial ecosystem. Entrepreneurial activity is an intermediary output of the system that is an early indicator of progress toward the desired outcome of value created within a social context. This entrepreneurial activity has many manifestations, such as innovative start-ups, high-growth start-ups, and entrepreneurial employees (Stam 2014). The term productive entrepreneurship refers to “any entrepreneurial activity that contributes directly or indirectly to net output of the economy or to the capacity to produce additional output” (Baumol 1993, p. 30); which we interpret as entrepreneurial activity that generates aggregate welfare. Productive entrepreneurship might also include failed enterprises that have provided a fertile breeding ground for subsequent ventures or inspired them, creating net social value (‘catalyst ventures’: Davidsson 2005).

### 3 Differences and Similarities with Related Concepts

How does the entrepreneurial ecosystem concept differ from related concepts such as clusters, industrial districts, innovation systems, innovation ecosystems, and the triple helix model of industry, government and university interactions? What are the similarities? What the entrepreneurial ecosystem approach has in common with the

other established concepts is the focus on the external business environment: that there are forces beyond the boundaries of an organization that can contribute to a firm's overall competitiveness, and that the firm contributes to a system larger than itself (see Table 1).

The industrial district approach emphasizes the local division of labour of an industry (Marshall 1920) and the interaction between the community of people and a population of firms within a socio-territorial entity (Becattini 1990) in order to be successful on international markets. The cluster approach focuses on 'geographic concentrations of interconnected companies, specialized suppliers, service providers, firms in related industries, and associated institutions (...) in particular fields that compete but also co-operate' (Porter 1998: 197). Innovation systems refer to the networks and institutions linking knowledge producing hubs such as universities and public research labs within a region and innovative firms. These linkages allow knowledge to spill over between different organizations, increasing a region's overall innovativeness (Cooke et al. 1997). The triple helix model is a specific approach to analysing the relationships between the broad categories of government, industry and the university sector that contribute to development of a territorial innovation system.

Unlike previous uses of the term 'ecosystem' in the management literature, such as business ecosystems (Moore 1993; Iansiti and Levien 2004) and innovation ecosystems (Adner and Kapoor 2010; Adner 2012) that focus on the organization of a single industry or value chain, entrepreneurial ecosystems are an inherently geographic perspective. That is to say, entrepreneurial ecosystems focus on the cultures, institutions, and networks that build up within a region over time rather than the emergence of order within global markets. It does not involve the strategic management of a firm or group of firms, but the strategic management of a place (Audretsch 2015). The entrepreneurial ecosystem approach differs from industrial district, cluster, and innovation system approaches by the fact that the entrepreneur, rather than the firm, is the focal point of analysis. The entrepreneurial ecosystem approach thus begins with the entrepreneurial individual instead of the company but also emphasizes the role of the social and economic context surrounding the entrepreneurial process. Most cluster studies focus on firms and industries, including their dynamics (Frenken et al. 2015). As opposed to the clusters, district, and innovation systems literature, the focus of entrepreneurial ecosystems research is placed firmly on the entrepreneur and the start-up rather than larger, more established firms or slower growing SMEs. The high-growth start-ups that make up the basis of entrepreneurial ecosystems are not necessarily included in all cluster and industrial district models (Markusen 1996). While frameworks of industrial districts, clusters, and innovation systems do include a role for entrepreneurs (e.g. Cooke 2001; Henry and Pinch 2000; Ylinenpää 2009), the focus is not specifically on them but rather the role of entrepreneurs and start-ups within larger systems of value creation and innovation. As a result, these existing approaches often see start-ups as smaller versions of larger, international firms rather than as unique organizational entities with different (and often more constrained) capabilities and resources.

**Table 1** Comparison with industrial district, cluster, and innovation system literature

|                                 | Key actors                            | Key concepts                                                                                                  | Input into entrepreneurial ecosystem approach                                                                                                                      | Key outcome                                                               | Key references                                            | Key references entrepreneurship                                          |
|---------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------|
| Marshallian industrial district | SMEs                                  | Labor market pooling; specialized goods and services; knowledge spillovers; market competition                | Talent (labor market pooling), intermediate services (specialized goods and services), knowledge (spillovers)                                                      | Regional economic growth (productivity)                                   | Krugman (1991), Markusen (1996), Marshall (1920)          | –                                                                        |
| Italianate industrial district  | SMEs; local government                | Flexible specialization, interfirm cooperation, trust (social embeddedness)                                   | Networks between entrepreneurs and enterprises                                                                                                                     | Regional economic growth (employment)                                     | Becattini (1990), Harrison (1992), Piore and Sabel (1984) | Johannisson et al. (1994), Lazerson and Lorenzoni (1999), Malecki (1997) |
| Cluster                         | Innovative firms                      | Factor conditions; demand conditions; related and supporting industries; firm structure, strategy and rivalry | Talent, finance, knowledge, physical infrastructure (factor conditions); demand (demand); support services/intermediaries (related and supporting industries); ... | National/regional competitiveness (productivity of particular industries) | Porter (1990, 1998)                                       | Delgado et al. (2010), Rocha (2004), Rocha and Sternberg (2005)          |
| Innovation system               | Innovative firms; national government | Networks, inter-organizational learning, system                                                               | Knowledge, finance, formal institutions, demand                                                                                                                    | Innovation                                                                | Braczyk et al. (1998), Freeman (1987), Lundvall (1992)    | Sternberg (2007), Ylinenpää (2009)                                       |

(continued)

Table 1 (continued)

|                      | Key actors                         | Key concepts                                                                             | Input into entrepreneurial ecosystem approach                     | Key outcome                                                             | Key references                                                     | Key references entrepreneurship                      |
|----------------------|------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------|
| Triple helix model   | Government, firms and universities | Interactions between university-industry-government, innovation, knowledge-based society | Knowledge transfer and interdependence of three sectoral actors   | Innovation system                                                       | Etzkowitz and Leydesdorff (2000), Leydesdorff and Etzkowitz (1998) | Etzkowitz et al. (2000)                              |
| Innovation ecosystem | Innovative firms                   | Co-innovation, adoption chain, shared value proposition                                  | Interdependence of actors involved in innovation; global networks | Value creation and capture by the firms in the ecosystem; firm survival | Adner (2012), Iansiti and Levien (2006)                            | Nambisan and Baron (2013), Zahra and Nambisan (2011) |

Based on: Stam and Spigel (2018)

An important contrast with other concepts is that the entrepreneurial ecosystem approach not only sees entrepreneurship as a result of the system, but also sees the importance of entrepreneurs as central players (leaders) in the creation of the system and in keeping the system healthy (Feldman 2014). This “privatization” of entrepreneurship policy diminishes the role of the state compared with previous policy approaches. However, Feld (2012) argues that this does not remove its role but rather shifts it to that of a ‘feeder’ of the ecosystem than as a ‘leader’. Entrepreneurs with a long-term commitment to the ecosystem are often best positioned to recognize the opportunities and restrictions of the ecosystem and to deal with them together with other ‘feeders’ of the ecosystem (such as professional service providers and the financial infrastructure). These successful businesspeople and philanthropists can act as ‘dealmakers,’ using their own social networks and capital to improve the entrepreneurial environment of their home region (Feldman and Zoller 2012). Increasingly these entrepreneurs are viewed as key co-creators in the ecosystem by convincing stakeholders (e.g., government, customers, non-financially involved local citizens) to legitimize their new ventures and establish new industry standards within domestic markets and across international borders (Alvarez et al. 2015). However, the government retains an important role as a ‘feeder’ who acts to create a conducive economic and social environment for

**Table 2** Differences and similarities between entrepreneurial ecosystems and related concepts

| Approach        | Industrial district, cluster, innovation system, triple helix                                                                                                                                                  | Innovation ecosystem                                                                                                                           | Entrepreneurial ecosystem                                                                                                                                                                                               |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main focus      | Economic and social structures of a place that influence overall innovation and firm competitiveness. In many cases, little distinction made between (fast growing) start-ups and other types of organizations | Creating customer value through a chain of interdependent organizations, with differential value capture by different players in the ecosystem | Start-ups explicitly at center of ecosystem. Seen as distinct from established large firms and (lower-growth) SMEs in terms of conceptual development and policy formation                                              |
| Locus of action | Private firms and state is primary locus of action in building and maintaining industrial district/cluster/innovation system. Little room for individual agency in their creation                              | One large firm as orchestrator of the ecosystem, with many other firms co-innovating or involved in the adoption of innovation                 | Entrepreneur is the core actor in building and sustaining the ecosystem. While state and other sources might support ecosystem through public investment, entrepreneurs retain agency to develop and lead the ecosystem |

Based on: Acs et al. (2017)

entrepreneurship, for example in adjusting laws and regulations or providing training and educational opportunities.

As illustrated in Table 2, there are significant differences between entrepreneurial ecosystems and allied economic development concepts such as industrial districts, clusters, innovation systems, the triple helix model, and the strategy concept of innovation ecosystems. This does not mean that work on ecosystems cannot draw on the decades of research underlying these concepts, but that the findings from this work must be reinterpreted through the lens of the (entrepreneurial) agent that is at the heart of the entrepreneurial ecosystem approach.

## 4 Outline of the Book

In this chapter we have introduced the entrepreneurial ecosystem concept, especially its definition and differences and similarities with other related concepts. In this section we will cover the scope of research compiled in this book and synthesize the contributions to exhibit how places are affected by and are affecting their entrepreneurial ecosystems. We shall then conclude the chapter with an appraisal of the opportunities for further research and suggestions on the empirical investigation of entrepreneurial ecosystems.

Chapter “[Deconstructing the Entrepreneurial Ecosystem Concept](#)” contributed by Daniel et al. (2017) highlights the pre-paradigmatic stage of development of the entrepreneurial ecosystem approach. The researchers reinforce the link between entrepreneurial ecosystems and place and in particular, find strong themes of dynamics and change observable among their data. The multi-disciplinary collaboration for this work saw two approaches to analysis contrasted and compared with the entrepreneurial ecosystem approach. The authors argue that research of place-based dynamics could adopt network and systems analytical frames of reference for analysing entrepreneurial ecosystems. The entrepreneurial ecosystem is focused on community context, multi-level content from individual through to macro-influences, a process focus on the ingredients for serendipity, opportunity, and circumstance to underpin entrepreneurship, its nature being relative to place, its purpose to stimulate enterprise and place development while in practice it is subjected to objective interventions such as may be introduced by individual actors. While the various elements that comprise an entrepreneurial ecosystem are influenced by the various individual actors, organizations and institutional settings, it is a multi-level process of serendipity, opportunity, and circumstance that characterizes transitions and the accumulated outcomes of enterprising and innovative activity that evolves the dynamic changes and transformation relevant to place.

The contribution of Chapter “[Institutional Dynamism in Entrepreneurial Ecosystems](#)” by Fuentelsaz et al. (2017) focuses on the institutional support available to increase a new venture’s chance of success and improve the efficiency of the resource base of ecosystems. The chapter considers new venture development stages and examines resource availability that responds to the dynamic needs of a

start-up business. While the entrepreneur is flagged as the primary actor in the ecosystem the authors also acknowledge the diverse set of actors representing and being embedded in institutions that influence a new ventures development. Their argument suggests that entrepreneurs define the 'place' boundary and that transitions are dependent on entrepreneurs being able to access resources that assist in the new ventures development. Transformation therefore is evidenced through the ability of firms to not only start, but for these firms to grow.

Chinta and Sussan (2017) take a different approach to place by concentrating their discussion on the university campus in Chapter "[A Triple-Helix Ecosystem for Entrepreneurship: A Case Review](#)". Mirroring the triple helix model, the authors consider the relationships among universities, government and business and the implications for entrepreneurial firms. Their argument suggests that the role of the university in this relationship has somewhat shifted over time to become more than a place of learning and new knowledge development, becoming an integral part of the supply and demand equation for seeding entrepreneurial firms. However, universities are just one part of the triad and they need government and business to contribute other elements within a supply and demand model for new ventures. For these authors transitions will be observable through the lens of increased and focused collaboration between the triad partners. Transformations will be perceived when entrepreneurial firms eventuate through a strong supply of entrepreneurs and a resource base oriented around the university campus that readily meets the needs of the entrepreneurs and their new venture demands.

Continuing the discussion on the university context, O'Connor and Reed (2017) adopt an approach that drew upon interviews with various ecosystem stakeholders to construct a view on the role of universities in an entrepreneurial ecosystem. These authors too suggest that the role of universities has expanded beyond the teaching and research expectations to include tighter integration within their communities, shared responsibilities for regional development, and a place at the table in considering governance matters in the regions within which they are located. The five roles they isolate suggest different dimensions of strategic decisions that universities should address and identify a position within. Interpreting O'Connor and Reed's stance suggests that transitions occur through increased activity in new knowledge, innovation based entrepreneurial firms. Transformations will be observed when a region evidences an economic renewal that leverages the university as a regional resource.

Chapter "[Regional Entrepreneurship Ecosystems Support: South East Queensland as Case Study](#)" marks a change in focus in the book from theory (this chapter and Chapter "[Deconstructing the Entrepreneurial Ecosystem Concept](#)") and particular actor and relationship functions (Chapters "[Institutional Dynamism in Entrepreneurial Ecosystems](#)"–"[Theorizing the University Governance Role in an Entrepreneurial Ecosystem](#)") to move to the regional level. The remaining chapters provide an international cross-section of various places and focuses on aspects of the entrepreneurial ecosystem. Place is variously defined by provincial boundaries in national contexts (the provinces of the Netherlands) to a grouping of sub-state regions in South East Queensland and through to particular cities, e.g. Brisbane,

Glasgow and Las Vegas. Perhaps notable though that in each case the entrepreneurial ecosystem of a place is defined more by spatial proximity and community populations than by national boundaries. This is particularly evident where sub-state regions and provinces are defined to demarcate differences and draw comparisons. Indeed throughout all the chapters none take a particularly national vantage point of the entrepreneurial ecosystem instead all tend to take community structures as place-based boundaries, be it cities, a province, local council areas, industry or university communities. This suggests a human cohesiveness and proximal interaction may be important dimensions in boundary definition. At the national level it may be that dimensions of human cohesion and proximal interaction are too weak to form a coherent sense of community although exceptions may be found with small nations in cases such as Singapore or the principality of Liechtenstein for instance.

The first spatially focused chapter provides a Regional Entrepreneurial Ecosystem Support framework that suggests the associated dimensions of support for entrepreneurs. The framework may be a useful device to understand the contributions of various actors and identify support gaps. The comparative research of three regions in relative close proximity (South East Queensland) highlights that entrepreneurial ecosystems can be perceived to differ over fairly short distances; an hour or two in driving terms. The chapter by Villiers Scheepers et al. (2017) can be interpreted to suggest that transitions are grounded by activities that grow technology and knowledge based businesses. However, the transformation is observed through the objectives of the specific regions in question. In other words the authors draw attention to the legacies of history, culture, regional values and institutional forces that each region will invariably have (unless it was previously completely uninhabited) and transformation is dependent on the point of departure to define and evidence that change has taken place.

Moving up to the northern hemisphere, Lassalle and Johnston (2017) consider Polish migrant entrepreneurs in Glasgow, Scotland in Chapter “[Where Are the Spiders? Proximities and Access to the Entrepreneurial Ecosystem: The Case of Polish Migrant Entrepreneurs in Glasgow](#)”. This definition of an ecosystem brings into focus the idea of sub-systems within entrepreneurial ecosystems. The authors discuss the distinctive practice of this sub-grouping of entrepreneurial activity and the cultural and geographic proximity that both enables entrepreneurial activity but at the same time isolates it from other entrepreneurial activities outside of the migrant group community. This too strengthens the case for community as a dimensional aspect in some form and makes apparent that different communities can co-exist within geographies. To create transitions it seems that approaches to reduce distance between actors in both a cultural and geographic sense are appropriate strategies. Transformation therefore would be witnessed at the point of increased business start-up and growth activities of the targeted community.

Moving around the globe, Chapter “[Is there a Path from Sin City to Tech City? The Case for Las Vegas](#)” by Sussan et al. (2017) takes us to Las Vegas to take a close look at how legacy industry and new industry development co-habit. The history of Las Vegas is partly driven by its isolation that has meant that the vibrancy

**Table 3** Relevant aspects and characteristics of the place-based studies

| Chapter | Primary actor(s) in consideration      | Relationships with/among...                                                                                                              | Transition dynamics                                               | Transformation effects                                                          | Place description      |
|---------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------|
| 2       | Individuals                            | Organization, institutions                                                                                                               | Multi-level process of serendipity, opportunity, and circumstance | Accumulated through enterprise and entrepreneurial activity                     | Community defined      |
| 3       | Entrepreneurs among diverse actors     | <i>Institutions</i>                                                                                                                      | Venture stage development                                         | Growing Businesses                                                              | Entrepreneur perceived |
| 4       | Entrepreneurial firms                  | <i>University-business-government (the triad)</i>                                                                                        | Triad research collaboration                                      | Strong supply and demand conditions for entrepreneurial firms through the triad | University campus      |
| 5       | <i>University executive management</i> | Industry and government stakeholders, students (past and present), global research networks                                              | Introducing knowledge-based innovation                            | Economic renewal                                                                | Regional               |
| 6       | Entrepreneurs                          | Universities, research centers, hospitals, incubators, accelerators, state, city and local councils, angel and venture capital investors | Growing knowledge and technology industries                       | Development goals as defined by regions                                         | <i>SE Queensland</i>   |
| 7       | <i>Migrant entrepreneurs</i>           | Migrant community, communities outside defined migrant population                                                                        | Developing cultural and geographic proximities                    | Starting, sustaining, and developing businesses at segmented ecosystem level    | <i>Glasgow</i>         |

(continued)

Table 3 (continued)

| Chapter | Primary actor(s) in consideration | Relationships with/among...                                                             | Transition dynamics                                                                                                                                 | Transformation effects                                  | Place description                   |
|---------|-----------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------|
| 8       | <b>Digitaltech entrepreneur</b>   | Legacy and new industry, financiers, deal-makers                                        | Developing university student entrepreneurs, importing entrepreneurs, attracting large tech firms, digital governance, citizenship, and marketplace | New industry emergence                                  | <b>Las Vegas</b>                    |
| 9       | <b>Gazelles/high growth firms</b> | Networks of entrepreneurs, leadership, finance, talent, knowledge, and support services | Entrepreneurship culture, talent and intermediate services                                                                                          | Structural change as an emerging property of the system | <b>Provinces in the Netherlands</b> |

**Bold and italics** indicates the perspective of the study where applicable

of the place has been developed by importing entrepreneurial talent and ideas to grow the entertainment industry that characterizes the identity of Las Vegas. Attempts to grow a technology business sector are reviewed in this chapter which makes apparent the limitations of talent movement between sectors and how history is tending to repeat with the importing of ideas and scale-up business opportunities. The demarcation between uptown (the Strip) and downtown also demonstrates that boundaries are easily created that can serve to define but also isolate communities. Nevertheless, the research suggests that transitioning a place can benefit by developing university student entrepreneurs, importing entrepreneurs, attracting large technology firms and improving technology or industry specific governance, citizenship, and marketplace. The transformation of place would be signaled through the emergence of a stable new industry sector diversified away from the legacy industry.

The chapter by Sussan et al. (2017) also points to issues of measurement and application of methods that are not wholly transferable between places. This issue with entrepreneurial ecosystem measurement is taken up by Stam (2017) in Chapter “[Measuring Entrepreneurial Ecosystems](#)”. Stam (2017) argues that identifying the entrepreneurial ecosystem as a system for measurement purposes gives the ‘ecosystem’ term more substance and makes clear the complex systems status of the phenomenon. By measuring entrepreneurial ecosystems and their outputs in this way we move from the ecosystem metaphor to a complex system model of the entrepreneurial economy. The ‘standard’ multivariate regression model of measurement does not seem to adequately account for the holistic ‘system’ and its quality for producing certain types of ventures, particularly those defined as ‘gazelles’. The transitions that seem to be important through this view are building an entrepreneurship culture, promoting talent development and providing intermediate services. The transformation would be evident as structural change as it emerges as a property of the system (Table 3).

## 5 Transitions and Transformations

By distinctly connecting the entrepreneurial ecosystem approach to questions of place-based transitions and transformations, the chapters contained within this work make clearly apparent the multi-level nature of the entrepreneurial ecosystem phenomenon. Daniel et al. (2017) in Chapter “[Deconstructing the Entrepreneurial Ecosystem Concept](#)” put it most pointedly by describing the entrepreneurial ecosystem as a multi-level process of serendipity, opportunity, and circumstance. By developing an entrepreneurial ecosystem the probability of entrepreneurs recognizing, discovering or emerging opportunities through the circumstances created and the serendipitous intersections between people and resources of all types, increases. Therefore it is not surprising to also observe the transitional patterns featured across the following chapters suggest activities are undertaken at multiple levels to influence individuals, firms, industry sectors and regions.

At the individual level there is clearly a need for entrepreneurs to be present within the ecosystem. Two approaches to transitions are suggested by Sussan et al. (2017) that include developing entrepreneurs through the university education system and attracting entrepreneurs to take up residence in a place. Before entrepreneurial activity takes place, entrepreneurs are needed.

At the firm level a further two approaches are highlighted. Supporting new firms through the various stages of development is suggested by Fuentelsaz et al. (2017). This transitional intervention is designed to shore up the firms that are created to assist in their survival and growth. Entrepreneurial firms are a characteristic of an entrepreneurial ecosystem and therefore ensuring new firm continuation appears to be a strategic option. O'Connor and Reed (2017) suggested a more directed approach toward facilitating the knowledge and technology sectors to inspire more new firms. However, Sussan et al. (2017) offer another firm level approach and that is to attract large and established firms into the region. This firm level intervention is specifically designed for spill over effects and importing new skills and capabilities into the region.

The industry sector level of transition strategies appear to be targeted more specifically at creating stronger bonds between the relationships of industry, research and government (Chinta and Sussan 2017). Creating the conditions for new knowledge and technology businesses to start and grow into thriving industry sectors seems to be the thrust for Villiers Scheepers et al. (2017). However, Sussan et al. (2017) are a little more specific by suggesting that developing governance, citizenship and marketplace for new knowledge and technology areas are important for attracting and developing a like-minded community around any particular technology sector.

To approach transition strategies through the lens of regions it would appear a concerted effort to create and enhance an entrepreneurship culture, develop talent and provide intermediate services bring a focus to institutional interventions (Stam 2017). Similarly Lassalle and Johnston (2017) encourage addressing the gaps in connectedness in both cultural and geographic terms will be important to enable transitions, again bringing institutional forces into focus.

While transition strategies are apparent across multiple levels, transformations on the other hand seem to be focused on a much narrower set of outcome indicators at either the firm or regional levels. Firm level indicators of transformation would include a notable accumulation of enterprise and entrepreneurial activity (Daniel et al. 2017), a greater number of growing businesses (Fuentelsaz et al. 2017) or increases in the rate of start-up, development and growth of businesses either sectorally or broadly (Lassalle and Johnston 2017).

At the regional level transformation indicators across government, industry and the research sectors would show strong supply and demand conditions for entrepreneurial firms to start and prosper (Chinta and Sussan 2017). Other indicators of new industry emergence (Sussan et al. 2017) or structural change (Stam 2017) may become apparent or more generally meeting specific regional development goals (Villiers Scheepers et al. 2017) and economic renewal targets (O'Connor and Reed 2017) would suggest transformational gains.

This book does not provide eternal truths about what entrepreneurial ecosystems are nor how they should be developed: other viewpoints on entrepreneurial ecosystems remain. However a cacophony of perspectives may constrain the accumulation of knowledge about how entrepreneurial ecosystems function, and how to improve them. By drawing attention of the phenomena to the specifics of place it highlights a particular perspective and reveals the array of approaches being adopted in various places. Some things may be common but other things may differ. In the next section we explore the research policy implications of this place-based approach to entrepreneurial ecosystems.

## 6 Conclusion: A Research Agenda

Throughout the set of works contributed to this volume, the multi-level nature of the entrepreneurial ecosystem becomes clearly apparent. Individuals, firms, industry sectors and regions are inter-related in the study of an entrepreneurial ecosystem. This multi-level conceptualization in some way responds to calls for further research that adopts a multi-level approach (Terjesen et al. 2013) but in other ways raises the same concerns as exists with any multi-level analysis. Concerns such as the complexity in chains of causation, embedded relationships that are structurally non-linear and fallacies that appear when moving between analysis of groups and the specifics of individuals (Diez Roux 2003) can be problematic. Multi-level modelling approaches may not easily account for the heterogeneity of cases but may be useful for understanding the bounds and norms of relationships and are therefore encouraged.

There is an apparent disparity among the objective outcomes of an entrepreneurial ecosystem also evident across the chapters. While some authors incorporate knowledge and technology as integral to the study of an entrepreneurial ecosystem others do not. While for some the account of the entrepreneurial ecosystem ends with firm survival and growth others are concerned with structural change. This suggests more theoretical work is still required to work through when particular viewpoints of the entrepreneurial ecosystem are best applied and under what conditions any analysis of an entrepreneurial ecosystem should vary the elements of inclusion.

Taking this disparity one step further brings into question the definition of an entrepreneurial ecosystem, and more fundamentally the ontological question of what an entrepreneurial ecosystem really is. The view expressed at the opening of this chapter that the definition of entrepreneurial ecosystems includes (entrepreneurial) agency and (human made) context that harbour humanly devised constraints and structures for human interaction that combine to shape the presence and form of important elements such as capital, labour and knowledge for entrepreneurship holds across the chapters. But even so, the actual conceptualization of the various studies of an entrepreneurial ecosystem varies with respect to scope, application, outputs and outcomes. This does support the idea that a systems

view is about recognizing patterns relevant to place but it also challenges the idea that there may be a universal portrayal that can be applied in cross-national or cross-regional studies. Therefore we are left with a serious question for further research that asks: How can knowledge on entrepreneurial ecosystems best be accumulated, and what should be prioritized?

Across the contributed studies there is reference to issues of leadership, governance, resources, and institutions but none clearly bring out how these particular aspects are accommodated, coordinated and arranged within an entrepreneurial ecosystem. Therefore the need for greater understanding of how these elements within an entrepreneurial ecosystem emerge and intervene in the development of entrepreneurial ecosystems is work still remaining to be done.

For policy-makers the relevance of this volume draws attention to the place specific needs of entrepreneurial ecosystems. Within the enclosed chapters various approaches to managing the transitions and defining transformations are discussed in more detail than outlined in this opening chapter. At a place specific level the history, culture, resources and institutional effects need to be taken into account. Entrepreneurship and value creation from entrepreneurial activity rests with the actions and agency of individuals. The job of the policy-maker in these circumstances is to identify the interventions that will encourage individuals into more purposeful actions to achieve place-based objectives. While a universal view of an entrepreneurial ecosystem may not be a feasible project, that is not to suggest that aggregated and multi-level 'systems' studies do not have a place. Indeed these will help to identify normal practices and potentially assist in identifying variation within datasets and provide insights into the causes of variation. Universally we already are aware that human capital, finance, labour, access to technology etc. are important. When they are important, under what conditions, what priorities exist among the options for developing elements and how may influence in the various systems best be exerted, are questions that the entrepreneurial ecosystems approach to understanding a regional economy are yet to discover.

## References

- Acs Z, Autio E, Szerb L (2014) National systems of entrepreneurship; measurement issues and policy implications. *Res Policy* 43(3):465–622. <https://doi.org/10.2139/ssrn.2008160>
- Acs Z, Stam E, Audretsch D, O'Connor A (2017) The lineages of the entrepreneurial ecosystem approach. *Small Bus Econ* 49(1):1–10. <https://doi.org/10.1007/s11187-017-9864-8>
- Adner R (2012) *The wide lens: a new strategy for innovation*. Penguin/Portfolio, New York
- Adner R, Kapoor R (2010) Value creation in innovation ecosystems: how the structure of technological interdependence affects firm performance in new technology generations. *Strateg Manag J* 31(3):306–333
- Alvarez SA, Young SL, Woolley JL (2015) Opportunities and institutions: a co-creation story of the king crab industry. *J Bus Ventur* 30(1):95–112
- Audretsch DB (2015) *Everything in its place: entrepreneurship and the strategic management of cities, regions, and states*. Oxford University Press, New York

- Audretsch DB, Belitski M (2016) Entrepreneurial ecosystems in cities: establishing the framework conditions. *J Technol Transf.* <https://doi.org/10.1007/s10961-016-9473-8>
- Baumol WJ (1993) Entrepreneurship, management, and the structure of payoffs. MIT Press. <https://doi.org/10.1080/08109029508629199>
- Becattini G (1990) The Marshallian industrial district as a socio-economic notion. In: Pyke F, Becattini G, Sengenberger W (eds) *Industrial districts and inter-firm co-operation in Italy*. International Institute for Labour Studies, Geneva
- Braczyk HJ, Cooke PN, Heidenreich M (1998) *Regional innovation systems: the role of governance in a globalized world*. Routledge, London
- Cabrera D, Colosic L, Lobdell C (2008) Systems thinking. *Eval Program Plann* 31:299–310
- Chinta R, Sussan F (2017) A triple-helix ecosystem for entrepreneurship: a case review. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume
- Cooke P (2001) Regional innovation systems, clusters, and the knowledge economy. *Ind Corp Change* 10:945–974
- Cooke P, Gomez Uranga M, Etzebarria G (1997) Regional innovation systems: institutional and organizational dimensions. *Res Policy* 26:475–491
- Daniel L, Medlin C, O'Connor A, Statsenko L, Vnuk R, Hancock G (2017). Deconstructing the entrepreneurial ecosystem concept. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume
- Davidsson P (2005) *Researching entrepreneurship*. Springer, New York
- Delgado M, Porter ME, Stern S (2010) Clusters and entrepreneurship. *J Econ Geogr* 10(4): 495–518
- Diez Roux AV (2003) Potentialities and limitations of multilevel analysis in public health and epidemiology. In: Courgeau Daniel (ed) *Methodology and epistemology of multilevel analysis: approaches from different social sciences*. Springer, The Netherlands, pp 93–119
- Etzkowitz H, Leydesdorff L (2000) The dynamics of innovation: from national systems and “mode 2” to a triple helix of university–industry–government relations. *Res Policy* 29(2):109–123
- Etzkowitz H, Webster A, Gebhardt C, Terra BRC (2000) The future of the university and the university of the future: evolution of ivory tower to entrepreneurial paradigm. *Res Policy* 29 (2):313–330
- Feld B (2012) *Start-up communities*. Wiley, Hoboken
- Feldman MP (2014) The character of innovative places: entrepreneurial strategy, economic development, and prosperity. *Small Bus Econ* 43:9–20
- Feldman M, Zoller TD (2012) Dealmakers in place: social capital connections in regional entrepreneurial economies. *Reg Stud* 46:23–37
- Freeman C (1987) *Technology and economic performance: lessons from Japan*. Pinter, London
- Freeman JH, Audia PG (2006) Community ecology and the sociology of organizations. *Annu Rev Sociol*:145–169
- Frenken K, Cefis E, Stam E (2015) Industrial dynamics and clusters: a survey. *Reg Stud* 49(1): 10–27
- Fuentelsaz L, Maicas JP, Mata P (2017) Institutional dynamism in entrepreneurial ecosystems. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume
- Harrison B (1992) Industrial districts: old wine in new bottles? *Reg Stud* 26(5):469–483
- Henry N, Pinch S (2000) Spatialising knowledge: placing the knowledge community of Motor Sport Valley. *Geoforum* 31:191–208
- Herrmann B, Gauthier J, Holtschke D, Berman R, Marmer M (2015) *The global startup ecosystem ranking 2015: technical report for compass*
- Iansiti M, Levien R (2004) *The keystone advantage: what the new dynamics of business ecosystems mean for strategy, innovation, and sustainability*. Harvard Business Press, Cambridge
- Iansiti M, Levien R (2006) Strategy as ecology. *Harvard Bus Rev* 82:68–81
- Isenberg D (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev*:41–50

- Johannisson B, Alexanderson O, Nowicki K, Senneseth K (1994) Beyond anarchy and organization: entrepreneurs in contextual networks. *Entrepreneurship Reg Dev* 6(4):329–356
- Krugman P (1991) *Geography and trade*. MIT Press, Cambridge
- Lassalle P, Johnston A (2017) Where are the spiders? Proximities and access to the entrepreneurial ecosystem: the case of polish migrant entrepreneurs in Glasgow. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume
- Lazerson MH, Lorenzoni G (1999) The firms that feed industrial districts: a return to the Italian source. *Ind Corp Change* 8(2):235–266
- Leydesdorff L, Etzkowitz H (1998) The triple helix as a model for innovation studies. *Science Publ Policy* 25(3):195–203
- Lundvall B (1992) *National systems of innovation. towards a theory of innovation and interactive learning*. Anthem Press, London
- Malecki EJ (1997) Entrepreneurs, networks, and economic development: a review of recent research. In: Katz JA (ed) *Advances in entrepreneurship, firm emergence, and growth*, vol 3. JAI Press, Greenwich, pp 57–118
- Markusen A (1996) Sticky places in slippery space: typology of industrial districts. *Econ Geogr* 72(3):293–313
- Marshall A (1920) *Principles of economics*. MacMillan and Co., London
- McKenzie BM, Sud M (2009) Prolegomena to a new ecological perspective in entrepreneurship. *Acad Entrepreneurship J* 15(1):43–60
- Moore J (1993) Predators and prey: a new ecology of competition. *Harvard Bus Rev*:76–86
- Nambisan S, Baron RA (2013) Entrepreneurship in innovation ecosystems: entrepreneurs' self-regulatory processes and their implications for new venture success. *Entrepreneurship Theor Pract* 37(5):1071–1097
- Neck HM, Meyer GD, Cohen B, Corbett AC (2004) An entrepreneurial system view of new venture creation. *J Small Bus Manage* 42(2):190–208
- O'Connor A, Reed G (2017) Deconstructing the entrepreneurial ecosystem concept. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume
- OECD (2013) *Entrepreneurial ecosystems and growth-oriented entrepreneurship workshop* (The Hague). <http://www.oecd.org/cfe/leed/entrepreneurialecosystemsandgrowth-orientedentrepreneurshipworkshop-netherlands.htm>
- Piore MJ, Sabel CF (1984) *The second industrial divide: possibilities for prosperity*. Basic Books, New York
- Porter ME (1990) *The competitive advantage of nations*. Macmillan, London
- Porter ME (1998) *On competition*. Harvard Business School Press, Boston
- Raven R, Schot J, Berkhout F (2012) Space and scale in socio-technical transitions. *Environ Innov Societal Transitions* 4:63–78
- Rocha HO (2004) Entrepreneurship and development: the role of clusters. *Small Bus Econ* 23(5):363–400
- Rocha HO, Sternberg R (2005) Entrepreneurship: the role of clusters theoretical perspectives and empirical evidence from Germany. *Small Bus Econ* 24(3):267–292
- Schumpeter J (1934) *The theory of economic development*. Harvard University Press, Cambridge
- Spigel B (2017) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theor Practice* 41(1):49–72
- Stam E (2014) The Dutch entrepreneurial ecosystem. Available at SSRN:<http://dx.doi.org/10.2139/ssrn.2473475>
- Stam E (2015) Entrepreneurial ecosystems and regional policy; A sympathetic critique. *Eur Plann Stud* 23(9):1759–1769
- Stam E (2017) Measuring entrepreneurial ecosystems. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) *Entrepreneurial ecosystems: place-based transformations and transitions*, this volume

- Stam E, Spigel B (2018). Entrepreneurial ecosystems. In: Blackburn R, De Clercq D, Heinonen J, Wang Z (eds) (2017) Handbook for entrepreneurship and small business. SAGE, London
- Stangler D, Bell-Masterson J (2015) Measuring an entrepreneurial ecosystem. Kauffman foundation research series on city, metro, and regional entrepreneurship
- Sternberg R (2007) Entrepreneurship, proximity and regional innovation systems. *Tijdschrift voor Economische en Sociale Geografie* 98(5):652–666
- Sussan F, Sloboda B, Hall R (2017) Is there a path from sin city to tech city? The case for Las Vegas. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) Entrepreneurial ecosystems: place-based transformations and transitions, this volume
- Terjesen S, Hessels J, Li D (2013) Comparative international entrepreneurship: a review and research agenda. *J Manag* 42(1):299–344. <https://doi.org/10.1177/0149206313486259>
- Villiers Scheepers MJ, Mealy E, Clements M, Lawrence A (2017). Regional entrepreneurship ecosystems support: South East Queensland as case study Concept. In: O'Connor A, Stam E, Sussan F, Audretsch D (eds) Entrepreneurial ecosystems: place-based transformations and transitions, this volume
- Ylinenpää H (2009) Entrepreneurship and innovation systems: towards a development of the ERIS/IRIS concept. *Eur Plan Stud* 17(8):1153–1170
- Zahra SA, Nambisan S (2011) Entrepreneurship in global innovation ecosystems. *AMS review* 1 (1):4–17

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# Deconstructing the Entrepreneurial Ecosystem Concept

Lisa Daniel, Christopher J. Medlin, Allan O'Connor ,  
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**Abstract** With innovation and entrepreneurship lauded as important contributors to economic futures, there is a pressing need to unravel the complexities of entrepreneurial ecosystems as a context for cultivating new businesses initiatives. This chapter reports on the deconstruction and analysis of the entrepreneurial ecosystem concept, through a hermeneutic reflection catalysed by a symposium of international scholars. We apply the theoretical fields of business networks and systems theory within our reflective method. This reflective comparison reveals parallels and divergences as well as consistencies and contrasts between these two fields and the concept of entrepreneurial ecosystems. The analysis revealed that the concepts of place and dynamics are specific to entrepreneurial ecosystems and so provide a path for guiding research and policy investigations.

**Keywords** Entrepreneurship · Ecosystems · Business network  
Systems · Hermeneutic phenomenology

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# 1 Introduction

With innovation and entrepreneurship in focus as a means for delivering sustainable economic futures, there is a need for researchers and policy makers to unravel the complexities of entrepreneurial ecosystems. Innovation and entrepreneurship rely on processes of human collective initiative, which implies contexts where resources and activities are brought together to cultivate artefacts as manifestations of innovation and entrepreneurship, for example as the emergence of new technologies, novel combinations of resources or creation of new business ventures. Over recent years the concept of the entrepreneurial ecosystem has evolved in some way to represent this phenomenon of synergistic action and elements seeking towards an innovation or an initiative within a context. Understanding the characteristics and attributes of the entrepreneurial ecosystem concept, in framing adaptive milieus for innovation and entrepreneurship, is the challenge discussed in this paper. The notion of the entrepreneurial ecosystem as a new concept is the subject of much debate by network researchers, systems theorists, cluster researchers, economic geographers and others questioning the significance and legitimacy of the notion.

This chapter reports on the conceptual deconstruction and an analysis of the entrepreneurial ecosystem concept undertaken through the hermeneutic reflections of the authors. This follows discussions and presentations by a group of international scholars brought together in a two-day symposium exploring 'The Entrepreneurial Ecosystem' in Adelaide, Australia. The paper is informed by the reflective interpretations and critical appraisals of the authors as a group of researchers and participant-observers in the intellectual discussions and presentations at the symposium. A significant, novel and timely contribution to academic literature and to the nomenclature of policy makers is offered here by drawing attention to key elements as well as highlighting consistencies, convergences and contrasts of entrepreneurial ecosystems with other theories of business development, start-up, entrepreneurship and innovation. The motivation for the analysis and critique of these conceptual positions, inform a process of conceptual deconstruction of entrepreneurial ecosystems that arose because of significant parallels with the other more established entrepreneurship and business development theories. In this paper two established approaches, business networks and systems thinking, provide the basis from which to compare and contrast the entrepreneurial ecosystem concept. Insights from these theoretical perspectives in conjunction with scrutiny and exploration of a current understanding of the entrepreneurial ecosystem concept revealed some significant cross-overs, parallels and differences in approaches and in what constitutes empirical evidence. The key conceptual themes underlying the discussion include; context, content, process dynamics, purpose and place.

This paper is structured in four parts. First, we present the two key informing theoretical fields. Second, a brief overview of the symposium, as the source of intellectual dialogue for subsequent hermeneutic analysis and interpretation follows. Third, the results of that reflective critique are presented. Fourth, a final discussion to synthesize and highlight key conceptual dimensions of the

entrepreneurial ecosystem as the significant characteristics for future theory development and policy advances.

## 2 Informing Theories

In the world of business, the term ‘ecosystem’ was coined in an article by Moore (1993) who argued that businesses do not evolve as isolated entities but rather develop through relationships and interaction with suppliers, financiers and customers. This definition highlights the interdependencies of entities as part of a business ecosystem, as well as their purposive nature. These characteristics are logical and substantial, and there are strong parallels with the characteristics of business networks and systems to which we turn as a foundation to further explore entrepreneurial ecosystems. This is in line with recent pleas for more use of network and systems approaches to the study of entrepreneurial ecosystems (Alvedalen and Boschma 2017; Stam 2017—this book).

For the purposes of this chapter and because of their specific focus on change and dynamics, business networks and systems thinking are the two theoretical perspectives which will briefly be presented to expose potential contributions and present constructive frameworks for developing the emerging theoretical concept of entrepreneurial ecosystems. Business networks and systems thinking have both been explored extensively as significant contexts supporting regional and business development, innovation, start-ups and entrepreneurial ventures (Huggins 2000; Lechner and Dowling 2003; Isenberg 2010; Kanter 2012; Munksgaard and Medlin 2014a, b). The question here is how do these theoretical viewpoints contribute to the discussion and foundations of a dynamic approach to the entrepreneurial ecosystem?

At this point it is pertinent to say that the consistencies across these and other start-up and business development frameworks inherently involve levels of analysis issues, such that (a) participating entities (individuals or businesses, and other organizations), (b) connect with each other or others (agencies and consumers) to achieve, (c) mutually beneficial but not necessarily agreed goals. In doing so, they source and share (d) resources which they transform through their, (e) dynamic processes, interactions and activities. In addition, (f) contexts are important because they can influence and impact the entities and their connections as well as the availability of resources and the purpose of various interactions and alternatively, (g) the entities can and do also influence and affect their surroundings and contexts so that new opportunities are created. Thus, multiple stakeholders and entities, as actors with varying degrees of interests and differing intentions, negotiate and interact within a particular context or understood space. There is no contention that details and nomenclature can be debated and it is appropriate to acknowledge up-front that no resolution or reconciliation of the conceptual debate is sought. The ontological distinctions likely preclude a full resolution. Rather, this paper offers an exposé of the differences, consideration of similarities and seeks to reify the arguments as constructive contributions to the paradigmatic evolution of entrepreneurial ecosystems.

## 2.1 *Business networks*

While there are other network conceptions that prioritize the nature of relationships between individuals, for instance the social network perspectives of Granovetter (1985) and Burt (2005), our research is focused on the dynamics of change that is inherent in business networks. The business network model emerged in the mid-1970s from a program of research focused on the functioning of business markets at the University of Uppsala, which spread to other research organizations mainly in Europe (Håkansson and Snehota 1989). The business network concept was the subject of multiple approaches but the focus was on the characteristics, structure and development of ‘organically evolved’ networks (Möller and Svahn 2003: 211, citing Moller and Halinen 1999). This approach became known as the Industrial Network Approach studied by the Industrial Marketing and Purchasing (IMP) Group which emphasized the long-term and evolving character of such networks (Möller and Svahn 2003; Möller et al. 2005). Since that time the IMP Group has provided conceptual models and empirical studies designed to improve understanding of the nature of relationships and the interrelatedness of businesses (Håkansson 1982; Håkansson and Johanson 1992; Håkansson and Snehota 1995). Concepts such as interdependence, trust, adaptation, investment, and mutuality have been explored in these studies, as have business relationships in various studies within different empirical settings. These studies of business networks suggest that multiple interactions, interdependence and connections between diverse participants together with linked processes occur within a networked context. In sum, the multiple, interdependent and connectedness represents a complexity that is similar to the view of Isenberg (2010: 43) who suggests “(t)he entrepreneurship ecosystem consists of a set of individual elements—such as leadership, culture, capital markets, and open-minded customers—that combine in complex ways.” Implied within that definition are the distinct and different interests and intentions to create value by various actors, and open is the question of how the elements are combined or connected.

Methodologies for examining business networks have spanned qualitative and quantitative approaches and involved case studies, interviews and surveys generating a wealth of empirical insights, from which the IMP Group produced the ‘Actors-Resources-Activities’ (ARA) model depicting business-to-business markets as interwoven networks of actors, resources and activities (Håkansson and Johanson 1992). In this model each interaction in the network is conceptualized as being composed of *actors* (firms, individuals, or groups) who perform *activities* using directly or indirectly managed *resources*, and explored through business *relationships* with other actors (Sousa 2010). Thus, the business relationship is laden with value from a number of perspectives, being either the nature of actors involved (content and intention), the available resources or factors of production they utilize (context) or the activities they undertake (process). Links between activities, ties between resources and bonds between actors constitute three dimensions of interactions contributing to the dynamic processes within business networks.

Each business relationship is unique, dynamic and connected to other relationships contributing to a network where the boundary is obscure but the interactions are a key. The network has no boundaries and the firm boundaries are permeable in different ways and are continuously changing. “Relationships have value for their participants beyond the immediate transactions that take place within them” (Ford and Håkansson 2006: 250–251). Rather than separate transactions, the relationships are continuous over time and are characterized by a complex and evolving set of interdependencies. In this way a ‘web of interactive relations’ is formed (Håkansson and Snehota 1989: 190–191). The complexity of multiple interactions that constitute business networks means no one actor or firm can manage the network but everyone can influence it through their exchanges, expectations and contributions.

The dynamics of an organic network arise through changes emerging among the interactions of each business relationship. Firms adapt within their business relationships (Hallén et al. 1991) and each relationship within the network spreads or absorbs changes to some degree (Easton and Lundgren 1992). Firms also proactively adjust their position within the network by changing their business relationships (Harrison et al. 2010; Aaboen et al. 2013; Medlin and Törnroos 2015). Evident in the dynamic business network view is a pragmatic approach to theory development, with the key concept of interaction involving subjective interpretation, joint action, interdependencies, relativity, and time (Ford and Håkansson 2006). This *insiders’ view* of change, purpose, interests and intentions within the network strongly contrasts with our next theoretical field.

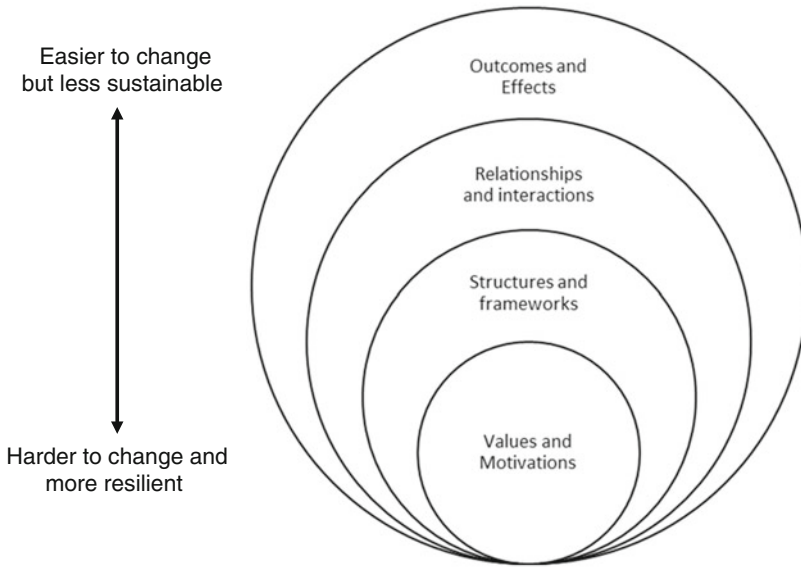
## 2.2 *Systems thinking*

Systems thinking views parts (elements—individuals, groups, businesses) and their relationships and interactions as a complex dynamic interconnected whole. That whole is recognized by its boundary, which is arbitrary and relative to the focus of investigation or discussion, because in reality systems are connected to and are part of other systems, just as networks are connected to and part of other networks. A boundary is an important integral system property, providing characteristics and purpose to the system, as well as informing processes and filtering inputs and outputs. Without boundaries there is no system or systems. Open systems thinking is grounded in the premise that the system responds to its environment—which must deliver it inputs (e.g. resources) and accept its outputs (Ashmos and Huber 1987)—and as such emphasizes the permeability of boundaries (Peery 1975). Further, the system is a set of inter-connected parts which affect and are affected by other elements in the system, both directly and indirectly, such that a change in one part of the system will influence changes in the rest of the system often in subtle or unexpected ways (non-linearity). Changes within systems arise through the continual adjustments of elements as they respond and adapt to each other through their internal and external connections (feedback). Thus, a system is considered to have properties greater than the sum of its parts (Sherwood 2002; Meadows and Wright

2008). Self-organization, or “autogenesis,” is the natural result of complex adjustments and feedback between system elements, where order arises spontaneously and inherently, as a dynamic and evolving equilibrium, because parts are partially, not fully connected (Anderson 1999).

Characteristics of complex systems include a large number of (1) elements (often diverse -complexity) which (2) combine and adjust in either routine or unpredictable ways (dynamic processes arise through inter-dependencies, non-linearities, tipping points, critical junctures) such that the nature of their connectedness is key to the evolution of the system. (3) Dynamics emerge from those evolving patterns and trends, resulting in a need for adaptations and adjustments, so introducing novelty as the system self-organizes in response to change. (4) Strange attractors are random, distinct events which emerge from within the system. Strange attractors can catalyse change and anchor the actions of entities around novel events providing zones of renewal and adaption which keep the system poised at the edge of chaos and thus stimulated, motivated and changing (Marion 1999; Dimitrov and Woong 2000; Connell 2001; Gilstrap 2005). Additionally, the communication that occurs between parts is (5) feedback, and additionally, (6) hierarchy is recognized in the order which emerges from interactions. The various activities in a system, from lower level of aggregation (dyads, groups) up to collective actions, occur through the connections between different sub-systems and collectively support (7) the purpose of the complex dynamic whole as well as the unique objectives of the parts (Stacey 1995; Anderson 1999). Consequently, change in a systems view is strongly theorized as dynamic patterns and novel perturbations of multiple parts, and their interactions and inter-dependencies evolving on the edge of chaos (Marion 1999). Boundaries are important in conceptualizing the elements which make up a whole system.

Applying systems thinking to social and organizational considerations means understanding how the dynamic whole actually works. This is important because the premise is that for a sustainable systemic change to occur system level drivers need to be identified and applied. One issue is the hierarchical structure of social systems. Thinking of the embedded dimensions of social systems helps delineate key levels for analysis, see Fig. 1. Drivers of system level change result in broad shifts in actions among entities rather than just discrete, local effects which eventually become subsumed and changed by systemic influences. The system at any time is informed fundamentally through (1) participant mental models (e.g. values, motivations, expectations and beliefs) which are expressed and must act through the (2) existing structures and systems (i.e. frameworks which establish how things are done e.g. social, regulatory, institutional, or community standards, conventions, frameworks and policies etc.). Those governance systems, organizational and community structures influence (3) the patterns of exchanges, opportunities and nature of relationships which ultimately (4) inform and are manifest in the events, activities, artefacts and outcomes evident in the experienced reality of those involved. The following illustration reveals the relationship of those various levels of influence in a complex social system.



**Fig. 1** A systems thinking view of interacting dimensions in social systems (adapted from Bosch et al. 2013)

### 3 Methods

The rich detail for the argument presented in this paper is derived from the reflective critique by the authors, achieved through a process of participant observation after a two-day symposium on the entrepreneurial ecosystem. The authors, who hosted the symposium, engaged two note-takers during the symposium, who with the permission of participants recorded the various academic approaches and conceptual perspectives discussed and presented. This occurred alongside the co-authors participating in the symposium, who themselves were simultaneously making reflective interpretations of the content and perspectives presented.

The authors are part of an emerging research group interested in the ‘Strategic Innovation of Place’ and who jointly initiated the symposium concept. The Entrepreneurial Ecosystem symposium was held in Adelaide in June, 2016 and involved nineteen researchers from around the world who held research interests in the entrepreneurial ecosystem from a variety of perspectives, including entrepreneurship, innovation, regional development, economic geography, knowledge, gender studies, business networks, business development, and strategic management. The two days of discussion, presentations and debate yielded a tapestry of rich in-depth perspectives about theory, concepts, practice, policy and empirical research into entrepreneurial ecosystems. The conversations and dialogue attended to and noted by the authors, as reflexive participant observers, was then interrogated and interpreted, consistent with a Socratic dialogue approach

(Skordoulis and Dawson 2007). The subsequent discussions, critiques and reflections, the extrapolations and insights distilled were next collated to inform the results and discussion for this paper. This experiential engagement and critical reflection is consistent with a hermeneutic phenomenology epistemology whereby the experienced reality of the participants was both considered and interpreted to facilitate intellectual extension and insight (Kafle 2011).

While acknowledging the intellectually catalytic contributions of symposium participants, this research and the insights it presents is the product of the reflective practice of the contributing authors and does not necessarily represent the views of the other symposium participants. Naturally, the data carries the bias of the observers which is to some extent ameliorated by the number of participant observers but not completely. In theory building research, bias is inescapable but acknowledged. Whetten (1989) described the issues of credibility in the theory building venture as pertaining to the explication of the underlying assumptions. Logic replaces data as the basis of evaluation and the logic of the theory and the underlying assumptions both need full explanation to satisfy the issue of credibility. The objective of this research is to compare and contrast between network and system thinking and provide a basis from which to inform theory development about entrepreneurial ecosystems. The analysis contained in this research explicates and opens for critique the underlying assumptions, as perceived through the of participant observers.

Leximancer software was utilised to identify and support the investigation of the key concepts. The notes from the authors and assistants at the symposium were codified and then assessed using Leximancer analysis to produce a broad conceptual framework (Smith and Humphreys 2006). The Leximancer software develops a multi-dimensional array of hierarchically ordered concepts based on the co-occurrence between high-frequency words in two sentence passes through the document set. Computer coding of textual data produces an objective conceptual framework based upon quantitative criteria, so providing a strong foundation for qualitative interpretation (Munksgaard and Medlin 2014a, b). An assumption made in using Leximancer is that each word has the same meaning regardless of the speaker's purpose or theoretical meaning in use. While this assumption avoids bias in constructing the framework it then is incumbent on the researcher(s) to interpret meaning(s) in accordance with the data collected. To ensure clarity of the input data the notes and transcriptions were examined to eliminate conventionally used words by adding them to the stop words list. Furthermore, over-connected indiscriminate concepts such as 'social capital', 'business networks' or 'complex systems' were substituted by single concepts in order to avoid distortion caused by negating the associations between less connected concepts by over-connected ones (Leximancer 2011). Leximancer hierarchically groups the concepts, depending on the weight of inter-concept connections. An issue here is that the interconnected concepts are held in a multi-dimensional database, which cannot be easily interpreted. Two solutions, the first provided by Leximancer and the second by the researcher are performed. First, Leximancer codes the degree of concept connection from high to low where warm red colours indicate higher dominance of the concepts and cooler blue tones

indicate less prevalence (Leximancer 2011). Next, the concepts and their groupings or themes are projected onto a two-dimensional display to allow analysis by a researcher (Smith 2007).

Since projection of a multi-dimensional array of concepts onto a two dimension plane to form a map necessarily results in a different representation for each occurrence, the researcher must perform the operation multiple times. The researcher then selects the map that is most commonly presented by Leximancer's projection to a two dimension representation. Leximancer automatically assigns the most frequently occurring concept as the theme name. The concentration or spread of concepts in a map is a researcher's decision. The researcher can choose a highly concentrated view with only one or two major themes through to a dispersed view, when each theme may represent a single concept. Ultimately the most representative map is decided by the researcher, based on working back and forth between a theoretical view and the many projections undertaken. In our research we found two useful representations. Our first representation provides a high order view, where concepts are concentrated into fewer mapped themes. The second representation, based on a theoretical decision that agency is an important concern, provides a view where the actor concept is brought into focus as a theme. Thus the different Leximancer figures presented in this research allow analysis at two levels of concept concentration.

## 4 Consistencies, Contrasts and Contributions

The hierarchy of concepts prepared by Leximancer, as presented in Table 1 below, reflects evidence of the key entrepreneurial ecosystem themes: entrepreneurship and place with the supporting economic, geographic and social dimensions of ecosystems (cf. Spigel 2017). The dominance of the ecosystem theme displays that the term operated as a boundary object, that is one which allows researchers to seemingly discuss the same concept but in different ways (Star and Griesemer 1989).

The hierarchy of themes also reveals place was considered to be conceptually closely linked to the idea of an entrepreneurial ecosystem. The strong ordering of ecosystem and place in Table 1 suggests that each concept relies on the other for its definition. What is intriguing is that the system concept is not found in Table 1. The likely reason is that ecosystem subsumes the system concept. Network alternatively was a clearly identifiable concept, as were the concepts of dynamic and relationships. The network concept was found within the ecosystem theme in both projections. Interestingly concepts ranked fourth to twelfth are all focused in some way on the dynamics theme. The policy concept, ranked at 22nd in Table 1, deserves attention because governments have been attending to how entrepreneurial ecosystems might further economic development.

Figure 2 presents the first Leximancer projection displaying a highly concentrated interpretation of concepts linked into fewer themes. The most prevalent

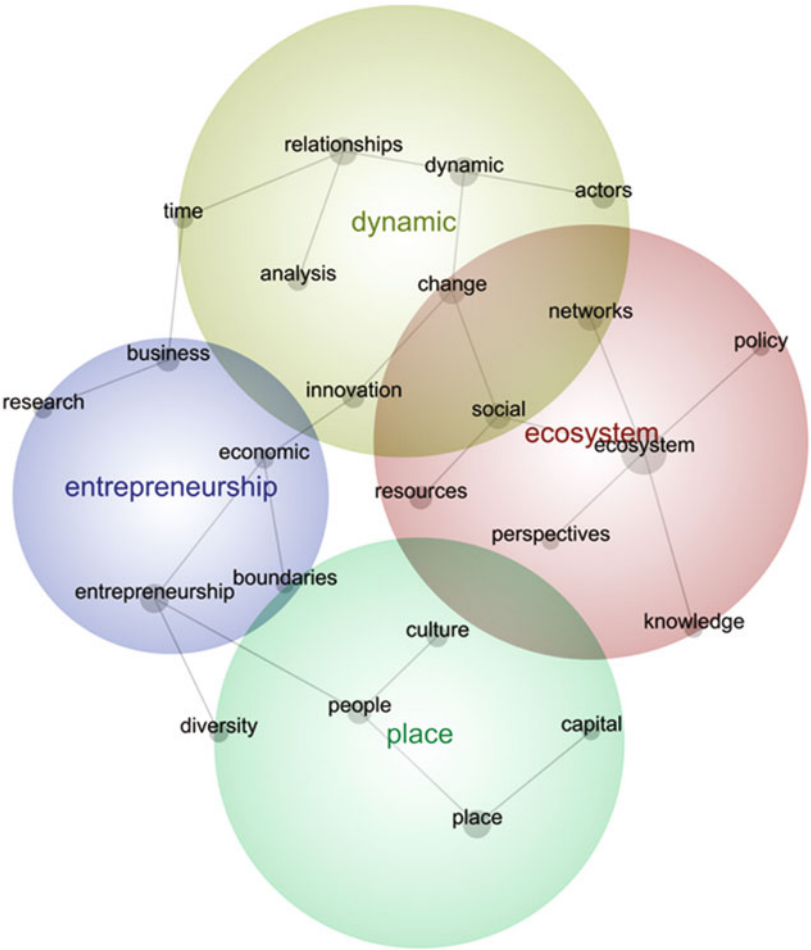
**Table 1** The Leximancer count and percentage of concepts discussed at the strategic innovation of place symposium

| Concept          | Count | Percentage of occurrence (%) |
|------------------|-------|------------------------------|
| Ecosystem        | 67    | 100                          |
| Place            | 39    | 58                           |
| Entrepreneurship | 29    | 43                           |
| Dynamic          | 22    | 33                           |
| Relationships    | 22    | 33                           |
| Actors           | 18    | 27                           |
| Change           | 17    | 25                           |
| Networks         | 17    | 25                           |
| People           | 17    | 25                           |
| Business         | 16    | 24                           |
| Social           | 15    | 22                           |
| Time             | 11    | 16                           |
| Research         | 11    | 16                           |
| Resources        | 10    | 15                           |
| Innovation       | 10    | 15                           |
| Culture          | 10    | 15                           |
| Knowledge        | 10    | 15                           |
| Analysis         | 9     | 13                           |
| Diversity        | 9     | 13                           |
| Economic         | 8     | 12                           |
| Capital          | 8     | 12                           |
| Policy           | 7     | 10                           |
| Perspectives     | 6     | 09                           |
| Boundaries       | 5     | 07                           |

themes evident in the entrepreneurial ecosystem discussion were according to an ordinal ranking: (i) ecosystems, (ii) place, (iii) entrepreneurship, and (iv) dynamics.

Interesting in Fig. 2 is how the first and second ranked themes of ecosystem and place are connected to each other by the themes dynamic and entrepreneurship. The doing themes (i.e. dynamic and entrepreneurship) connect the object themes (ecosystem and place). Looking inside each theme, entrepreneurship captured the concepts of research, business, economic and boundaries, while place captured capital, people, culture and diversity (see Fig. 2). Also interesting are the long chains of concepts connecting themes, for example ecosystem is connected to place by the concept chain: ecosystem, social, change, innovation, economic, entrepreneurship, people and place. Another chain of concepts connects the entrepreneurship theme to itself via concepts within the dynamic theme: entrepreneurship, research, business, time, relationships, dynamic, change, innovation, economic, and entrepreneurship. Clearly dynamics and relationships have a high value in the way the symposium participants conceptualized innovation and entrepreneurship.

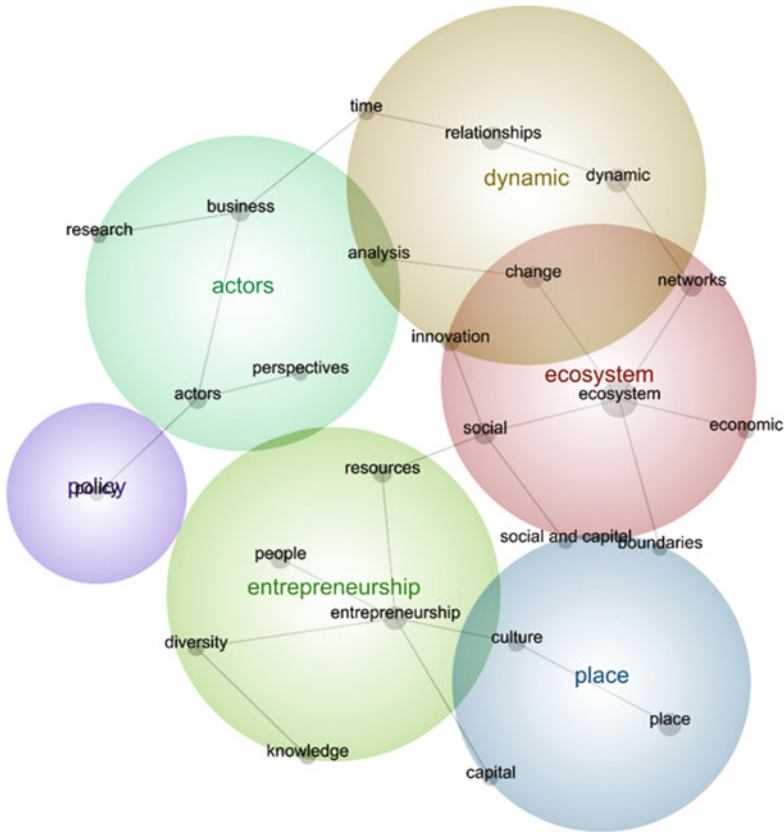
In Fig. 3 a larger number of themes are projected from the multi-dimensional hierarchy of concept connections. Important in selecting this projection is the



**Fig. 2** The most prevalent concepts from discussions of strategic innovation of place

bringing forth of the actors theme, so that questions concerning agency are brought into focus. Interesting is that this projection also brought into focus the policy theme. Policy is connected to entrepreneurship through the themes of actors, dynamic and ecosystem. According to a single view of the symposium participants the dynamics of actors in an ecosystem appears as the key to developing entrepreneurship.

In Fig. 3 the theme actors captured the concepts of business, research, and perspectives, which in Fig. 2 are dispersed within the themes of entrepreneurship, dynamic and ecosystem. In Fig. 2 the concept of actors is neither related to business and research nor to perspectives. Interestingly, in both projections, business and research concepts hold together. Our interpretation is that agency within an entrepreneurial ecosystem is located in the activities of business and research.



**Fig. 3** A less concentrated view of related concepts from discussions of strategic innovation of place

Turning now to the chain of concepts connecting the policy and entrepreneurship themes we find: policy, actors, business, time, relationships, dynamic, networks, ecosystem, social, resources, and entrepreneurship. The series of concepts suggests policies should support business actors in a network of business relationships that forms an ecosystem of resources and social connections. Since policies must be bounded in some way the concept of an entrepreneurial ecosystem seems to make a logical concept suitable for influencing a social system. The close connection of place to the entrepreneurship theme offers a means to define a boundary for an entrepreneurial ecosystem.

Although limited by diversity, there was rich and descriptive discussion across the two-day symposium about theories, context, challenges and paradigms which informed a more robust understanding of the concept of entrepreneurial ecosystems. The strong debate illustrated a pre-paradigmatic field (Nicholls 2010), with conceptual linkages into the research domains of entrepreneurship, business networks,

systems, clusters, economic geography and regional development to name a few. Thus, the term has yet to achieve a recognized level of consensus and legitimacy among 'normal' academic science (Nicholls 2010). Following Kuhn (1962) a pre-paradigmatic state is recognized in a field when a concept lacks an established and agreed (a) epistemology (framework of understanding) and (b) rules (methods and approaches to research) (cited in Nicholls 2010; p. 613). This was the basis of discussions and presentations at the symposium as those present sought to extend understanding and explore the concept to consider if there was sufficient paradigmatic substance for advancing the field.

Turning now to consistencies and tensions, it was evident from the notes and reflections among the participant observers that there was consistency across a variety of issues about entrepreneurial ecosystems and, to the extent that there was consensus, it revolved around there being no apparent formula for creating, sustaining or examining them. There were evident and observable clear analytical tensions in considering the multiple academic perspectives, temporal dynamics, nebulous boundaries, and definitional variance across levels of analysis and nomenclature. There were clear consistencies and considerable cross-overs in concepts like entities, resources, purpose, connections, and levels of analysis; indicating a basis for further exploration and development of an acceptable paradigmatic basis, but again there were also continuing differences and distinctions in how these terms were to be understood. However, some opportunities were found for conceptual refinement as the tension between theoretical domains highlighted that place, the permeability and morphing of boundaries and the dynamics of interactions, change and time were important considerations in thinking about entrepreneurial ecosystems.

The issue of purpose raised a distinction and point of delineation for entrepreneurial ecosystems as reflections revealed a clear difference in its role of contextual reframing to that of networks or systems. In business networks purpose was viewed as originating from actors inside their exchanges and interactions while alternatively from a systems perspective a collective purpose is defined by the whole, frequently in contradiction to that articulated or intended by the elements i.e. entities and actors involved. The dynamics of networks and systems are emergent in nature emanating from the changing basis of relationships. For entrepreneurial ecosystems the notion of purpose was primarily concerned with place, complex collectives of venture building actors and the resources and connections that constrain or enable them in pursuit of independent albeit entrepreneurial purposes. Important to the entrepreneurial ecosystem purpose and dynamic is the declaration for growth in the number and size of entrepreneurial ventures and consequently socioeconomic development of place. Different to the emergence view of dynamics in networks and systems, the dynamics in the entrepreneurial ecosystem are suggested to be more purposeful and prescribed by political and collective interventions to achieve place-based outcomes by changes in entrepreneurial behaviour. Notable in Fig. 3, is that policy and actors are connected to place and entrepreneurs by the dynamics of an ecosystem. This suggests that the purpose and intention of a political actor is at least one way to focus entrepreneurial ecosystem analysis.

**Table 2** Conceptual consistencies and cross-overs apparent in the symposium discussions and entrepreneurial ecosystem reflections

| Concept                                                             | Describes                   | Consistent themes                                                                                                                           |
|---------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Entities                                                            | <b>Content</b>              | Stakeholders, participants, SME's, actors, entrepreneurs, microbusinesses, elements                                                         |
| Resources                                                           | <b>Content</b>              | People, finance, assets, inputs, factors, place-based advantage                                                                             |
| Levels of analysis                                                  | <b>Context</b>              | Sub-systems, hierarchy, value chain, networks, businesses, business relationships, leaders, local, regional, global, community, individuals |
| Connections                                                         | <b>Process/<br/>context</b> | Inter-dependence, relational capital, interactions, relationships, within and with external entities/resources                              |
| Boundary (systems) actor (networks)                                 | <b>Purpose</b>              | Agenda, goal, objectives, intent, place-based purpose                                                                                       |
| Environmental context (social, geographic, economic, institutional) | <b>Place/<br/>context</b>   | Network, region, community, place-specific resources                                                                                        |
| Time                                                                | <b>Dynamics</b>             | Change, evolution, development, progress, transitions, adaption, advancement                                                                |

Table 2 presents the conceptual consistencies, parallels, and cross-overs apparent in the entrepreneurial ecosystem discussions.

The notion of place was considered as a space where humans construct meaning i.e. ‘place’ is relative to specific human interactions, while space, exists as a more general human construction (Törnroos et al. 2016). Unlike systems thinking where the integrity of the whole is integral to the concept and relative to the sum and synergy of the parts, albeit informed by context, for entrepreneurial ecosystems the integrity is ultimately about a place (actual or virtual).

Place is interpretive, relative, historic, opaque and needs to be shared to be known (Pred 1984). Place-based knowledge was highlighted as a valuable resource for entrepreneurial ecosystems where actors or entities are embedded in the informing context of their interactions. Characteristics of place (environmental/inherent assets) and nature of the place (process dynamics) were considered to be key influencing factors to the arrangement of entities, availability of resources and the nature of processes and interactions therein. Consistent with Storper (1995), capital, culture, people, identity and history were all mentioned relative to place (see Fig. 2), as features and factors that matter. Nonetheless place alone doesn’t indicate potential for entrepreneurial ecosystems but rather place-specific elements and influences may facilitate entities seeking wider connections for potential advantage, be that value, resilience, survival or sustainability. Thus, place appears integral to the concept of entrepreneurial ecosystems.

Consistent with the fundamentally human oriented nature of entrepreneurial ecosystems, business network thinking suggests a need to look closely at the

permeable nature of boundaries and their continually changing nature. Relationships that connect actors and places are thus a key conduit of change for entrepreneurial ecosystems. The permeability of entrepreneurial ecosystem boundaries is also consistent with systems thinking, suggesting again that this deserves further research attention.

The temporal nature of entrepreneurial ecosystems was a topic which emerged to demarcate dynamics as a contributing concept in entrepreneurial ecosystem analysis. The nature of progress, time and change in relationships, the potential for re-emergence over time, the recycling of resources and actors, and the transitions of activities as relationships change over time all framed the notion of dynamics. These are all concepts strongly present in recent elaborations of dynamic business networks (Chou and Zolkiewski 2012; Möller 2013; Medlin and Törnroos 2015; Olaru and Purchase 2015; Andersen and Medlin 2016; Fonfara et al. 2016). Stability, evolution and change were understood to co-exist in entrepreneurial ecosystems as consistent with the earlier view for business networks. Systems thinking also brought to the forefront the dynamic aspect of entrepreneurial ecosystems, although conceptually change is considered differently with respect to what is changed and how. Still, the emergence and development of businesses was seen as inherent to the fundamental purpose of entrepreneurial ecosystems and an objective dynamic.

Recognition of the implicit requirement for the emergence and evolution of entrepreneurial ventures in an entrepreneurial ecosystem gave rise to discussion about resources and context in supporting that process. In particular, the availability and appropriateness of resources for the start-ups and initiatives emerging from that context led to conversation about entrepreneurial ecosystems as incubators or accelerators. The importance of diversity and flexibility of connections to facilitate fortuitous serendipity, opportunities and circumstance for development as well as strategic positioning of new firms for optimum growth was also discussed.

## 5 Implications for Research

There are implications here for research as the debate about what constitutes an entrepreneurial ecosystem continues to develop through the confounding and confusing comparisons of other business paradigms. The term ‘ecosystem’ in a business context was coined by Moore (1993) in an article where he argued that businesses do not evolve as isolated entities but develop through interaction with suppliers, financiers and customers. This idea was earlier elaborated in business network thinking (Håkansson and Snehota 1989). This chapter shows that at issue is how to treat boundaries and intent and/or purpose. Stam (2015) proposed entrepreneurial ecosystems are “a set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship”. These perspectives are consistent with the views shared in the discussions reported above.

A comparison of key elements of entrepreneurial ecosystem is presented with business network and systems thinking in Table 3.

According to Moore (2006) ecosystems are intentional communities of economic actors whose individual business activities share in some large measure the fate of the whole community. Moore proposed in a (non-spatial) business ecosystem that companies coevolve capabilities as they work cooperatively and competitively to support new products, satisfy customer needs, and eventually incorporate the next round of innovations (Moore 1993: 76). Similarly Zahra and Nambisan (2012: 220) suggest that a business ecosystem is a group of companies—and other entities including individuals—that interacts and shares a set of dependencies. Indeed, Moore (2006: 53) proposes the ecosystem is a ‘collaboration’ to ‘create a system of complementary capabilities’ and to support innovation. In doing so he implies a clear suggestion of self-organization and indeed some degree of self-determination. These definitions are consistent with the issues discussed from a variety of perspectives and so reveal a clear foundation for entrepreneurial ecosystem understanding, where the complementarity of capabilities within a permeable boundary (i.e. place) is sought by actors adopting collective political intentions. The boundary thus defines the ways capabilities are judged to be complementary and how resources should be applied to enhance future capabilities. The concept of political intentions is never separate from actor self-interest, and so the tension between entrepreneurial ecosystem collectives and actors notes a continuing dynamic.

However, in contrast to networks where the relationship is based on interaction, and systems where the whole is the focus, for entrepreneurial ecosystems the intent is relative to the interactive community supporting the various ventures. Thus, entrepreneurial ecosystems are a collective of diverse actors supported by strong institutional stakeholders. A contribution of this paper is the suggestion that place and place specific resources, especially those of place specific stakeholders, are relevant and fundamental to an entrepreneurial ecosystem’s purpose, which contrasts with the other discussed theoretical framings used in our analysis. A further implication was reflected in discussions on measurement which often focuses on elements rather than connections, relationships or process. For entrepreneurial ecosystems it may be better to examine what elements are involved and the processes that influence them given that simple relocation of elements, relationships, resources or activities will not produce an identical entrepreneurial ecosystem.

The major contribution to entrepreneurial ecosystem theory of this exploration of network and systems thinking is to highlight the objective nature of entrepreneurial ecosystem dynamics which draws attention to a potential driving force, the politically persuaded actor. In system theory, dynamics are found and emerge inherently from the inter-connected nature of the parts and whole, whereas the evolution of business networks is emerging and more intentional. By contrast the entrepreneurial ecosystem subsumes and depends on actor interaction in a context of network and/or system and therefore to achieve an objective, intervention to drive change needs to be targeted to particular and interdependent actors/levels. By contrast in a systems approach these levels integrate fundamental values and motivations, structures

**Table 3** Table of common frames of reference and how they exist in the different theoretical perspectives

| Concept                    | Context       | Content     | Process                                    | Place             | Purpose                             | Dynamics                                |
|----------------------------|---------------|-------------|--------------------------------------------|-------------------|-------------------------------------|-----------------------------------------|
| Business networks          | Relationships | Multi-level | Relational interaction                     | Nebulous          | Actor/firm interests and intentions | Emergent and changing relationships     |
| Systems thinking           | Relative      | Collective  | Dynamic                                    | Integral/relative | Systemic                            | Evolutionary interactions               |
| Entrepreneurial ecosystems | Communities   | Multi-level | Serendipity, opportunity, and circumstance | Relative          | Enterprise/place development        | Organic evolution; policy interventions |

and frameworks, relationships and interactions to bring about outcomes and effects. Therefore interventions must make sense given a specific system understanding. Alternatively in a network approach, the key old and new relationships are the means for intervention selected by actors relevant to specific business objectives. By contrast entrepreneurial ecosystems may themselves be considered an intervention instrument whereby a particular actor can mediate to achieve a collective outcome as motivated by individual actor objectives. Hence politically motivated actors (in terms of influencing collective action) can act to define the boundary of place and seek to create compatible and complementary capabilities across the encompassed public and private entities to alter the trajectory of the ecosystem.

Finally, while business networks are about relationships connected in an unending net, and systems are about holism, perhaps research on entrepreneurial ecosystems may consider participant relations more generally, within the context of inter-dependencies around connections between actors and differences of intent or purposes. Our analysis of the data also suggests that intent and purpose of actors in entrepreneurial ecosystems may be embedded in or related to the concepts of dynamics and change. Notwithstanding the limitations of a purposive and focused data collection and the particulars of the methodology, our intent was to suggest new areas of investigation as a theory building exercise. Many questions still remain. Understanding entrepreneurial ecosystem's as unique, diverse, complex phenomenon, grounded in place as well as emergent, dynamic, evolving and contingent on context and relative perspectives, opens new research opportunities. What has been presented here is a small start.

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## References

- Aaboen L, Dubois A, Lind F (2013) Strategizing as networking for new ventures. *Ind Mark Manage* 42(7):1033–1041
- Andersen PH, Medlin CJ (2016) Transient commitments and dynamic business networking. *Ind Mark Manage* 58:11–19
- Anderson P (1999) Complexity theory and organization science. *Organ Sci* 10(3):216–232
- Ashmos DP, Huber GP (1987) The systems paradigm in organization theory: correcting the record and suggesting the future. *Acad Manag Rev* 12(4):607–621
- Bosch O, Nguyen N, Sun D (2013) Addressing the critical need for new ways of thinking. In: *Managing complex issues in a socially responsible way. Business systems review* 2, doi:<https://doi.org/10.7350/BSR.B-2013>
- Burt RS (2005) *Brokerage and closure: an introduction to social capital*. Oxford University Press, Oxford
- Chou H-H, Zolkiewski J (2012) Decoding network dynamics. *Ind Mark Manage* 41(2):247–258

- Connell DJ (2001) Re-formulating community as a complex social system. Qualifying Exam Paper. U. o. Guelph
- Dimitrov V, Woong R (2000) Making sense of social complexity through strange attractors. In: Altmann G, Lamp J, Love P et al (eds) International conference on systems thinking in management, ICSTM2000, Geelong, Australia. CEUR workshop proceedings 72
- Easton G, Lundgren A (1992) Changes in industrial networks as flow-through nodes. In: Axelsson B, Easton G (eds) Industrial networks: a new view of reality. London, Routledge
- Fonfara K, Ratajczak-Mrozek M, Leszczyński G (2016) Change and transformation of networks and relationships—fundamental aspects of business reality. *Ind Mark Manage* 58:187–189
- Ford D, Håkansson H (2006) The idea of business interaction. *IMP J* 1(1):4–27
- Gilstrap D (2005) Strange attractors and human interaction: leading complex organizations through the use of metaphors. *Complicity Int J Complex Educ* 2
- Granovetter M (1985) Economic action and social structure: the problem of embeddedness. *Am J Soc* 91(3):481–510
- Håkansson H (1982) International marketing and purchasing of industrial goods: an interaction approach. Wiley, Chichester
- Håkansson H, Johanson J (1992) A model of industrial networks. In: Axelsson B, Easton G (eds) industrial networks: a new view of reality. New York, Routledge, pp 28–34
- Håkansson H, Snehota I (1989) No business is an island: the network concept of business strategy. *Scand J Manage* 5(3):187–200
- Håkansson H, Snehota I (1995) Developing relationships in business networks. International Thomson Press, Boston
- Hallén L, Johanson J, Seyed-Mohamed N (1991) Interfirm adaptation in business relationships. *J Mark* 55(2):29–37
- Harrison D, Holmen E, Pedersen A-C (2010) How companies strategies deliberately in networks using strategic initiatives. *Ind Mark Manage* 39(6):947–955
- Huggins R (2000) The success and failure of policy-implemented inter-firm network initiatives: motivations, processes and structure. *Entrepreneurship Reg Dev* 12(2):111–135
- Isenberg DJ (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):40–50
- Kafle NP (2011) Hermeneutic phenomenological research method simplified. *Bodhi Interdisc J* 5:181–200
- Kanter RM (2012) Enriching the ecosystem. *Harvard Bus Rev* 90(3):140–147
- Kuhn T (1962) The structure of scientific revolutions. The University of Chicago Press, Chicago
- Lechner C, Dowling M (2003) Firm networks: external relationships as sources for the growth and competitiveness of entrepreneurial firms. *Entrepreneurship Reg Dev* 15(1):1–26
- Leximancer (2011) Leximancer manual. Version 4
- Marion R (1999) The edge of organization: chaos and complexity theories of formal social systems. Sage Publications, Thousand Oaks
- Meadows DH, Wright D (2008) Thinking in systems: a primer. Chelsea Green Publishing
- Medlin CJ, Törnroos JA (2015) Exploring and exploiting network relationships to commercialize technology: a biofuel case. *Ind Mark Manage* 49:42–52
- Möller K (2013) Theory map of business marketing: relationships and networks perspectives. *Ind Mark Manage* 42(3):324–335
- Möller K, Rajala A, Svahn S (2005) Strategic business nets—their type and Manage. *J Bus Res* 58(9):1274–1284
- Möller K, Svahn S (2003) Managing strategic nets: A capability perspective. *Mark Theory* 3(1):209–234
- Moore J (1993) Predators and prey: a new ecology of competition. *Harvard Bus Rev*:76–86
- Moore J (2006) Business ecosystems and the view from the firm. *Antitrust Bull* 51(1):31–75
- Munksgaard K, Medlin CJ (2014a) Self- and collective-interests: using formal network activities for developing firms' business. *Ind Mark Manage* 43(4):613–621
- Munksgaard KB, Medlin CJ (2014b) Self-and collective-interests: using formal network activities for developing firms' business. *Ind Mark Manage* 43(4):613–621

- Nicholls A (2010) The legitimacy of social entrepreneurship: reflexive isomorphism in a pre-paradigmatic. *Entrepreneurship Theory Pract* 34(4):611–633
- Olaru D, Purchase S (2015) Innovation network trajectories: the role of time and history. *J Bus Ind Mark* 30(3/4):342–353
- Peery NSJ (1975) General systems theory approaches to organizations: some problems in application. *J Manage Stud* 12(3):266–275
- Pred A (1984) Places a historically contingent process: structuration and the time-geography of becoming places. *Ann Assoc Am Geogr* 74(2):279–297
- Sherwood D (2002) Seeing the forest for the trees: a managers guide to applying systems thinking. Nicholas Brealey Publishing, London
- Skordoulis R, Dawson P (2007) Reflective decisions: the use of socratic dialogue in managing organizational change. *Manage Decis* 45(6):991–1007
- Smith A (2007) Leximancer manual. Version 2.23
- Smith AE, Humphreys MS (2006) Evaluation of unsupervised semantic mapping of natural language with Leximancer concept mapping. *Behav Res Methods* 38(2):262–279
- Sousa F (2010) Markets-as-networks theory: a review. In: *Organizational culture, business-to-business relationships, and interfirm networks. Advances in business marketing and purchasing*. Emerald Group Publishing Ltd., pp 405–454
- Spigel B (2017) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory Pract* 41:49–72
- Stacey RD (1995) The science of complexity: an alternative perspective for strategic management. *Strateg Manage J* 16(6):477–495
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plan Stud* 23(9):1–11
- Star SL, Griesemer JR (1989) Institutional ecology, translations' and boundary objects: amateurs and professionals in Berkeley's museum of vertebrate zoology, 1907–39. *Soc Stud Sci* 19(3):387–420
- Storper M (1995) The resurgence of regional economies, ten years later: the region as a nexus of untraded interdependencies. *Eur Urban Reg Stud* 2(3):191–221
- Törnroos J.-Å, Halinen A, Medlin CJ (2016) Dimensions of space in business network research. *Ind Mark Manage*
- Whetten DA (1989) What constitutes a theoretical contribution? *Acad Manage Rev* 14(4):490–495
- Zahra S, Nambisan S (2012) Entrepreneurship and strategic thinking in business ecosystems. *Bus Horiz* 55:219–229

## Author Biographies

**Lisa Daniel** brings a cross-disciplinary background to the UoA Business School with a Bachelor of Science with a first class Honours in the field of molecular plant pathology (UQ) and PhD in Technology and Innovation Management (UQ). Her current focus on agri-business eco-system research takes Lisa the full circle from researching the attributes and contributions of food crops to society, to researching agribusiness, regional communities and the people involved in the processes. This research circle demonstrates a profound connection agriculture, rural communities and food production issues.

Lisa's research focuses on the relational, institutional and system dynamics that support innovation, knowledge and sustainability in businesses and organisations. Her emphasis is systems research seeks understanding of the complexities and synergies in organisations and regional

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**Dr. Christopher J. Medlin** is an Associate Professor of business-to-business marketing at the Adelaide University Business School, Australia. He researches the dynamics of business relationships and industrial networks. He has a deep interest in the role of time and timing in business interactions. He has published in the journal of Industrial Marketing Management, the Journal of Business-to-Business Marketing, the Journal of Business & Industrial Marketing, the Journal of Business Research and the International Journal of Project Management.

**Dr. Allan O'Connor** is the Academic Director Innovation and Entrepreneurship Government Relations at the Entrepreneurship, Commercialisation and Innovation Centre at the University of Adelaide. Allan combines his extensive industry experience in confronting the growth issues of small and medium enterprises and business start-up to inform and guide his teaching and research. Allan's main teaching interests are in the assessment of business opportunities, entrepreneurial strategy and developing research skills. Since 2012 he has co-authored the leading Asia Pacific text, 'Entrepreneurship: Theory, Process and Practice', with Professors Howard Frederick and Donald Kuratko. His research examines the intersection between entrepreneurship, innovation and socioeconomic development which has led to the development of the Australian Cluster Observatory and an in-depth study of entrepreneurial ecosystems. His portfolio of over fifty research publications in international peer reviewed journals, books and research reports is testament to the significance of Allan's research agenda to his field. In application, his research is designed to inform policy-makers, regional development agencies and the practicing entrepreneur with respect to creating and managing the resources necessary to foster and develop innovation and entrepreneurship in response to the strategic challenges of economic change.

**Larissa Statsenko, PhD** is a project manager of the Research Consortium at the Institute of Mining and Energy Resources, the University of Adelaide, and Assistant Professor at the Rudny Industrial Institute, Kazakhstan. Her research interests are in Complex Systems and Supply Chain Management. She has an academic, research and consultancy experience, and works on university—industry—government collaboration projects related to mining industry development in Australia and Kazakhstan.

**Rowena Vnuk** is a current PhD candidate with the University of Adelaide. Building upon her previous business experience and recent Masters of Applied Innovation and Entrepreneurship, Rowena is researching further how intermediary organisations intermediate industry transitions under dynamic conditions and how entrepreneurial endeavours by these organisations may influence industry specific innovation systems. Her particular interests in regional development originates from her knowledge and experience in regional development practice in the field of town planning and regional transport systems; in turn, these resources are now being extended by her research on entrepreneurship and innovation at the industry sector level.

Rowena enjoys research, particularly when she uses her questioning abilities to achieve sound conclusions, based on sound qualitative or mixed methods research methodologies. Past experience in community consultation and project management within transport planning fields enable Rowena to achieve high standards in research and project management. She looks forward to completing and publishing research papers to high level research journals as time progresses.

**Gary Hancock** is lecturer and undergraduate Bachelor of Innovation and Entrepreneurship program director for the Entrepreneurship, Commercialisation, and Innovation Centre in The University of Adelaide. Involved in the university's e-Challenge competition, Commercialisation Accelerator Program, and ThincLab, the University's business incubator Gary, has taught and developed courses in the areas of small business management, new venture creation, business consulting, and entrepreneurship (both face to face and on-line education modes).

Gary has worked in a large telecommunications organisation both in a technical and senior management role. He spent over sixteen years starting, operating, and harvesting growth-oriented ventures in franchise and non-franchise environments across the entertainment, telecommunications, business consulting, vocational education, and child care industries. Gary also assisted in developing the Innovation Foundation Program, a joint project between the Department of Transport and Regional Services, the City of Salisbury, and UniSA.

Gary provides mentor support and advice to young entrepreneurs via the South Australian Young Entrepreneur Scheme (SAYES). He has developed and run courses about planning and starting a business for Business SA (the South Australian Chamber of Commerce) and for a local not-for-profit education organisation (WEA) and is the past President and Chair of the Board of the WEA.

Gary's research is in the field of early stage business financing and start-up. He is interested in the phenomenon of how and why people engage in new venture formation.

# Institutional Dynamism in Entrepreneurial Ecosystems

Lucio Fuentelsaz, Juan P. Maicas and Pedro Mata

**Abstract** Entrepreneurs are involved in entrepreneurial ecosystems. These ecosystems incorporate institutions that interact with entrepreneurs in different ways depending on the stage of the business life cycle the entrepreneur is in. By incorporating institutional theory and business life cycle theory to the analysis of ecosystems, we propose which kind of institution will answer the entrepreneurial question addressing who is there to help their venture depending on the stage of the life cycle it is in. This paper seeks to provide guidance on the institutional support available to increase a venture's chance of success and improve the efficiency of the resource base of ecosystems.

**Keywords** Entrepreneurial ecosystems · Institutions · Business life cycle  
High growth firms

**JEL Codes** L26 · M21 · M38

## 1 Introduction

Entrepreneurs—or those who seek to generate value through the creation or expansion of economic activity and through identifying and exploiting new products, processes, or markets (Ahmad and Seymour 2008)—devise, create, and develop their projects in a social and geographical context. This context is known in

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the literature as the entrepreneurial ecosystem (Brown and Mason 2017; Spigel 2017; Stam 2015). The literature has revealed that the personality of the entrepreneur and the nature of the activity are not the only reasons behind the success of a venture; there is an ecosystem that helps these ventures to prosper (Autio and Thomas 2013).

In this paper, we argue that there is a series of institutions in these ecosystems that facilitate the relationship between the different actors. These institutions include the law and established procedures that normally generate *tangible* agencies (usually labelled as formal institutions), and also the values, beliefs, and culture of a given community (informal institutions; North 1990). Given the relationship between entrepreneurial ecosystems and institutions, we integrate both of these areas of literature to explore the nature of the kinds of institutions that are involved in an entrepreneurial ecosystem. Our first objective is to propose a taxonomy of the institutions that are present in an entrepreneurial ecosystem. We suggest that there are four main types of institutions: (a) institutions linked to venture creation; (b) support organizations; (c) financing; and (d) infrastructure elements, awards, and grants.

Entrepreneurs look for what the ecosystem can provide them and consequently search for the institutions that can give them and their ventures what they are seeking. However, the nature of these institutions may differ depending on the stage the venture is in. This is why in studying the relationship between entrepreneurial ecosystems and institutions, we go one step further and analyse the relationship from a dynamic perspective. We propose that the role of institutions within an entrepreneurial ecosystem is contingent on what stage of the business life cycle the venture is in.

The life cycle theory of a firm is used to show that successful entrepreneurs grow through different stages.<sup>1</sup> This theory suggests that an entrepreneur's activities, attitudes, and requirements change from stage to stage. Consequently, after defining a taxonomy of institutions that operate in an entrepreneurial ecosystem, we put this taxonomy into a dynamic context to show how, depending on the stage an entrepreneur is in and on the typology of the institution that operates in the ecosystem, the entrepreneur will look for and eventually find different kinds of institutions. The purpose of the paper is not to depict how firms' external knowledge or support requirements change as firms grow, but rather to show how the different types of institutions address the specific needs that firms have in the various stages of the firm's growth cycle.

The contributions of this paper are twofold. First, we help to integrate the literature on entrepreneurial ecosystems and institutions with the aim of contributing to the development of the theoretical framework of the former. The idea is that institutional theory may provide a basis to define and categorize the different actors operating in entrepreneurial ecosystems. Institutions play a determining role

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<sup>1</sup>Broadly speaking, the literature identifies five main stages in the business life cycle: birth, growth, maturity, revival, and decline (Miller and Friesen 1984), although the number of stages is not standardized. Other classifications include only three stages: growth, maturity, and decline (Miles et al. 1993) or other numbers depending on different circumstances (Levie and Lichtenstein 2010).

in the entrepreneurial ecosystem as they are there to help the entrepreneur's venture. Better guidance on available institutional support will increase a venture's chance of success and improve the efficiency of the resource base of ecosystems. Second, and more importantly, we analyse entrepreneurial ecosystems from a dynamic perspective, suggesting that each stage of the business life cycle needs a specific treatment to determine the types of institutions that are the most appropriate. It is important to note that the so-called, 'one size fits all' strategies cannot be applied to firms without taking the context into account. On the contrary, it is necessary to consider the specificities of each stage and act accordingly. The link between institutions and the stage of the firm's lifecycle facilitates the identification of the institution that best fits entrepreneurs' interests and may provide them with the necessary support to launch and develop their business.

## 2 Entrepreneurial Ecosystems

Spilling (1996, 91) introduces the concept of entrepreneurial ecosystem as 'the complexity and diversity of actors, roles, and environmental factors that interact to determine the development of entrepreneurship in a region or locality'. An entrepreneurial ecosystem is a set of individual elements including leadership, culture, capital markets, and open-minded consumers that interact in complex ways (Isenberg 2010) and in a specific geographic area (Cohen 2006), thus influencing the formation and final trajectory of the different actors and potentially the whole economy (Spilling 1996; Iansiti and Levien 2004). There is an emerging body of literature in the field of entrepreneurship dedicated to examining the interaction between entrepreneurs, firms, and the context in which they operate (see Chap. 1), that is, to examining entrepreneurial ecosystems.

Although they have common characteristics (Neck et al. 2004; Isenberg 2010), each entrepreneurial ecosystem is different (Mason and Brown 2014). Based on these differences, we should examine the multiple actions conducted by entrepreneurs and their contribution to the dynamism, performance, and long-term transformation of regional economies (Spilling 1996). Therefore, and for the purpose of this paper, it seems plausible to analyse regions with entrepreneurial activity as a system instead of examining the actions of the actors and organizations involved in the region individually and separately. The latter approach is usually tracked in the entrepreneurship literature (Neck et al. 2004). However, it would be a mistake to believe or think that entrepreneurial ecosystems are the domain of only a few highly economically productive places.

Van de Ven (1993) explains that entrepreneurial ecosystems evolve through a set of interdependent elements that interact to generate the creation of new businesses. The existence of entrepreneurial ecosystems makes sense if they facilitate the development of new businesses and ensure their success. Therefore, governments need to create an ecosystem that supports entrepreneurs and facilitates their growth and the creation of new businesses (Isenberg 2010). Additionally, if we focus on a

special type of company such as start-ups, which traditionally experience difficulties with prospering, entrepreneurship ecosystems facilitate their development, especially in the case of ‘high-growth’ companies (Bahrami and Evans 1995). The literature suggests that entrepreneurial ecosystems provide competitive advantages to partners within the ecosystem (Clarysse et al. 2014).

Entrepreneurial ecosystems are often facilitated by external action, especially by the government (Van de Ven 1993). In this sense, entrepreneurial ecosystems are the result of a ‘smart evolution’ (Isenberg 2011), meaning that the market’s invisible hand joins the government’s deliberate hand. According to Isenberg (2010), government authorities can design entrepreneurial ecosystems that later become self-sufficient. At this point, the presence of government agencies would be significantly reduced, although they would not disappear completely. Policy makers seek to stimulate knowledge ecosystems in hotspots of technology with the aim of generating value networks that help participating companies gain a competitive advantage (Clarysse et al. 2014).<sup>2</sup>

Ács et al. (2014) redefine the concept of entrepreneurial ecosystems, creating the new concept of National Systems of Entrepreneurship (NSEs) and providing an approach to characterize them. An NSE is the ‘dynamic, institutionally embedded interaction by individuals between entrepreneurial attitudes, abilities, and aspirations, which drives the allocation of resources through the creation and operation of new ventures’ (17).

The basis for these NSEs is the availability of resources to provide opportunities at the individual level with the purpose of creating new companies, regulated by the specific institutional characteristics of each country. In these NSEs, institutions generate and regulate the activity.

### 3 Elements Within an Entrepreneurial Ecosystem

Having verified the existence of environmental elements that facilitate the work of the entrepreneur, we can now go on to analyse what they are. Reviewing several studies on environmental factors that affect entrepreneurship in different geographical areas, Bruno and Tyebjee (1982) draw up a list of the key factors most commonly cited: the availability of capital, the presence of experienced entrepreneurs, a technically skilled workforce, accessibility to suppliers, access to customers or to new markets, favourable governmental policies, proximity to universities, availability of land or buildings and offices, availability of support services, and attractive living conditions in the area.

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<sup>2</sup>Nonetheless, there are various cases of the lack of effectiveness of public policies. This ineffectiveness is caused by corruption (Estrin et al. 2013), by the way policy is formulated (Arshed et al. 2014), or by the lack of connection between knowledge and business (Clarysse et al. 2014).

Later, in their description of Silicon Valley, Bahrami and Evans (1995) talk about the existence of six elements in the entrepreneurial ecosystem: universities and research institutes, a private investment market, support infrastructure, a base of talent, an entrepreneurial spirit, and a leading group of users. Universities and research institutes not only provide the scientific and technological training, but also generate links between people that facilitate the creation of new companies. A private investment market whose investment profile differs depending on the phase at which the project is located, delivers funds in different rounds of financing, renders management skills that facilitate the sustainability of projects and provides networking, monitoring of the funding, and even guidelines for the entrepreneur. Support infrastructures are necessary for entrepreneurs to be focussed on their projects and not dissipated in peripheral activities; support include law firms, human resources services, and engineering and design firms to develop prototypes, a base of talent, including both local and foreign talent, is critical to launch and sustain the venture. An entrepreneurial spirit is described as a critical although intangible ingredient of the ecosystem that ends up creating a culture of hard and well-done work, and behaves as a source of inspiration that accepts failure as part of the process. Finally, it is important to have access to a leading group of users—early adopters who are eager to incorporate new technologies and products.

Similarly, in their study of the entrepreneurship ecosystem of Boulder County in Colorado, Neck et al. (2004) also find a number of interrelated elements that facilitate, support, and enhance entrepreneurship. These elements are incubators, spin-offs, formal and informal networks, physical infrastructure, and culture. Incubators are defined as the organizations where entrepreneurs were employed before creating their own company (Cooper 1986). Spin-offs are new companies formed by an individual or group of individuals who leave an existing company and start a new one in the same industry (Garvin 1983) or related sectors [this is the version used by Neck et al. (2004)]. Formal networks include universities, government agencies, professional support services, financial sources, access to qualified personnel, or the presence of large companies. Informal networks comprise, among others, friends, family, and co-workers. Physical infrastructure refers to tangible infrastructure components in the county: communications, access to offices, housing, etc. Finally, Neck et al. (2004) mention the existence of a specific culture: one that makes an organization, an industry, or even a nation, unique (Mintzberg et al. 1998).

Isenberg (2010, 2011) identifies 50 components involved in an ecosystem and groups them into six domains: a culture of entrepreneurship, policies that facilitate entrepreneurship and leadership, access to adequate funding, appropriate human resources, a market receptive to innovative products, and support from infrastructures and institutions. A culture of entrepreneurship creates an attitude that rewards innovation, creativity, and experimentation. A suitable legislation enhances entrepreneurship. Access to funding includes microcredit and private and public investment. Human resources provide skilled or unskilled workers. A market receptive to innovative products facilitates the creation of the appropriate distribution

channels and entrepreneurial networks. Support from infrastructures and institutions (incubators, accelerators, etc.) includes legal, economic, and financial support.

3.1 The Entrepreneurial Ecosystem and Institutions

Institutional theory claims that societies have institutions that facilitate market relations (North 1990). These institutions appear because there is a need to generate market rules and regulations. Although institutions limit actors’ behaviours, they facilitate the progress of society.

Institutional theory is being increasingly used in the academic literature, and its application has proven to be helpful to entrepreneurial research. For instance, Bruton et al. (2010) conducted a thorough review of the previous research that had used institutional theory in entrepreneurship and found up to 80 recent articles published in top business journals.

Several classifications of institutions have been presented. However, the taxonomies of North (1990) and Scott (2007) are the most widely accepted. North (1990) distinguishes between formal and informal institutions while Scott (2007) offers the idea of three supportive pillars: regulative, normative, and cognitive. Peng et al. (2009) integrate both these taxonomies and suggest that they are equivalent (see Table 1).

For simplicity, our approach will follow North’s (1990) terminology. North defines institutions as ‘the rules of the game in a society (...) they structure incentives in human exchange, whether political, social or economic. Institutional change shapes the way societies evolve through time and hence is the key to understanding historical change’ (3), and as mentioned distinguishes between formal and informal institutions. The former refer to established laws and procedures while the latter refer to a specific community’s values, beliefs, and culture. Formal and informal factors constitute the rules of the game in a community and affect the birth and the development of new ventures.

The institutional framework influences the decision to become an entrepreneur. For instance, the existence of entrepreneurial activity in a society is inversely

Table 1 Dimensions of Institutions

| Degree of formality (North 1990) | Examples    | Supportive pillars (Scott 2007) |
|----------------------------------|-------------|---------------------------------|
| Formal institutions              | Law         | Regulative (coercive)           |
|                                  | Regulations |                                 |
|                                  | Rules       |                                 |
| Informal institutions            | Norms       | Normative                       |
|                                  | Cultures    | Cognitive                       |
|                                  | Ethics      |                                 |

Source Peng et al. (2009)

proportionate to the size of the government and is significantly related to the absence of corruption (Aidis et al. 2012). Additionally, the chances of becoming an entrepreneur increase in countries that are more entrepreneurship-oriented and have higher levels of trust and education (De Clercq et al. 2013).

As we said, previous literature supports the existence of a number of similar factors or elements within an entrepreneurial ecosystem. In what follows, we introduce institutional theory into the analysis, studying these environmental factors from the point of view of institutions. More precisely, we analyse whether there are different types of institutions involved depending on which of these factors we are considering. Ecosystems consist of formal and informal institutions (Mason and Brown 2014; Stam 2015).

As we have already mentioned, Bahrami and Evans (1995), Neck et al. (2004), and Isenberg (2010, 2011) have reached similar conclusions; that is, that there are six different kind of institutions involved in the entrepreneurial ecosystem. As can be seen in Table 2, these authors agree on the existence of four groups of factors with similar characteristics within what we have called the formal ecosystem: (a) the existence of institutions related to the process of creating the idea, the generation of the concept, the actual launch of the innovative project, and the existence of a market *prepared* to adopt the product; (b) support agencies (i.e., formal networks dedicated to supporting the creation of new businesses); (c) institutions related to the financing of entrepreneurial projects; and (d) other entrepreneurship support infrastructures not included in the rest of the groups. From this similarity in the nature of the institutions, we suggest classifying formal institutions within the entrepreneurial ecosystem into four groups based on those previously cited. We will use the following labels: (a) institutions linked to venture creation; (b) support organizations; (c) financing; and (d) other infrastructure elements, awards, and grants.

Regarding the informal ecosystem, Bahrami and Evans (1995), Neck et al. (2004), and Isenberg (2010, 2011) identify several factors clearly linked to the informal dimension of institutions, and agree on the existence of two kinds of

**Table 2** Elements within an entrepreneurial ecosystem

| Institutional theory  | Bahrami and Evans (1995)               | Neck et al. (2004)      | Isenberg (2010, 2011)                   |
|-----------------------|----------------------------------------|-------------------------|-----------------------------------------|
| Formal Institutions   | Leader users                           | Incubators              | Innovative products                     |
|                       | Universities and research institutions | Formal networks         | Facilitating policies                   |
|                       | Private financing market               | Spin-offs               | Financing                               |
|                       | Support infrastructure                 | Physical infrastructure | Support infrastructure and institutions |
| Informal Institutions | A talent base                          | Informal networks       | Human resources                         |
|                       | Entrepreneurial spirit                 | Culture                 | Culture                                 |

Source Self-made

informal institutions. The first is related to people and their ties and relationships (determined by human resources and networks), and the second is embedded in the culture of the ecosystem.

In this paper, we concentrate on the formal institutions and leave the informal institutions for future research. The reason for this is that informal institutions are more difficult to systematize, strongly depend on people's attitudes and beliefs, and would require an empirical approach that is beyond the scope of this paper.

### ***3.2 Start-Ups and High Growth Firms***

At this point, a distinction should be made between the business ventures that interact within the entrepreneurial ecosystems in terms of their speed of development. We can basically differentiate between high growth firms (HGF) and traditional start-ups.

The Organization for Economic Co-operation and Development (Wennberg 2013) defines HGFs as those 'enterprises with average annualised growth greater than 20% per annum, over a three year period (...). Growth can be measured by the number of employees or by turnover'. Therefore, the main characteristic of an HGF is its rapid growth. This growth can be measured in many ways: as relative sales growth over time, in relative sales in the sector or the country, or in absolute terms as growth in the number of employees or sales.

Another distinguishing feature of HGFs is their innovative capability. A difference between more traditional start-ups and HGFs lies in the importance of innovation (usually measured but not limited, via R&D investment and intellectual property registration) for their growth (Coad and Rao 2008; Stam and Wennberg 2009). Despite the existence of numerous ways of measuring innovation, Wennberg's (2013) scientific literature review shows that there is a proven difference between the use of the diverse forms of innovation (process, market, and product) when it comes to distinguishing between an HGF and a traditional start-up of more moderate growth.

### ***3.3 The Business or Organizational Life Cycle***

The development and progression of the firm as a linear sequential process through a number of stages is generally known as the corporate, business, or organizational life cycle model, or organizational growth model (Miller and Friesen 1984, among others).

Numerous authors (e.g., Adizes 1979; Greiner 1972; Scott 1971 in Miller and Friesen 1984; Lyden 1975) have investigated the fact that corporations go through different sequential stages that are progressive, make them change their activities

and attitude, modify their needs, and transform the company and industry structure. The different growth models share three main propositions (Levie and Lichtenstein 2010):

- (1) Just like a growing organism, an organization goes through differentiated stages;
- (2) These stages are sequential, predetermined, and predictable; and
- (3) There is an implicit sequence of progress that leads organizations from a primitive stage to a differentiated, more developed and mature stage (Van de Ven and Poole 1995).

This similarity with the life cycle of organisms has a direct influence on the name given to the process: the business ‘life cycle’ (Kroegeer 1974; Lippitt and Schmidt 1967).

Organizational literature has traditionally identified five common stages in this corporate life cycle: birth, growth, maturity, revival, and decline (Miller and Friesen 1984). Quinn and Cameron (1983) synthesize nine previous papers, and generate their own model, concluding that there are four phases in the organizational life cycle: (a) entrepreneurial phase; (b) collectivity phase; (c) formalization and control phase; and (d) elaboration of structure stage. Although the labels used were not particularly appealing, their integration of previous models has a lot of merit. Nevertheless, these models were all theoretical and there were no empirical longitudinal studies to support them (Quinn and Cameron 1983; Miller and Friesen 1984). To overcome this problem, Miller and Friesen (1984) analyse data from 36 firms over 161 periods, assigned characteristics to each of the five stages of the corporate life cycle, and have drawn three interesting conclusions:

- (1) These stages have a certain configuration, meaning that there is a pattern of different phenomena with properties not derivable from the summation of their parts;
- (2) Each of the phases are in many ways unique; and
- (3) While the stages of the life cycle are internally coherent and very different from one another, they are by no means connected to each other in any deterministic sequence.

Yet, what is involved in these stages? In the first, or birth phase (Scott 1971 in Miller and Friesen 1984), the new firm is ‘attempting to become a viable entity’ (Miller and Friesen 1984, 1162). Here, entrepreneurs face difficult times; firms are young and have simple and informal structures. In the growth stage (Lyden 1975), the company is starting to grow, their distinctive competence has been established, and there is some kind of market success (Miller and Friesen 1984). Structures are larger, and the business aims to grow to an even larger scale. In the maturity phase, there is more formalization and control (Quinn and Cameron 1983), and bureaucratic structures are established. The fourth stage—revival—is a phase of diversification and expansion of the product market scope (Miller and Friesen 1984) and is normally the last stage of the life cycle for many authors. There is a final stage—decline—that happens when markets dry up and firms consequently decline.

Recent literature on firm growth criticises the prominence and widespread acceptance of this life-cycle view. There is a stream of research that currently refutes the linear growth of firms. Garnsey et al. (2006), for example, determine that the study of the growth of new companies suffers from a lack of conceptual models that can filter and assimilate the evidence that enterprises change over time, and conclude that there is no sequence of linear stages in the growth of firms. Additionally, in their investigation of up to 104 articles about the life cycle theory published between 1962 and 2006, Levie and Lichtenstein (2010) conclude that in the models analysed, there was no general agreement as to what a stage is, the number of existing stages, or on how they relate. Furthermore, in their empirical study of a database of Scottish HGFs, Mason and Brown (2013) show that HGFs do not necessarily follow the standard evolutionary path. In this case, growth is episodic and random and does not follow a sequential pattern.

There should then be a distinction in entrepreneurial ventures between small business start-ups, and the more innovative and high-growth entrepreneurship. Small business start-ups would tend to follow a more classical sequential version of the life cycle model, whereas in HGFs the linear and sequential stage-by-stage growth of firms would not be directly applicable.

Despite the critiques mentioned, there is one aspect of the growth model that has been supported by empirical evidence: company growth shows different stages or configurations at different times in its history (Levie and Lichtenstein 2010). In other words, the existence of the life cycle model may be questioned, but the existence of the different stages is empirically proven. This is the basis of our argument: that there are different institutions ready to assist companies in each of these differentiated stages. Consequently, institutions are dedicated to act in the ecosystems to support the dynamism of various stages of new venture creation and growth, regardless of the order of interaction needed by the entrepreneur or the moment in time such interaction is required. Additionally, it is important to note that the existence of these stages does not imply that all companies will necessarily move over time from one stage to the next in a sequential way. This is of special importance for HGFs that, as we have noted, do not necessarily follow a standard evolutionary path.

Having established the stages of the organizational life cycle, we should bear in mind that even before the organization is born, entrepreneurs have to carry out many activities. This is of special importance in the context of entrepreneurial ecosystems because they are not only dedicated to existing firms, but are also involved in the venture creation process. In other words, when discussing entrepreneurial ecosystems, we should remember that ‘the first success of a firm is its birth’ (van Gelderen et al. 2005, 365), and that there are four additional stages, known as ‘pre-start-up phases’, prior to the actual existence of the firm that should be considered. In these four phases, those that are involved in creating businesses are called ‘nascent entrepreneurs’ (Reynolds and White 1992 in van Gelderen et al. 2005). The ‘pre-start-up phases’ should be added to the phases of the firm growth model when considering entrepreneurial ecosystems, as there are institutions in these ecosystems dedicated to giving support to nascent entrepreneurs.

### ***3.4 The Entrepreneurial Ecosystem and the Business Life Cycle***

To our knowledge, there is no attempt in the literature to link entrepreneurial ecosystems with the different stages of the business life cycle (with the exception of Stam and Bosma 2015). Borrowing from Miller and Friesen's (1984) birth, growth, maturity, revival, and decline stages and from van Gelderen et al. (2005), who incorporate previous research to conclude that there are four pre-start-up phases, we propose the following entrepreneurial phases in order to characterize an entrepreneurial ecosystem: (a) intentional; (b) conceptual; (c) seed; (d) start-up; (e) growth; and (f) maturity. That is, we propose to incorporate the four pre-start-up phases (the last coincides with the birth stage of the business life cycle) with the first three stages of the business life cycle. We consider that entrepreneurial ecosystems are not involved, at least in a significant manner, in attending firms in the last two stages of the business life cycle proposed by Miller and Friesen (1984; i.e., revival and decline).

The first or intentional phase concerns the development of an intention to start an enterprise (Krueger et al. 2000). In the second phase, an entrepreneurial opportunity is recognized, and a business concept is developed. In this conceptual stage, the project is therefore basically an idea or concept. Entrepreneurs usually start by finding an opportunity and begin to seek solutions in the form of a product, service, or both, assessing what outcomes to expect. Entrepreneurs in this phase are usually found in research centres and universities, need very little funding, and are basically seeking initial advice on whether their idea has potential (van Gelderen et al. 2005). As time goes by, ideas incorporate a business concept, and projects enter the seed stage. In this third phase, resources are assembled and the organization is created. There are still no sales, and most of the time is spent on research and development. Entrepreneurs continue seeking advice about the most appropriate business model for the company by exploring new alternatives and investigating the potential viability of their venture. Financing needs are growing but are not excessive and are usually covered with insider finance. Funds are dedicated to initial operating expenses. It is the pre-launch stage of a new venture.

In the final phase of the nascent entrepreneur, the organization starts to interact with the market (van Gelderen et al. 2005) and blends into the first phase of the business life cycle. This start-up phase is usually defined as a temporary stage where the company seeks to achieve a scalable business model. The company has already started selling its products or services, although in a restricted manner. Entrepreneurs begin to see the results of market tests about their products. Financing needs for the business grow significantly, especially to obtain the working capital needed to meet sales growth and to cover marketing expenses to increase customer awareness and initial operating expenses. Moreover, market experience forces entrepreneurs to continually adapt their business models to customer requirements. Consequently, entrepreneurs involved in this phase are looking to grow as fast as possible while they improve their offer and obtain financing for their operations.

In the growth stage, the venture is relatively consolidated and seeks to scale-up its activities. While it may still need some financing for working capital, financing is usually allocated to long-term investments that seek the company's consolidation, its expansion, or the restructuring of its operations and growth in local markets.

In the maturity phase, companies seek their final development. This phase usually begins with international growth (in recent times, internationalization often starts even before this phase) and the acquisition of another company to integrate its operations or markets, and culminates with an IPO, or ultimately the sale of the company and replacement of the original shareholders.

### 3.5 *The Entrepreneurial Ecosystem, Institutions, and the Business Life Cycle*

As we have discussed in previous sections, we can find different institutions within the entrepreneurial ecosystem dedicated to facilitating the different needs of an entrepreneur over time. For example, an established firm that is thinking about an IPO would not consider visiting an incubator to develop its business plan. Similarly, a group of young entrepreneurs considering the initiation of their business venture would not visit a bank to issue corporate bonds.

In this section, we build a grid (Fig. 1) that incorporates all the pieces of the puzzle previously described. We argue that depending on the stage an entrepreneur is in and on the typology of the formal institutions that operate in the ecosystem, the entrepreneur will find different institutions. We will consider which institution has a greater influence on the entrepreneur, depending on the stage the entrepreneur is in. We will now describe in detail the type of institutions we can find.

**Institutions Linked to Venture Creation.** These institutions are mainly found in the first stages of the business life cycle (i.e., when the concept is being generated and the business plan is being developed). Among the institutions dedicated to facilitating the creation and launch of new ventures, we can find the following:

- *Nurseries*, usually publicly owned, are aimed at new projects in the very first stages of their life cycle that need space at low cost. The objective is the growth of the project to its next phase.
- *Incubators*, similar to nurseries, facilitate the creation and development of technological entrepreneurial projects prior to their actual launch onto the market and through to early survival (Peters et al. 2004). They are often located in technology centres and have close relationships with local universities.
- *Accelerators* provide rapid growth to projects already launched, shortening the period needed to transform a company in the start-up stage into a profitable and successful company (Kim and Wagman 2014).

| Type of institution                         | Life cycle stage |                                                                                                                                             |                                                  |                                                                                                                                |                           |                                             |
|---------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------|
|                                             | Intentional      | Conceptual                                                                                                                                  | Seed                                             | Start-up                                                                                                                       | Growth                    | Maturity                                    |
| Institutions linked to venture creation     |                  | Nurseries<br>Incubators<br>Co-working centres<br>Networking areas<br>Networking initiatives                                                 |                                                  | Accelerators                                                                                                                   |                           |                                             |
| Support organizations                       |                  | Universities<br>Professional associations<br>Social agents<br>Business schools<br>Research and technology centres<br>Other support agencies | Employment and local development agents          | Clusters<br>Industry and trade chambers<br>Technological parks                                                                 |                           |                                             |
| Financing                                   |                  | Insider finance<br>Crowdfunding                                                                                                             | Angel financing<br>Financial institutions        | Venture capital<br>Family office<br>Other investment companies<br>Mutual guarantee societies<br>Financial support institutions | Private equity            | Alternative stock markets<br>Stock exchange |
| Infrastructure elements, awards, and grants |                  | Awards                                                                                                                                      | International institutions<br>Other institutions | Support lines<br>International institutions                                                                                    | Internationalisation aids |                                             |

Fig. 1 Institutions and the Business Life Cycle (Source Self-made)

- *Co-working centres* are places where entrepreneurs in the seed and start-up stages share certain common areas, generating synergies between projects in diverse sectors.
- *Networking areas* are places to generate networking to improve business models and companies' commercial activities. Finally,
- *Networking initiatives* are like start-up weekends in which entrepreneurs from different fields and with different abilities and skills work collaboratively to create a start-up, and where networking among entrepreneurs, coaches, mentors, and specialists in different fields facilitates innovation and the development of high-impact ventures.

**Support Organizations.** These institutions facilitate the entrepreneurial process, giving support to entrepreneurs (Ferrary and Granovetter 2009; Bresnahan and Gambardella 2004). They include the following:

- *Universities:* Public and private universities normally have various kinds of advisory services and training in entrepreneurship for projects in the conceptual and seed stages (Swamidass 2013; Levie 2014).
- *Employment and local development agents,* who are workers of local corporations or related to local authorities, and whose main mission is to collaborate in the promotion and implementation of active employment policies related to venture creation.
- *Clusters* are a group of companies, institutions, and universities that share an interest in a particular economic and strategic sector located in the same geographical area. Therefore, clusters within the ecosystem help companies that are entering the growth stage.
- *Professional associations,* which are organizations of employers, are usually grouped by industry and carry out actions aimed at certain groups of entrepreneurs, helping them from the seed to the growth stage.
- *Social agents* (employer associations and unions) are organizations representing the social and collective interests of economic agents. They provide advisory services and training programmes for entrepreneurs and unemployed people normally in the creation stages.
- *Business schools:* Academic institutions that cover the training needs of managers and senior professionals and provide specialized courses aimed at entrepreneurs and business creation.
- *Industry and trade chambers:* Public corporations whose mission is to represent, promote, and defend the general interests of trade and industry and provide services supporting entrepreneurs in the seed, start-up, and growth stages.
- *Research and technology centres* are usually non-profit entities created for the purpose of contributing to the overall benefit of society and improving the competitiveness of enterprises through the generation of technological knowledge in the early stages of the business cycle.
- *Technological parks* (or science and technology parks) are physical spaces and facilities that stimulate and manage knowledge and technology flows between universities and research institutions, companies, and markets. They promote the

creation and growth of innovative companies, generating knowledge, collaborative networks, and technology transfer.

- *Other support agencies*: Those not included in the above categories that also participate in the entrepreneurship ecosystem. Some examples include public institutions dedicated to helping young entrepreneurs, those specialized in specific industries, or municipality- owned agencies dedicated to entrepreneurs.

**Financing Institutions.** As mentioned above, the type of financing that can be achieved by an entrepreneur is also different depending on the stage the venture is in (Berger and Udell 1998).<sup>3</sup> We can find the following:

- *Insider finance*, also known as ‘Friends, Family, and Fools’: These investors support entrepreneurs in their first steps. In these early days, investors have to be found among relatives, friends, or some ‘fools’ who trust entrepreneurs.
- *Angel finance* or *Business angels*: These are investors in the capital stock of the company. The term ‘angels’ refers to the fact that they contribute in the early stages, especially in the start-up phase, with something more than capital (with contacts, experience, management, etc.) and facilitate the expansion of the company to the next phase (Shane and Cable 2002; Mason 2011; Mason and Harrison 1997).
- *Venture capital* are funds of companies that invest in the initial stages of a start-up and participate from the early stages to the growth phase (Carter and van Auken 1994).
- *Private Equity*: These are private equity funds that invest from the growth to the maturity stage. There are four types of private equity funds: (1) growth funds that invest in relatively mature companies that seek financing to restructure their operations or to enter new markets; (2) buyout funds, or funds specialized in acquisition financing (thus, they often need additional leverage); (3) rescue funds specialized in the redirection of a company that is going through problems; and (4) substitution funds that substitute an existing investor in a company (a private investor, a fund, etc.).
- *Alternative stock markets* are dedicated to small cap companies looking to expand. These normally have less liquidity than the stock markets and provide funding, visibility, liquidity, and valuation to SMEs.

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<sup>3</sup>There is a general consensus that depending on the stage of the life cycle they are in, small firms present different capital structures and sources of financing. The most widely accepted view (Gregory et al. 2005) of small-business capital structure is that of Berger and Udell’s (1998) financial growth cycle model. This model explicitly suggests that the optimal capital structure evolves with changes in firm size, age, and the information available. According to this model, very small firms (among which we find start-ups) are financed with insider and angel finance; small firms with a limited track record are financed by venture capital companies; firms of medium size with some track record are also financed with venture capital and medium-term financial institutions; and large firms with a known track record are financed with public equity and long-term financial institutions.

- *Stock exchanges* are regulated markets where securities are traded. Subject to regulation (such as the Securities Market Act), they are supervised by specialized agencies (like the SEC).
- *Family offices*: These are companies that manage part of the equity of private families. They provide investment management and other services (e.g., tax or legal services) to families. When the company manages the assets of more than one family, it is known as a multi-family office. These family offices participate in companies and are present from the seed to the growth stage.
- *Financial institutions*: These are companies that offer financial services to businesses and consumers. Two types of financial institutions can be distinguished: those that give credit (banks, financial institutions, etc.), and those that manage risks (insurance company).<sup>4</sup> Due to the uncertainty and risk of the early stages, these institutions prefer to finance ventures from the start-up stage onwards.
- *Other investment companies* are entities that focus on investing funds, especially in the growth stage, rather than on lending.
- *Mutual guarantee societies*: These societies are business entities whose primary function is to grant guarantees and other collateral to facilitate SME operations management (i.e., to help access to bank financing, import and export guarantees, guarantees in tenders and auctions, etc.).
- *Financial support institutions*: These institutions are governmentally-owned companies that aim to participate in private companies to finance their operations. These institutions have different funds and investment programs depending on the stage companies are in and can participate as investors or by granting loans (usually in the form of equity loans).
- *Crowdfunding*: This is a financing system consisting of many investors pooling funds for a project (Vachelard et al. 2016; Vismara 2016). In crowdfunding, a large number of investors contribute with small amounts of money to facilitate the achievement of a certain objective of the entrepreneur, normally in the seed stage. There are several types of crowdfunding platforms: (1) capital or equity crowdfunding, where investors receive a percentage of the company's stock for their money; (2) reward crowdfunding, where a CD, book, etc., is received in exchange for the money provided, depending on the type of project being financed; (3) pre-purchase crowdfunding, where investors finance the actual production of goods and receive these products in exchange for their investment once the production is completed; (4) donation crowdfunding, where the money is donated for nothing; and (5) loan crowdfunding, where investors lend money to the entrepreneurs.

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<sup>4</sup>In this paper, we will only consider financial institutions that provide specific services to entrepreneurs.

**Infrastructure Elements, Awards, and Grants.** Within these institutions we can find the following (Hsu 2006):

- *Awards* are a term used for institutions that give awards to projects related with entrepreneurship, usually in the early stages.
- *Support lines* are requested and processed by the banks. These are special financing lines that fund investment projects in the early stage.
- *International institutions aids* include EU aids to entrepreneurs in all stages, for example.
- *Internationalization aids* are institutions that give grants and support to promote the internationalization of enterprises.
- *Other institutions* that are not included in the previous categories, like local government aids to increase competitiveness, or subsidies to develop knowledge and technology in a certain area.

Figure 1 shows the organizations representing formal institutions. We should make an additional distinction between the organization and the programs or support services they provide. The reason is that different organizations may provide similar services. For instance, universities that are situated in the conceptual model distinctly to accelerators, incubators, or business schools can offer any or all of the support programs and services these other institutions provide. Similarly, nurseries and incubators offer open spaces to brainstorm, access to networks of business partners, one-on-one or group coaching, access to executive education programs, or support for funding through seed funds, business angels' networks, and so on, just like universities do. Likewise, banks can offer services from early establishment through IPO depending on the banks' market orientation. In conclusion, although formal institutions are very different as organizations, some of the services provided can be more similar.

## 4 Conclusions

This paper has analysed the relationship between entrepreneurial ecosystems and institutions in order to propose a taxonomy that identifies which institutions are more suitable at the different stages of the venture life cycle the firm is in. In recent years, we have witnessed the explosion of a myriad of public and private institutions aimed at stimulating entrepreneurship all around the world. It is true that these institutions may constitute valuable assistance for individuals or groups that decide to start a new venture. However, it is not easy for them to identify the institution that best fits their interests and may provide them with the support necessary to launch their venture. As a consequence, any guidance in this matter will be helpful and will facilitate an adequate assignment of resources.

Our research has identified the different agents that provide support to entrepreneurs within entrepreneurial ecosystems, taking into account that the level of

development of the venture usually determines the value that these agents may provide. With this aim in mind, we depart from the concept of company life cycle. The firm usually evolves through different stages, and we match these stages with the institutions that participate in the entrepreneurial ecosystem. Even if company growth did not follow a linear sequential process, stages would exist, and firms in these stages would still share common characteristics that make them suitable to receive similar aid from formal institutions.

Our taxonomy is valuable both for the individuals that have created (or are in the process of creating) a new venture and for policymakers. From the entrepreneurs' point of view, an adequate knowledge of the whole set of public or private agents that constitute the ecosystem may provide a clear picture of the possibilities available for receiving different types of help (technical or administrative support, financial resources, or even new shareholders that assist in the development and growth of the firm).

From a public policy perspective, the information that we provide in this research is helpful for two reasons. First, a detailed knowledge of the entrepreneurial ecosystem may facilitate policymakers' action, paying special attention and providing resources to these stages less covered by private or other public institutions. Second, it is widely known that public resources are scarce. Thus, it is necessary to decide where to allocate these resources. It is also known that entrepreneurship is not always equally desirable (Baumol 1990), and governments may be more interested in promoting certain kinds of ventures (e.g., quality or high-impact entrepreneurship that is considered to generate higher levels of employment or economic growth).

This paper opens several promising avenues for further research. First, it portrays a theoretical view of how organizations representing formal institutions interact with entrepreneurs within the entrepreneurial ecosystem, and how institutions are dedicated to helping entrepreneurs depending on the stage the venture is in. However, it is not grounded in empirical evidence. Future research should incorporate the empirical component to test how accurate our theoretical predictions are. For example, we may find that only a small proportion of high-tech firms use incubators or accelerators even if they are in the start-up stage, or that firms not only use or interact with universities in their conceptual/seed stages—which is true for university spin-offs.

Second, the life cycle view has been questioned in recent literature. Nonetheless, the existence of the stages as moments in the growth of companies where firms share common characteristics and needs is empirically proven. Therefore, more research is needed on the evolution of firm growth from this non-sequential perspective.

Finally, informal institutions are not included in the framework. Their inclusion could represent the icing on the cake for the link between institutions, ecosystems, and the business stage. In this research, a more sociological approach could be useful.

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## References

- Ács ZJ, Autio E, Szerb L (2014) National systems of entrepreneurship: measurement issues and policy implications. *Res Policy* 43(3):476–494
- Aldizes I (1979) Organizational passages—diagnosing and treating lifecycle problems of organizations. *Org Dyn* 8(1):3–25
- Ahmad N, Seymour R (2008) Defining entrepreneurial activity: definitions supporting frameworks for data collection. OECD Statistics Working Paper
- Aidis R, Estrin S, Mickiewicz TM (2012) Size matters: entrepreneurial entry and government. *Small Bus Econ* 39(1):119–139
- Arshed N, Carter S, Mason C (2014) The ineffectiveness of entrepreneurship policy: is policy formulation to blame? *Small Bus Econ* 43(3):639–659
- Autio E, Thomas L (2013) Innovation ecosystems: implications for innovation management. In: Dodgson M, Phillips N, Gann DM (eds) *The Oxford handbook of innovation management*. Oxford University Press, Oxford
- Bahrami H, Evans S (1995) Flexible re-cycling and high-technology entrepreneurship. *Calif Manag Rev* 37:3
- Baumol WJ (1990) Entrepreneurship: productive, unproductive, and destructive. *J Polit Econ* 98(5):893–921
- Berger AN, Udell GF (1998) The economics of small business finance: the roles of private equity and debt markets in the financial growth cycle. *J Bank Finance* 22(6):613–673
- Bresnahan T, Gambardella A (eds) (2004) *Building high-tech clusters: Silicon Valley and beyond*. Cambridge University Press
- Brown R, Mason C (2017) Looking inside the spiky bits: a critical review and conceptualisation of entrepreneurial ecosystems. *Small Bus Econ*. doi:<https://doi.org/10.1007/s1187-017-9869-3>
- Bruno AV, Tyebjee TT (1982) The environment for entrepreneurship. In: *Encyclopedia of entrepreneurship*. Prentice-Hall, Inc., Englewood Cliffs, pp 288–315
- Bruton GD, Ahlstrom D, Li HL (2010) Institutional theory and entrepreneurship: where are we now and where do we need to move in the future? *Entrepreneurship Theory Pract* 34(3):421–440
- Carter RB, van Auken HE (1994) Venture capital firms' preferences for projects in particular stages of development. *J Small Bus Manage* 32(1):60
- Clarysse B, Wright M, Bruneel J, Mahajan A (2014) Creating value in ecosystems: crossing the chasm between knowledge and business ecosystems. *Res Policy* 43(7):1164–1176
- Coad A, Rao R (2008) Innovation and firm growth in high-tech sectors: a quantile regression approach. *Res Policy* 37(4):633–648
- Cohen B (2006) Sustainable valley entrepreneurial ecosystems. *Bus Strategy Environ* 15(1):1–14
- Cooper AC (1986) The role of incubator organizations in the founding of growth-oriented firms. *J Bus Ventur* 1(1):75–86
- De Clercq D, Lim DS, Oh CH (2013) Individual-level resources and new business activity: the contingent role of institutional context. *Entrepreneurship Theory Pract* 37(2):303–330
- Estrin S, Korosteleva J, Mickiewicz T (2013) Which institutions encourage entrepreneurial growth aspirations? *J Bus Ventur* 28(4):564–580
- Ferrary M, Granovetter M (2009) The role of venture capital firms in Silicon Valley's complex innovation network. *Econ Soc* 38(2):326–359
- Garnsey E, Stam E, Heffernan P (2006) New firm growth: exploring processes and paths. *Ind Innov* 13(1):1–20

- Garvin DA (1983) Spin-offs and the new firm formation process. *Calif Manag Rev* 25(2):3–20
- Gregory BT, Rutherford MW, Oswald S, Gardiner L (2005) An empirical investigation of the growth cycle theory of small firm financing. *J Small Bus Manage* 43(4):382–392
- Greiner L (1972) Evolution and revolution as organizations grow. *Harvard Bus Rev* 50(4):37–46 (July–August)
- Hsu DH (2006) Venture capitalists and cooperative start-up commercialization strategy. *Manage Sci* 52(2):204–219
- Iansiti M, Levien R (2004) Strategy as ecology. *Harvard Bus Rev* 82(3):68–81
- Isenberg D (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):40–50
- Isenberg D (2011) The entrepreneurship ecosystem strategy as a new paradigm for economy policy: principles for cultivating entrepreneurship. Babson Entrepreneurship Ecosystem Project. Babson College, Babson Park
- Kim JH, Wagman L (2014) Portfolio size and information disclosure: an analysis of startup accelerators. *J Corp Finance* 29:520–534
- Kroeger CV (1974) Managerial development in the small firm. *Calif Manag Rev* 17(1):41–47
- Krueger NF, Reilly MD, Carsrud AL (2000) Competing models of entrepreneurial intentions. *J Bus Ventur* 15(5):411–432
- Levie J (2014) The university is the classroom: teaching and learning technology commercialization at a technological university. *J Technol Transfer* 39(5):793–808
- Levie J, Lichtenstein BB (2010) A terminal assessment of stages theory: introducing a dynamic states approach to entrepreneurship. *Entrepreneurship Theory Pract* 34(2):317–350
- Lippitt GL, Schmidt WH (1967) Crises in a developing organization. *Harvard Bus Rev* 47:102–112
- Lyden FJ (1975) Using Parsons' functional analysis in the study of public organizations. *Adm Sci Q* 20(1):59–70
- Mason C (2011) Business angels. In: *World encyclopedia of entrepreneurship*. Elgar original reference. Edward Elgar, Cheltenham
- Mason C, Brown R (2013) Creating good public policy to support high-growth firms. *Small Bus Econ* 40(2):211–225
- Mason C, Brown R (2014) Entrepreneurial ecosystems and growth oriented entrepreneurship. Paper prepared for a workshop of the OECD LEED Programme and the Dutch Ministry of Economic Affairs, The Hague, Netherlands, 7 November 2013
- Mason CM, Harrison RT (1997) Business angel networks and the development of the informal venture capital market in the UK: is there still a role for the public sector. *Small Bus Econ* 9(2):111–123
- Miles G, Snow CC, Sharfman MP (1993) Industry variety and performance. *Strateg Manag J* 14(3):163–177
- Miller D, Friesen PH (1984) A longitudinal study of the corporate life cycle. *Manage Sci* 30(10):1161–1183
- Mintzberg H, Ahlstrand BW, Lampel J (1998) *Strategy safari: the complete guide through the wilds of strategic management*. Financial Times, Harlow
- Neck HM, Meyer GD, Cohen B, Corbett AC (2004) An entrepreneurial system view of new venture creation. *J Small Bus Manage* 42(2):190–208
- North DC (1990) *Institutions, institutional change and economic performance*. Cambridge University Press
- Peng MW, Sun SL, Pinkham B, Chen H (2009) The institution-based view as a third leg for a strategy tripod. *Acad Manage Perspect* 23(3):63–81
- Peters L, Rice M, Sundararajan M (2004) The role of incubators in the entrepreneurial process. *J Technol Transfer* 29(1):83–91
- Quinn RE, Cameron K (1983) Organizational life cycles and shifting criteria of effectiveness: some preliminary evidence. *Manage Sci* 29(1):33–51
- Scott WR (2007) *Institutions and organizations: ideas and interests*. Sage Publications, Thousand Oaks
- Shane S, Cable D (2002) Network ties, reputation, and the financing of New Ventures. *Manage Sci* 48(3):364–381

- Spigel B (2017) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory Pract* 41(1):49–72
- Spilling OR (1996) The entrepreneurial system: on entrepreneurship in the context of a mega-event. *J Bus Res* 36(1):91–103
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plann Stud* 23(9):1759–1769
- Stam E, Bosma N (2015) Local policies for high-growth firms. In: *Oxford handbook of local competitiveness*. Oxford University Press, pp 286–305
- Stam E, Wennberg K (2009) The roles of R&D in new firm growth. *Small Bus Econ* 33:77–89
- Swamidass PM (2013) University startups as a commercialization alternative: lessons from three contrasting case studies. *J Technol Transfer* 38(6):788–808
- Vachelard J, Gambarra-Soares T, Augustini G, Riul P, Maracaja-Coutinho V (2016) A guide to scientific crowdfunding. *PLoS Biol* 14(2):e1002373
- Van de Ven H (1993) The development of an infrastructure for entrepreneurship. *J Bus Ventur* 8(3):211–230
- Van de Ven AH, Poole MS (1995) Explaining development and change in organizations. *Acad Manag Rev* 20(3):510–540
- van Gelderen M, Thurik R, Bosma N (2005) Success and risk factors in the pre-startup phase. *Small Bus Econ* 24(4):365–380
- Vismara S (2016) Equity retention and social network theory in equity crowdfunding. *Small Bus Econ* 46(4):579–590
- Wennberg K (2013) Managing High-Growth Firms: a literature review. International Workshop on Management and Leadership Skills in High-Growth Firms, Warsaw, 6 May 2013. OECD

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# A Triple-Helix Ecosystem for Entrepreneurship: A Case Review

Ravi Chinta and Fiona Sussan

**Abstract** The collaborations among universities, businesses, and governments leading to entrepreneurial activities have received a lot of attention. Successful examples of such endeavours are plenty. However, how this triple-helix ecosystem may have changed in the digital economy remains unexplored. Specifically, there is a lack of research that addresses the changing roles of universities from acting as places for conducting R&D to become marketplaces for customers. The purpose of this chapter is to investigate the triple helix ecosystem of entrepreneurship in the digital economy. First, we use a historical lens to uncover the antecedents and drivers of university-business research collaborations with support from governments, and offer a few historical examples. Second, we highlight some recent digital businesses from the perspective of triple helix ecosystem of entrepreneurship. Third, we offer a new conceptual framework depicting the new roles of the three triadic players that have expanded from the supply side (i.e., R&D, funding) to the demand side (i.e., customer base) in enabling entrepreneurial activities.

**Keywords** Digital entrepreneurs • China mass entrepreneurship policy  
Government funded unicorns • Productive entrepreneurship • Digital ecosystem

## 1 Introduction

The term ecosystem in management literature refers to a network of interconnected entities that are linked in a goal-oriented pursuit that draws resources from all the collaborators (Adner and Kapoor 2010; Autio and Thomas 2016; Moore 1993; Teece 2007). Our focus here is university-based research which is well known to be

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collaborative with funding from businesses and government agencies. Universities are increasingly viewed as an actor of entrepreneurship (Etzkowitz et al. 2008; Hsu et al. 2007) or a facilitator of entrepreneurship (Rasmussen and Borch 2010). Leading edge scientific innovations such as the Human Genome project, stem-cell applications, alternative energies, advanced materials, social networking, cloud computing and even business process reengineering have emerged as a result of university research endeavours that are often conducted in collaborations with businesses and funding from government agencies. One outgrowth of these collaborative inventions is the spate of start-up firms that have emerged from these collaborations. However, there are wide variations in how the *university-business-government* collaboration manifests in reality, and a closer scrutiny reveals that such triadic collaboration is highly context-specific. This chapter aims to provide insights into this phenomenon after a review of current literature that is relevant to the topic.

In the following, we begin with the history of the triadic collaboration in the U.S. As the U.S. is widely viewed as a country that has always held strong triadic relationships (e.g., Silicon Valley), it is appropriate to use it as a benchmark for our global discussion. From the U.S. case, we uncover the drivers of these collaborations, followed by three historical global samples (Stanford, Cambridge, Sophia Antipolis) of productive entrepreneurship that are the results of the triple helix model. To add recency to this triadic relationship, we investigate recent digital businesses and the roles of universities and governments in these businesses. Based on the results of our investigation, we offer a new conceptual framework for the triad ecosystem in the digital economy. We conclude with limitations and future outlook.

## 2 The U.S. Case

In the U.S., the collaboration between universities and industry has a long history. Despite the fact that universities traditionally focused on the development of basic knowledge, academic research has evolved over time to become more pragmatic in terms of new technology development, new products development, new processes development and new business development. To promote more academic research, the Hatch Act of 1887 in the United States established the land grant state universities with a clear intent to develop and disseminate knowledge that resulted from academic research. There have been several recorded entrepreneurial efforts by the universities that helped farmers and manufacturers through extension services in the late 19th and early 20th century (Rosenberg and Nelson 1994). However, the pace of university-industry collaboration had been quite slow until the end of World War II. Shane (2004) documents the accelerated pace of technology commercialization from universities during 1945–1980, but those efforts were still limited to a few universities and the phenomenon of university-industry collaboration was not widespread. Commercialization of academic research was viewed with scepticism by academics during this period (Lee 1998). Despite this scepticism, research collaborations between businesses and universities accelerated over time partly aided by government regulations conducive to collaborations (Chinta and Culpan 2014).

In the history of university-industry collaboration in the USA, the Bayh-Dole Act of 1980 was a major paradigm-shifting action undertaken by the government to promote commercialization of academic research. The Bayh-Dole Act, an amendment to the Patent and Trademark Law, essentially gave away the federal rights to intellectual property (IP) emanating from federally funded research to the universities that conduct the research. The idea was that the new IP policy would foster greater university-industry collaboration with collaborative actions initiated by universities which would get new streams of revenues for universities. While universities that own the IP rights would earn royalties, the firm that licenses the IP would negotiate exclusive rights to the IP and also have further access to university resources. Within 10 years after the passage of the Bayh-Dole Act, more than 1000 university-industry research centres were established within universities (Cohen et al. 1994). Since 1980, the incentives provided by the Bayh-Dole Act resulted in a five-fold growth in university patenting and a significant rise in spin-offs from universities (Shane 2004). A self-reinforcing virtuous cycle that Shane (2004) calls the “*contagion effect*” has accelerated the knowledge discovery, transfer, and utilization from universities to industry. Such a contagion effect has become so ubiquitous that ecosystems for collaborative research, open innovation, multiple project collaboration agreements, multiple partner consortia and other new forms of governance are receiving increased attention in both academia and corporate world. Based on the U.S. case, we offer our explanations of what is driving the current trends toward collaboration below.

### 3 Drivers for Collaboration

Historically, innovation was parochial and suffered from geo-temporal separation of knowledge development, invention, and the commercialization process (Culpan 2014). As a result, innovation was manifest as an intra-firm activity with corporate laboratories (e.g., Bell Labs, Xerox, IBM) leading the charge. However, over time as firms realized the benefits of inter-firm partnerships, innovation grew to be more collaborative in nature. By necessity, “...the knowledge monopolies built by the centralized R&D organizations of the twentieth century have ended” (Chesbrough 2003: 45). Reasons for this change are simple. Over time the costs and risks of the innovation process within the closed system increased dramatically forcing the closed system to open up. A good example is the huge costs associated with new drug development. On average, the cost of bringing an idea from lab to the markets for a new drug is greater than one billion dollars (Siew 2013).

In modern business environments, innovation has become a major source of competitive advantage and the *sine-qua-non* for firm survival. Rapid technological change and sophisticated customer expectations have caused firms to innovate faster and forced firms to revisit the traditional notion of firm-level competences (Barney 2001a, b). Firms, nowadays, have an expanded view of competences that go beyond those resident within the firm and scout globally for partnerships with

other firms which may possess competences needed for innovation. Thus, the first driver is the power of synergies that reduce the cost and time of bringing new products to market.

Digitalization is another major driver of collaboration. The advent of the Internet dramatically changed the way we communicate and work. Both affordable and ubiquitous communications enable workers or entrepreneurial actors to exchange information and knowledge effectively, efficiently, and speedily. In an Internet-enabled entrepreneurial ecosystem, or a digital entrepreneurial ecosystem, entrepreneurial actors embrace the new culture of ‘sharing economy’, user-participatory governance, and open source digital architecture (Sussan and Acs 2017), contrasting entrepreneurial activities in the past that were mainly conducted in closed-form within clusters (Autio et al., forthcoming). The major change that differentiates an entrepreneurial ecosystem (in the digital economy) from a cluster is that knowledge spillover among entrepreneurial actors are voluntary and within horizontal surrounding platform and thus not subject to industry boundary (Autio et al., forthcoming). Extending this to the triad collaboration context, entrepreneurial actors within a single organization (e.g., the business) can no longer develop the requisite level of innovation demanded by the market. Thus firms have to develop dynamic capabilities using intellectual capital or open innovation outside of the firm in order to adapt to market needs (Chesbrough 2010; Dodgson et al. 2006). Indeed, companies like IBM, Nokia, Qualcomm, Oracle, Microsoft and many other hi-tech firms have found ways to collaborate with other firms to extend the innovation reach of their own internal R&D. Another example is Google, an output of productive entrepreneurship based on triadic collaborations. Google continuously works with various universities across the globe [e.g., with Oxford University on Artificial Intelligence (AI), with Carnegie Mellon on Internet of Things (IOT)] to leverage new intellectual capital or collect data or develop new products (Cuthbertson 2014; Spice 2015).

The third driver comes from universities broadening their scope of engagement in research which was earlier focused primarily on new knowledge creation. In the U.S., the Bayh-Dole Act of 1980 essentially gave away the federal rights to the intellectual property (IP) to universities for almost all federally funded research projects at universities. Royalties from licensing became a new revenue stream for universities; and royalties come only from commercialization of IP and realization of revenues. After 1980, universities increasingly started collaborating with business, not only through the licensing of existing IP, but also in the funding of new research projects by businesses which have rights of first refusal on the new knowledge (IP) created for the funding firms. Enkel et al. (2009) reported that the extensive links between universities and businesses are bidirectional, i.e., firms engage universities in creating new knowledge and universities engage businesses in commercializing existing IP. This practice was common as experienced by the first author of this chapter—he was in the venture capital industry in late 1980s and early 1990s and was involved in funding several research projects with leading researchers at various universities to develop new biotechnology-based innovations.

The fourth driver is the development of new structures called innovation ecosystems. Adner (2006: 98) defines *innovative ecosystems* as “collaborative arrangements through which firms combine their individual offerings into a coherent customer facing solution”. These new structures are more than transient inter-firm partnerships that end after the alliance objectives are met. Increasingly interdependence of external partners are key to value co-creation (Adner and Kapoor 2010). Indeed, instead of organizing a focal firm, innovation ecosystems could be viewed as being organized around a platform that incorporates both production, use side participants, and focuses on the development of new value through innovation (Autio and Thomas 2016). Etzkowitz (2008) describes industrial parks and innovation hubs involving the “The Triple Helix”—universities, businesses and government—to kick start new business startups. Even as startup businesses graduate from these structural support systems after a certain incubation period, the structures remain as new knowledge flows through structures from the initial ideation to the final commercialization stages of entrepreneurship. Such structural drivers of collaborative innovation have been so successful that some firms are bringing such structures in-house to accelerate open innovation by the firm. For example, Proctor and Gamble (P&G) headquartered in Cincinnati, Ohio launched an online website called “Connect+Develop” through which anyone in the world can make a suggestion for new product development and incubate the idea in collaboration with P&G (Brown and Anthony 2011).

The fifth driver for collaborative research is the diversification of sources of funding new knowledge creation. From an intra-firm self-funding model, research funding is benefitting from multiple sources of funding. Federal government grants (research grants from NSF, NIH; business development loan guarantees from SBA), venture capital funding of research, joint research partnerships, multiple party research consortia, specialized projects funding for targeted problems, government subsidies (for plant locations; jobs creation and poverty alleviation, etc.) among other sources are diversifying the sources of funding for new knowledge creation in the long chain of activities from ideation to the commercialization stages of the entrepreneurial process. Such diversification of funding sources also means that the external engagement of universities has to expand to interact with their appropriate stakeholders around intellectual, educational, social, cultural, economic and technological development (Gutteridge 2007).

#### **4 Notable Historical Examples of Triple Helix Ecosystem of Entrepreneurship**

University-business-government collaborations imply that three distinct systems of governance come together synergistically that are effective and efficient. A boundary should be an edge that connects rather than divides (Girvan and Newman 2002). This edge is thus double edged: by exposing separation, it also

effaces it each time one encounters it, relishing each encounter with diversity while keeping its own identity. It's like a vacation in a foreign country. Universities become more like business with focus on commercial R&D while not forsaking basic research; businesses become more like universities seeking non-commercial and long term R&D; governments become more focused on tangible gains for society at large via innovations facilitated by grants and subsidies. While there are many examples of multiple systems coming together which can be detailed, Chinta and Culpán (2014) describe four university-led collaborative efforts in three nations as noted below.

Stanford and University of California Berkeley in the US: Both Stanford University and UC Berkeley played significant roles in the development of Silicon Valley into a success story which today we all recognize. Stanford University started in 1891 and right from its beginning university-industry collaboration was emphasized. For example, in 1908 Stanford research commercialized vacuum tubes as a way to intensify electrical signals. Firms such as Hewlett Packard, SUN Microsystems, Varian, Litton Engineering, Fairchild Semiconductors and Cisco were born at Stanford University. UC Berkeley with 20 Nobel Prize winners was also a leading world center for knowledge creation. UC Berkeley alumni, Andy Grove and Gordon Moore, founded Intel. In 1951, Stanford University opened the Stanford Industrial Park—a decision that was unique for its time.

Cambridge in the UK: Cambridge University is one of the most renowned universities in the world that has made conscious decision to nurture university-industry collaboration. A number of high-tech spin-offs and consulting firms emerged from Cambridge University's competences in the areas of electronics, instrument development, and computing. Not only have new ventures sprung out of the university, but also many entrepreneurs from other parts of the UK chose to locate in the university area to collaborate with Cambridge University. The university thus became a magnet with its knowledge spillovers.

Sophia Antipolis University in France: In 1965, Sophia Antipolis University was founded in Nice, France. In 1974, an incubator promoting university-industry collaboration, the Sophia Antipolis Science Park was set up. Large firms such as France Telecom, IBM, Texas Instruments located on the site. Of the firms located on the site, Information, Computers and Telecommunications (ICT) sector accounted for 65% and health sciences sector accounted for 20% with more than 50% coming out of the university as spin-offs (Sophia Antipolis 2013). Today, the university campus benefits from the presence of 1300 companies in the near vicinity with co-location as its main driver of the open innovation. Sophia Antipolis is a laboratory of 21st century companies and an exceptional knowledge community.

The corporate landscape is also fast changing in that corporate venture capital is now undertaking outside R&D that in an earlier era would have been too risky to be undertaken within a firm. Colombo and Grilli (2010) empirically show, in a study of 439 new technology based firms in Italy, that venture capital financing was a key success factor for these firms many of which would not have survived as intra-corporate R&D projects. Not too long ago, corporate venture capital funds (e.g., Intel, IBM, Alphabet, Microsoft, Apple) may account for the funding of

nearly 13% of all new technology based startup companies (Benson and Ziedonis 2009). In most cases, excluding corporate spinouts, corporate venture capital funds invest equity in business plans that do not originate from within the firm. Croce et al. (2013) that this is a global phenomenon though they find stronger evidence in US and Europe.

Procter and Gamble (P&G) was a consumer products company that is a pioneer in market-driven innovation with more than 20 brands each over \$1 billion in revenues. Xerox was a technology-driven company whose technology historically followed science-led innovations. DeFillippi et al. (2014) describe how these two firms have formed the Xerox-P&G co-innovation partnership that is also open to participation from universities and governmental funding. Co-creation of value expanded from the initial dyadic B2B supply chain relationship model to engaging multiple stakeholders such as universities and governmental funding sources. The co-creation mix varies by the distinctive contexts in which co-creation takes place, a point empirically stated by Roser et al. (2012) in their study of Xerox and P&G whose historical intra-corporate R&D evolved into diverse extra-corporate R&D initiatives of wide variety. Such diversity defies any generalization without any reference to the context of co-creation.

## **5 Examples of Triple Helix Ecosystem of Entrepreneurship in the Digital Economy**

While Google may be considered the classic productive output of a triad ecosystem in the digital economy, many newer digital businesses were started by entrepreneurs who were college students who did not conduct R&D at their universities. Google's Larry Page and Sergey Brin were both doctoral students at Stanford when Google was conceptualized and funded by the U.S. intelligence agencies—NSA and the CIA (Szoldra 2016). Before the incorporation of Google in 1998, Page and Brin (supported by an NSF graduate student fellowship) worked on The Stanford Integrated Digital Library Project funded by U.S. federal (NASA, DARPA) grant (NSF 2004). During this project, Page and Brin developed an a prototype algorithm that crawled from page to page by following links and mapped out a vast family tree that reflected the Web links among pages, which is known as PageRank and being part of the core component of Google's search service (NSF 2004). Google thus is an example of the emergence of an entrepreneurial firm within the context of university-government positive relationship. Facebook's Mark Zuckerberg used Harvard in a different way. Zuckerberg leveraged Harvard's group listserv to populate his user base (Kirkpatrick 2010). In other words, the university was the initial market for the new product, or the digital marketplace in a digital entrepreneurial ecosystem mainly populated by platforms (Sussan and Acs 2017). Yahoo was founded by Jerry Yang and David Filo while attending Stanford. Ofo, a bike sharing Unicorn in China, was founded by a student, Dai Wai and his classmates, at

Peking University. Again, they used their university and the customers within it as the basecamp and market to launch their “Campus Bicycle Sharing” new product (Ofo bicycle 2015). Ele.me, a food delivery Unicorn, was founded by college students at Shanghai Jiaotong University (Crunchbase 2017).

As for the role of governments, apart from Google receiving funding from the U.S. government as a startup, the intelligence agencies in the U.S. have supported many other entrepreneurial activities via their own venture capital firm In-Q-Tel (Szoldra 2016). Recently National Science Foundation (NSF) and Small Business Innovation Research (SBIR) have teamed up together to ‘incubate’ entrepreneurial adventures (NSF website 2017). The role of federal and local governments in funding entrepreneurs and their innovations is more explicit in China. In 2015, China announced its “Mass Entrepreneurship and Innovation” economic policy (Li 2015). In the same year, it raised a massive US\$231 billion to fund start-ups, in addition to the US\$6.5 billion in government grants, tax breaks and subsidised technology parks pledged for start-ups by the Chinese premier (Mittal and Lloyd 2016). Even prior to 2015, local government, specifically the Wuxi government, was notorious in funding Suntech, a solar-power firm founded by a returnee Chinese academic who had worked as a solar energy research director at the University of New South Wales, and commercialized its products into an eventually led to a successful IPO. The Wuxi government acted essentially as an investment banker and at the same time provided manufacturing support to enable Suntech to become the first mover in solar energy business (Liu et al. 2017). Recent fintechs in China are generously funded by state-controlled entities. Examples include Alibaba’s Ant Financial. In fact, the government is leading fintech’s rise in China in investing many fintech projects via its state-owned funds (Mittal and Lloyd 2016).

These current trends point toward a new phenomenon we are seeing in the triadic relationships in terms of the changing roles of each partners. Universities, which used to be the place mainly to ‘develop’ new products, now increasingly become the test markets for new products (e.g., Facebook, Ofo), providing a demand side of the market that the entrepreneurship literature has yet to address (Acs and Sussan 2017). Governments’ role in investing also seems to be different, especially in a new economy like China. In China, state-owned companies invest and gain from these entrepreneurial firms (e.g., Suntech) and at the same time providing manufacturing assistance and regulatory (e.g., fintech in China) push. These new trends provide us with insights that we consolidated in Fig. 1.

Figure 1 depicts the different roles of each partner (i.e., university, business, government) in enabling the demand side and/or the supply side of entrepreneurial activities. In this Figure, universities can provide the supply side of the equation when supporting R&D of a new product (e.g., Stanford incubates and also takes equity). At the same time, universities can provide the demand side of the equation serving as a customer base (e.g., Facebook obtains group listserv from within the university but Harvard did not have a stake in the company). Businesses also work with universities and governments on both the supply and demand sides of the equation. Government supports the supply side of the equation by providing funding (e.g., NSA and CIA funded Google). However, governments can be on the

|            | Supply                                                                                                                                                                                    | Demand                                                                                                                                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| University | <b>R&amp;D</b><br>(e.g., Google was started by PhD students at Stanford)                                                                                                                  | <b>Customer base</b> (Facebook at Harvard; Ofo at Peking University)                                                                                                                                                                           |
| Business   | <b>Funding</b> (Alibaba, Baidu, Tencent are successful digital businesses that are funding many Unicorns in China)                                                                        | <b>Customer</b> (Alibaba is also customer of many Unicorns it funds)]                                                                                                                                                                          |
| Government | <b>Funding</b><br>(e.g., NSA CIA funded Google ; CIA funds Palantir, Fuel3d; Singapore government funds Xiaomi in China; Chinese government funds Alibaba affiliate Alibaba Ant Financial | U.S. intelligence agencies as <b>customer</b> of entrepreneurs they fund (Palantir, Fuel3d)<br><br>Chinese government as <b>customer</b> of entrepreneurs they fund (Alibaba Ant Financial build national credit rating system for government) |

**Fig. 1** Productive triadic entrepreneurial activities in the digital economy

demand side of the equation given that new entrepreneurial firms were funded for government use also (e.g., Palantir, Fuel3d). For example, Alibaba affiliate Alibaba Ant Financial is partially funded by the Chinese government for customer online payment and at the same time developing a national consumer credit rating system.

In addition to the more active roles that the three players participate in on the demand side of entrepreneurship, we increasingly find that universities have become more entrepreneurial and actively solicit funds from the industry with at least an estimated 10% of all academic funds generated from industry (Chinta and Culpán 2014). Entrepreneurial universities will increasingly shape the entrepreneurial orientation of a place or region - collective influence even though universities do not have an entrepreneurial focus by nature (Cunningham et al. 2017). Stanford University (Silicon Valley) and MIT (Route 128) are well known for noting this phenomenon, but there are several other universities that are also as effective (e.g., University of San Diego and the biotech startups in and around San Diego; industrial parks in IITs in India). In essence, the entrepreneurial orientation of universities will become major differentiators for universities in the highly competitive higher education marketplace (Guerrero et al. 2016).

6 Conclusion

We began the chapter by tracing the formation of the triple helix ecosystem of entrepreneurship among university, business, and government. As the history of the Bayh-Dole Act 1980 in the U.S. favoured U.S. universities for their IP ownership when innovation was funded by the government, university-based innovation proliferated. The subsequent commercialization process largely depended on collaborating with industry. We provided historical examples of Stanford, Cambridge,

and Sophia Antipolis to substantiate these triad ecosystems around the globe resulting in productive entrepreneurship as seen in Stanford Industrial Park and Sophia Antipolis Science Park. Similar models are followed around the globe (e.g., China-Singapore Suzhou Industrial Park). We also examined the triadic relationships among some digital businesses that emerged in the last two decades. As we chronicled the birth of entrepreneurial firms such as Google, Facebook, and Yahoo, we uncovered a new pattern in that instead of providing the supply side of factors of production (i.e., money, R&D, product knowledge), the triple helix ecosystem is providing the demand side of the equation (i.e., the customers). This leads to the insight of a new framework in Fig. 1 in which the three players in the triple helix ecosystem is providing support for both the supply and the demand sides of entrepreneurial activities.

The introduction of the new role of triadic partners in providing a market for entrepreneurial activities is thus a major contribution of this article to the entrepreneurial ecosystem literature. In this process, we raised important questions about the future roles of campuses as a place not only for developing products but as a place to accumulate user base (e.g., Ofo in Peking University, Facebook in Harvard). User base is the core competency for platform businesses. Campuses should thus re-strategize their roles in the digital economy beyond offering of online courses such as MOOC or HarvardX to instead become a major player in leveraging and harvesting productive entrepreneurship.

As campus-based digital entrepreneurship mushrooms, universities are however faced with a different set of problems: “What is public property and what is private property?” The example of University of Illinois (UI) and Netscape is instructive in this regard. UI wanted to restrict the use of its Mosaic patents, while Netscape wanted to freely build on it. The irreconcilable conflict became so acute that when Jim Clark became the CEO of Netscape, the firm completely eradicated all code pertaining to the original UI Mosaic code. Also, Marc Andreessen, founder of Netscape and UI alumnus, publicly stated that he will not give a single dime of donation to UI. This question of public versus private property remains the most perplexing question in multi-stakeholder collaborative ecosystems. While ecosystems comprise multiple participants bound by complex relations and contracts that often involve mutual interdependence (Thomas and Autio 2013), socializing the costs and privatising the profits remains a commonly observed phenomenon in these multi-stakeholder collaborative ecosystems (Mazzucato 2015).

As digital entrepreneurs are a prominent addition to the digital economy, we have in this chapter provided a new framework and a new lens to examine university-industry-government collaboration. Thus far, our framework is a static one examining the relationship of campus-based entrepreneurship in broad strokes. Future research should use a process-based approach to unpack the chronological and complex relationships among student-user-entrepreneurs, businesses, and governments. In particular, understanding the process of how student-entrepreneurs obtain knowledge about opportunities provided by the government is important.

Major questions remain on how to manage the processes of entrepreneurial ecosystem emergence when multiple partners bring differing goals into their collaboration causing significant agency problems (Hove and Clarysse 2017). Recognizing the potential and actual value created by the partners relative to their contributions is laden with complexities that defy any generalizations. A deeper understanding of the determinants of success for different types of triple helix ecosystem of entrepreneurship structures is crucial, and we believe that our chapter sheds some light on the subject matter in calling for continued theory development, empirical analyses and dialogue between academics and practitioners.

The global dimension of entrepreneurship ecosystems presents another unique dilemma. Although we included a few examples from China, to a large extent the empirical evidence on triple helix ecosystem of entrepreneurship comes from the Western hemisphere (and predominantly from the USA). Macro-environmental differences such as market-driven capitalism and state-driven capitalism or the stage of technological development, population demographics would shape the context (place) differently for the emergence of triple helix ecosystem of entrepreneurship in both their evolutionary trajectory and design. Prior research established institutional influence as a key macro-environmental factor such as infrastructure development (Bergara et al. 1998; North 1990) and VC investments (Da Rin et al. 2006; Samila and Sorenson 2010). Formal institutions in an economy define the legal, political, and economic rules and conventions that become part of the heritage we call culture. For example, in some countries not all entrepreneurs register their businesses. This is more true with respect to need-driven entrepreneurs and less so with opportunity-driven entrepreneurs (Chinta 2017). Future research in this area should consider the cultural context as a primary factor in triad ecosystem of entrepreneurship (Hayton et al. 2002).

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## References

- Acs Z, Sussan F (2017) Knowledge spillover theory of entrepreneurship in the digital age. Working Paper Presented at Oxford Residency of Entrepreneurship, June
- Adner R (2006) Match your innovation strategy to your innovation ecosystem. *Harvard Bus Rev* 84(4):98
- Adner R, Kapoor R (2010) Value creation in innovation ecosystems: how the structure of technological interdependence affects firm performance in new technology generations. *Strateg Manag J* 31(3):306–333
- Autio E, Thomas L (2016) Innovation ecosystems: implications for innovation management. In: Dodgson M, Phillips N, Gann DM (eds) *The Oxford handbook of innovation management*. Oxford University Press
- Autio E, Nmbisan S, Thomas LDW, Wright M (forthcoming) Digital affordances, spatial affordances, and the genesis of entrepreneurial ecosystems. *Strateg Entrepreneurship J*

- Barney JB (2001a) Is the resource-based “view” a useful perspective for strategic management research? Yes. *Acad Manag Rev* 26(1):41–56
- Barney JB (2001b) Resource-based theories of competitive advantage: a ten-year retrospective on the resource-based view. *J Manag* 27(6):643–650
- Benson D, Ziedonis R (2009) Corporate venture capital as a window on new technologies: implications for the performance of corporate investors when acquiring startups. *Organ Sci* 20(2):329–351
- Bergara ME, Henisz WJ, Spiller PT (1998) Political institutions and electric utility investment: a cross-nation analysis. *Calif Manag Rev* 40(1):18–35
- Brown B, Anthony SD (2011) How P&G tripled its innovation success rate. *Harvard Bus Rev* 89(6):64–72
- Chesbrough H (2003) *Open innovation: the new imperative for creating and profiting from technology*. Harvard Business School Press, Boston
- Chesbrough H (2010) *Open services innovation: rethinking your business to grow and compete in a new era*. Wiley
- Chinta R (2017) Empirical study of need-driven and opportunity-driven women entrepreneurship in the state of Alabama. Working paper, Auburn University, Montgomery
- Chinta R, Culpán R (2014) The role of open innovation in business-university R&D collaborations. In: Culpán R (ed) *Open innovation through strategic alliances*. Palgrave MacMillan, New York, pp 145–165
- Cohen W, Florida R, Goe WR (1994) University-industry research centers in the United States. Unpublished paper, Carnegie-Mellon University, Center for Economic development, H. J. Heinz III School of Public Policy and Management, Carnegie Mellon University, Pittsburgh, PA
- Colombo MG, Grilli L (2010) On growth drivers of high-tech start-ups: exploring the role of founders’ human capital and venture capital. *J Bus Ventur* 25(6):610–626
- Croce A, Marti J, Murtinu S (2013) The impact of venture capital on the productivity growth of European entrepreneurial firms: ‘screening’ or ‘value added’ effect? *J Bus Ventur* 28(4): 489–510
- Crunchbase (2017) <https://www.crunchbase.com/organization/ele-me#/entity>. Accessed 7 July 2017
- Culpán R (2014) Open innovation business models and the role of inter-firm partnerships. In: Culpán R (ed) *Open innovation through strategic alliances*. Palgrave MacMillan, New York, pp 18–39
- Cunningham JA, Guerrero M, Urbano D (2017) Entrepreneurial universities: overview, reflections and future research agendas. In: Siegel D (ed) *The world scientific reference on entrepreneurship. Volume 1: Entrepreneurial universities: technology and knowledge transfer*. World Scientific Publishing Co., Hackensack, pp 3–19
- Cuthbertson A (2014) Google partners with Oxford University to develop artificial intelligence. <http://www.ibtimes.co.uk/google-partners-oxford-university-develop-artificial-intelligence-1471437>. Accessed 5 July 2017
- Da Rin M, Nicodano G, Sembenelli A (2006) Public policy and the creation of active venture capital markets. *J Public Econ* 90(3):1699–1723
- DeFillippi R, Dumas C, Bhatia S (2014) Genesis and evolution of the Xerox-P&G co-innovation partnership. In: Culpán R (ed) *Open innovation through strategic alliances*. Palgrave MacMillan, New York, pp 121–144
- Dodgson M, Gann D, Salter A (2006) The role of technology in the shift towards open innovation: the case of Procter & Gamble. *R&D Manag* 36(3):333–346
- Enkel E, Gassmann O, Chesbrough H (2009) Open R&D and open innovation: exploring the phenomenon. *R&D Manag* 39(4):311–316
- Etzkowitz H (2008) *The triple helix: university-industry-government innovation in action*. Routledge, London and New York
- Etzkowitz H, Ranga M, Benner M, Guarany L, Maculan AM, Kneller R (2008) Pathways to the entrepreneurial university: towards a global convergence. *Sci Public Policy* 35(9):681–695

- Girvan M, Newman ME (2002) Community structure in social and biological networks. *Proc Natl Acad Sci* 99(12):7821–7826
- Guerrero M, Urbano D, Fayolle A, Klofsten M, Mian S (2016) Entrepreneurial universities: emerging models in the new social and economic landscape. *Small Bus Econ* 47(3):551–563
- Gutteridge T (2007) Outreach and engagement: afterthought or strategic priority? *Mid-Am J Bus* 22:5–6
- Hayton JC, George G, Zahra SA (2002) National culture and entrepreneurship: a review of behavioral research. *Entrepreneurship Theory Pract* 26(1):33–45
- Hove JV, Clarysse B (2017) The ecosystem approach: rethinking academic entrepreneurship in Belgium. In: Siegel D (ed) *The world scientific reference on entrepreneurship. Volume 4: Process approach to academic entrepreneurship: evidence from the globe*. World Scientific Publishing Co., Hackensack, pp 335–352
- Hsu DH, Roberts EB, Eesley CE (2007) Entrepreneurs from technology-based universities. Evidence from MIT. *Res Policy* 36(50):768–788
- Kirkpatrick D (2010) Excerpt from The Facebook Effect. [http://www.ft.com/cms/s/0/f2619600-ca37-11df-87b8-00144feab49a.html?ft\\_site=falcon&desktop=true#axzz4n7i1VY3l](http://www.ft.com/cms/s/0/f2619600-ca37-11df-87b8-00144feab49a.html?ft_site=falcon&desktop=true#axzz4n7i1VY3l). Accessed 5 July 2017
- Lee YS (1998) University-industry collaboration on technology transfer: views from the ivory tower. *Policy Stud J* 26:69–84
- Li K (2015) A blue print for global growth at the ninth annual meeting of the new champions, Summer Davos opening ceremony. [http://news.xinhuanet.com/english/2015-09/11/c\\_134615607.htm](http://news.xinhuanet.com/english/2015-09/11/c_134615607.htm). Retrieved 10 Mar 2016
- Liu Y, Sussan F, Krueger N (2017) The tale of two entrepreneurial cities: top-down versus bottom-up entrepreneurial ecosystem building. Working Paper
- Mazzucato M (2015) *The entrepreneurial state: debunking public vs. private sector myths*, vol 1. Anthem Press
- Mittal S, Lloyd J (2016) The rise of Fintech in China. Ernst and Young, Development Bank of Singapore (DBS). [http://www.ey.com/Publication/vwLUAssets/ey-the-rise-of-fintech-in-china/\\$FILE/ey-the-rise-of-fintech-in-china.pdf](http://www.ey.com/Publication/vwLUAssets/ey-the-rise-of-fintech-in-china/$FILE/ey-the-rise-of-fintech-in-china.pdf). Accessed 5 July 2017
- Moore JF (1993) Predators and prey: a new ecology of competition. *Harvard Bus Rev* 71(1):75–86
- North DC (1990) *Institutions, institutional change and economic performance*. Cambridge University Press, Cambridge, New York and Melbourne
- NSF (2004) Discovery: on the origins of Google. [https://www.nsf.gov/discoveries/disc\\_summ.jsp?cntn\\_id=I00660](https://www.nsf.gov/discoveries/disc_summ.jsp?cntn_id=I00660). Accessed 15 Aug 2017
- NSF website (2017) <https://www.nsf.gov/eng/iip/sbir/topics.jsp>. Accessed 5 July 2017
- Ofo bicycle (2015) “ofo bicycle”: riding bicycles anytime and anywhere. [http://english.pku.edu.cn/news\\_events/news/campus/4059.htm](http://english.pku.edu.cn/news_events/news/campus/4059.htm). Accessed 5 July 2017
- Rasmussen E, Borch OJ (2010) University capabilities in facilitating entrepreneurship: a longitudinal study of spin-off ventures at mid-range universities. *Res Policy* 39(5):602–612
- Rosenberg N, Nelson R (1994) American universities and technical advances in industry. *Res Policy* 23:323–348
- Roser T, DeFilippi R, Samson A (2012) Managing your co-creation mix: co-creation ventures in distinctive context. *Eur Bus Rev* 25(1):20–41
- Samila S, Sorenson O (2010) Venture capital as a catalyst to commercialization. *Res Policy* 39(4):1348–1360
- Shane S (2004) *Academic entrepreneurship: university spin-offs and wealth creation*. Edward Elgar, Northampton
- Siew A (2013) Pharma embraces open innovation. *Pharm Technol* 25:6–8
- Spice B (2015) Cmu leads Google expedition to create technology for “internet of things” campus will be living lab for interconnected sensors, gadgets. <https://www.cmu.edu/news/stories/archives/2015/july/google-internet-of-things.html>. Accessed 5 July 2017
- Sussan F, Acs Z (2017) Digital entrepreneurial ecosystem. *Small Bus Econ* 49(1):55–73
- Szoldra P (2016) 14 cutting edge firms funded by the CIA. <http://www.businessinsider.com/companies-funded-by-cia-2016-9>. Accessed 5 July 2017

- Teece DJ (2007) Explicating dynamic capabilities: the nature and microfoundations of (sustainable) enterprise performance. *Strateg Manag J* 28(3):1319–1350
- Thomas L, Autio E (2013) Emergent equifinality: an empirical analysis of ecosystem creation processes. In: Proceedings of the 35th DRUID Celebration Conference, Barcelona, Spain

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# Theorizing the University Governance Role in an Entrepreneurial Ecosystem

Allan O'Connor  and Gerard Reed

**Abstract** This chapter examines the role of higher education institutions among the various programmes, community and private sector initiatives within a regional milieu and specifically an entrepreneurial ecosystem. We use data collected from a set of sixteen interviews with service providers to the entrepreneurship community in Adelaide, Australia, to construct a framework of the perceived roles and contributions of the university sector to an entrepreneurial ecosystem. The contribution of this chapter relates to its focus on what universities are expected to contribute within a regional entrepreneurial ecosystem. The findings suggest five specific roles for universities which opens a discussion around the strategic choices for university executive management in approaching regional engagement in economic development issues through contributions to entrepreneurship. We also discuss the issue of governance that arises through our analysis and the implications for place-based and regional entrepreneurial ecosystem development.

## 1 Introduction

This research engaged with key actors in an emerging entrepreneurial ecosystem and posed the question: what is the role of higher education institutions in an entrepreneurial ecosystem among the various public programmes, community and private sector initiatives? This question responds to calls for further research into the different characterizations of university engagement and diverse patterns of university-society interactions across various locational settings (Perkmann et al. 2013).

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Adopting an ecosystems approach to the study of entrepreneurship, invites an examination of particular actors and institutional influences on the productive entrepreneurship system. In this chapter we examine how universities may shape positive entrepreneurship effects as perceived by other regional stakeholders that participate in influencing the development of entrepreneurs in the ecosystem. Universities are commonly recognized to have a dual role in regional development both to develop human capital and provide a knowledge and technological resource to the region (Lindqvist et al. 2013). A better understanding of the expected roles of universities in serving entrepreneurial ecosystems is critical for both governments and universities if robust models of policies and practices are to be built.

This chapter is structured by first portraying the rise of the entrepreneurial ecosystem concept before outlining the relationship between universities and economic development and in particular in regional place-based contexts. We use the Triple Helix Model (THM) literature to examine the studies of others with respect to the university sector contributions to regional development. It includes a discussion on the contribution of entrepreneurship education as a specific measure that universities may adopt to influence economic development. The subsequent section then addresses Adelaide, Australia, as a case from which a purposive sample of entrepreneurial ecosystem stakeholders was drawn for the research. The analysis of sixteen interviews is then developed to reveal a diverse range of expectations with respect to the contributions universities can make to an entrepreneurial ecosystem. The conclusion proposes the implications for governance and suggests a way forward to conducting research that captures the dynamics and complexities of university offerings in an entrepreneurial ecosystem for regional development.

## 2 The Emerging Concept of Entrepreneurial Ecosystems

The process of entrepreneurship is centrally concerned with the recognition, discovery and/or creation, and exploitation of opportunity (Alvarez and Barney 2004; Schendel and Hitt 2007; Shane and Venkataraman 2000). Importantly for this research, the pursuit of an opportunity by an entrepreneur has been linked not only to such issues as entrepreneurial alertness (the propensity of a population of entrepreneurs to spot opportunities) but also to a positive perception of entrepreneurial munificence (the availability of assistance and support services that facilitate the start-up process) and, particularly where self-efficacy of the entrepreneur is high (Tang 2008). The combination of entrepreneurs and the elements of that contribute to entrepreneurial munificence alludes to the entrepreneurial ecosystem concept.

The notion of an entrepreneurial ecosystem has been gaining prominence since the early nineteen nineties when ecological concepts started to filter into entrepreneurship research (see Aldrich 1990). A variety of definitions of an entrepreneurial ecosystem have since been developed with most agreeing that it consists of institutionally embedded elements influencing actors who bring about entrepreneurial performances. (Isenberg 2010; Spigel 2015; Stam 2015; Mason and Brown 2014;

Acs et al. 2014). By introducing a systems perspective into the study of entrepreneurship, it acknowledges that entrepreneurship does not start and stop with the actions of an entrepreneur or their firm but instead it includes the ecosystems that thrive when multiple sectors and actors work together to create a supportive environment for entrepreneurship (Nadgrodiewicz 2013). The entrepreneurial ecosystem operationalizes the regional milieu of resources, institutions, firms, actors, social, political and economic factors and elements that provide the setting for place-based entrepreneurial performances.

Isenberg (2010) has described the elements—those ‘things’ that surround the entrepreneur—within an entrepreneurial ecosystem, by classifying them into six main categories: a conducive culture; enabling policies and leadership; availability of appropriate finance; quality human capital; venture friendly markets for products and services; and a range of institutional and infrastructure support. These elements are institutional in nature meaning that the entrepreneur is embedded among elements that impose rule-making structures (Scott 1987) including formal, e.g. regulatory and statutory authorities and organizations (Mason and Brown 2014), and informal, e.g. political, cultural and economic socially conditioned drivers (Stam 2015; Spigel 2015). By implication then, actors within the ecosystem will respond to the changes imposed by the actions of others within the ecosystem to maintain their own survival, growth and health of the ecosystem within their relevant areas of activity.

Entrepreneurship is an accepted dynamic of any economy (Minniti and Lévesque 2008) and increasingly, quality or ambitious and productive entrepreneurship is being recognized as a vital element to consider when planning for regional development (Stam 2015). Universities have emerged as one of the essential levers of entrepreneurial development in an ecosystem driving toward a knowledge economy (Andersson et al. 2010). As the appreciation of the relative importance of entrepreneurship deepens it becomes increasingly urgent to question what universities can contribute to the entrepreneurial ecosystem to determine what their specific impact might be.

## ***2.1 Universities and Regional Economic Development***

Recent studies have illustrated the need and benefits of regional level stakeholder relationships through such models as the university engagement model (UEM) and, more predominantly, the Triple Helix Model (THM) (Gunasekara 2006). The THM was originally described by Etzkowitz and Leydesdorff (1997) who sought to argue for a ‘new social contract between the university and the larger society’. Etzkowitz (2003) points out that ‘...the interaction in university-industry-government is the key to improving the conditions for innovation in a knowledge-based society’ (p. 295). Further, analysing the THM can be informative with respect to revealing the extent to which innovation is systemic within particular defined geographic boundaries (nations or regions) (Leydesdorff 2012). Because these models are relatively new and informed by limited cases, there is still much to learn about how universities engage in regional performance.

Application of the THM interaction demonstrates the power of building consensus between industry, government and the university sectors particularly in terms of revitalizing declining economies. The idea of creating triple-helix “spaces” demonstrated in Sweden reveals that cross-institutional entrepreneurship that aggregates both regional and national resources focusing around a local, triple helix generated strategy was key to successful industrial transition (Svensson et al. 2012). In application, the Swedish case appears more aligned to ‘economic gardening’ principles that use long-term approaches to economic development designed to strengthen local businesses and provide new growth opportunities through support and assistance with market and strategic analysis (Desplaces et al. 2009).

From an education perspective, the influence of universities on regional economic development can be associated to issues of human capital development through higher levels of general education attainment (Millán et al. 2013). Due to the very natures of education and entrepreneurship, it is difficult to make explicit connections between these and any form of business or entrepreneurial success. The case remains that many factors influence entrepreneurial success and isolating the extent to which education specifically contributes to the success of a particular entrepreneur is contentious (O'Connor 2013). However, university education in the general sense also does not address the specific case of entrepreneurship education nor entrepreneurship education's specific contribution to the stimulus for entrepreneurship in a region.

A second view of the role of universities in regional economic development is gained from the perspective of universities as public research providers. From this perspective knowledge spill-over is indicated as the primary source of economic contribution and universities are assumed to be the *engines of economic growth* (Smith and Bagchi-Sen 2012). This view suggests that the research knowledge generated by universities promotes entrepreneurship and universities act as a pivotal node in a regional network of innovation activity. This occurs through such mechanisms as science and research parks, knowledge exchange with external organizations, outreach activities and research partnerships and forming new spin-out firms. However, the OECD report (OECD 2011) on skills for innovation and research suggests that more attention is needed on the ‘soft skills’ such as entrepreneurship for those in academic research careers in order to improve the relationships between the university and industry sectors. Therefore, we next consider more specifically the issues of entrepreneurship education's contribution to economic development.

Laukkanen (2000) views universities as a ‘regional evolution model’ generating business and defines this stating “... by conceptualizing the university as a regional evolution mechanism, a different yet parallel educational strategy may be suggested, called a business generating model” (p. 25). Laukkanen suggests “... that a university can be conceptualized as a societal (regional) innovation system, and that entrepreneurial education, when embedded in such a system, could be regarded, not only as a task of producing entrepreneurially oriented competent individuals, but also as reproducing the social mechanisms that underpin and facilitate the birth and growth of businesses and firms” (p. 26).

Higher Educational Institutions (HEI) also have an important role to play in regional areas that involves the support of small and medium sized (SME) enterprises (Johnston et al. 2008). With respect to entrepreneurship educational training for SME

owner/managers Gordon et al. (2012) found that "... the evolution of strong networks ... endured beyond the programme and the learning that took place in these networks" (p. 768) was vital to stimulating entrepreneurial activity. Others note that 'reflection' (Cope 2003; Gordon et al. 2012; Kempster and Cope 2010) by the participants in entrepreneurship education is essential for building mutual trust. Building strong networks appears to be a primary outcome if an SME entrepreneurship education and its structure provide the right environment.

Universities have also been considered as an entrepreneurial ecosystem in their own right built upon historic precedents. For example Miller and Acs (2017) describe the case of the University of Chicago and the departure from the traditional focus on research by the Booth Business School to instead focus on the creation of high-growth firms. This reflected the university's history and ideals established at its foundation and was supported by a mix of student action, inherent demand for entrepreneurship, and university leadership. Furthermore, as an entrepreneurial ecosystem, universities play a vital role in creating new knowledge that is instrumental in catalysing new companies, jobs and industry in a region which is thought to be the hallmark of an entrepreneurial university (Walshok and Shapiro 2014).

Leydesdorff and Deakin (2011) provide a co-evolution view that universities, government and individuals can present an alternative to 'entrepreneurship-based and market dependent representation of knowledge-intensive firms' (p. 59). Benneworth (2004) notes 'densification' (Fontes and Coombes 2001, p. 84) as a role for universities that can increase the development of techno-economic networks and activities in regional areas through entrepreneurship and industry engagement. Järvi (2012) further strengthens the case for regional engagement by universities through entrepreneurship education explaining that '... [t]he analysis indicates that entrepreneurship education should be assessed and applied from the perspective of field and region' (p. 365). While this background reveals an awareness of the relative importance of entrepreneurship education, we find that we know relatively little about how this teaching should be embedded in regional ecosystems in order to best influence actual short-term or potential long-term economic objectives (O'Connor 2013).

Clearly universities are an important part of the entrepreneurial ecosystem and particularly with respect to realizing innovation. A dominant part of the literature focuses on the university through the lens of entrepreneurship education, research commercialization, technology transfer, new firm spin-outs and spin-offs with some studies taking into account the systems view of innovation and entrepreneurship that interconnects the university agenda with external influencing factors (Walshok and Shapiro 2014). However, most studies focus on the United States and selected European countries (Perkmann et al. 2013) and this current study reveals viewpoints from an Australian context. Universities have been identified for various contributions to regional economic development but none of the studies covered in this extant literature review spell out the range of options available to university executive management in positioning their universities to maximize their contribution to the regional economic development agenda.

### 3 Methodology

This research deploys an exploratory and interpretive approach to examine perceptions of the entrepreneurial ecosystem programs and service providers about the role of universities. From this perspective the research operates from Schein's (1985) point of view that suggests that research should identify what participants themselves perceive to be important. The data collection included 16 targeted interviews to inform this chapter. The analysis involves the researcher in looking for patterns in the data (Schwandt 1998; Taylor and Bogdan 1984) as well as exploring rich contextual evidence and meanings to processes that have not been previously explored in depth (Howorth et al. 2005). Each of the invited respondents participated in a semi-structured interview which ranged in duration from 60 to 90 min.

The adopted method comprised data collected through interviews with key program and service providers identified from Adelaide's Entrepreneurial Ecosystem map of programmes and service providers. Due to time and resource limitations, a purposive sampling strategy was adopted to identify and invite two participants from each of the eight categories of ecosystem service providers provided by the map (discussed in more detail in the next section). In order to canvass a broad spectrum of views, one participant from each category was selected for diversity; a long-standing program actor and the other a relatively new actor. Notably, some respondents' organizations contribute to multiple program categories and some respondents, that may represent newer or older programs, may either be new to or quite experienced with the ecosystem and these nuances were also considered when choosing participants. Considerations were also given to geographic representation and a cross-section of types of entrepreneurial ventures that the providers seek to support.

The analysis for this chapter extended to coding the notes recorded independently by the two interviewers in consultation with the audio recordings of the interview. The codes were then analysed for key conceptual themes and related to observations from the literature to isolate commonalities and differences through comparative analysis.

#### 3.1 *Overview of the Entrepreneurial Adelaide Project*

Australia faces the prospect of a sustained high-cost economy. Smaller and relatively isolated regions, once supported by the value chains of scale-based manufacturing firms, are vulnerable and face a challenging future unless the region transitions to a sustainable industrial base. In response, our case study—the South Australian state government and Adelaide's local community and local government—are turning their attentions to entrepreneurship as a means to stimulate regional development through new economic activity. This response is consistent with the argument that shifts in the economic base toward knowledge requires a context and policy environment supportive of entrepreneurship (Audretsch and Thurik 2004).

In Australia, the engagement of universities in matters of regional development has been historically low and business and university relationships are not generally strong (Gunasekara 2006). This is particularly so because the history and cultural roots of the university system in Australia are characterized by dependence on government funding, weak private sector funding and, a history of a two tier (vocational and university) higher education system (Gunasekara 2006). Regional governance and leadership participation has not generally been a feature of university strategic management in Australia (Garlick 2000) and consequently the Australian context differs to the international cases, often European or US based, upon which the models of university engagement and/or the THM are frequently based.

Entrepreneurial Adelaide is a movement within the city of Adelaide, South Australia, Australia which ran under a working title of ‘Towards a City of Entrepreneurs—The Emergence of Adelaide as a Recognized Startup Community’ (Daly 2013). It owes its genesis to a catalytic question posed within a discussion between two people at one of the many entrepreneurial events and activities being run around Adelaide in December, 2012. That question asked: ‘[with] so much happening to support entrepreneurship in Adelaide, how ... do you keep track of it?’

The question inspired one of the discussants to adopt a self-directed research project to pursue an answer and an understanding of what he and others had also wondered. Subsequently, a meeting was held on the 4th of January 2013 with seven out of the ten invitees from the entrepreneurial community in Adelaide attending the meeting. Preliminary research into the programs that supported entrepreneurial activity in Adelaide was tabled at this meeting that represented four categories of entrepreneurship support in Adelaide. The meeting participants developed this further adding programs and using coloured ‘sticky notes’ with yellow to denote ‘Education Programs’, pink for ‘Networking Programs and Events’, blue for ‘Incubators, Accelerators, and Co-Working Spaces’, and green was used to identify ‘Government Support Programs and Investors’. This mapping exercise was the first iteration of what has now become known as the map of the Adelaide Entrepreneurial Ecosystem (AEE).

The lack of a device to interpret the landscape of an ecosystem, in this case within the context of the city of Adelaide and its environs in the state of South Australia, was perhaps indicative of the very nature of an ecosystem. This is due, in part, to the fact that the constituent contributors of an ecosystem (Hwang and Horowitz 2012), being participants, elements or subsections, are not always equipped with the ability to view the ecosystem in its entirety. Further meetings were held with working groups being established on five topic areas, namely; Ecosystem Governance, Marketing and Communications, Early-stage Funding, Education, and Social Media and the Internet. Two public forums were also convened on the 25th June 2013 and the 3rd of September 2013. All those within the structure of what has become known as the AEE, have continued to contribute to the map and the latest iteration of the ‘Adelaide Entrepreneurial Ecosystem Map’ (Waterhouse 2013, p. 57) now contains eight categories and the colour themes have been extended. The map is regularly updated according to new entrants and recent exits with Draft 7.6 February 2017, being the latest at the time of writing with 134

**Table 1** Matrix of participant respondents to date (*Source* Authors)

|                           | Networking and start-up events | Formal education | Industry education | Co-working spaces | Incubators and accelerators | Advisory services | Government assistance | Investors |
|---------------------------|--------------------------------|------------------|--------------------|-------------------|-----------------------------|-------------------|-----------------------|-----------|
| Newer Entrant             | ✓                              | ✓                | ✓                  | ✓                 | ✓                           | ✓                 | ✓                     | ✓         |
| Long-standing participant | ✓                              | ✓                | ✓                  | ✓                 | ✓                           | ✓                 | ✓                     | ✓         |

programs being identified. In addition a website supports the program view of the AEE and can be found at: <http://www.startupadelaide.com.au/entrepreneurial-ecosystem-map/>.

It is from this map and the eight categories of programs and services that the respondents for this research were drawn as described earlier. Table 1 summarizes the categories and the respondent sampling design that have contributed to the data collection.

## 4 Analysis and Findings

The interview guide used for this research canvassed a range of issues dealing with the market place, industry and competitive environment for the ecosystem program and service providers. It included questions on the customers and entrepreneurial behaviour of the program and service providers and probed the working of the ecosystem and the existence of gaps and opportunities along with querying the benefit of developing the Adelaide Ecosystem Map and a specific question on the role of the university. The analysis involved listening through the recorded interviews and scanning the written notes to identify any reference to university roles, opportunities for universities to contribute to the ecosystem, observed gaps or inadequacies mentioned with respect to university contributions to the ecosystem or any references to what universities do or how they are engaged in the ecosystem. These references were then brought together and themed into five role descriptions being: Regional Governance; Human Capital Development; Intellectual Resource; Network Facilitator; and, Entrepreneurial Node (refer Table 2). While on the surface these roles may seem unremarkable, the contribution of our work can be found in the depth and diversity of the activities that are encompassed in these roles.

Perhaps the most unsurprising roles that emerged from this analysis are the two roles of 'Human Capital Development' and 'Intellectual Resource'. Both of these areas are readily and easily identified in the academic literature (Lindqvist et al. 2013). However, with respect to Human Capital Development what is significant is that the analysis surfaces an explicit expectation that the graduates will be entrepreneurially prepared and not just academically or discipline prepared. Furthermore, the analysis suggested that entrepreneurially prepared goes beyond just referring to the entrepreneur role and surfaced both entrepreneurial team roles and entrepreneurial leadership more generally.

The 'Intellectual Resource' role of the university is also apparent as would be expected although, notably, the expectations of this role surface the tension between 'near to market' research and fundamental research that leads to new discoveries that may underpin new market/industry opportunities. However, the expectations associated with this division were not exposed further from within the interview data.

The next two roles, Regional Governance and Network Facilitator that emerged also have a ring of familiarity. However, in the context of the Australian market, these two roles are less articulated and developed. The expectations of and

**Table 2** Key university roles and contributions (*Source* Authors)

| Role                      | Contribution/Expectation                                                 | Examples of types of activities                                                                                                                                                                                         |
|---------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Human Capital Development | Education Industry                                                       | Provide employment opportunities and career pathways in the education sector                                                                                                                                            |
|                           | Education Touch-point                                                    | Provide education opportunities post-secondary school for lifelong learning                                                                                                                                             |
|                           | Prepare 'excellent' graduates technically competent in chosen discipline | Deliver high quality education                                                                                                                                                                                          |
|                           | Prepare graduates for entrepreneurial activity                           | Develop entrepreneurial career opportunities; develop entrepreneurial team skills among graduates; develop entrepreneurial attitudes, behaviours and 'spirit' among graduates; develop graduate leadership capabilities |
| Intellectual resource     | Industry problem solver                                                  | Provide responsive technical research skills and capabilities to industry                                                                                                                                               |
|                           | New 'knowledge' commercialization opportunities                          | Invest in research and new knowledge creation and provide commercialization opportunities to/with regional actors                                                                                                       |
| Network facilitator       | Connect with Alumni                                                      | Maintain relationships with alumni to anchor with the region and contribute to regional growth and development                                                                                                          |
|                           | Global connections in fields of expertise                                | Make global connections with expertise and facilitate knowledge exchange with region                                                                                                                                    |
|                           | Connected academic staff                                                 | Expect academic staff engagement with regional industry and business                                                                                                                                                    |
|                           | Connected students                                                       | Provide opportunities for students to work with industry through internships, research projects and short-term industry/business placements                                                                             |
|                           | Provide open access to physical meeting spaces                           | Provide plazas, venues and other physical places for industry and community to interact with the university and each other                                                                                              |
|                           | Community connected advocate                                             | Diverse interactions and activities among various community segments, industry groups, organizations, other institutions and individuals advocating entrepreneurship                                                    |
| Entrepreneurial node      | Outreach                                                                 | Provide outreach programs and support to facilitate the dissemination and development of entrepreneurial knowledge and expertise to the ecosystem                                                                       |
|                           | In-reach                                                                 | Provide in-reach programs and support to facilitate the dissemination and development of entrepreneurial knowledge and expertise within the university                                                                  |

(continued)

**Table 2** (continued)

| Role                | Contribution/Expectation | Examples of types of activities                                                                                                                                                                                    |
|---------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regional governance | Long term commitment     | Coordinating ecosystem data and information collection and distribution regionally on an ongoing basis                                                                                                             |
|                     | Strategy Development     | Expert informant and advisor on regional development issues based on international intelligence                                                                                                                    |
|                     | Providing Leadership     | Partnering in innovation and risk bearing; leading new projects and experiments of regional significance; acting as lead customer; lending brand credibility; facilitating and supporting student entrepreneurship |
|                     | Operational Informant    | Advisor on ecosystem performance issues and advances in international trends                                                                                                                                       |
|                     | Self-regulating Steward  | Through self-regulation, maintain focus on best use of public knowledge resources and maximize internal and external efficiencies among community assets                                                           |

contributions from universities under the Regional Governance role include a long term commitment, strategy development, providing leadership, being an operational informant and being self-regulated to maximize community resource efficiencies. The long term commitment has emerged from the ideas that developing an entrepreneurial ecosystem will take time and the history of longevity in the university system raises expectations that universities are well positioned to ‘stay the course’ and provide a steady hand as neither government nor industry are positioned to fulfil this role. Government political persuasion shifts more regularly with short-term election cycles and leadership changes while industry has a primary concern with commercial viability that is at best secondary to regional matters if a focus at all. These views are consistent with the Swedish study by Svensson et al. (2012) and the ‘economic gardening’ principles outlined by Desplaces et al. (2009).

The strategy development expectation reflects the views that universities should have a ‘seat at the table’ when the development of regional entrepreneurial ecosystems are being determined. The role emerges from the data suggesting that universities have a legitimate contribution to make and possibly a specific expertise on entrepreneurship to offer. The expectation of providing leadership is observed from the data that suggests universities need to model the behaviours necessary for entrepreneurship and endorse and support entrepreneurial activities. In this way universities show leadership by taking the active initiating role and promoting the accumulated benefits that may accompany entrepreneurship. The data also suggests the idea that universities should be active in monitoring and sharing information about the regional entrepreneurial ecosystem which gives rise to the expectation of Operational Informant. Critical also was the idea that universities should be

Self-regulating Stewards of the knowledge assets and resources residing in the community with the universities actively coordinating efficiencies in a non-competitive way to maximize the advantage and minimize the costs to the community.

The fourth category of university role is Network Facilitator. The data underpinning this area pointed to the fact that universities have the opportunity to make connections. Some of these connections are directly through students and academic staff and relate to connecting industry and international expertise. The university alumni also were raised as a point of indirect connection in that universities can keep expatriate alumni informed and connected back to their region. Locally, alumni also can be connected to the entrepreneurial community through participation in university entrepreneurial events. Another point of connection highlighted the value of the physical assets and spaces the universities have that can serve as attractors and public meeting places to increase the chances of knowledge spill-over. Lastly, universities by their nature connect with a broad array of public and private organizations and institutions and are in a position to act as advocates for entrepreneurship.

The final role is one of being an Entrepreneurial Node. In this role the university acts as the 'go to place' either in the physical or virtual sense for all things entrepreneurial. This role is likely to manifest as an enterprise or entrepreneurship center, institute or hub. As a node in the entrepreneurial ecosystem, entrepreneurial and innovative ventures will be attracted to the node from either inside or outside of the university. The activities of the node motivate, re-energise or sustain momentum for entrepreneurial ventures. The node also acts as a resource center that serves and supports entrepreneurial activities again either internal or external to the university.

Table 2 details more explicitly these various roles and exposes greater detail. The examples in column three are drawn from synthesis of the interview participant responses to illustrate concrete examples of each of the coded expected contributions. The emergence of these roles adds depth and breadth to the more common ideas of the university role as an education and research institution. Although at the high level there is little by way of radical new insight, the composite idea of the various roles paints a more detailed picture of how universities can serve an entrepreneurial ecosystem.

## 5 Discussion

In Australia at least, entrepreneurship is only just emerging as a legitimate part of Australian policy as evidenced by the Australian Government's recent announcement of the formation of the 'Entrepreneurs Infrastructure Program' (Commonwealth of Australia 2014). Up until this time, innovation and commercialization have been the focus of previous governments, often neglecting the extent of the challenge that faces new and established firms to build local commercially thriving businesses from the world of innovation, research and new knowledge.

In Adelaide, South Australia, particularly, the urgency for economic renewal and transition is focusing the attention of policy-makers and others on the role that entrepreneurship can play in re-shaping an economy and introducing innovation and hence the entrepreneurial ecosystem has been highlighted as an area for policy attention.

This research set out to examine what universities are expected to contribute in raising the prospects of innovation from entrepreneurship contained within an entrepreneurial ecosystem. From this standpoint we utilized the resource mapping of service providers to the entrepreneurial ecosystem as was being promulgated in the South Australian Adelaide region. During our analysis we encountered a problem with the analogy of an ecosystem in that in contrast to biological ecosystems that are self-preserved, self-organizing and self-directed with no artificial interference, the human entrepreneurial ecosystem is prone to interventions by governance models. We therefore instead recognized, as have others, that the ecosystem analogy was to some extent imperfect for our purposes (Holling 2001; Peltoniemi 2006).

In our analysis and discussion of these ideas with key stakeholders, a further complication was recognized as it was also pointed out that the ecosystem rainforest analogy (Hwang and Horowitz 2012) that was frequently drawn upon in discussions of Adelaide's entrepreneurial ecosystem did carry some weight if it were acknowledged that rainforests vary according to the climatic conditions. This interpretation meant that tropical rainforests differ to temperate rainforests for instance. In following the thread of this idea it also suggests then that the drawing in or adoption of elements from rainforests that do not share the same conditions, climate and population of species is potentially a flawed idea. Extending the analogy suggests at the extreme that the policies, practices and inherent activities of one ecosystem should not be expected to be easily or readily supplanted into another ecosystem and expected to survive with the same vigour.

This draws attention to a governance model and an alternate analogy that we drew was that the 'ecosystem' could be treated more like a botanic garden than a natural rainforest. In a botanic garden certain areas and plots are cultivated and specifically designed to deliberately resemble or approximate the ecosystems of other places to sustain the life of plants that may be foreign to the local environment. However, the local conditions provide limits to the extent that foreign ideas can take root and flourish in a non-sustaining environment and therefore the careful selection of ideas that can be sustained within the limits of cost and artificial infrastructure is important. In practice, building the entrepreneurship ecosystem more closely mimics the concept of 'economic gardening' that recognizes the resource base, incumbent firm attributes, matters of market proximities and infrastructure and emerging trends to build and nurture new opportunities as they are relevant to the region within which the entrepreneurial ecosystem is to be cultivated.

The evidence from the data also supported this idea that the term 'ecosystem', and its associated analogy, had limited applicability. It was clearly viewed that universities had a service obligation and a leadership role to play that we placed under the heading of regional governance adopting the terminology of the literature.

Natural biological ecosystems do not have a regional intervening or conscious supra governance mechanism. At the same time there is an element of 'borrowing' and 'planting' the ideas of entrepreneurship from other places, in Adelaide's case, and in particular, from the USA with Silicon Valley being the iconic example. This approach to the local ecosystem endowed with a governance model renders the ecosystem analogy less like a rainforest concept and closer to the 'botanic garden' concept. The implication for policy-makers and universities is that their roles within the frame of regional governance is to help carefully select and cultivate the entrepreneurship seeds and ideas that can adapt to the local conditions. Furthermore, they can prepare the ground, provide the infrastructure and induce a habitat that will support new ideas and entrepreneurial ventures

The contribution of this chapter relates to its focus on what universities are expected to contribute within a regional and entrepreneurial milieu. The findings suggest five specific roles. As one may expect Human Capital Development through the education of highly skilled graduates is a key contribution and expectation, although, the expectation extended to graduates who are not only highly and well educated but also prepared and capable of operating in an entrepreneurial way, within an entrepreneurial team and providing entrepreneurial leadership. This supports the idea of a broad entrepreneurial education across the disciplines in university.

The Intellectual Resource contribution on its own is perhaps a second unremarkable finding and is quite consistent with the literature. However, the tension between the 'near to market' research and 'fundamental' research is unresolved by the data. There is no question that universities have a role to play in providing both research functions and while expectations may go to both strands of activity the investment, training and development, strategic positioning and reputational influences of both positions can cause internal conflicts and tensions within the university system. As the higher education system in Australia undergoes further de-regulation and opening to free market principles, individual universities may need to make clear strategic choices to leverage differentiation.

The Network Facilitation expectation and contribution adds a refinement to the community engagement concept that universities generally define as part of their role. An explicit statement of network facilitation highlights the important function of infrastructural facilities, information technology usage (apart from educational platforms), service provision and contract management issues that are critical to businesses that are designed to operate on a network facilitation business model (Stabell and Fjeldstad 1998). This too increases the complexity of the issues facing the university executive management team and will force specific trade-offs and difficult strategic choices.

The role of Entrepreneurial Node is on one hand obvious but on the other potentially obscure. The Entrepreneurial Node role is obvious when universities host a specific entrepreneurship or enterprise center or institute. It should be noted though that these centers are not homogenous in their activities and while some may play a role as incubator, accelerator and facilitator of new venture initiatives, others may be research oriented and knowledge based nodes (Mazzarol 2014). On the

other hand the Entrepreneurial Node role may be quite obscure observable in the absence of an enterprise or entrepreneurship center within the university. In this instance the role of the university as an entrepreneurial node may not be immediately evident or definable as a specific place and activity. Instead the entrepreneurial node function and role may take place as discipline specific activities of industry engagement or commercialization found in various locations distributed across the university.

The fifth expected contribution is Regional Governance. This arguably has the most applicability to the issues of innovation within a regional milieu. Within this role there are some points that universities are likely to have already adopted and others that may be foreign concepts, particularly in Australia. The idea of contributing to Regional Governance also surfaces issues that involve the extent to which the other expectations and contributions are embraced. Quite aside from the strategic and competitive choices that need to be made for the survival of the university, the regional governance choices endow the university with added responsibilities that may conflict with or, at the very least, be incompatible with the resources and any given strategic direction. In a region such as Adelaide, there are three major universities and a number of smaller national and international university outposts. This presents a risk whereby no one university assumes the role of regional governance leaving the region devoid of university engagement with regional entrepreneurial ecosystem governance as each university leaves it to the other to play any specific role. Bringing the university into account in these issues should be the task of policy-makers although it is unlikely that any one mechanism for doing so would be effective in all cases as the milieu histories, cultures, regulatory frames, economic strengths, capabilities and competencies to name a few factors will all vary between different regions and nations.

The expectation of Regional Governance role for the entrepreneurial ecosystem is perhaps the most intriguing and enlightening perspective to emerge from the data. The literature on the contribution of universities focuses on what universities do in particular with respect to their educational and research functions. Case studies of universities within ecosystems are predominantly on the iconic success stories (see for example Saxenian 1994) or those with US or European histories. Our entrepreneurial ecosystem case draws upon an aspirational view for the region rather than evidence from iconic success. It draws upon a substantially different history to cases that are represented by Europe or the US. Notably this reveals a role for universities that are actively engaged in the leadership and stewardship of a region to provide insight and develop coordinated strategy. Rather than a region benefiting from the natural or residual endowments of the investments in a university, our case suggests that universities can coordinate with industry and government to lead and facilitate building complementary strengths, provide an enduring strategic trajectory for a region, it can evidence progress and develop research potential. Our case suggests that universities can and are expected to be a strategic resource to the regional community inputting into and building the capacity of the entrepreneurial

ecosystem. Essentially, in the absence of natural or endowed entrepreneurial legacies, universities can be the beating heart of entrepreneurial ecosystem governance.

## 6 Limitations and Future Research Opportunities

The presentation of this research is currently based on analysis of a small pool, of purposive respondents which draws attention to an obvious limitation. Given that the data collection of this research is small and drawn from a single region a much broader sample would be beneficial. However, as the work is intended to be a theory building exercise it is fit for purpose and is designed in a way that leads to further research opportunities to explore and examine the findings in different regions and contexts and with different methodologies.

The five roles elaborated by this research though also raises the question as to what extent should all universities contribute to every aspect of these roles and under what circumstances should the roles vary in scope? In a region such as Adelaide, there are three major universities and a number of smaller international and niche universities operating. Is it feasible for each university to adopt all roles and how is it that specific roles can be elevated to be of more strategic importance than others? By contrast, what happens in regions where there is only one university? How does a region manage the demands upon its university while still allowing the university autonomy and freedom of strategic choice?

From the perspective of the regional milieu, it is easily recognizable that milieus can vary enormously due to a myriad of factors related to history, culture, regulations, regional resources, access to capital etc. In particular it is apparent that a university's history is important in defining its role in ecosystem building (Miller and Acs 2017). The questions this research raises therefore is, if the roles for universities are accepted, what are the specific practices that have greatest effect within an entrepreneurial ecosystem and under what conditions can these practices be readily adopted in different types or forms of milieu?

These types of questions establish a set of difficult conditions that research will need to make some allowances if a set of experimental conditions are to shed light on these issues. A solution we propose is to establish an international regional governance network of government and university partnerships that are able to cooperate in research with strategic coordinated interventions and the gathering of data that can be shared and accessed across the network. While this method of data collection and comparison will not allow complete isolation of variables equivalent to experimental conditions, it will facilitate research through relationships that can be coordinated in such a way that some factors can be controlled on specific interventions. A feature of this network would be the learning opportunities that ultimately arise through the network which in effect will allow universities to be informed curators of regional knowledge, networks and education that can be crafted to most effectively meet the innovation and entrepreneurship needs of their region.

## 7 Conclusion

The findings of this research suggest the development of a refined set of ideas about how universities can engage in the entrepreneurial ecosystem assisting with innovation in differing regional milieus. In our analysis we suggest the analogy of an ecosystem has limitations with respect to human systems that are imbued with governance characteristics that do not faithfully reflect natural ecosystems. Instead we suggest that interventions in entrepreneurial ecosystems are more similar to developing a botanic garden whereby entrepreneurship is one segment area within the milieu that serves a specific function of regional development and industrial transition. Governments without question are drawn to attending to this section of the garden and we propose that universities also have at least five potential roles to play as defined by this research.

The five potential roles for a university also imply service obligations that suggest to some extent that universities are knowledge curators with respect to engaging in matters of regional governance. In terms of governance of an entrepreneurial ecosystem, this implies the usefulness of the economic gardening concept whereby regional collaborative strategies are developed with a long-term view among engaged and diverse stakeholders. In response to the difficulties in accounting for milieu variation and the research and policy-making that can most effectively develop an entrepreneurial milieu we call for the establishment of a research community among a network of regional governance partnered universities and regionally based governments. In proposing this idea, the authors invite discussion and feedback.

## References

- Acs ZJ, Estrin S, Mickiewicz T, Szerb L (2014) The continued search for the solow residual: the role of national entrepreneurial ecosystem. The institute for the study of labor (IZA), Bonn, Germany. Available at <ftp.iza.org/dp8652.pdf>. Accessed on 10 May 2015
- Aldrich HE (1990) Using an ecological perspective to study organizational founding rates. *Entrepreneurship Theory and Practice* Spring: 7–24
- Alvarez SA, Barney JB (2004) Organizing rent generation and appropriation: toward a theory of the entrepreneurial firm. *J Bus Ventur* 19:621–635
- Andersson T, Curley MG, Formica P (2010) Knowledge-Driven entrepreneurship, innovation, technology, and knowledge management 102, doi: [10.1007/978-1-4419-1188-9\\_11](https://doi.org/10.1007/978-1-4419-1188-9_11), Springer, Science + Business Media, LLC
- Audretsch D, Thurik R (2004) A model of the entrepreneurial economy. Discussion paper series on Entrepreneurship, Growth and Public Policy. Max Planck Institute for Research into Economic Systems, Jena, Germany
- Benneworth P (2004) In what sense ‘regional development?’ entrepreneurship, underdevelopment and strong tradition in the periphery. *Entrepreneurship Reg Dev: Int J* 16(6):439–458
- Commonwealth of Australia (2014) Budget 2014–15, Budget Measures, Budget Papers No. 2 2014–15. Available at [www.budget.gov.au](http://www.budget.gov.au). Accessed on 15 May 2014

- Cope J (2003) Entrepreneurial learning and critical reflection: discontinuous events as triggers for “higher-level” learning. *Manag Learn* 34(4):429–450
- Daly P (2013) Towards a city of entrepreneurs—The emergence of Adelaide as a recognised startup community. Paper presented at the ‘towards a city of entrepreneurs’ forum, Adelaide Convention Centre, 25 June 2013
- Desplaces DE, Wergeles F, McGuigan P (2009) Economic gardening through entrepreneurship education: A service-learning approach. *Ind High Educ* 23(6):1–12
- Etzkowitz H (2003) Innovation in innovation: the triple helix of university-industry-government relations. *Soc Sci Infor* 42(3):293–337
- Etzkowitz H, Leydesdorff L (1997) Universities and the global knowledge economy: a triple helix of university-industry-government relations. Cassell Academic, London
- Fontes M, Coombs R (2001) Contribution of new technology based firms to the strengthening of technological capabilities in intermediate economies. *Res Policy* 30:79–97
- Garlick S (2000) Engaging universities and regions: knowledge contributions to regional economic development in Australia. Department of Education, Training and Youth Affairs (ed), Commonwealth of Australia, Canberra, Australia
- Gordon I, Hamilton E, Jack S (2012) A study of a university-led entrepreneurship education programme for small business owner/managers. *Entrepreneurship Reg Dev* 24(9–10):767–805
- Gunasekara C (2006) Universities and associative regional governance: Australian evidence in non-core metropolitan regions. *Reg Stud* 40(7):727–741
- Holling CS (2001) Understanding the complexity of economic, ecological and social systems. *Ecosystems* 4:390–405
- Howorth C, Tempest S, Coupland C (2005) Rethinking entrepreneurship methodology and definitions of the entrepreneur. *J Small Bus Enterp Dev* 12(1):24–40
- Hwang VW, Horowitz G (2012) The rainforest—the secret to building the next silicon valley. Regenwald, California
- Isenberg D (2010) How to start an entrepreneurial revolution. *Harvard Business Review*, June. Available at <http://www.frederickentrepreneurship.net/app/download/745698/Entrepreneurial+Revolution+Isenberg,+2010.pdf>. Accessed on 8 April 2014
- Järvi T (2012) Teaching entrepreneurship in vocational education viewed from the regional and field perspectives. *J Vocat Educ Training* 64(3):365–377
- Johnston L, Hamilton E, Zhang J (2008) Learning through engaging with higher education institutions. *Int Small Bus J* 26(6):651–660
- Kempster S, Cope J (2010) Learning to lead in the entrepreneurial context. *Int J Entrepreneurial Behav Res* 16(1):5–34
- Laukkanen M (2000) Exploring alternative approaches in high-level entrepreneurship education: creating micromechanisms for endogenous regional growth. *Entrepreneurship Reg Dev* 12:25–47
- Leydesdorff L (2012) The Triple Helix, Quadruple Helix..., and an N-Tuple of Helices: explanatory models for analyzing the knowledge-based economy? *J Knowl Econ* 3:25–35
- Leydesdorff L, Deakin M (2011) The Triple-Helix model of smart cities: A neo-evolutionary perspective. *J Urban Technol* 18(2):53–63
- Lindqvist G, Ketels C, Sölvell O (2013) The cluster initiative greenbook 2.0, Ivory Tower Publishers, Stockholm
- Mason C, Brown R (2014) Entrepreneurial ecosystems and growth oriented entrepreneurship. Organisation for Economic Co-operation and Development. Available at <https://www.oecd.org/cfe/leed/Entrepreneurial-ecosystems.pdf>. Accessed on 10 May 2015
- Mazzarol T (2014) How do Australia’s universities engage with entrepreneurship and small business? Centre for entrepreneurial management and innovation. Available at <http://www.cemi.com.au/sites/all/publications/CEMI-DP1401-Mazzarol-2014.pdf>. Accessed on 6 October 2014
- Millán JM, Congregado E, Román C, van Praag M, van Stel A (2013) The value of an educated population for an individual’s entrepreneurship success. *J Bus Venturing*. Available at <http://dx.doi.org/10.1016/j.jbusvent.2013.09.003>. Accessed on 22 October 2013

- Miller DJ, Acs ZJ (2017) The campus as entrepreneurial ecosystem: the University of Chicago. *Small Bus Econ* 49(1):75–95
- Minniti M, Lévesque M (2008) Recent developments in the economics of entrepreneurship. *J Bus Ventur* 23:603–612
- Nadgrodskiewicz A (2013) Building Entrepreneurship Ecosystems. Available at <http://www.cipe.org/publications/detail/building-entrepreneurship-ecosystems-0>. Accessed on 8 April 2014
- O'Connor A (2013) A Conceptual framework for entrepreneurship education policy: meeting government and economic purposes. *J Bus Ventur* 28:546–563
- O'Connor A, Reed G (2015) Promoting regional entrepreneurship ecosystems: the role of the university sector in Australia. Presented at the Australian Centre for Entrepreneurship Research Exchange (ACERE), Adelaide, SA, Australia, February 2015
- OECD (2011) Skills for innovation and research, OECD Publishing. Available at <http://dx.doi.org/10.1787/9789264097490-en>. Accessed on 21 April 2011
- Peltoniemi M (2006) Preliminary theoretical framework for the study of business ecosystems. *Emergence: Complexity and Organisation* 8(1):10–19
- Perkmann M, Tartari V, McKelvey M, Autio E, Broström A, D'Este P, Fini R, Geunae A, Grimaldi R, Hughes A, Krabel S, Kitson M, Llerena P, Lissoni F, Salter A, Sobrero M (2013) Academic engagement and commercialisation: a review of the literature on university–industry relations'. *Res Policy* 42:423–442
- Saxenian A (1994) *Regional advantage: culture and competition in Silicon Valley and Route 128*. Harvard University Press, Cambridge
- Schein EH (1985) *Organizational culture and leadership*. Jossey-Bass, San Francisco
- Schendel D, Hitt MA (2007) Introduction to volume 1. *Strateg Entrepreneurship J* 1:1–6
- Schwandt T (1998) Constructivist, interpretivist approaches to human inquiry. In: Denzin N, Lincoln Y (eds) *The landscape of qualitative research*. Sage, Thousand Oaks, CA
- Scott WR (1987) The adolescence of institutional theory. *Adm Sci Q* 32(December):493–511
- Shane S, Venkataraman S (2000) The Promise of entrepreneurship as a field of research. *Acad Manag Rev* 25:217–226
- Smith HL, Bagchi-Sen S (2012) The research university, entrepreneurship and regional development: Research propositions and current evidence. *Entrepreneurship & Reg Devel: Int J* 24(5–6):383–404
- Spigel B (2015) The relational organization of entrepreneurial ecosystems. *Entrepreneurship theory and practice*, June: 7–24
- Stabell CB, Fjeldstad ØD (1998) Configuring value for competitive advantage: On chains, shops and networks. *Strateg Manag J* 19:413–437
- Stam E (2015) Entrepreneurial ecosystems and regional policy: A sympathetic critique. *Eur Plan Stud* 23(9):1759–1769. doi: [10.1080/09654313.2015.1061484](https://doi.org/10.1080/09654313.2015.1061484)
- Svensson P, Klofsten M, Etzkowitz H (2012) An entrepreneurial university strategy for renewing a declining industrial city: the norrköping way. *European Planning Studies* 20(4):505–525. doi: [10.1080/09654313.2012.665616](https://doi.org/10.1080/09654313.2012.665616)
- Tang J, (2008) Environmental munificence for entrepreneurs: entrepreneurial alertness and commitment. *Int J Entrepreneurial Behav Res* 14(3):128–151
- Taylor SJ, Bogdan R (1984) *Introduction to qualitative research methods: the search for meanings*, 2nd edn. John Wiley, New York
- Walshok ML, Shapiro JD (2014) Beyond tech transfer: a more comprehensive approach to measuring the entrepreneurial university. In: Corbett AC, Siegel DS, Katz JA (eds) *Academic Entrepreneurship: creating an entrepreneurial ecosystem* 16:1–36
- Waterhouse T (2013) Entrepreneurial Adelaide—an ecosystem', in *Business South Australia*, August–September, 72: 56–57

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**Gerard Reed** completed a Bachelor of Arts degree at the University of New South Wales where he studied Theatre, History, Politics, Film, and Religion. Gerard holds a Master of Arts degree from the University of The Arts, London where he further developed his interest in the history and production of independent film and documentary.

Gerard Reed co - founded Remo Media with Vanna Morosini a film and digital production company based at Adelaide, South Australia. Remo Media has developed a range of documentary and drama projects over several years.

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Gerard received a scholarship from the University of Adelaide and completed a Master of Entrepreneurship degree with the Entrepreneurship, Commercialisation, and Innovation Centre (ECIC) and recently a PhD research study investigating the ways in which film and digital production businesses from regional areas develop in the global market place.

# Regional Entrepreneurship Ecosystems Support: South East Queensland as Case Study

M. J. de Villiers Scheepers, E. Mealy, M. Clements  
and Anne Lawrence

**Abstract** The literature on support within entrepreneurship ecosystems is in emergence, characterized by multiple levels of analysis, different stages of development, and different conceptualizations of regionality, without the ability to extend a successful model from one region to another. Given the emergent stage of development and the need regions have to locally develop their own ecosystems, this chapter provides a framework to study the support provided by different role players within the system and proposes dimensions to study the interdependencies in providing support. The proposed framework was illustrated and revised using a case study from South East Queensland, Australia, drawing on documents, interviews and observations. This paper contributes to the literature by providing a framework, with associated dimensions of support for entrepreneurs that is useful to understand the contributions of various players to identify support gaps and develop interventions to meet the development goals of regions of varying densities.

## 1 Introduction

Policy makers today prioritize supportive systems that enable the entrepreneurial economy to grow, contributing to regional prosperity, with a focus to nurture entrepreneurship in their region (Audretsch and Thurik 2001). Productive entrepreneurship that contributes to economic output, yielding innovative, growth

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firms is of national importance (Stam 2015), and knowledge-based innovation and entrepreneurship is high on countries' agendas (NISA 2015; Kinner 2015). Regions play an enabling role to foster productive entrepreneurship and innovative activity (Huggins and Williams 2011), due to the social interdependencies and economic assets within the entrepreneurship ecosystem (Feldman 2014). These interdependencies facilitate innovation and lead to cooperation with other economic actors, not only to explore new sources of knowledge, but also to exploit existing ones within an ecosystem of entrepreneurship (Gronum et al. 2012; Porto Gómez et al. 2016).

Entrepreneurial ecosystems represent a conceptualization of entrepreneurial stakeholders from a geographic, systems perspective, attempting to explain the emergence of sustainable communities of entrepreneurs (Spigel 2015). Regional entrepreneurship support aims to provide entrepreneurs with access to various forms of capital such as social, human, financial and technological capital by government and universities. This manifests as human capital development, collaborative funding, grants and investment, and incubators and accelerators (Feldman and Francis 2004; Spigel 2015). While there is agreement that regional entrepreneurial ecosystems are critical for creating resilient economies (Ács et al. 2014; Feldman et al. 2005; Isenberg 2010; WEF 2013) research on the support within these systems are in its infancy with a limited evidence base for regions beyond the United States and Europe.

Spigel (2015) warns against the tendency to '*import best practices* from thriving ecosystems *without regard to the underlying local economic and cultural attributes on which their success depends*'. The underlying relational *interdependencies* between system elements (e.g. firms, policy-makers and universities) are critical for innovation and economic growth (Motoyama and Watkins 2014). These dynamics are home-grown within regional systems and cannot easily be replicated elsewhere. Each region is a unique configuration of ecosystem elements and there is little value in trying to measure the impact of entrepreneurship support programs without taking the contextual nature of the system into account (O'Connor and Reed 2015). Therefore, the question arises: what type of role players offer support in ecosystems, and what dimensions can be used to understand the interdependencies between various forms of role-player support.

This chapter addresses this gap by examining the support available in regional entrepreneurship ecosystems by developing a conceptual framework and illustrating this framework through the case of South East Queensland (SEQ). Three embedded case studies examine the support and dimensions of interdependencies between the entrepreneurial role players and distinctions across the three regions, focussing on productive entrepreneurship rather than small business activity. Productive entrepreneurship contributes to economic welfare in an aggregate sense and creates net social value, allowing unviable ventures to fail and those meeting market needs to thrive. In this way, we contribute to this emergent body of knowledge by focusing on the support within the system and how different configuration of support elements influence the interdependencies in different regions, emphasizing the policy and practical implications for the development of regional entrepreneurship ecosystems (REES), and further research opportunities.

The chapter is organized as follows: first, we locate this study on support in regional entrepreneurship ecosystems within the literature, showing the research gap it fills. Second, we outline the conceptual model of the regional entrepreneurship ecosystem role players, the support they provide and the differentiating dimensions, and illustrate this framework in our case study protocol for the embedded cases. This framework highlights the function of role players as system actors, knowledge generation and exchange processes, proximity, permeable boundaries, degree of industry specialization, and leadership of the system. Subsequently the methodological approach is outlined and descriptive findings compare the embedded regional entrepreneurship ecosystems, contrasting metropolitan and smaller regions at different stages of economic development. These cases demonstrate how different configurations has implications for the interdependency of support provided within these communities. Finally, this chapter concludes with a discussion of the implications for role players, with the purpose of creating an attractive, enabling environment for entrepreneurs within a region and suggests future research opportunities.

## **2 Regional Entrepreneurship Ecosystem Conceptualizations**

Local entrepreneurial actors, embedded in regional entrepreneurship ecosystems, play a leading role to use and exploit knowledge productively for customer benefits (Audretsch and Thurik 2001). These actors draw from and contribute to regional resources, integral to knowledge based activities and innovation (Stam 2015). The tacit knowledge related to innovation, embedded within local networks, is not as easily and costlessly transferable across geographic boundaries as explicit, codified information and knowledge. To facilitate innovation, local actors cooperate beyond contractual specifications. Cooperation facilitates knowledge creation and the returns from knowledge spill-overs, due to inter-industry network contact, crucial to maintaining regional competitiveness (Murdock 2012). These socio-cultural and economic considerations have policy implications in terms of regional incentives, tax incentives, collaboration with local universities, capital attraction and associated regional growth and jobs (Audretsch and Thurik 2001).

Productive, regional entrepreneurship ecosystems are fostered by a strong business community, high in social cohesion, knowledge sharing, innovation, growth and job creation. The relationships between the system role players, who provide support to local entrepreneurs and facilitate knowledge creation and exchange, are based on trust, and should be nurtured as a basis for business interactions and interdependencies (Porto Gómez et al. 2016). There is a risk however that in smaller, early stage regional entrepreneurship ecosystems (REES), vacuums might develop, where there is a lack of sufficient actors and resources (e.g. certain types of financing, regulatory barriers, or experienced mentors), or less than

optimal alignment between actors (e.g. bureaucratic processes within larger organizations hindering cooperation with smaller, flexible actors) (Stam 2015). This type of system failure could mean such regions risk losing their high-potential entrepreneurs to regions with higher endowed resources, or a larger diversity of actors.

The emergent literature on entrepreneurship ecosystems provide insight into the types of role players needed in REES, and the roles they play in these systems. This emergent literature focuses on the critical agentic role entrepreneurs play within the system (Ács et al. 2014) as depicted in Table 1, in contrast to the innovation system literature which tends to focus on the firm and institutional level of analysis (Watkins et al. 2015; Porto Gómez et al. 2016). Entrepreneurial activity within the system is focused on productive entrepreneurship, which is entrepreneurial activity that contributes directly or indirectly to the economic output, or increases the capacity to produce additional output (Baumol 1993). The notion of pillars as components of the ecosystem, proposed by the World Economic Forum (2013) depicts a static, supportive perspective, without an explicit depiction of the interdependencies between these components. Isenberg (2010) argues for a principled policy approach, where entrepreneurial actors take the lead to create a thriving entrepreneurial economy and ecosystem, however policy makers find it challenging to measure entrepreneurial aspirations. The work of Ács et al. (2014) addresses these shortcoming by examining entrepreneurship at a national level, with interactions between individual entrepreneurial attitudes, abilities and aspirations and macro-level variables, while Spiegel (2015) views an entrepreneurship ecosystem as a socio-cultural system where cultural, social and material attributes interact within a specific regional context. Stam (2015) synthesizes the popular literature (Feld 2012) and WEF (World Economic Forum 2013) pillars with a systems view and specifies framework conditions, systemic conditions, entrepreneurial activity as an output and aggregate value creation as an outcome. The national perspective of an entrepreneurial ecosystem is beneficial to see how the interrelations between system actors influence entrepreneurial activity, however given that entrepreneurial activity is often concentrated regionally (Isenberg 2010), a more focused REES perspective is needed.

Regions differ in terms of their resource endowments (Audretsch et al. 2012) and the support available to gain or leverage access to critical resources. Give the focus of this chapter on support available for entrepreneurs within regions, the diversity of population size, nature of economic activity and stage of development of a region should be accounted for. This case study does not focus on a single industry, such as tourism or technology, as that tends to be more typical of cluster or innovation ecosystem approach. Entrepreneurial ecosystems favour diversity of industrial specialization to facilitate the exchange of new ideas and greater innovative activity, as inter-industry knowledge spillover enables innovation in another industry (Audretsch and Thurik 2001). These processes are dynamic and the knowledge base, entrepreneurial capital stock available (Kibler et al. 2014) and regional resources influence the character and support within the regional entrepreneurship ecosystem (Feldman 2014).

Table 1 Summary of entrepreneurship ecosystem models

| Authors              | Pillars of entrepreneurial ecosystem                                                                                                                                                                                                                               | Domains of the entrepreneurship ecosystem                                                                                                                                                                                                                                                                                | National entrepreneurship system (GED)                                                                                                                                                                                                                                                                          | Relational organization of entrepreneurial ecosystem                                                                                                                                                                                                                                                               | Synthesis of elements, outputs and outcomes of entrepreneurial ecosystem                                                                                                                                                             |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Elements/ Components | WEF (2013)<br>Pillars<br>Accessible markets<br>Human capital<br>workforce<br>Funding and finance<br>Mentors, advisors, support systems<br>Regulatory framework and infrastructure<br>Education and training<br>Major universities as catalysts<br>Cultural support | Isenberg (2010)<br>Policy<br>Leadership and government<br>Finance<br>Financial capital<br>Culture<br>Success stories<br>Societal norms<br>Supports<br>Non-government institutions<br>Support professions<br>Infrastructure<br>Human capital<br>Educational institutions<br>Labour<br>Markets<br>Networks early customers | Ács et al. (2014)<br>Entrepreneurial attitudes and abilities<br>Opportunity perception<br>Startup skills<br>Risk acceptance<br>Entrepreneurial aspirations<br>Networking<br>Cultural support<br>opportunity Startup<br>Gender<br>technology Sector<br>Quality of human resources<br>Competition<br>Risk capital | Spigel (2015)<br>Cultural attributes<br>Supportive culture<br>Histories of entrepreneurship<br>Social attributes<br>Worker talent<br>Investment capital<br>Networks<br>Mentors and role models<br>Material<br>Policy and governance<br>Universities<br>Support services<br>Physical infrastructure<br>Open markets | Stam (2015)<br>Framework<br>Conditions<br>Formal institutions<br>Culture<br>Physical infrastructure<br>Demand<br>Systemic conditions<br>Networks<br>Leadership<br>Finance<br>Talent<br>Knowledge<br>Support services/ Intermediaries |

(continued)

Table 1 (continued)

|                         | Pillars of entrepreneurial ecosystem | Domains of the entrepreneurship ecosystem                                                           | National entrepreneurship system (GEDI)                                                                                            | Relational organization of entrepreneurial ecosystem                                                | Synthesis of elements, outputs and outcomes of entrepreneurial ecosystem |
|-------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Relational interactions |                                      | National context, regional and local activity, principle-driven policies                            | National context, stage of economic and macro-level variables influence individual entrepreneurial behaviour, outputs and outcomes | Regionally material and social; and social and cultural attributes reinforce and support each other | Mechanisms not specified, other than systemic conditions                 |
| Outputs                 |                                      | Repeated, sustaining high-potential entrepreneurial ventures                                        | Product innovation<br>Process Innovation                                                                                           | Entrepreneurial activity varies in profit and community focus                                       | (High-growth/Productive) Entrepreneurial Activity                        |
| Outcomes                |                                      | Entrepreneurship ecosystem strategy<br>Small numbers of high-potential ventures, poised for success | High growth<br>Internationalization<br>GEDI on national level                                                                      | Economic and social value creation                                                                  | Aggregate Value Creation                                                 |

### 3 Support Within Regional Entrepreneurship Ecosystems (REES)

The focus of this chapter is on support within regional entrepreneurship ecosystems (REES), where a REES is a combination of macro-, regional-, organizational and individual elements that facilitate the growth and development of productive entrepreneurship through regional resources and beyond, linked to other national and global systems (Spigel 2015). Support, in this study, is seen broadly as enabling and linking entrepreneurial actors to resources, such as social, human, financial and technological capital. By taking a broad view of support, rather than a narrow view of peripheral supporting service providers (e.g. Isenberg 2010), it allows REES role players to see the interdependencies between them. By taking a systems perspective, three sub-systems are identified within the larger REES by their function (Kenney and Von Burg 1999), namely the productive or knowledge exploitation subsystem, the knowledge generation subsystem, and socio-institutional organizations (as explained in Sect. 3.1). The REES is influenced by existing path dependency or history of a region (Spigel 2015) which informs the socio-institutional and cultural setting, the regional policy attractiveness (Robson et al. 2009), social capital generated through interactions, based on trust (Aarstad et al. 2010), and the availability of financial capital (Samila and Sorenson 2010). The support in REES is influenced by dimensions such as the character and roles of multi-level role players, proximity and regional culture, degree of industry specialization, permeable boundaries and the leadership role and autonomy norms (Feldman and Zoller 2012), as shown in Table 2, provides an initial overview of what will be investigated in this study.

#### 3.1 Multi-Level Role-Player Roles

Role players within the entrepreneurial ecosystem can be categorized into three subsystems, namely the productive or knowledge exploitation subsystem, knowledge generation subsystem and social and political subsystems.

- *Productive or knowledge exploitation subsystem*: regarded as the core of a REES, consists of existing firms and experienced entrepreneurs as leaders, as well as emerging entrepreneurial firms. This subsystem is shaped by growth-orientated firms that produce goods and services, exploiting the knowledge developed within and outside the system. The focal actors are entrepreneurs focused on value creation for existing and potential customers through knowledge exploitation (Stam 2015). Serial entrepreneurs or organizations take on leadership roles within the system and is dedicated to shaping the system's development and connecting other actors with opportunities (Feldman and Zoller 2012).
- *Knowledge generation subsystem* focuses on developing scientific and technological outcomes, characterized by (i) universities, who educate students and

**Table 2** Overview of Regional Entrepreneurship Ecosystems Model

| Dimensions                                                                                                                                                                      | Stakeholder Subsystems                                                                                                                                                                                                                                      | Support—Resources                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Multi-level role players</li> </ul>                                                                                                    | <b>Knowledge exploitation</b><br>Entrepreneurial actors<br>Serial and lead entrepreneurs                                                                                                                                                                    | Social capital                                               |
|                                                                                                                                                                                 | <b>Knowledge generation</b><br>Universities, research laboratories, large firms' R&D, education and training providers                                                                                                                                      | Human Capital<br>Knowledge<br>Talent                         |
|                                                                                                                                                                                 | <b>Socio-political subsystem</b><br>National, State and Local Government—supportive policies<br>Venture capitalists, angel investors, intermediaries<br>Specialists, incubators, and intermediaries                                                         | Financial Capital<br>Technological capital<br>Infrastructure |
| <ul style="list-style-type: none"> <li>• Proximity</li> <li>• Permeable boundaries</li> <li>• Degree of industry specialization</li> <li>• Leadership and Governance</li> </ul> | <ul style="list-style-type: none"> <li>• Geographic, cognitive, social, organizational, institutional</li> <li>• Openness</li> <li>• More industry specialization, more support and entrepreneurial activity</li> <li>• Type of leadership model</li> </ul> |                                                              |

work in cooperative innovation projects with firms and other role players; (ii) vocational and technology centers who develop technical skills and actively collaborate with local firms; and (iii) research centers, focused on basic research activities and the commercialization of this research. These institutions serve as knowledge brokers, generate knowledge and cultivate human capital. Their knowledge generation work may accidentally or deliberately create seeds for entrepreneurship through technology transfer.

- Socio-institutional organizations: consisting of private and public organizations which create an enabling environment to strengthen the social capital and promote a culture of collaboration. Facilitators or intermediaries which specialize in facilitating new firm formation, such as incubators or accelerators lawyers, public relations agencies, and capital providers fulfil functions with this system to ensure productive entrepreneurial startups are nurtured (Autio et al. 2014). These organizations are supported by policy-making institutions, regional development agencies, and associations (Mathews and Stokes 2013).

### 3.2 Proximity

The importance of proximity and a dense urban population has been emphasized in the literature as an asset for entrepreneurs (Salvador et al. 2013; Audretsch et al. 2012). Cases like the garment district in New York (Rantisi 2002), science parks in

Europe and agglomeration industrial districts are examples of micro-regional environments, devoid of political autonomy (Audretsch et al. 2012). Geographical proximity, related to living in the same regional space, is more advantageous in more densely populated areas where venues which enable networking and unplanned collisions in social venues encourage the development of social capital through more frequent interactions in certain key ‘nodes’ (Nijkamp 2003). Economic geographers have emphasized that learning is enhanced, and knowledge spillovers and innovation facilitated through geographic and other proximities, such as cognitive, organizational, social and institutional proximities (Salvador et al. 2013).

In a REES, all five types of proximity facilitate entrepreneurial activity. Cognitive proximity refers to knowledge bases of firms, and how a firm will seek and share knowledge close to their current capabilities to enable knowledge creation, access to new ideas and information sources and the potential to establish learning communities of practice (Porto Gómez et al. 2016). Organizational proximity refers to compatible organizational practices that enhance the relationships between existing firms, institutions and organizations due to having worked together on projects, participants in the same supply chain, or in a similar industry group (Watkins et al. 2015). Beyond the organization institutional proximity refers to the relationships between national, state and regional policy makers, institutions and agencies focused on developing an enabling entrepreneurial environment with the power to provide legitimacy to new firms (Porto Gómez et al. 2016), while social proximity refers to the norms, values and trust relationships between different role players in the system (Spigel 2015). Boschma (2005) argue that there are optimal levels of proximity, and too much proximity can inhibit novelty, decrease openness of the system, and lead to lock-in, therefore proximity dimensions should be balanced to ensure learning and sustainable entrepreneurial activity within the system.

### **3.3 *Permeable Boundaries***

The demarcation of a REES by geographical or location boundaries does not mean that all resources or knowledge are within-system attributes. Due to the mobility of role players and collaborative networks with external actors, the boundaries of REES need to be permeable to enable knowledge flows and knowledge spillovers (Audretsch and Thurik 2001). Empirical evidence confirms that collaboration and networking have a positive impact on the innovation performance of firms (Nieto and Santamaría 2010) and supports the notion of open innovation. Furthermore, systems should foster interactions between diverse role players, as diverse human and social capital have positive and significant relationships with regional innovation and entrepreneurial outcomes (Lee et al. 2004; Stam 2015).

### ***3.4 Degree of Industry Specialization***

Industry specialization refers to a group of firms who offer similar products and services by performing competitive and complementary activities. Regions who have technologically different but related sectors and activities benefit from geographical, cognitive, organizational, institutional and social proximities (Porto Gómez et al. 2016). As such firms might share knowledge and best practices as they have a common goal to advance the technological trajectory of a base technological innovation, for example in Waterloo for the tech industry (Spigel 2015). Diverse industries are more likely to develop new technological processes, markets or sectors due to knowledge spill-overs (Audretsch and Thurik 2001). A high degree of industry specialization and collaborative projects enable a region's endowment of entrepreneurship capital to grow, encouraging individuals willing to take the risk of starting a new firm within a supportive industry value chain (Audretsch and Keilbach 2004).

### ***3.5 Leadership and Governance***

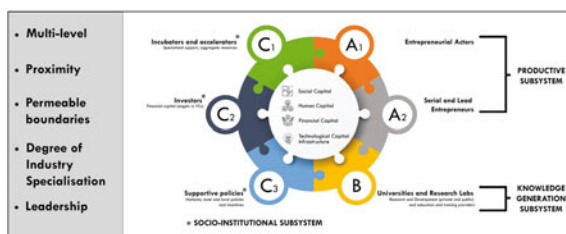
In national entrepreneurship systems, policy and governance is of central importance, as national policy and incentives can be manipulated to influence the outcomes of these systems (Trippel 2006; Isenberg 2010). However, in regional entrepreneurship ecosystems, all role players take action under the same national conditions and the emphasis of support is focused on growing and developing the system. Therefore, leadership is more likely to be bottom-up from community leaders, and does not fit a top-down governance model (Porto Gómez et al. 2016). Leaders, such as dealmakers, garner regional social capital and play an active stewardship role within entrepreneurial communities. Dealmakers are defined as individuals with valuable social capital and deep fiduciary ties within regional economies, enabling them to make new connections and facilitate new firm formations. Feldman and Zollner (2012) empirically found that dealmakers strongly influenced start-up rates in 12 US regions. While there are cultural differences between countries in terms of who should take the leadership role in REES, it is a vital role to ensure a vibrant system (Isenberg 2010). In the US, the private entrepreneurial leader is favoured in a leadership role, supported by a strong capitalist tradition, while the UK's favours economic development agencies to organize enterprising projects, social events and mobilizing others, supported by a more socialistic tradition (Morgan 2007). Regardless of who plays this role, it appears to be an important component of a REES.

### 3.6 Modelling Support in a Regional Entrepreneurship Ecosystem

The conceptual framework shown in Fig. 1 illustrates how certain dimensions of support within a REES are valuable, identifying linkages that enable entrepreneurs to access resources (social, human, financial, and technological capital and infrastructure), through formal and informal means by various role players within the local and regional system.

- **Support enablers:** The resources entrepreneurs require are varied, such as financial, social, human and technological capital, as well as infrastructure. Entrepreneurial actors often cite financial capital as the most important and scarce resource, yet the need for financial capital varies over the stages of firm development. The entrepreneurship literature strongly emphasizes the critical role of social capital for entrepreneurs (Kim and Kang 2014). For knowledge based innovation, human capital can provide a competitive advantage for a firm, both in terms of its technological base, but also in terms of talented employees. Technological capital referring to the ability to physically transform inputs to services and outputs and finally material infrastructure resources and supportive policies can provide entrepreneurs with access to the resources they need.
- **Multi-level:** Support within the REES emanates from various role players which, depending on the context is focused on multiple levels, from individual, organizational, regional, state, as well as on a national-level.
- **Proximity:** While geographic proximity within the REES can facilitate knowledge flow and innovation, other forms of proximity such as cognitive, organizational, social and institutional also influence entrepreneurial activity, however proximity has an optimum level to ensure there is sufficient openness between loosely coupled actors. Some role players within the system may identify strongly with a region and experience a sense of belonging to the region, as this relates to the cultural capital of a region (Gill and Larson 2014; Spigel 2015).
- **Permeable boundaries** ensure openness, enabling support from other systems not geographically bound within a REES. Openness ensures role players also interact with other state, national and global subsystems to acquire resources that

**Fig. 1** Dimensions related to support in Regional Entrepreneurship Ecosystems



lie beyond regional borders, such as knowledge generated in other subsystems, or attracting human or financial capital.

- Degree of industry specialization: Value chains of industries offer complimentary opportunities, when there is strong regional industry specialization. In the absence of these, entrepreneurial actors might find it difficult to access industry specific resources.
- Leadership and Governance: The literature is divided on who takes the lead within the entrepreneurship system, as the US experience suggests a ‘privatization’ approach where experienced entrepreneurs and dealmakers lead and shape the system (Stam 2015), while the European experience bestows this role upon Development Agencies (Morgan 2007).

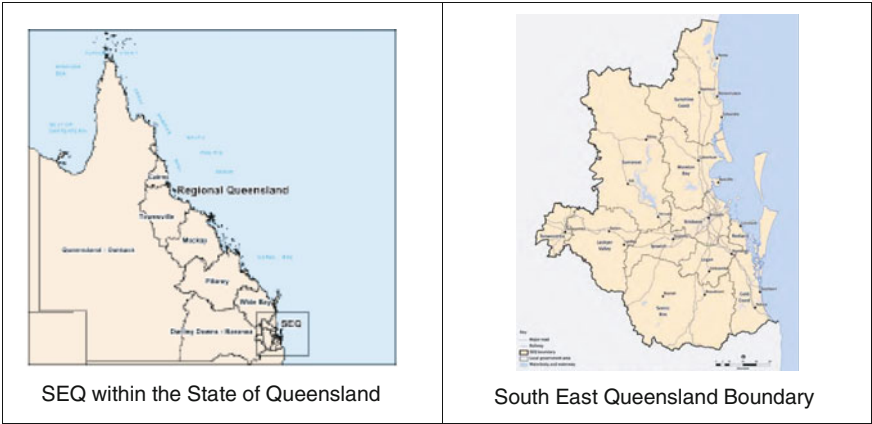
The dimensions depicted in Fig. 1 were investigated through the case of South East Queensland (SEQ) and embedded case studies.

## 4 Methodology

This study uses a single case study design, with three embedded cases to address the research questions. The single case is SEQ, an emerging entrepreneurial region, covering a territory of 22,420 km<sup>2</sup> with a population of 3.27 million inhabitants, as the most populous region within Queensland, as shown in Fig. 2. The unemployment level amounts to 6.3% and the GDP per capita in this region is AU\$269,866 million (ABS 2011). Three embedded cases within this region were studied: Brisbane, as the capital city of Queensland; the Gold Coast, 78 km south of Brisbane, as the fastest growing region in Queensland at 3.8% population growth; and the Sunshine Coast, 80 km north of Brisbane, as the second fastest growing region in Queensland with 3.4% population growth.

The research design is a suitable methodological approach to explore the relationships among regional role players in each context, as the regions experience similar national and state policies, with variations in regional population and entrepreneurial capital. Descriptive analysis is an effective way to study and understand entrepreneurship and innovation practices in emerging research areas, where limited studies have been conducted (Huizingh 2011). This qualitative approach was used to bring insights regarding the support and dimensions of interdependencies between entrepreneurial role players and distinctions across the three regions’ REES.

The REES dimensions and support depicted in Fig. 1 was used as the theoretical frame for the case study protocol. The protocol specified the documentation process, data collection procedures and analysis of findings. Multiple data sources were used to gain insight into the structure, relationships, support and dimensions within each embedded ecosystem. These data sources include: existing documentation about support within local REES (including 30 policy, community and governmental documents); 36 semi-structured interviews with five types of key informants



**Fig. 2** Regional Queensland and South East Queensland. *Source* Queensland Government Statistician’s Office 2015

in three REES, and observations to augment and support the data, as shown in Table 3. In this way ‘rich’ evidence was gathered of different forms of entrepreneurship support and relevant comparisons can be made.

Interviews took place between June and October 2016 with key informants in Brisbane, the Gold Coast and the Sunshine Coast. Interviews were recorded and transcribed to enable thematic analysis. On average interviews lasted 90 min, with a total of 54 h of transcribed interviews. To address the multiple dimensions of REES, interviews were open and focused on the structure, relationships and dimensions in Fig. 1. Interview questions covered the following themes, dependent on the role of the key informant:

- Role of stakeholder, support provided and demographics of entrepreneurs, as well as stage of development
- The entrepreneurship support available and accessed in the region and observable benefits of support
- Barriers to support and suggestions for improvement
- Culture of collaboration between role players in the region and how opportunities are exploited
- Regional attraction and leadership roles within REES.

Table 3 provides a comprehensive list of the data sources, 36 interviewees and their profiles.

Seven growth orientated entrepreneurs, with a track record of success and multiple start-ups were interviewed as key informants of the knowledge exploitation subsystem. These entrepreneurs operate in diverse industries, namely mobile technology, health, clean energy and the gaming industries. All entrepreneurs grew their firms, with the smallest firm employing 7 employees. Sixty percent of these firms have customers overseas. In addition, seven serial, male entrepreneurs who

**Table 3** Data sources and key informants interviewed in three embedded regions

| Data source                                              | Scope of data                                                                                                                                                                                                             |                                                                                                                                                                                                                                                     |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Written documents                                        | 30 regional policy, community and governmental documents                                                                                                                                                                  |                                                                                                                                                                                                                                                     |
| Observations                                             | 45 h of observations and notes                                                                                                                                                                                            |                                                                                                                                                                                                                                                     |
| Key informants                                           | Sub-region                                                                                                                                                                                                                | Profile                                                                                                                                                                                                                                             |
| Active entrepreneurs<br>(n = 7)                          | Brisbane (n = 2) [EB <sub>1</sub> ; EB <sub>2</sub> ] <sup>a</sup> ,<br>Sunshine Coast (n = 3) [ESC <sub>1</sub> ;<br>ESC <sub>2</sub> ; ESC <sub>3</sub> ], Gold Coast<br>(n = 2) [EGC <sub>1</sub> ; EGC <sub>2</sub> ] | Growth orientated<br>entrepreneurs with track record<br>of success and multiple<br>startups, with minimum 7<br>employees in technology,<br>health, clean energy and the<br>gaming industries, 3 female and<br>4 male                                |
| Lead entrepreneurs<br>(n = 7)                            | Brisbane (n = 3) [LEB <sub>1</sub> ; LEB <sub>2</sub> ,<br>LEB <sub>3</sub> ]<br>Sunshine Coast (n = 3)<br>[LESC <sub>1</sub> ; LESC <sub>2</sub> , LESC <sub>3</sub> ]<br>Gold Coast (n = 1) [LEGC <sub>1</sub> ]        | Diverse industries, international<br>and local commercialization<br>experience, all male serial<br>entrepreneurs, received<br>investment and have exit<br>experience. Play leadership<br>roles within sub-regions                                   |
| Universities,<br>knowledge-production<br>systems (n = 5) | Brisbane (n = 2) [UB <sub>1</sub> ; UB <sub>2</sub> ]<br>Sunshine Coast (n = 2)<br>[USC <sub>1</sub> ; USC <sub>2</sub> ]<br>Gold Coast (n = 1) [UGC <sub>1</sub> ]                                                       | University professors engaged<br>with entrepreneurship<br>ecosystem and academic<br>entrepreneurship in business<br>faculties, with research and<br>teaching focus                                                                                  |
| Local government<br>(n = 5)                              | Brisbane (n = 1) [LGB <sub>1</sub> ]<br>Sunshine Coast (n = 3)<br>[LGSC <sub>1</sub> ; LGSC <sub>2</sub> , LGSC <sub>3</sub> ]<br>Gold Coast (n = 1) [LGGC <sub>1</sub> ]                                                 | Local government economic<br>development managers<br>responsible for regional<br>economic development growth<br>plan in sub-regions                                                                                                                 |
| Incubator managers<br>(n = 7)                            | Brisbane (n = 3)<br>[InB <sub>1</sub> ; InB <sub>2</sub> , InB <sub>3</sub> ]<br>Sunshine Coast (n = 3)<br>[InSC <sub>1</sub> ; InSC <sub>2</sub> , InSC <sub>3</sub> ]<br>Gold Coast (n = 1) [InGC <sub>1</sub> ]        | Incubators within each<br>sub-region, with<br>entrepreneurial experience and<br>a track record. Differentiation in<br>Brisbane refined, Sunshine<br>Coast emerging, Gold Coast<br>incubator closed down, now<br>only co-working space and<br>events |
| Investors (n = 5)                                        | Brisbane (n = 2)<br>[AB <sub>1</sub> ; AB <sub>2</sub> ]<br>Sunshine Coast (n = 2)<br>[ASC <sub>1</sub> ; ASC <sub>2</sub> , ASC <sub>3</sub> ]<br>Gold Coast (n = 1) [AGC <sub>1</sub> ]                                 | Angel investors successfully<br>exited at least three ventures,<br>affiliated with local investor<br>networks, with more than<br>25 years shared experience                                                                                         |

<sup>a</sup>Key informants are identified by their stakeholder group, such as E for Entrepreneur, and then region, such as B for Brisbane, result in the code EB<sub>1</sub>

fulfil leadership roles in their regions were interviewed. All of these entrepreneurs had experience of receiving investment and exiting from one or more ventures. They had a holistic perspective of entrepreneurial activities within their region.

In the knowledge generation subsystem, five university professors, who are extensively engaged in their local entrepreneurial ecosystem were interviewed. All five interviewees had a research, teaching and engagement focus related to entrepreneurship.

In the socio-institutional subsystem local government, incubator managers and investors were interviewed. Five key informants across the three regions were interviewed, as they were responsible for enabling the regional economic development strategy for their regions, often working closely with entrepreneurs. Seven incubator managers were interviewed, with the highest number in Brisbane, then the Sunshine Coast and a co-working space in the Gold Coast. These individuals had unique insights into the entrepreneurial development of their regions. Finally, five investors were interviewed who had successfully exited at least three ventures and were active angel investors.

## **5 Regional Entrepreneurship Ecosystem in South East Queensland**

South East Queensland (SEQ) produces two thirds of the state's gross regional product (ABS 2014). The regional areas of Brisbane, Gold Coast and Sunshine Coast are endowed with human capital and have nine research universities with research clusters, attracting large numbers of international students. These embedded cases within SEQ range in population size and density, with Brisbane City being the largest, most economically developed area, with a population of 1.16 million people and a population density of 8.76 persons per hectare. The mid-size region is the Gold Coast, with a population of 555,608 people and a population density of 3.71 persons per hectare. In contrast, the smallest of the three regions is the Sunshine Coast, with a population of 287,539 people and a population density of 1.26 persons per hectare, as shown in Table 4.

An overreliance on the mining industry in Queensland has resulted in the call nationally for a more diversified economy from industries such as mining, construction, tourism and retail, to other high-value industries (NISA 2015). Significant investment in several infrastructure projects in the SEQ area is expected to create economic growth opportunities for entrepreneurs in associated industries in the value chain. SEQ is characterized by the highest employment in health care and social assistance (12.2%), followed by retail trade (10.7%), construction (9.1%), manufacturing (8.7%), education and training (7.9%), professional services (7.7%) and accommodation and food services (6.9%).

Policy makers are committed to national innovation and growth during the past three years, emphasizing the importance of growing the knowledge and technology

**Table 4** Demographic Data of Brisbane, Gold Coast, and Sunshine Coast regions

|                                     | Brisbane                                   | Gold Coast                                                | Sunshine Coast                                               | SEQ                                               |
|-------------------------------------|--------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------|
| <i>2011 Census data descriptive</i> |                                            |                                                           |                                                              |                                                   |
| Population 2015                     | 1,162,186                                  | 555,608                                                   | 287,539                                                      | 3.27 million persons<br>(70.2% of Qld population) |
| Land area                           | 132,618 hectares/<br>km <sup>2</sup>       | 133,372 hectares/<br>km <sup>2</sup>                      | 229,072 hectares/<br>km <sup>2</sup>                         | 22,420 km <sup>2</sup>                            |
| Population growth last 10 years     | 1.9%                                       | 3.8%                                                      | 3.4%                                                         |                                                   |
| Population density                  | 8.76 persons/<br>hectare                   | 3.71 persons/<br>hectare                                  | 1.26 persons/<br>hectare                                     | 1.29 persons/<br>hectare                          |
| Gross regional product              | AU\$129 billion<br>\$64 500 GRP per capita | AU\$26.89 million<br>9.3% GRP of Qld<br>\$43,706 p/capita | AU\$12.36 billion<br>4.3% of GRP of Qld<br>\$39 747 p/capita | 2.3 GDP growth (0.6% per capital growth)          |
| <i>Age Distribution</i>             |                                            |                                                           |                                                              |                                                   |
| 18–24 years (tertiary)              | 216,152 (10.5%)                            | 47,507 (9.6%)                                             | 18,853 (7.3%)                                                |                                                   |
| 25–34 years (young workforce)       | 306,627 (14.84%)                           | 68,464 (13.8%)                                            | 26,514 (10.3%)                                               |                                                   |
| 35–49 years                         | 447,197 (21.6%)                            | 106,332 (21.5%)                                           | 54,503 (21.1%)                                               |                                                   |
| <i>Birth place</i>                  |                                            |                                                           |                                                              |                                                   |
| New Zealand                         | 4.8%                                       | 8.6%                                                      | 4.9%                                                         |                                                   |
| UK                                  | 5.3%                                       | 6.6%                                                      | 7.3%                                                         |                                                   |
| South Africa                        | 0.9%                                       | 1.2%                                                      | 0.8%                                                         |                                                   |
| Japan                               | 0.2%                                       | 0.7%                                                      | Germany—0.7%                                                 |                                                   |
| China                               | 1.0%                                       | 0.7%                                                      | Netherland—0.5%                                              |                                                   |
| Bachelor or higher degree           | 20.1%                                      | 14.3% (57,643)                                            | 14% (29,266)                                                 | 18.4% (429,047) 2011 data                         |
| Unemployment                        |                                            | 2016 data—June                                            | 2016 data June                                               | 2011 data                                         |
|                                     | 62 895 (5.9%)                              | 319,941 (5.39%)                                           | 149,793 (4.94%)                                              | 92,749 (6.3%)                                     |
| Industry                            |                                            |                                                           |                                                              | 2011 data                                         |
| Health care and social assistance   | 12.4%                                      | 11.1%                                                     | 13.6%                                                        | 12.2%                                             |
| Retail trade                        | 10.1%                                      | 12.4%                                                     | 12.6%                                                        | 10.7%                                             |
| Construction                        | 8.3%                                       | 11.3%                                                     | 11.9%                                                        | 9.1%                                              |
| Manufacturing                       | 7.36%                                      | 7.6%                                                      | 6.7%                                                         | 8.7%                                              |

(continued)

**Table 4** (continued)

|                                                 | Brisbane | Gold Coast | Sunshine Coast | SEQ       |
|-------------------------------------------------|----------|------------|----------------|-----------|
| Education & training                            | 8.1%     | 7.1%       | 8.3%           | 7.9%      |
| Professional, scientific and technical services | 8.3%     | 6.3%       | 5.9%           | 7.7%      |
| Accommodation and food services                 | 6.0%     | 9.6%       | 8.5%           | 6.9%      |
| Occupation                                      |          |            |                | 2011 data |
| Professionals                                   | 22.2%    | 17.4%      | 18.0%          | 20.9%     |
| Clerical and admin                              | 16.2%    | 14.5%      | 13.9%          | 15.6%     |
| Technicians and trade                           | 13.5%    | 15.4%      | 16.3%          | 14.1%     |
| Managers                                        | 12.3%    | 12.3%      | 12.0%          | 11.8%     |
| Sales                                           | 12.7%    | 12.7%      | 11.6%          | 10.2%     |
| Community and personal services                 | 10.9%    | 10.9%      | 10.7%          | 10.0%     |

Source Australian Bureau of Statistics (2011–2015)

industries. Nevertheless, a large number of small businesses dominate the SEQ business population. Immigrant and Australian entrepreneurs contribute to the entrepreneurial capital of the region. Immigrants are attracted to Queensland due to its desirable climate, high standard of living and associated opportunities. In SEQ immigrants predominantly come from the United Kingdom (6.4%), New Zealand (6.1%), South Africa (0.97%), and China (0.6%). Between 14 and 20% of the population have a Bachelor degree or higher.

Regarding financial capital, venture capital investment in Queensland and Australia as a whole does not compare favourably with the rest of the world. Venture capital investment per capita in SEQ is AU\$5, \$1 more than the Australian average. However, in the UK, venture capital investment per capita is AU\$15; in Israel, AU\$183; and in Silicon Valley, AU\$4241 (Markham et al. 2016). That being said, venture capital and private equity investments are growing to support the transitioning Australian economy (AVCAL 2016).

The number of start-ups in the technology and knowledge industry sectors are growing (Kruger and Cacioppe 2014), and significant State government investment and investment attraction programs are focused on encouraging this growth. Brisbane, as the capital city, is home to more than 50 listed companies, however in the Gold Coast and Sunshine Coast employment opportunities provided by larger companies are limited (Daquino and Rickert 2016).

South East Queensland provides a rich context to examine regional entrepreneurial ecosystems in three stages of development, examining the diversity of role

**Table 5** Configuration of entrepreneurial role players and support in three REES

|                             | Brisbane                                                                                                                                                                                                                                                                                                                                                                       | Gold Coast                                                                                                                                                                                                                             | Sunshine Coast                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-level<br>role players | Higher population<br>Large number of entrepreneurial actors in all age cohorts, mix of lead entrepreneurs                                                                                                                                                                                                                                                                      | Mid-level population and growing<br>Entrepreneurial actors, immigrant and experienced entrepreneurs                                                                                                                                    | Smallest population<br>Smaller number of entrepreneurial actors, international market focus for growing firms                                                                                                                                                                                                                                         |
|                             | Three main campus-based world-class universities and research centers, established university hospitals                                                                                                                                                                                                                                                                        | Three main campus-based universities, new world-class hospital                                                                                                                                                                         | Two regional universities with one campus, recent opening of world-class hospital                                                                                                                                                                                                                                                                     |
|                             | Numerous incubators and growth in accelerator programs                                                                                                                                                                                                                                                                                                                         | No incubators, only co-working spaces                                                                                                                                                                                                  | Two incubator spaces with one sector-specific accelerator program, growing number of co-working spaces                                                                                                                                                                                                                                                |
|                             | Well-endowed city council and state government, located in same region<br>Supportive policies, investment in entrepreneurship-related infrastructure.                                                                                                                                                                                                                          | Active city council, focused on event attraction<br>Regional support for entrepreneurship                                                                                                                                              | Supportive, connected local council, emerging hub of innovation and collaboration                                                                                                                                                                                                                                                                     |
|                             | Formal angel investment network                                                                                                                                                                                                                                                                                                                                                | Some level of private sector investment                                                                                                                                                                                                | Very limited finance network                                                                                                                                                                                                                                                                                                                          |
| Support                     | Social capital in strong and weak tie relationships with specific networks, concentrated in micro geographies<br>Human capital talent developed from local universities, developed by local industry<br>Financial capital accessible through formal angel investor networks, limited venture capital<br>Technological capital, accessible from knowledge generation sub-system | Social capital through meetups in co-working spaces<br>Human capital talent from local universities and immigrant inflow<br>Limited financial capital, often from outside the region<br>Emerging technological capital drawn to region | Social capital through multiple networking events and local collaborations, facilitated by incubator and accelerator spaces<br>Human capital talent from local universities and immigrant inflows in region<br>Very limited financial capital within the region, seek to attract outside region investors.<br>Limited technological capital available |

(continued)

**Table 5** (continued)

|                                   | Brisbane                                                                                                            | Gold Coast                                                                          | Sunshine Coast                                                                                          |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Proximity                         | Higher population density—higher geographical proximity in precinct areas                                           | Medium population density, social and cognitive proximities                         | Low population density, meetups generate social proximity                                               |
| Permeable boundaries              | Outside the region, limited to GC and SC, aims to attract entrepreneurial actors from GC and SC to support programs | Draw resources from Brisbane and immigrant networks provide international access    | Draw resources from Brisbane, and serial and immigrant entrepreneurs provide international access       |
| Degree of industry specialization | High<br>Established larger firms<br>Emerging tech industry, and Emerging creative industry                          | Developing<br>Mostly tourism and sport events, retail and health care industries    | Limited industries, diverse with growing firms in relation to population density, developing industries |
| Leadership                        | Distributed, visible investor network, state government and local city council<br>Prominent entrepreneurs           | Distributed leadership within role-player networks e.g. universities, local council | Distributed, collaborative leadership focused on the promotion and growth of high potential ventures    |

players who provide support, proximity and regional culture, permeable boundaries, degree of industry specialization, and leadership, as shown in Table 5.

### 5.1 Role Players Providing Support Within SEQ

The following section present the three subsystems providing support to entrepreneurs in SEQ, and the three embedded cases/regions, summarized in Table 5.

**Productive or knowledge exploitation subsystem:** Entrepreneurs active in the SEQ region participate in diverse industries, as the region transitions from a retail, tourism and construction economy to a knowledge economy. In this subsystem entrepreneurs in the early and growth stages play an important role to exploit opportunities, while a number of serial entrepreneurs play a leadership and mentoring role within their communities. Across SEQ key informants viewed entrepreneurs in three age and development cohorts. Key informants viewed the young cohort (<25 years) as having limited experience of industry problems, but typically involved in a large number of activities and events, which stimulate early interest among this group. Those who achieve success in this cohort are small in numbers, but achieve significant media attention, presenting *‘role models for others to demonstrate the benefits of entrepreneurship, however the impact of this cohort’s*

*entrepreneurial ventures is limited*' [InSC<sub>3</sub>]. In contrast the older cohort (25–35 years and 35–45 years) have more industry experience and are able to identify and address specific problems. A large number of startups in this cohort are in the incubation and early trading stage. *'Their ideas have higher potential and founders have better developed networks, however this cohort may lack the technical (and technological) skills'* [InSC<sub>3</sub>] to move their ventures forward. The last cohort are older than 45 years, have achieved initial customer success and have earned revenue, however they *'need support to grow with their ventures'* [LEB<sub>2</sub>], which have higher potential.

In Brisbane 25.3% of the population has tertiary education, seek independence and are part of the young workforce (18–25 year olds). This age group is smaller on the Gold Coast, with 23.4% of the population, with only 17.6% of this population group on the Sunshine Coast. This suggests that more entrepreneurial activity within this age cohort would be expected in Brisbane. Comparatively, the older cohort (35–49 years) make up one fifth of the population (21%) in all three areas. Thus, proportionately similar entrepreneurial activity is expected from these groups, across the three regions.

Serial entrepreneurs within SEQ play a critical role, as they have experience of founding, growing and exiting multiple ventures. They often serve as role models within their communities in Brisbane, Gold Coast and Sunshine Coast and *'play a leadership role'* [EB<sub>2</sub>] to develop the entrepreneurial capacity of these regions. There are a higher number of serial entrepreneurs in Brisbane and the Gold Coast, compared to the Sunshine Coast, who support and nurture entrepreneurial talent within this region. However, the experienced, immigrant entrepreneurs who settle on the Sunshine Coast should not be overlooked, as their *'expertise is under-utilized'* [ESC<sub>2</sub>] and their networks could potentially provide an opportunity for international expansion.

For many entrepreneurial ventures, especially those based on the Sunshine Coast, the local market is too small to enable growth, therefore they need to access markets outside the region, and some early stage ventures therefore focus on internationalization. Despite this realization, an early stage entrepreneur states that *'the fact that we run digital businesses should mean that we don't need to be near our clients, however there are still a lot of traditional ways of doing business and deals and working close with your first clients hands-on'* [EB<sub>1</sub>]. Others claim that Australian ventures are still *'struggling to grasp globalization'* [EGC<sub>2</sub>] and *'that it has never been easier to start a start-up and more difficult to sustain one'* [LESC<sub>3</sub>]. For many entrepreneurial ventures in smaller regions such as the Sunshine Coast and Gold Coast, this requires extensive travel to meet clients, suppliers and investors.

**Knowledge generation subsystem:** Two types of knowledge organizations are identified across the region. On the one hand, vocational training centers develop human capital through education and teaching, as well as technological optimization of firms through innovation projects. In SEQ Education and Training is a

substantial industry (8%), with public and private providers of training. Brisbane is home to three universities with world-class reputations, namely the University of Queensland, Queensland University of Technology and Griffith University, and a large number of vocational training centers, including TAFE Brisbane. On the Gold Coast graduates come from Griffith University, Bond University, Southern Cross University as well as TAFE Gold Coast, while on the Sunshine Coast, the University of the Sunshine Coast (USC) and Central Queensland University, as well as TAFE East Coast deliver skilled graduates. Most universities offer programs that stimulate entrepreneurial activity among students and with the community, with student start-ups, projects and internships with entrepreneurial ventures.

On the other hand, research centers focused on advancing technological (science-based) knowledge through radical and/or incremental innovations are also located in the region. Research centers are typically associated with universities, and the research is commercialized in some cases. In particular, the University of Queensland, Brisbane has the longest history of this for example its cervical cancer vaccines and its mining research center. The university hospitals in the Gold Coast and Sunshine Coast, both have associated research centers, which should develop research to further knowledge in future. In addition, the Commonwealth Scientific and Industrial Research Organization (CSIRO) has a branch in Brisbane and offers its ON innovation accelerator program, focused on converting research knowledge to beneficial, commercial outcomes for society. These knowledge and research resources provide a solid foundation for knowledge based entrepreneurship.

**Socio-institutional subsystems providing support:** This subsystem involves a multilevel public and private support network, located with the Queensland state, supported by Federal nation-wide innovation initiatives, as shown in Table 5. The National Science and Innovation Agenda and Advance Queensland have made significant investments to drive economic growth, through innovation and entrepreneurship in the science and technology fields. This strategy is further enacted on a local level through economic development strategies in the Councils of Brisbane City, Gold Coast and Sunshine Coast favouring infrastructure development and investment attraction, with selected programs providing opportunities for entrepreneurial ventures.

In addition, incubators, accelerators and co-working spaces provide SEQ entrepreneurs with a mix of physical spaces, events, structured programs and networking events. Brisbane has experienced a rapid growth in these spaces and programs, with iLab, associated with the University of Queensland, and River City Labs, a co-working space with mentors and structured programs, offering these services initially. In addition, the last three years has seen the rise of Creative Enterprise Australia, an initiative targeting creative startups by Queensland University of Technology, Little Tokyo Two, The Coterie, Fishburners and Blue Chilli to name but a few. This growth in facilitating institutions has led to a higher degree of differentiation between providers, and more refined value offering to entrepreneurs. The Gold Coast has a clearer delineation between co-working spaces

such as Wotso Workspace, Club Orange and the Exchange, among others, and supportive meetup groups such as Silicon Beach and Young Professionals, yet there is no organisation devoted to incubation. The Gold Coast Innovation Center closed in June 2013, after five years of operation (Griffith University 2013).

The Sunshine Coast, in contrast, has two dedicated incubators, with a variety of value-adding offerings, namely the Innovation Center Sunshine Coast, associated with the University of the Sunshine Coast (USC) and Spark Bureau, offering co-working and structured programs and events. Beyond these incubators, co-working spaces have also grown rapidly in the area, with more than nine currently open, with some established players, such as Noosa Boardroom and many others in the early stages. These facilitators provide valuable programs and offerings, but the biggest value entrepreneurs derive from these facilitator spaces and events is networking with like-minded individuals, building social capital and collaboration within a region.

A number of associations and meetup groups advocate for the needs of entrepreneurs and provide interactive hubs for learning, sharing of resources, social proximity and social capital. The large number of professional services occupations also suggest that financial, legal and marketing services are outsourced by entrepreneurial firms. In addition, the Australian Government provides Research and Development Tax Incentives for innovative firms, and Advance Queensland support grants, events and mentoring for growth initiatives. While State and Federal programs are accessible to all in different regions, the specialist intellectual property, financial and legal advice early stage entrepreneurial ventures need are more easily accessible in Brisbane, compared to the Sunshine Coast. Therefore, entrepreneurs in smaller regions often need to integrate with the expertise in other areas.

## 5.2 Proximity

SEQ as a region *is not generally perceived as an entrepreneurial region* [LGGC<sub>1</sub>], compared to Sydney and Melbourne, seen to offer more attractive ecosystems to entrepreneurs and investors [EB<sub>1</sub>]. For SEQ this means that the priority of the region is to establish its credentials and build its *reputation as an entrepreneurial hub* [LGB<sub>1</sub>], through developing the regional culture. One way to do this is by promoting role-models through *'telling the stories'* [LGB<sub>1</sub>] of successful entrepreneurs in the region, however *'Brisbane and surrounds is still seen by those outside Queensland as a small country town'* [LGB<sub>1</sub>], not a place where entrepreneurial ventures can thrive. This means that facilitators and local development agencies face the challenge of *'raising the self-perception of entrepreneurs'* [LGB<sub>1</sub>], to increase their growth ambitions as *'many in Australia are still struggling to grasp globalization'* [LESC<sub>3</sub>]. These sentiments are supported by angel investors from SEQ, who claim to have 75% of their investments in startups in Melbourne and Sydney, and only 25% in Brisbane [ASC<sub>1</sub>]. A serial entrepreneur also acknowledges this cultural attitude as problematic locally, saying: *'If you go into a*

*bar in San Francisco everyone is talking about start-ups, they're talking about their ideas, and it is socially acceptable to be doing a start-up. Here I still can't even tell my friends [what I do] because they get bored or laugh me out of the room. It's not something you can discuss in a social environment. Even for the younger generation it is just not socially acceptable to be talking about it' [LEB<sub>3</sub>].* Therefore the SEQ region face not only perceptual barriers from outside the region, but also the critical mass and entrepreneurial attitudes within the region.

Local Councils in Brisbane, the Gold Coast and the Sunshine Coast focus on promoting their regions as desirable places to live and work. A high level of sophistication is evident in Brisbane City Council's marketing of the region through its website, with targeted messaging at different groups including entrepreneurial talent. This message is amplified through the Queensland State Government's presence and collaboration in close proximity. The Gold Coast's economic development message is part of the Council's website and investment projects: growth, a supportive approach to industry, a strong workforce and lifestyle are promoted as part of the value proposition of this region, with messaging targeted at different stages of the development of a venture, namely early stage start-up and competitive growth. The Sunshine Coast Council's message is also part of its website and is formulated first on what the region offers geographically, and then through infrastructure and education.

Regional proximities of firms, knowledge creators and intermediaries is advantageous for entrepreneurs and firms as it enables networking and unplanned collisions, facilitating social capital and the sharing of knowledge (Nijkamp 2003). The higher density in Brisbane (8.76 persons/hectare), compared to the Sunshine Coast (1.26 persons/hectare) suggest the beneficial outcomes of geographic and social proximity. The advantage of smaller regions, however is that it is easier for new migrants and immigrants to become part of the entrepreneurial networks. Proximity in Brisbane, the Gold Coast and Sunshine Coast can be analysed more specifically by cognitive, organizational, institutional and social proximity, as summarized in Table 5.

- i. Cognitive proximity in the SEQ region is justified by the productive relationship among local firms within industries, and between local intermediaries, especially incubators and knowledge generators, such as universities and TAFEs. The co-location of firms, intermediaries and regular networking events between role players facilitate cognitive proximity.
- ii. Organizational proximity refers to working relationships between role players on larger projects. The economic development branches of local councils in all three regions demonstrate a track record in this area. Similarly, incubator and structured accelerator programs bring together diverse organizations, such as start-up and growth ventures, investors, professional service firms and knowledge generators in some cases. Collaboration on projects has a direct influence on the dissemination of local knowledge and provide access to external networks and knowledge.

- iii. Institutional proximity refers to the relationships between national, state and local development departments to develop an enabling entrepreneurial environment. Brisbane shows the strongest institutional proximity, with the State Government and Brisbane City Council located in the same region.
- iv. Social proximity refers to the values and norms embedded in the relationships between role players. The comparably smaller Sunshine Coast region's entrepreneurial ecosystem shows a high degree of social proximity, with many entrepreneurs finding intermediaries and the local council supportive as trial users of new innovations. While the Gold Coast Council is generally seen as supportive, there is *'a lack of connection between people and programs ... [there is a] large number of initiatives, but little umbrella branding'* [AGC<sub>1</sub>]. Brisbane has a higher density with a large number of events, but similar to the Gold Coast, while the collaboration is there in spirit, there is a *'high level of fragmentation, with support distributed across hubs and nodes'* [LGB<sub>1</sub>].

### 5.3 Permeable Boundaries

Permeable boundaries and the degree of openness reflects the flow of sources of capital (human, financial, intellectual) across geographical boundaries, and industry boundaries where best practice is shared to diffuse knowledge and innovations (see Table 5). While there is evidence that a moderate degree of openness occurs in the SEQ region, some entrepreneurs express their concerns about the low number of high growth firms. *'In Sydney there were three unicorns in a room from the most famous accelerators in Silicon Valley and that was really interesting because they were in their late twenties and the advice they were giving....and what they were learning...it was so different to Brisbane'* [EB<sub>2</sub>]. Access to such role models facilitate the flow of knowledge, in the words of a Sunshine Coast entrepreneur: *'We need to get out ...tap into what's happening around the world...not just isolating ourselves here... we need to be travelling a lot and making connections and bringing back ... that wealth of knowledge'* [ESC<sub>2</sub>]. These comments reflect that the degree of openness can be improved internationally, nationally and also between regions. As noted by a lead entrepreneur: *'Sunshine Coast needs more cross-pollination with Brisbane ... [which] could be a two-way street with opportunities. We need to nurture the natural flow of talent between the ecosystems.'* [LESC<sub>1</sub>]. These shared learnings would improve the innovativeness of entrepreneurial activities in different regions.

### 5.4 Degree of Industry Specialization

There is not one dominant industry in SEQ and the current top seven industries are health care and social assistance (12.2%), retail trade (10.7%), construction (9.1%),

manufacturing (8.7%), education and training (7.9%), professional services (7.7%) and accommodation and food services (6.9%). Nationally, there has been a strong emphasis on innovation and growth the past two years from policy-makers, emphasizing the importance of growing the knowledge and technology industries, and within SEQ these knowledge industries are promoted, especially those linked to university research centers and CSIRO. The growth in emerging industries is linked to infrastructure development and investment projects, with the importance of a globally competitive broadband infrastructure a major issue for Australian ventures.

## 5.5 Leadership and Governance

Supported empirically, the entrepreneurs and lead entrepreneurs in the productive sub-system agree that the leadership role in REES should be undertaken by experienced entrepreneurs with valuable social capital and fiduciary ties to the community (Feldman and Zoller 2012). Entrepreneurs and incubator managers concur that: *‘whoever leads should have built a successful start-up and exited... credible, successful entrepreneurs’* [ESC<sub>3</sub>]; and *‘Ultimately leadership or the catalyst for these ecosystems is a very small number of successful entrepreneurs who have decided to invest back into their ideas and foster the next generation’* [ASC<sub>1</sub>]. The knowledge generators and socio-institutional subsystems are seen as too slow and unresponsive to fit the entrepreneurial way of working on many early stage ventures. However, as the supply of dealmakers may be limited in regional areas, leadership of the entrepreneurial ecosystem as a whole supports a distributed leadership model with a unifying vision, led by champions in different industries, as shown in Table 5. Leadership roles should focus on providing value to the community first, as self-interest could damage the credibility of community building in this area. In Brisbane and the Sunshine Coast incubators and co-working spaces play a leadership role, aggregating valuable information, mentors and resources. These organizations provide visibility to the REES.

## 6 Discussion

The model used in this chapter applied to the SEQ REES and three embedded regions was useful to the key informants who participated in the study to gain a more refined understanding of their roles, as well as the cumulative dimensions of support within the entrepreneurial ecosystems. Table 5 illustrates how population, economic and demographic dimensions are interdependent in influencing entrepreneurial outcomes, similar to Isenberg’s (2010) view that regions should aim to cultivate one high-potential venture per 100,000 residents.

An aspect that is sometimes overlooked in REES models is market access (Isenberg 2010; Porto Gómez et al. 2016). In regional communities, with smaller

markets, entrepreneurs often need to cultivate clients in other regions or consider trading internationally. This is the reality for many Australian ventures, as the domestic market is relatively small and growth orientated ventures then seek Asian and other international markets. The implication is that internationalization often form a cornerstone to enable growth, and the capacity, competitiveness and skills to enter international markets should be offered as part of the support in the ecosystem. In growing entrepreneurial talent among early stage entrepreneurs, the value trade delegations and trade shows should form part of the support provided.

For SEQ and embedded regions there are few larger firms, which play a critical role as knowledge generator and as a potential client. In cities such as Brisbane, this is less of a problem, compared to smaller regions such as the Sunshine Coast. In smaller regions, the involvement of larger firms with entrepreneurial ventures should be cultivated as entrepreneurial ecosystem partners.

The limited degree of industry specialization and R&D facilities in smaller regions such as the Gold Coast and Sunshine Coast limit the trajectory of innovations. Therefore, the three subsystems of knowledge exploiters, knowledge generators and socio-institutions should collaborate to develop strong industries. The health sector in both regions could form a hub for healthcare innovations, given the medical research centers associated with university research hospitals.

Empirically the value of leadership in entrepreneurial communities has been shown to be beneficial, however given the diversity of role players and industries in which entrepreneurship occurs, it is more likely that REES will be facilitated by a distributed leadership model. However, these leadership efforts should be coordinated as there is a tension to navigate fragmentation at the cost of collaboration.

Finally, REES should nurture entrepreneurial talent of its younger entrepreneurs, as it could focus all support on stimulating early stage and growth ventures, neglecting the skills and attitudes of potential younger entrepreneurs. A number of educational initiatives and events can cultivate a passion for entrepreneurial activities.

Future research opportunities abound in this area. First the model with contextual descriptors, dimensions, and support should be tested in different REES settings to improve its usefulness an applicability in different settings, and develop heuristic measures to guide policy development. As there is no one ideal type of REES, the model in this chapter provides a useful framework to understand the current stage of development of the regional system, and develop interventions accordingly. Second the leadership dimensions in REES seem to be understudied, lacking in the type of collaborative leadership models that might be more effective in one regional context to another, as well as the complimentary roles of private and public sector leaders in more socialistic settings. Third the potential of REES to contribute towards community building and development should be investigated, as many informal initiatives, driven by entrepreneurs for entrepreneurs, are often overlooked, yet these initiatives form the social capital in entrepreneurial communities, which often precedes access to other forms of capital. Finally, the influence of pro-entrepreneurship policies in different regions on entrepreneurial activities and outcomes should be studied in longitudinally, as the current state of

the field provides only anecdotal evidence for policy makers, failing to guide them to develop policy that enables entrepreneurship that is self-sustaining and generates beneficial economic and social outcomes.

## 7 Conclusion

The aim of this chapter was to examine the support available in regional entrepreneurship ecosystems, using a conceptual framework to study the role players, the dynamics of the relationship using dimensions and support and access to resources. The framework identified the interdependence of role players in providing support and how configurations of system elements differ depending on the regional context and stage of economic development. Using South East Queensland as a case study, with three embedded regions which experience similar national and state-wide policy conditions, helped to show how the system elements are configured. This paper provides a useful systems framework to understand the roles and contributions of different role players, but it should be used to describe REES in other contexts. Using the framework also helps to identify support gaps, and to determine whether limited resources should be sourced from other systems or developed locally. The practical value of the framework is its relative simplicity to use, enabling diverse role players to gain a deeper understanding of their local context, develop a common language and to co-create the desired REES.

As regional entrepreneurship ecosystem models are path-dependent, role players should consider the history, culture and prevailing values of regions and how the knowledge exploitation, knowledge generation and socio-institutional sub-systems have developed over time. These legacies influence the ability of the REES to adapt and change in response to technological and market changes.

## References

- Aarstad J, Haugland SA, Greve A (2010) Performance spillover effects in entrepreneurial networks: assessing a dyadic theory of social capital. *Entrepreneurship Theory Pract* 34 (5):1003–1019
- Ács ZJ, Autio E, Szerb L (2014) National systems of entrepreneurship: measurement issues and policy implications. *Res Policy* 43(3):476–494
- Audretsch DB, Falck O, Feldman MP, Heblich S (2012) Local entrepreneurship in context. *Reg Stud* 46(3):379–389
- Audretsch D, Keilbach M (2004) Entrepreneurship capital and economic performance. *Reg Stud* 38(8):949–959
- Audretsch DB, Thurik AR (2001) What's new about the new economy? Sources of growth in the managed and entrepreneurial economies. *Ind Corp Change* 10(1):267–315
- Australian Bureau of Statistics (2011–2015) Regional Population Growth Australia. Available <http://www.abs.gov.au/>. Accessed 15 Sept 2016

- Australian Private Equity & Venture Capital Association Limited (2016) Funding the new economy: a 'scale-up' policy blueprint. Available <http://www.avcal.com.au/documents/item/1265>. Accessed 10 Sept 2016
- Autio E, Kenney M, Mustar P, Siegel D, Wright M (2014) Entrepreneurial innovation: the importance of context. *Res Policy* 43(7):1097–1108
- Baumol WJ (1993) Formal entrepreneurship theory in economics: existence and bounds. *J Bus Ventur* 8(3):197–210
- Boschma RA (2005) Proximity and innovation. A critical assessment. *Reg Stud* 39(1):61–74
- Daquino L, Rickert K (2016) Brisbane top listed companies. Available <http://www.businessnewsaus.com.au/articles/2016-brisbane-top-listed-companies—21-30.html>. Accessed 15 Oct 2016
- Feld B (2012) Startup communities: building an entrepreneurial ecosystem in your city. Wiley, Hoboken, NJ
- Feldman MP (2014) The character of innovative places: entrepreneurial strategy, economic development, and prosperity. *Small Bus Econ* 43(1):9–20
- Feldman MP, Francis JL (2004) Homegrown solutions: fostering cluster formation. *Econ Dev Q* 18(2):127–137
- Feldman M, Francis J, Bercovitz J (2005) Creating a cluster while building a firm: Entrepreneurs and the formation of industrial clusters. *Reg Stud* 39(1):129–141
- Feldman M, Zoller TD (2012) Dealmakers in place: social capital connections in regional entrepreneurial economies. *Reg Stud* 46(1):23–37
- Gill R, Larson GS (2014) Making the ideal (local) entrepreneur: place and the regional development of high-tech entrepreneurial identity. *Hum Relat* 67(5):519–542
- Griffith University (2013) Sun sets on gold coast innovation centre. <http://app.griffith.edu.au/news/2013/06/07/sun-sets-on-gold-coast-innovation-centre/>. Accessed 12 Sept 2016
- Gronum S, Verreyne ML, Kastelle T (2012) The role of networks in small and medium-sized enterprise innovation and firm performance. *J Small Bus Manage* 50(2):257–282
- Huggins R, Williams N (2011) Entrepreneurship and regional competitiveness: the role and progression of policy. *Entrepreneurship Reg Dev* 23(9–10):907–932
- Huizingh EK (2011) Open Innovation: state of the art and future perspectives. *Technovation* 31(1):2–9
- Isenberg DJ (2010) The big idea: how to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):40–50
- Kenney M, Von Burg U (1999) Technology, entrepreneurship and path dependence: industrial clustering in Silicon Valley and Route 128. *Ind Corp Change* 8(1):67–103
- Kibler E, Kautonen T, Fink M (2014) Regional social legitimacy of entrepreneurship: Implications for entrepreneurial intention and start-up behaviour. *Reg Stud* 48(6):995–1015
- Kinner C (2015) Boosting high-impact entrepreneurship in Australia: a role for universities. Australian Government: Office of the Chief Scientist. Available <http://www.chiefscientist.gov.au/wp-content/uploads/Boosting-High-Impact-Entrepreneurship.pdf>. Accessed 18 July 2016
- Kim BY, Kang Y (2014) Social capital and entrepreneurial activity: a pseudo-panel approach. *J Econ Behav Organ* 97:47–60
- Kruger M, Cacioppe J (2014) South East Queensland 2014 Startup ecosystem report. Boundlss. Available <https://www.qld.gov.au/dsiti/reports/>
- Lee SY, Florida R, Acs Z (2004) Creativity and entrepreneurship: a regional analysis of new firm formation. *Reg Stud* 38(8):879–891
- Markham N, Barker T, Walker J, Van Dalen J, Kruger M, Cacioppe J (2016) Startup Ecosystem Report: Western Australia 2015–16. Boundlss. Available: [https://www.commerce.wa.gov.au/sites/default/files/atoms/files/wa\\_startups2015.pdf](https://www.commerce.wa.gov.au/sites/default/files/atoms/files/wa_startups2015.pdf). Accessed 21 July 2016, 14 Aug 2016
- Mathews M, Stokes P (2013) The creation of trust: the interplay of rationality, institutions and exchange. *Entrepreneurship Reg Dev* 25(9–10):845–866
- Motoyama Y, Watkins K (2014) Examining the connections within the startup ecosystem: a case study of St. Louis (Kauffman Foundation Research Series on City, Metro, and Regional Entrepreneurship). Kansas City, MO: Kauffman Foundation. Accessed 19 August 2016

- Morgan K (2007) The learning region: institutions, innovation and regional renewal. *Reg Stud* 41 (1):147–159
- Murdock KA (2012) Entrepreneurship policy: trade-offs and impact in the EU. *Entrepreneurship Reg Dev* 24(9–10):879–893
- National Innovation and Science Agenda (NISA) (2015) Welcome to the ideas boom. <http://www.innovation.gov.au/>. Accessed 10 Oct 2016
- Nieto MJ, Santamaría L (2010) Technological collaboration: bridging the innovation gap between small and large firms. *J Small Bus Manage* 48(1):44–69
- Nijkamp P (2003) Entrepreneurship in a modern network economy. *Reg Stud* 37(4):395–405
- O'Connor A, Reed G (2015) South Australia's entrepreneurial ecosystem: voice of the customer research report. Report prepared for the South Australian Department of State. Available: <http://ecic.adelaide.edu.au/docs/ecosystem-research-project-13-08-2015.pdf>. Accessed 15 July 2016
- Porto Gómez I, Otegi Olaso JR, Zabala-Iturriagagoitia JM (2016) ROSA, ROSAE, ROSIS: modelling a regional open sectoral innovation system. *Entrepreneurship Reg Dev* 28(1–2): 26–50
- Rantisi NM (2002) The local innovation system as a source of 'variety': openness and adaptability in New York City's garment district. *Reg Stud* 36(6):587–602
- Robson PJ, Wijbenga F, Parker SC (2009) Entrepreneurship and policy challenges and directions for future research. *Int Small Bus J* 27(5):531–535
- Salvador E, Mariotti I, Conicella F (2013) Science park or innovation cluster? similarities and differences in physical and virtual firms' agglomeration phenomena. *Int J Entrepreneurial Behav Res* 19(6):656–674
- Samila S, Sorenson O (2010) Venture capital as a catalyst to commercialization. *Res Policy* 39 (10):1348–1360
- Spigel B (2015) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory Pract* 41(1):49–72
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plan Stud* 23(9):1759–1769
- Trippel M (2006). Cross-border regional innovation systems. SRE—Discussion Papers, 2006/05. Institut für Regional- und Umweltwirtschaft, WU Vienna University of Economics and Business, Vienna. Available <http://epub.wu.ac.at/1110/1>. Accessed 10 Oct 2016
- Watkins A, Papaioannou T, Mugwagwa J, Kale D (2015) National innovation systems and the intermediary role of industry associations in building institutional capacities for innovation in developing countries: a critical review of the literature. *Res Policy* 44(8):1407–1418
- World Economic Forum (2013) Entrepreneurial ecosystems around the globe and company growth dynamics (industry agenda). World Economic Forum, Geneva

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# Where Are the Spiders? Proximities and Access to the Entrepreneurial Ecosystem: The Case of Polish Migrant Entrepreneurs in Glasgow

Paul Lassalle  and Andrew Johnston

**Abstract** Entrepreneurship research is increasingly taking into account external factors in order to provide context for the conditions under which new firms are created. Thus, the entrepreneur is increasingly recognized as a constituent part of the ecosystems in which they operate. In addition, a strong and vibrant ecosystem should be host to diversity—the presence of migrant entrepreneurs is a sign of this diversity, contributing to the ecosystem at the city level. This chapter focusses on a particular group of entrepreneurs, Polish migrant entrepreneurs based in the city of Glasgow, UK, and provides details of exploratory research that examines the influence of the entrepreneurial ecosystem on their new venture creation process. In order to examine which external factors are of importance developing vibrant ecosystems, and to draw attention to the role of proximities in facilitating their use by Polish migrant entrepreneurs, this chapter synthesizes the current entrepreneurial ecosystem literature with that discussing opportunity structure and proximity. The results suggest that both geographic and cultural proximity are important factors in accessing market and resources within the local migrant community. However, it also appears that despite positive effects in the start-up phase, the high level of proximity displayed between entrepreneurs and their market base can constrain future business growth potential—leading to a lack of diversity and suggesting a lack of local diversity within the community based sub-ecosystem.

**Keywords** Entrepreneurial ecosystems • Migrant entrepreneurship  
Proximity • New venture creation

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## 1 Introduction

This chapter draws on developments in the entrepreneurship literature that have witnessed the broadening out of the field to examine entrepreneurship in terms of both migration and ethnicity, for example transnational entrepreneurship (Drori et al. 2009), ethnic minority entrepreneurship (Aldrich and Waldinger 1990; Zhou 2004), (im)migrant entrepreneurship (Hjerm 2004; Kloosterman 2010), and diversity (Sepulveda et al. 2011; Smallbone et al. 2010). The focus of this chapter is on migrant entrepreneurship, an area that has been widely examined in the context of the resources available to the entrepreneurs both from demand and supply-side perspectives (Aldrich and Waldinger 1990; Kloosterman and Rath 2001; Kloosterman 2010), and builds on the recent developments in the entrepreneurial literature suggesting that the new venture creation process is increasingly contextualized (Garud et al. 2014; Stam 2015; Zahra 2007; Zahra et al. 2014). This approach typically views entrepreneurship as influenced by its social (Spigel 2015; Welter 2011) and institutional contexts (Ács et al. 2014; Welter and Smallbone 2011), and comprising of multiple components (Isenberg 2011; Stam 2015).

In light of these developments, a broader range of factors require consideration when examining the entrepreneurship process, particularly taking into account the influence of what has been termed the *entrepreneurial ecosystem* (EE) (Isenberg 2011; Moore 1993). To develop understanding of the interplay between place-specific EE and migrant entrepreneurship, this chapter chooses a micro approach concerned with the individual level of entrepreneurial activity and action within the contexts in which they operate. It therefore calls for a focus on the final user of public policy: the entrepreneur within their ecosystem. By doing so and focussing on the individual entrepreneur (the agent) as the focal point of analysis, the approach adopted also matches with the original concept of ecosystem (an analogy borrowed from the ecology literature) and therefore draws attention to a population particularly sensitive to external conditions and change; such as spiders (a specific population) in bio-diverse ecosystems (Paetzold et al. 2011; Pearce and Venier 2006). Furthermore, this chapter also builds on the argument of Smallbone et al. (2010: 174) that ethnic diversity is a ‘source of competitiveness’ for a specific place (the EE), to propose that a strong ecosystem should host diversity, and that spiders (here migrant entrepreneurs) are a sign (or an indicator) of that diversity (Stangler and Bell-Masterson 2015).

In this chapter, we present an empirical analysis of Polish migrant entrepreneurs and the components of their entrepreneurial ecosystem within the city of Glasgow in order to examine the influence of external components of the entrepreneurial ecosystem on entrepreneurial action (Klein 2008), and more specifically on the new venture creation process among micro-business owners. These entrepreneurs should still be considered worthy of investigation due to their capacity to contribute to the economy (Baumol 1993; Burke et al. 2002), especially at the city level (Smallbone et al. 2010). This group of migrant entrepreneurs were new entrants to the UK who have started their businesses since arrival.

The chapter is organized as follows, Sect. 2 presents the conceptual and theoretical framework, and integrates the entrepreneurial ecosystem framework with the concept of opportunity structure developed in the migrant entrepreneurship literature. Section 3 outlines the contextual background and research methods employed in undertaking this exploratory research. Section 4 presents the findings and reveal the importance of geographic and cultural proximities in market seeking and input seeking activities among Polish migrant entrepreneurs (MEs). Finally, Sect. 5 outlines the conclusions, and highlights the chapter's key contributions around the role of proximity as an important facilitator of interaction with the entrepreneurial ecosystem among migrant entrepreneurs. In addition, considering the bounded nature of the community-based sub-ecosystem, findings also highlight the constraining effect of certain proximities on high growth venture creation. From this revelatory contribution we highlight avenues of future research in terms of replicatory research in different locations and communities as well as the policy implications regarding the support of entrepreneurial diversity within entrepreneurial ecosystems.

## 2 Theoretical and Conceptual Framework

### 2.1 *Migrant Entrepreneurship*

The extant literature identifies significant differences between these and 'native entrepreneurs' when it comes to the access to resources, the ability to identify or create opportunities, or business strategies, giving an interesting new perspective on the role of the ecosystem (Aldrich and Waldinger 1990; Kloosterman 2010; Portes and Sensenbrenner 1993; Ram and Jones 2008; Zhou 2004). Specifically, migrant entrepreneurs suffer from a lack of access to the host country's labour market and formal sources of finance compared to native entrepreneurs; consequently their ability to mobilize community-specific resources are considered to be important determinants of the new venture creation process (Deakins et al. 2007; Jones et al. 2000; Jones and Ram 2010; Ram et al. 2013; Smallbone et al. 2003). As such, migrant entrepreneurs provide an interesting context in which to examine the potential influence of entrepreneurial ecosystems on the new venture creation process, especially in their ability to access market and resources. Empirical studies of entrepreneurship within the migrant communities of entrepreneurs examine the external influences on their entrepreneurial activity, contributing to the increased diversity within the local economy, and by that ultimately on competitiveness in a specific location (Smallbone et al. 2010). This question is tackled through examining not only the importance of external factors but also the entrepreneurs' relationship to the different elements in terms of proximity. Here proximity is examined considering both spatial and non-spatial dimensions (Bathelt and Glückler 2003;

Boschma 2005) to understand which elements are important but also the mechanisms through which they are accessed.

As a subset of the ethnic minority entrepreneurship literature, migrant entrepreneurship has focussed on the question of migrants' access to markets (Kloosterman 2010; Rusinovic 2008a; Waldinger 2005), in addition to the crucial issue of access to resources; including finance, labour, and advice (Jones and Ram 2010; Ram and Jones 2008). The entrepreneurial process is not considered in isolation; instead, the concept of opportunity structure, introduced by Aldrich and Waldinger (1990), and later developed by authors such as Kloosterman and Rath (2001), provides an explicit link between context and migrant entrepreneurs' responses to local conditions, through focussing initially on two dimensions, market conditions in the host country, and access to ownership. The former relates to what could be referred to as the industry structure (production and demand conditions) as well as to the idea of *abandoned niche* shunned by mainstream entrepreneurs (Jones and Ram 2010). The latter refers to the ability to create a new venture, influenced by opportunities for new firms in the markets, and by government policies. As such, the opportunity structure can be considered to be situational, i.e. non-generic, and varies depending on the locality and the community considered; thus it is shaped by social and institutional factors, as well as particular circumstances (Kloosterman and Rath 2001; Kloosterman 2010). However, the concept of opportunity structure is not another term to refer to the *context*, but rather a combination of different contextual dimensions, which vary across different localities and communities.

In the case of migrant entrepreneurship, the ability to mobilize resources from within their own communities has been highlighted as one of the distinctive features of migrant entrepreneurship compared to other populations of entrepreneurs (Aldrich and Waldinger 1990; Deakins et al. 2007; Portes and Sensenbrenner 1993). As such, this consideration has motivated scholars to pay attention to the structure of social networks to provide a better understanding of migrant entrepreneurs' activity (Engelen 2001). In fact, community networks not only give access to resources to migrant entrepreneurs (Foley and O'Connor 2013; Hjerm 2004; Waldinger 2005; Zhou 2004), but also provide opportunities in terms of labour recruitment and financial support, especially at the regional and neighbourhood level (Jones and Ram 2010; Kloosterman and Rath 2001; Zhou 2004), thereby highlighting the existence of specific community-bounded systems in host countries.

## 2.2 *The Entrepreneurial Ecosystem*

The EE concept provides both an explicitly networked and spatial approach to the entrepreneurship process at national or regional levels (Ács et al. 2014; Malecki 2011). Thus, the entrepreneurial process takes place within a community of inter-dependent actors (Stam 2015) with the act of new venture creation dependent on first, what exists within these communities, and, second, how individuals interact

and draw on these to further their ventures (Spigel 2015). The EE has been conceptualized as multi-faceted, i.e. it focuses on a variety of local factors, which to some extent, influence the new venture creation process (Mason and Brown 2014). It can provide a useful systematic view to look at contextualized entrepreneurship; however, in light of a lack of definitive empirical work on the concept, the approach is required to be broad and cover many bases as it is as yet relatively unexplored (Stam 2015). A number of factors have been proposed as important components of the ecosystem; including the local leadership, availability of venture capital, presence of potential customers, specialist supporting businesses, networks, knowledge generators, relevant infrastructure, and, finally, the extent to which a location can be considered as an entrepreneurial milieu that is rich in start-ups (Isenberg 2010), contributing to its *vibrancy* (Stangler and Bell-Masterson 2015).

In short, the EE extends the focus through examining a set of external characteristics that may be of importance in the new venture creation process, giving a broader view of the process through an explicit focus on the locational dimension. The act of entrepreneurship remains the same, that is, individuals recognize opportunities for the creation of new goods and services and act upon them (Shane and Venkataraman 2000), but considering the drivers of this activity. There are, however, a number of criticisms of the concept, which has been described elsewhere as conceptually ad hoc and incoherent (Stam 2015; Thomas and Autio 2013), calling (among others) for an in-depth understanding of the role played by those components on local entrepreneurial action.

### ***2.3 Opportunity Structure and the Entrepreneurial Ecosystem***

Thus, through focussing on the opportunity structure, it is recognized that a range of external influences operating over differing spatial scales influence the migrant entrepreneurship process (Kloosterman 2010), with an empirical focus on the urban level (Sepulveda et al. 2011; Smallbone et al. 2010). This approach takes into account both hard and soft factors for example, local demand, institutions and social networks are all considered to be important external influences (Engelen 2001, 2006). Yet, the concept of opportunity structure is conceptually much narrower than the entrepreneurial ecosystem, which embraces a broader set of external factors as influences on the new venture creation process. In addition, whereas the concept of opportunity structure used in the migrant entrepreneurship literature only takes into account location in a general sense, acknowledging the fact that different processes occur at differing spatial levels, national, regional or neighbourhood level, the entrepreneurial ecosystem brings location into consideration as a key factor. Thus, while the opportunity structure provides a starting point for the analysis of a set of external factors that influence the entrepreneurship process it is weakened by a lack of explicit focus on location specific factors. Indeed, despite being more

place-focussed, the EE concept has been criticized as lacking a prescribed formula (Isenberg 2010), in that empirical work has not systematically demonstrated which factors are important. Thus, through moving on from a focus on the opportunity structure as presented in the migrant entrepreneurship literature to a wider focus on the entrepreneurial ecosystem as a *systemic*—yet agent-centered approach to entrepreneurial activity, a more in-depth understanding of the new venture creation process among migrant entrepreneurs can be gleaned. This focus on migrant entrepreneurs contributes to an understanding of diversity in an entrepreneurial ecosystem. Nonetheless, the lack of empirical work and *prescribed formula* suggest an exploratory approach to this phenomenon, leading to our first research question; how does the ecosystem influence the new venture creation process of migrant entrepreneurs?

## 2.4 Proximity and the Entrepreneurial Ecosystem

So far, however, one important area is not addressed in the conceptual framework; the importance of *proximity* to these external factors. As noted earlier, the ecosystem concept brings a specific locational element to the contextual approach, yet very little has been said about any potential advantages or disadvantages of co-location on entrepreneurial activity of agents, especially at the urban level. There exists a wealth of literature on the influence of proximity to interaction between actors and their external environment, which is briefly summarized here. Focussing on geographic proximity, this is seen to allow for effective knowledge transfer between, especially in the case of more tacit knowledge (Gertler 2003; Morgan 2004; Sonn and Storper 2008), and allows actions to be observed fostering trust between the two parties (Wood and Parr 2005). Geographic proximity is viewed as especially important in gaining more tacit types of knowledge as these rely on the co-located actors in order to understand context (Gertler 1995, 2003).

While proximity is frequently taken to mean *geographic* proximity, this focus ignores other types such as cultural proximity which may also be of importance (Boschma 2005; Lagendijk and Lorentzen 2007). Thus, physical closeness may be supplanted by other types of non-spatial proximity manifested in terms of social or cultural factors, where sharing a common set of norms, values and through institutional affinity or sharing a language, or codebook, facilitate interaction between actors (Bathelt and Glückler 2003). Similarities in language or shared culture promote interaction and facilitate reaction to opportunities for new venture creation as this ensures that an actor effectively understands the community in which they operate (Cowan et al. 2000; Håkanson 2005; Howells 2002). As such, networks may be more global in scope, highlighting the possibility that the entrepreneurial ecosystem may be geographically dispersed (Bathelt and Glückler 2003; Bathelt et al. 2004; Maskell et al. 2006; Teixeira 2001), or based around weaker ties (Granovetter 1973). Those networks enable entrepreneurs to enter foreign markets (as identified within the internationalization literature) or to benefit from

transnational links for sourcing or market access as well as for early internationalization (Cavusgil and Knight 2015; Coviello 2006; Rusinovic 2008b; Sharma and Blomstermo 2003).

The conceptual framework set out here does not assert the primacy of either type of proximity in terms of the entrepreneurial ecosystem. Instead, we highlight the duality inherent within the entrepreneurial ecosystem; EE are comprised of factors that are spatial, based on location, and cultural, based on membership of a group or groups. Consequently, the concept is viewed as a mixture of location and community, with access to these factors based on different types of proximity. Thus, the ability to utilize external factors in the new venture creation process may be based on either the location of an entrepreneur or their membership of a particular group or community. As such, these arguments lead to our second research question, how does proximity to external dimensions influence access to EE?

### 3 Research Methods and Data Source

Focussing on individual entrepreneurs' activities, this research is designed to capture and understand the influence of contextual dimensions on entrepreneurial behaviour of Polish MEs who started-up their business in the UK after the EU-enlargement to A8 countries in May 2004. Over 400,000 Polish workers entered the UK in the three years following the 2004-EU enlargement, significantly exceeding the numbers forecast by government officials (Home Office 2009). Thus, within a short space of time the Polish community became one of the most significant migrant communities within the UK, worthy of investigation when studying diversity issues in entrepreneurship.

An interpretive and phenomenological approach is adopted in this chapter which better captures the Polish MEs' account of their temporarily and situated entrepreneurial experience (Cope 2005) and which takes into account their entrepreneurial activity within their social and institutional context (Engelen 2006; Garud et al. 2014). Results of the case study is presented in this chapter, focussing on Polish migrant entrepreneurs who arrived in the UK following the EU enlargement of 2004, and who have since started their business in Glasgow between 2005 and 2008. By adopting a focus on the City of Glasgow as an urban center, the research is bound within one specific area, and by doing so, we are able to better control for local conditions.

Because this research is exploratory in nature and examines an understudied phenomenon, a qualitative approach with a specific population provides a clear lens for looking at the relevant phenomena, especially in the absence of reliable and precise statistical data. For theorizing, the environmental factors (family, networks, business environment, institutions) that shape and influence perceptions and actions are analytically central to the interpretivist stance (Crozier and Friedberg 1977; Patton 1990). Likewise, Leitch et al. (2010) argue that *verstehen* (understanding), is

bounded by the contextual nature of entrepreneurship (Welter 2011), understood here in spatial and social terms.

Importantly, interpretive research on entrepreneurial phenomena aims to identify conceptual insights that emerge from the empirical data collected. The data analysis process in exploratory and discovery-oriented research starts with a rich (or thick) description of data (Cope 2005; Eisenhardt and Graebner 2007; Geertz 1973) but is necessarily theory-laden (Klag and Langley 2013). The findings of this research provide an understanding of entrepreneurial activity of Polish MEs understood within the specific context of the Polish community in Glasgow. The single case study presented considers the spatial and social boundaries of the sub-ecosystem within the broader EE of Glasgow. Although there is no claim of empirical generalizability, the formulation of theory emerging from empirical data is to be presented as a theoretical proposition for future research across other communities and locations.

### **3.1 Data Collection**

For the case study, semi-structured interviews in Polish were conducted with 21 Polish MEs in Glasgow. The focus was on entrepreneurs who started-up and were sole-owners of their business, and these interviews were complemented with interviews with key informants (such as support agencies), spouses and employees when applicable, as well as with field notes following additional informal discussions with entrepreneurs, agents from the Polish community networks, and clients (that were all coded). These helped capturing the diversity of the lived-world experience of Polish entrepreneurs in the entrepreneurial ecosystem of Glasgow (Cope 2005). To ensure a greater quality of data on aspects such as perception, or entrepreneurial behaviour, all interviews (45–120 min) and discussions were conducted in Polish language by the lead author. This approach ensured greater depth to their answers as respondents were unconstrained by language barriers, as well as being an approach that maximized the ability of the interviewer to capture cultural nuances (Crick et al. 2001; Welch and Piekkari 2006).

As seen in Table 1, Polish MEs in Glasgow operate in service sectors (delicatessen, hairdressers, IT services, car repair, etc.) and can be defined as micro-businesses, employing between one and nine people (Commission of the European Communities 2003). Most of the interviewees arrived in Scotland between the late months of 2004 and the early months of 2005. The median age of the Polish entrepreneur in Scotland was 30 years old at the time of fieldwork. Regarding gender, the sample of respondents included 14 men and 7 women entrepreneurs. All of the entrepreneurs interviewed had previously worked in Scotland in low skilled and low paid occupations (as factory workers, butchers, cleaners, etc.) prior to starting-up their venture, despite the fact that most have achieved a degree level qualification from Poland. However, all but two of the respondents were not fluent in the English language.

**Table 1** Entrepreneur respondents details

| Respondent ID | Business             | Target market | Employees (incl. the entrepreneur) FTE | Gender |
|---------------|----------------------|---------------|----------------------------------------|--------|
| S             | Computer-shop        | Poles         | 2                                      | M      |
| M             | Garage               | Poles         | 2                                      | M      |
| L1            | Travel agency        | Poles         | 1                                      | F      |
| B             | Delicatessen         | Poles         | 4                                      | M      |
| U             | Book-shop            | Poles         | 1                                      | M      |
| M             | Hairdresser          | Poles         | 3                                      | F      |
| P             | Exercise Supplements | All           | 2                                      | M      |
| A1            | Construction         | All           | 1                                      | M      |
| K             | Hairdresser          | All           | 3                                      | M      |
| M&I           | Restaurant           | All           | 2                                      | F      |
| L 2           | Legal adviser        | Poles         | 1                                      | F      |
| H             | Legal adviser        | Poles         | 2                                      | M      |
| P2            | Construction         | All           | 1                                      | M      |
| I             | Hairdresser          | Poles         | 3                                      | F      |
| A2            | Boxing school        | Poles         | 2                                      | M      |
| K2            | IT                   | All           | 1                                      | M      |
| R             | Garage               | All           | 2                                      | M      |
| A3            | Hairdresser          | Poles         | 4                                      | F      |
| M2            | Driving school       | Poles         | 1                                      | M      |
| M3            | Delicatessen         | Poles         | 1.5                                    | F      |
| P3            | IT                   | Poles         | 1                                      | M      |

### 3.2 Data Analysis

The following analysis utilized an abductive logic (Klag and Langley 2013), emphasizing the importance of the iterative and reflective process of theorization, also referred to as *enfolding* in the literature (Eisenhardt 1989). Although pre-existing theoretical frameworks and concepts on migrant entrepreneurship and entrepreneurial ecosystems were used to initially formulate the research question on the importance of perception of contextual elements in understanding new venture creation among Polish MEs in Glasgow, as well as in the design of starting codes, those existing frameworks are combined with theories and models emerging from collected data (as for instance the notion of proximity).

Thus, the researchers sought emergent themes by reading, highlighting and grouping the data collected, using starting and emerging codes along the process. Klag and Langley (2013) refer to this as *bricolage* where this process of data structuring and reduction is the basis for theorization (Miles and Huberman 1994). Open coding and axial and coding were both used first by the lead researcher and then sent to the other authors to allow comparative analysis of the findings. The iterative

nature of the process pointed out by Klag and Langley (2013) requires a constant reference to the data to let themes emerge. The analysis of organized action proposes to look at the interplay of the actor (here; Polish MEs) and of the environment (here; the entrepreneurial ecosystem) as part of the process of analysis, as both cannot be separated (Crozier and Friedberg 1977), especially while adopting a contextualized view of entrepreneurship. In this contextualized view, entrepreneurship involves interaction between agent and structure, in which opportunities can be created within the constraints and enablers of the context (Garud et al. 2014; Mole and Mole 2010; Sarason et al. 2006; Shane and Venkataraman 2000).

## 4 Findings

### 4.1 Market Seeking

This first section of the analysis examines demand-side factors, i.e. those related to market opportunities. We find that, as the MEs are predominantly creating service sector firms, geographic proximity to their intended market appears to be an important factor in the new venture creation process. Indeed, the respondents tend to base their start-up in areas with high levels of Polish migrant households in the neighbourhood; i.e. Hillhead, the City Center, and Dennistoun<sup>1</sup> (Glasgow City Council 2012; Kelly and Ashe 2014). These ventures rely on the Polish community as their market; hence they are based where their customers are located the majority of the start-ups are designed to serve the Polish community markets with specialist goods and services, such as Polish food, Polish computer programs, or with services matching specific needs of the community (e.g. legal advice, car repair, hairdressing).

I chose to rent this local here. I was shopping at the retail park and I saw that this local was available. I thought: OK, this is a good place. Many Poles are shopping here you know.

U. Book-shop owner.

As such, Polish entrepreneurs see their community as their market and are able to spot opportunities within this niche. Indeed, for a large majority of the start-ups, the Polish community is the primary or even the only market targeted. Although non-Polish clients are potentially available, the businesses studied strongly rely on the Polish community as customers;

Almost all our clients are Poles. I think it is because of the language barriers. Sometimes I have to ask three or four times to understand a question with the locals [...] We are trying to reach more and more locals with the advertising, the flyers.

M. Garage owner.

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<sup>1</sup>But they did not start up in all inner city areas, such as Govan or Ibrox, with high numbers of Polish migrants..

The lack of proficiency in the English language among the entrepreneurs in some ways forced their hand in terms of target market. As such, they were focussed on seeking Polish markets, but also recognized that this was a wider problem within the community, restricting the number of products and services on offer. Thus, the entrepreneurs saw this gap as an opportunity to provide these services, such IT, legal services and hairdressing, to a Polish clientele. Thus, this cultural proximity through shared experiences and understanding allowed them to recognize the opportunities for new venture creation they presented rather than merely a barrier to everyday life.

Despite the reliance on their own community as a market and its ability to sustain these new ventures the Polish MEs were aware of the limited growth potential due to the size of the community, and are looking to serve individuals from outside to maintain growth:

We had some issues at the beginning with advertisement. At the beginning we were looking for Polish clients. There are now more and more Scottish people, men and women. But in general, I would say that most of our clients are Poles.

M. Hairdresser.

Thus, some entrepreneurs were looking to diversify their activities through the broadening of their products and clientele, breaking out of the *abandoned* niche (Jones and Ram 2010). Consequently, while the existence of a potential market is an important part of the ecosystem it is also important to consider whether it will also support a broadening of activities and whether there exists the presence of sufficient customers to foster future expansion. This potential may be more important in the context of MEs, who tend to rely more heavily on their own communities for the market (Jones et al. 2014), and demonstrated in these findings regarding the Polish community, where any significant expansions will require them to move beyond their own communities.

Geographic proximity to the Polish community was not always important for market access. For example, the use of social media and of Internet Portals (such as *emito.net* or *glasgow24.pl*) among the Polish migrants was also a source of information for those starting a new venture.

I posted on *emito*. Some guys wanted to do it. I advertised the class on the web. For example on *emito*, and people were asking questions about the class and showing interest. *Emito*. *Emito.pl* I think [note: *emito.net*]. Well I advertised about the class I was giving for the other school and people were interested. Many Poles are looking at *emito*.

K, Boxing School.

These portals are in Polish language and were highlighted by the respondents as the main *fora* for the interaction of Polish migrants in Glasgow to both meet and exchange information. While these media are location specific (at the city level of Glasgow), cultural proximity makes them accessible through the shared use not only of the Polish language, but also a shared cultural background and a similar experience of migration. These sites assist the Polish ME in identifying market opportunities within the community as these portals provide information for Polish

migrants (jobs, companionship, tips, or social life) and also serve as advertising platforms for Polish businesses. These portals appear to be important factors within the entrepreneurial ecosystem as they provide both access to the community and information on community needs and wants. Consequently, this existing infrastructure played an important role in the start-up process as platforms in the EE, allowing virtual access to the potential market and acting as a portal for the existing community to identify themselves to one another. As such, these findings highlight that existing network infrastructures, both hard and soft, are another important component of the EE in which Polish MEs operate. Importantly, however, cultural proximity to the market played an important part in facilitating the entrepreneurs' ability to effectively utilize these components of the ecosystem.

In terms of market access, the respondents suggested that it is more straightforward for a Polish entrepreneur to trade within their community because they have a better understanding of the needs of Polish migrant clientele than those from other communities. Again, this ability to access specific market highlights the importance of cultural proximity, which provides the entrepreneurs with the ability to recognize and/or create their own opportunity in Glasgow; a key element is the fact that they share a common language and this removes an important cultural barrier for Polish customers.

As I said, I do most of my work in Polish. Actually, it is only in Polish. I have a Polish company, it does everything. There are also a few Slovaks and Czechs but they only come for the driving part, they cannot understand the test in Polish. Concerning the Scots [missing words] they all are Poles, I mean that they have Polish roots. I have some students from Lithuania and Slovenia as well.

M. Driving school.

However, as shown above, the cultural proximity is not only bounded within the Polish migrant community but also extends to other former Central and Eastern European soviet bloc countries such as Czechs and Slovaks, slightly increasing the size of the market base.

## ***4.2 Supply-Side Key Factors; Networks and Perceptions***

The second part of our analysis focuses on inputs, or with respect to supply-side factors. The analysis shows that Polish entrepreneurs mobilize resources through their access to networks in terms of both sourcing inputs and recruiting labour.

All Polish MEs in Glasgow hire staff from within the Polish community exclusively, due to proximity reasons. Polish MEs find Polish workers available in the city of Glasgow. As those employees are generally working in low skilled and low paid occupations due to barriers to enter the UK labour market at the appropriate qualification level, the offer to work for a Polish business is attractive to them. Moreover, cultural proximity explains the choice of relying on the Polish migrant workforce. Indeed, Polish MEs and their employees share the same experience of

migration, share the same social networks, use the same social media, and share the same language. This cultural proximity generates trust between the entrepreneur and the employee. Trust is generated through social networks and thence links with the idea of reputation in social relationship (Light and Dana 2013). Following Granovetter (1985), more intimate relations between individuals make behaviour more predictable and reduce transaction costs. Polish MEs prefer to recruit fellow Poles because they can conduct the interview, understand them fully (in terms of language), and trust their skill level.

In addition, Polish MEs combine their knowledge of Polish products with their ability to network with Polish suppliers to source their products from Poland. They know where and what to order from central purchasing agents or directly from suppliers in Poland.

For example, if they are asking for some help. When they are buying, they want everything from Poland; the salt, and the sugar. Some of them want everything from Poland. I have to buy for the clients although it does not make any difference to me whether the sugar comes from Poland or from Scotland. Some clients are helping me with the orders because they can tell me what they want.

M1. Delicatessen.

Likewise, another entrepreneur running a delicatessen makes direct contact with Polish producers and signs partnership contracts with suppliers. This came after a year of cooperation with an intermediary. The ability to avoid the reliance on the middleman was made possible due to their awareness of suppliers, knowledge of products and of valuable partners in Poland. Thus, it would appear that a level of cultural proximity allows them to build and possess the right knowledge.

Conversely, with respect to inputs, the findings suggest that the entrepreneurial ecosystem was of little importance when it came to new venture creation among Polish migrants. While they utilized a broad network to obtain inputs they also showed a marked reluctance to access local institutions of business support and finance, effectively shunning the ecosystem in this respect (Smallbone and Welter 2006; Welter and Smallbone 2006). Yet, it must be noted that they were still able to set up and run a business, thanks to the favourably perceived elements of the ecosystem and the opportunities within the community market.

The respondents have little knowledge of what regional or local public support institutions such as Business Gateway<sup>2</sup> can provide, and they are reluctant to contact banks or such agencies in any way for their business (even though they formally register their activity). Polish MEs appear to lack not only awareness of these institutions, but also display a lack of willingness to engage with any formal institutions (which can partly be explained by the historical reasons in Poland, whereby formal institutions are not trusted). Many of the Polish MEs approached for this research were still using their personal account for business purposes. When asked about Business Gateway:

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<sup>2</sup>Business Gateway is a publicly funded business support institution in Scotland. Along with other services it provides advice and training for new businesses..

No, is there such a thing? Here they do not give me credit. Why? We do not know. I don't have a credit card.

P. Exercise supplement shop owner.

These findings on the lack of access to both formal and informal sources of finance within the EE and their communities run contrary to other evidence presented in the literature, which shows that other groups of migrant entrepreneurs tend to seek for finance or business support among their community (Smallbone et al. 2003; Wilson and Portes 1980). This emphasizes the contextualized relevance of networks, which are activated by Polish MEs for specific aspects of their entrepreneurial activity (namely accessing the community market, and labour). As such, broader components of the ecosystem do not appear to be important for this aspect of the new venture creation process, as this group of entrepreneurs relied solely upon their own means for support. The absence of efforts to access the domestic business support and financial sector can possibly be explained by lack of cultural proximity to these institutions among this group; yet, the absence of accessing support among the Polish community is more complex and relates to the ambivalent trust relationship within the community (Ryan 2011).

As previously noted, the Polish MEs' ventures are mostly micro-businesses (self-employed, or 1–3 employees) in the service sector, thus they only required a small amount of financial capital to create; they typically invested an average £5000 from their own savings, gathered during prior employment in the UK. In fact, the respondents overwhelmingly noted the fact that they found starting and running a business in the UK a particularly simple endeavour. In addition, they were also aware of the fact that small businesses benefit from discounts in their business rates through the Small Business Bonus Scheme, VAT exemption, and that self-employed status also provides the individual with income tax credits. Moreover, all interviewees highlighted the ease of the start-up process in Scotland, including the small amount of paperwork required:

Yes, it is straight forward. At the beginning, there is only the language barrier. Only the language barrier has prevented me from starting up earlier or doing it faster.

I. Hairdresser.

## 5 Conclusions and Implications

This chapter examines the role of the entrepreneurial ecosystem on the new venture creation activities of Polish MEs. This exploratory work set out to establish which areas of the ecosystem concept were of relevance to the entrepreneurs and also examines the importance of geographic or cultural proximity in accessing the components of the ecosystem.

This *duality* in the ecosystem is apparent in the findings, with geographic proximity to the market opportunities found to be an important mechanism through which market access is physically possible, but also cultural proximity to the

community found to be important for recognizing the opportunities, for instance to identify fellow migrants' needs, contact specialized suppliers of Polish goods, recruit individuals with appropriate skills from within the Polish community to work in the new ventures. Also, MEs positively perceived the broader institutional framework, as respondents felt that this made the process of business formation relatively smooth. Despite these perceived favourable conditions within the EE, the respondents shunned the support services on offer as well as opportunities for outside assistance/investment in the new venture from both inside and outside their communities.

Importantly, by contextualizing migrant entrepreneurs' activity within the boundaries of their community market, this research calls for a sub-system approach of the EE in Glasgow, in which entrepreneurs operate, draw resources and access markets through proximities. In fact, these types of proximity play a positive role at the start-up stages but also constrain the ability of migrant entrepreneurs to create high-growth and innovative ventures in the longer run. In spite of being relatively highly skilled, migrant entrepreneurs engage in micro-business start-up (among those few only will aim for growth), both enabled and constrained by the sub-ecosystem in which they operate, and that they contribute to co-create through their entrepreneurial activities, their network relations (for business and for social purposes), and their narratives (see for instance Garud et al. 2014).

Whereas geographic proximity to the market opportunities allows the entrepreneurs to physically place their businesses within the Polish community, as the ventures are mainly small service sector firms and because of language limitations, market access is restricted mostly to focussing on their own communities for market and labour. Being members of the Polish community themselves gives them an insight into the apparent opportunities, tastes and requirements of this market, highlighting the importance of cultural proximity in the process of new venture creation within the sub-entrepreneurial ecosystem of the Central and Eastern European community in Glasgow. Although geographically co-located with the 'mainstream' EE, migrant entrepreneurs operate within a sub-set of the EE, accessed through the cultural proximity with the labour force and customers. They are (partly) working independently of the entrepreneurial ecosystem of the host city of Glasgow and operate within a sub-ecosystem, of which they are constitutive and contribute to co-create through their business and social relations.

However, geographic proximity to the market served may ultimately restrict the entrepreneurs and prevent them from expanding beyond the niche (Kloosterman 2010; Zhou 2004). It is noted that the ventures created by the entrepreneurs in this study tend to be locally-focussed micro-businesses. In fact, these firms rely on just a sub-set of domestic demand –constrained geographically but also by the size of the community of migrants at the urban level of Glasgow City. Consequently, the results suggest that proximity, both geographic and cultural are important additional elements contributing to the entrepreneurial ecosystem as they facilitate access to market, workforce, in relation to contextual favourable regulatory frameworks (Stam 2014, 2015), extending the understanding of the system through explaining the relational mechanisms through which entrepreneurs have accessed them.

Moreover, the results highlight the relevance of both locational and social dynamics in understanding economic action (Granovetter 1985; Ulhoi 2005) and especially within EE (Spigel 2015). This suggests that a local view of the new venture creation process is important, taking into account the *local buzz* (Bathelt et al. 2004; Storper and Venables 2004), cultural attributes (Spigel 2015) and networks of untraded interdependencies that may exist at a more local/urban level (Storper 1995). In addition, research on competitiveness at the city level insists on the importance of entrepreneurial diversity, including ethnic or sector diversity, thereby requiring local level evidence.

Furthermore, the findings of this study are location and community specific, offering a finer, more detailed, examination of the ecosystem than that afforded by the regional or national level (e.g. Ács et al. 2014; Szerb and Trumbull 2015). As noted by Boschma (2005) and Aguiléra et al. (2012), non-spatial proximity highlights the importance of common experience, codes and language to analyse economic action in a relational perspective (Bathelt and Glückler 2003), starting from the individual. As such, cultural proximity constitutes a relational element of economic activity that operates at the local level, and thus “emerge under a unique set of conditions and circumstances” (Mason and Brown 2014: 5). It thus requires an understanding of the contextual conditions of entrepreneurial action at the local level, as a pre-requisite for the development of a micro-centered (but still co-created) approach to EE. This is an importance contribution to research on relational factors in EE as developed by authors such as Spigel (2015) as a lens to look at the interplay between a specific population of entrepreneurs and the wider structures and networks in which they operate (see also recent debate in Garud et al. 2014; Sarason et al. 2006; Shane and Venkataraman 2000). As highlighted earlier, proximities also play a constraining role on the growth potential of the new ventures locally. In fact, geographical and cultural proximity between entrepreneurs and their market base lead to risks of *lock-ins* (e.g. Boschma 2005), stressing the lack of opportunities for growth within community-based markets (Rusinovic 2008a; Zhou 2004). This reduces entrepreneurial diversity within a specific EE, which is a detrimental feature to vibrant and competitive ecosystems (Smallbone et al. 2010; Stangler and Bell-Masterson 2015).

The chapter also enriches the ecosystem literature through considering the role of geographic and cultural proximity in accessing elements of the ecosystem. The title of this chapter (“Where are the spiders?”) emphasizes the importance for research on entrepreneurial ecosystem to focus on the micro level of entrepreneurial activity to uncover the interplay of entrepreneurial activity (agency) and environment by focussing on entrepreneurial action and perceptions within the contexts in which the population operate. The literature on bio-diversity and ecosystems has pointed out that changes in diverse ecosystems impact strongly on the population of spiders, making them a very useful ecological indicator (Paetzold et al. 2011; Pearce and Venier 2006; Rainio and Niemelä 2003) of the sustainability of a given place-based environment, therefore stressing the importance of diversity on ecosystems’ vibrancy and competitiveness. The focus on the population of entrepreneurs as key agents of the EE supports the claim that an entrepreneurial

ecosystem approach cannot ignore the final implication of entrepreneurial research: providing support to entrepreneurs (the spiders) in starting-up, sustaining and developing their business activity though supporting better access to resources, encouraging innovation or internationalization (Agarwal et al. 2007; Cavusgil and Knight 2015; Coviello 2006). The existence of sub-ecosystems for migrant entrepreneurs, bounded by geographical and cultural proximity, needs to be understood by policy makers. The nature of the migrant businesses in this study (micro-businesses in the service sectors) reveals the enabling and constraining effects of those proximities in accessing relevant resources for new venture creation and development and ultimately the effect on the (lack of) diversity.

However, in spite of recent studies on internationalization among ethnic minority entrepreneurs (Johnston et al. 2016; Rusinovic 2008b) the effect on first generation migrant on internationalization is yet to be further explored. By following the analogy with literature concerned with bio-diversity and sustainability of ecosystems, this chapter proposes that the focus on a specific population of entrepreneurs at the micro level should focus on enabling and encouraging migrant entrepreneurs to engage in mainstream and high growth activity even among first generation (Rusinovic 2008b). We also argue that the use of transnational networks (activated through cultural but not geographical proximity) could enable migrant entrepreneur to overcome the barriers of the locally and community bounded sub-ecosystem, therefore contributing to the diversity and the competitiveness of a specific EE. We follow Baumol (1993) in stressing the importance of non-growth oriented micro/small businesses in their contribution to the *vibrancy* of a EE (Stangler and Bell-Masterson 2015), especially at the city level.

Moreover, previous work on migrant entrepreneurship lacked a specific geographical and locational focus, looking solely at their communities (Kloosterman 2010; Zhou 2004) as a sub-set of the local EE. These findings highlight the fact that both location and community factors are important and could be considered as part of sub-ecosystems. The focus on proximity offers an additional lens through which to examine the interaction of entrepreneurs with their external environment; it therefore captures the specific processes of resource acquisition and their ability to exploit these. Thus, it is not just the existence of community or location factors that is important; it is the entrepreneurs' relationship to these which is facilitated through proximity, nonetheless stressing its constraining ambivalent role in future developments. Here we suggest that proximity in its broadest sense is important; it may be geographic proximity to an external factor, i.e. being co-located, or it may be cultural proximity to an external factor, i.e. sharing similar, language, identity and experiences, that allow entrepreneurs to identify or create opportunities in a given environment.

Finally, in terms of policy implications, the findings have relevance to enterprise policy as there may be scope to increase levels of new venture creation among migrant groups through understanding which factors are important when starting a new venture to encourage entrepreneurial diversity within an entrepreneurial ecosystem. As such, greater levels of entrepreneurial activity may be possible if cultural barriers could be bridged increasing trust within and beyond the

community, including for transnational or international entrepreneurial activities. Once this has been developed, the findings suggest that there is scope for creating a champion from within the culture as a stepping stone between the support organizations and the migrant community. Furthermore, these findings have relevance to debate around place-based economic development and entrepreneurial diversity at the city level (Smallbone et al. 2010); where ecosystems may be specific to a particular community or location, entrepreneurship education and policy should aim to harness these resources (O'Connor 2013; Stangler and Bell-Masterson 2015).

## References

- Ács ZJ, Autio E, Szerb L (2014) National systems of entrepreneurship: measurement issues and policy implications. *Res Policy* 43(3):476–494. doi:[10.1016/j.respol.2013.08.016](https://doi.org/10.1016/j.respol.2013.08.016)
- Agarwal R, Audretsch D, Sarkar M (2007) The process of creative construction: knowledge spillovers, entrepreneurship, and economic growth. *Strateg Entrepreneurship J* 1(3–4):263–286
- Aguiléra A, Lethiais V, Rallet A (2012) Spatial and non-spatial proximities in inter-firm relations: an empirical analysis. *Industry and Innovation* 19(3):187–202
- Aldrich HE, Waldinger R (1990) Ethnicity and entrepreneurship. *Annu Rev Sociol* 111–135
- Bathelt H, Glückler J (2003) Toward a relational economic geography. *J Econ Geogr* 3(2): 117–144
- Bathelt H, Malmberg A, Maskell P (2004) Clusters and knowledge: local buzz, global pipelines and the process of knowledge creation. *Prog Hum Geogr* 28(1):31–56
- Baumol WJ (1993) Formal entrepreneurship theory in economics: existence and bounds. *J Bus Ventur* 8(3):197–210
- Boschma R (2005) Proximity and innovation: a critical assessment. *Reg Stud* 39(1):61–74
- Burke A, FitzRoy F, Nolan M (2002) Self-employment wealth and job creation: the roles of gender, non-pecuniary motivation and entrepreneurial ability. *Small Bus Econ* 19(3):255–270. doi:[10.1023/A:1019698607772](https://doi.org/10.1023/A:1019698607772)
- Cavusgil ST, Knight G (2015) The born global firm: an entrepreneurial and capabilities perspective on early and rapid internationalization. *J Int Bus Stud* 46(1):3–16
- Cope J (2005) Researching entrepreneurship through phenomenological inquiry philosophical and methodological issues. *Int Small Bus J* 23(2):163–189
- Coviello NE (2006) The network dynamics of international new ventures. *J Int Bus Stud* 37 (5):713–731
- Cowan R, David PA, Foray D (2000) The explicit economics of knowledge codification and tacitness. *Ind Corp Change* 9(2):211–253
- Crick D, Chaudhry S, Batstone S (2001) An investigation into the overseas expansion of small Asian-owned UK firms. *Small Bus Econ* 16(2):75–94
- Crozier M, Friedberg E (1977) *L'acteur et le système: les contraintes de l'action collective*. [The actor and the system: the constraints of collective action]. Paris: Éditions du Seuil
- Deakins D, Ishaq M, Smallbone D, Whittam G, Wyper J (2007) Ethnic minority businesses in Scotland and the role of social capital. *Int Small Bus J* 25(3):307–326
- Drori I, Honig B, Wright M (2009) Transnational entrepreneurship: an emergent field of study. *Entrepreneurship Theory and Practice* 33(5):1001–1022
- Eisenhardt KM (1989) Building theories from case study research. *Acad Manag Rev* 14(4): 532–550
- Eisenhardt KM, Graebner ME (2007) Theory building from cases: opportunities and challenges. *Acad Manag J* 50(1):25–32

- Engelen E (2001) 'Breaking in' and 'breaking out': a Weberian approach to entrepreneurial opportunities. *J Ethn Migr Stud* 27(2):203–223. doi:[10.1080/13691830020041570](https://doi.org/10.1080/13691830020041570)
- Engelen E (2006) Towards an explanation of the performance differences of Turks in the Netherlands and Germany: the case for a comparative political economy of integration. *Tijdschrift voor economische en sociale geografie* 97(1):69–79. doi:[10.1111/j.1467-9663.2006.00497.x](https://doi.org/10.1111/j.1467-9663.2006.00497.x)
- Foley D, O'Connor AJ (2013) Social capital and the networking practices of indigenous entrepreneurs. *J Small Bus Manage* 51(2):276–296. doi:[10.1111/jsbm.12017](https://doi.org/10.1111/jsbm.12017)
- Garud R, Gehman J, Giuliani AP (2014) Contextualizing entrepreneurial innovation: a narrative perspective. *Res Policy* 43(7):1177–1188
- Geertz C (1973) The interpretation of cultures: selected essays, vol 5019. Basic books
- Gertler MS (1995) "Being there": proximity, organization, and culture in the development and adoption of advanced manufacturing technologies. *Econ Geogr* 1–26
- Gertler MS (2003) Tacit knowledge and the economic geography of context, or the undefinable tacitness of being (there). *J Econ Geogr* 3(1):75–99
- Glasgow City Council (2012) Population and households by ethnicity in glasgow estimates of changes 2001–2010 for community planning partnership areas and neighbourhoods
- Granovetter M (1985) Economic action and social structure: the problem of embeddedness. *Am J Sociol* 481–510
- Granovetter MS (1973) The strength of weak ties. *Am J Sociol* 1360–1380
- Håkanson L (2005) Epistemic communities and cluster dynamics: on the role of knowledge in industrial districts. *Industry and Innovation* 12(4):433–463
- Hjerm M (2004) Immigrant entrepreneurship in the swedish welfare state. *Sociology* 38(4): 739–756. doi:[10.1177/0038038504045862](https://doi.org/10.1177/0038038504045862)
- Home Office (2009) Accession monitoring report. UK Border Agency, London
- Howells JR (2002) Tacit knowledge, innovation and economic geography. *Urban Stud* 39(5–6): 871–884
- Isenberg D (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):40–50
- Isenberg D (2011) The entrepreneurship ecosystem strategy as a new paradigm for economic policy: principles for cultivating entrepreneurship. Presentation at the Institute of International and European Affairs
- Johnston A, Prokop D, Crone M, Masango S, Lassalle P (2016) Assessing the characteristics, determinants and spatial variations of internationalised new ventures in the UK. Enterprise Research Council
- Jones T, Barrett G, McEvoy D (2000) Market potential as a decisive influence on the performance of ethnic minority business. In: *Immigrant businesses: the economic, political and social environment*, pp 37–53
- Jones T, Ram M (2010) Review article: ethnic variations on the small firm labour process. *Int Small Bus J* 28(2):163–173
- Jones T, Ram M, Edwards P, Kiselinchev A, Muchenje L (2014) Mixed embeddedness and new migrant enterprise in the UK. *Entrepreneurship & Reg Dev* 26(5–6):500–520. doi:[10.1080/08985626.2014.950697](https://doi.org/10.1080/08985626.2014.950697)
- Kelly B, Ashe S (2014) Local dynamics of diversity: evidence from the 2011 Census
- Klag M, Langley A (2013) Approaching the conceptual leap in qualitative research. *Int J Manage Rev* 15(2):149–166
- Klein PG (2008) Opportunity discovery, entrepreneurial action, and economic organization. *Strateg Entrepreneurship J* 2(3):175–190
- Kloosterman R, Rath J (2001) Immigrant entrepreneurs in advanced economies: mixed embeddedness further explored. *J Ethnic and Migr Stud* 27(2):189–201
- Kloosterman RC (2010) Matching opportunities with resources: a framework for analysing (migrant) entrepreneurship from a mixed embeddedness perspective. *Entrepreneurship & Reg Dev* 22(1):25–45. doi:[10.1080/08985620903220488](https://doi.org/10.1080/08985620903220488)

- Legendijk A, Lorentzen A (2007) Proximity, knowledge and innovation in peripheral regions. On the intersection between geographical and organizational proximity. *Eur Plan Stud* 15(4): 457–466
- Leitch CM, Hill FM, Harrison RT (2010) The philosophy and practice of interpretivist research in entrepreneurship: quality, validation, and trust. *Organ Res Methods* 13(1):67–84. doi:[10.1177/1094428109339839](https://doi.org/10.1177/1094428109339839)
- Light I, Dana L-P (2013) Boundaries of social capital in entrepreneurship. *Entrepreneurship Theory and Pract* 37(3):603–624. doi:[10.1111/etap.12016](https://doi.org/10.1111/etap.12016)
- Malecki EJ (2011) Regional social capital: why it matters. *Reg Stud* 46(8):1023–1039. doi:[10.1080/00343404.2011.607806](https://doi.org/10.1080/00343404.2011.607806)
- Maskell P, Bathelt H, Malmberg A (2006) Building global knowledge pipelines: the role of temporary clusters. *Eur Plan Stud* 14(8):997–1013
- Mason C, Brown R (2014) Entrepreneurial ecosystems and growth oriented entrepreneurship. Final Report to OECD, Paris
- Miles MB, Huberman AM (1994) *Qualitative data analysis: an expanded sourcebook*. Sage
- Mole KF, Mole M (2010) Entrepreneurship as the structuration of individual and opportunity: a response using a critical realist perspective: comment on Sarason, Dean and Dillard. *J Bus Ventur* 25(2):230–237
- Moore JF (1993) Predators and prey: a new ecology of competition. *Harvard Bus Rev* 71(3):75–83
- Morgan K (2004) The exaggerated death of geography: learning, proximity and territorial innovation systems. *J Econ Geogr* 4(1):3–21
- O'Connor A (2013) A conceptual framework for entrepreneurship education policy: meeting government and economic purposes. *J Bus Ventur* 28(4):546–563
- Paetzold A, Smith M, Warren PH, Maltby L (2011) Environmental impact propagated by cross-system subsidy: chronic stream pollution controls riparian spider populations. *Ecology* 92(9):1711–1716
- Patton MQ (1990) *Qualitative evaluation and research methods*. SAGE Publications, inc
- Pearce JL, Venier LA (2006) The use of ground beetles (Coleoptera: Carabidae) and spiders (Araneae) as bioindicators of sustainable forest management: a review. *Ecol Ind* 6(4):780–793
- Portes A, Sensenbrenner J (1993) Embeddedness and immigration: notes on the social determinants of economic action. *Am J Sociol* 98(6):1320–1350
- Rainio J, Niemelä J (2003) Ground beetles (Coleoptera: Carabidae) as bioindicators. *Biodivers Conserv* 12(3):487–506
- Ram M, Jones T (2008) Ethnic-minority businesses in the UK: a review of research and policy developments. *Environ Plann Gov Policy* 26(2):352
- Ram M, Jones T, Edwards P, Kiselinchev A, Muchenje L, Woldesenbet K (2013) Engaging with super-diversity: new migrant businesses and the research–policy nexus. *Int Small Bus J* 31(4):337–356. doi:[10.1177/0266242611429979](https://doi.org/10.1177/0266242611429979)
- Rusinovic K (2008a) Moving between markets? Immigrant entrepreneurs in different markets. *Int J Entrepreneurial Behav Res* 14(6):440–454. doi:[10.1108/13552550810910997](https://doi.org/10.1108/13552550810910997)
- Rusinovic K (2008b) Transnational embeddedness: transnational activities and networks among first- and second-generation immigrant entrepreneurs in the Netherlands. *J Ethnic and Migr Stud* 34(3):431–451
- Ryan L (2011) Migrants' social networks and weak ties: accessing resources and constructing relationships post-migration. *The Sociol Rev* 59(4):707–724. doi:[10.1111/j.1467-954X.2011.02030.x](https://doi.org/10.1111/j.1467-954X.2011.02030.x)
- Sarason Y, Dean T, Dillard JF (2006) Entrepreneurship as the nexus of individual and opportunity: a structuration view. *J Bus Ventur* 21(3):286–305. doi:[10.1016/j.jbusvent.2005.02.007](https://doi.org/10.1016/j.jbusvent.2005.02.007)
- Sepulveda L, Syrett S, Lyon F (2011) Population superdiversity and new migrant enterprise: the case of London. *Entrepreneurship & Reg Dev* 23(7–8):469–497. doi:[10.1080/08985620903420211](https://doi.org/10.1080/08985620903420211)
- Shane S, Venkataraman S (2000) The promise of entrepreneurship as a field of research. *The Acad Manage Rev* 25(1):217–226. doi:[10.2307/259271](https://doi.org/10.2307/259271)

- Sharma DD, Blomstermo A (2003) The internationalization process of born globals: a network view. *Int Bus Rev* 12(6):739–753
- Smallbone D, Kitching J, Athayde R (2010) Ethnic diversity, entrepreneurship and competitiveness in a global city. *Int Small Bus J* 28(2):174–190. doi:[10.1177/0266242609355856](https://doi.org/10.1177/0266242609355856)
- Smallbone D, Ram M, Deakins D, Baldock R (2003) Access to finance by ethnic minority businesses in the UK. *Int Small Bus J* 21(3):291–314
- Smallbone D, Welter F (2006) Conceptualising entrepreneurship in a transition context. *Int J Entrepreneurship and Small Bus* 3(2):190–206. doi:[10.1504/IJESB.2006.008928](https://doi.org/10.1504/IJESB.2006.008928)
- Sonn JW, Storper M (2008) The increasing importance of geographical proximity in knowledge production: an analysis of US patent citations, 1975–1997. *Environ Plann A* 40:1020–1039
- Spigel B (2015) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory and Pract*, Early view [online] doi:[10.1111/etap.12167](https://doi.org/10.1111/etap.12167)
- Stam E (2014) The Dutch entrepreneurial ecosystem. Available at SSRN 2473475
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plan Stud* 23(9):1759–1769. doi:[10.1080/09654313.2015.1061484](https://doi.org/10.1080/09654313.2015.1061484)
- Stangler D, Bell-Masterson J (2015) Measuring an entrepreneurial ecosystem. Kauffman Foundation Research Series on City, Metro, and Regional Entrepreneurship 1–16
- Storper M (1995) The resurgence of regional economies, ten years later the region as a nexus of untraded interdependencies. *Eur Urban and Reg Stud* 2(3):191–221
- Storper M, Venables AJ (2004) Buzz: face-to-face contact and the urban economy. *J Econ Geogr* 4(4):351–370
- Szerb L, Trumbull WN (2015) Entrepreneurship and entrepreneurial ecosystem in the V4 countries: the global entrepreneurship index perspective
- Teixeira C (2001) Community resources and opportunities in ethnic economies: a case study of Portuguese and black entrepreneurs in Toronto. *Urban Stud* 38(11):2055–2078. doi:[10.1080/00420980120080934](https://doi.org/10.1080/00420980120080934)
- Thomas LD, Autio E (2013) The fifth facet: the ecosystem as an organizational field. *Innovation and Entrepreneurship Group Working Papers*, pp 1–40
- Ulhøi JP (2005) The social dimensions of entrepreneurship. *Technovation* 25(8):939–946. doi:[10.1016/j.technovation.2004.02.003](https://doi.org/10.1016/j.technovation.2004.02.003)
- Waldinger R (2005) Networks and niches: the continuing significance of ethnic connections. In: Loury G, Modood T, Teles S (eds) *Ethnicity, social mobility, and public policy: comparing the USA and UK*. Cambridge University Press, Cambridge, pp 342–362
- Welch C, Piekkari R (2006) Crossing language boundaries: qualitative interviewing in international business. *Manage Int Rev* 46(4):417–437
- Welter F (2011) Contextualizing entrepreneurship—conceptual challenges and ways forward. *Entrepreneurship Theory and Pract* 35(1):165–184. doi:[10.1111/j.1540-6520.2010.00427.x](https://doi.org/10.1111/j.1540-6520.2010.00427.x)
- Welter F, Smallbone D (2006) Exploring the role of trust in entrepreneurial activity. *Entrepreneurship Theory and Pract* 30(4):465–475. doi:[10.1111/j.1540-6520.2006.00130.x](https://doi.org/10.1111/j.1540-6520.2006.00130.x)
- Welter F, Smallbone D (2011) Institutional perspectives on entrepreneurial behavior in challenging environments. *J Small Bus Manage* 49(1):107–125
- Wilson KL, Portes A (1980) Immigrant enclaves: an analysis of the labor market experiences of Cubans in Miami. *Am J Sociol* 86(2):295–319
- Wood GA, Parr JB (2005) Transaction Costs, Agglomeration Economies, and Industrial Location\*. *Growth and Change* 36(1):1–15
- Zahra SA (2007) Contextualizing theory building in entrepreneurship research. *J Bus Ventur* 22(3):443–452
- Zahra SA, Wright M, Abdelgawad SG (2014) Contextualization and the advancement of entrepreneurship research. *Int Small Bus J* 32(5):477–500. doi:[10.1177/0266242613519807](https://doi.org/10.1177/0266242613519807)
- Zhou M (2004) Revisiting ethnic entrepreneurship: convergencies, controversies, and conceptual advancements I. *Int Migrat Rev* 38(3):1040–1074. doi:[10.1111/j.1747-7379.2004.tb00228.x](https://doi.org/10.1111/j.1747-7379.2004.tb00228.x)

## Author Biographies

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**Dr. Andrew Johnston** is a Principal Lecturer in International Business and Economics at Sheffield Hallam University (UK) and is director of the International Business and Economics Research Group (IBERG). His research interests focus on the broad areas of innovation, entrepreneurship and regional economic development, specifically collaborative ties between firms, universities and entrepreneurial ecosystems.

# Is There a Path from Sin City to Tech City? The Case for Las Vegas

Fiona Sussan, Brian Sloboda and Richard Hall

**Abstract** Many cities aspire to have vibrant entrepreneurial ecosystems that are relevant to the digital economy—can they? To answer the question, we examine the economic history of Las Vegas, a land of gambling and entertainment, and report the recent entrepreneurial activities in the city along with the data that measures the vibrancy of the ecosystem and growth of the place. While there is no indication that the various data points converge, our analysis leads us to three insights. One, legacy industry is disconnected from the new tech hub in an entrepreneurship ecosystem when the economy has shifted from the ‘Main street’ mode to the ‘digital’ mode. Two, while there are two recent success stories of Zappos and Switch SUPERNAIP in Las Vegas, entrepreneurship activities remain moderate in spite of a bottom-up effort to build Downtown as a tech hub. Three, cluster advantages and positive network externalities do not seem to happen in unrelated industry in Las Vegas.

**Keywords** Digital economy · Entrepreneurial ecosystem · Tech hub

## 1 Introduction

Many cities in the U.S. aspire to have vibrant entrepreneurial ecosystems (EE) that are presumed will give birth to giants like Google, Facebook, and Amazon that are important players in a digital economy. Vibrant EEs exist in places like Boulder, Stockholm, and Silicon Valley (Feld 2012). Not so vibrant EEs also exist in cities

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like Phoenix (Mack and Mayer 2016). Extending the work of Mack and Mayer (2016) which uses a historical lens to capture the evolutionary dynamics of the EE of a city, the purpose of this paper focuses on the EE in Las Vegas today from the perspective of the digital economy. In other words, we seek to understand if there exists a vibrant ecosystem in Las Vegas today that supports entrepreneurs in the digital economy, and, given the investment to build a digital business presence in the city, if not, why not?

Situating the study of EE of a place within the digital economy is important. Digital economy is built on digitization and the advent of the Internet. Digitization and its infrastructure for communications and networking changes the way we live and work. The digital economy was born two decades ago and is now contributing to a substantial portion of the U.S. economy, an amazingly rapid and exponential value creation within a short period of time and thus a ‘growth multiplier’ for both matured and developing economies (Knickrehm et al. 2016). The companies born out of the digital economy have displaced traditional firms (e.g., Wikipedia—Encyclopedia Britannica; Instagram—Eastman Kodak; Netflix—Blockbuster), introduced new business models (e.g., Uber, Google), cannibalized iconic retailers (e.g., online giant Amazon threatening brick-and-mortar retailers such as Sears and Macy’s), invented new currencies (e.g., Bitcoin), and many more. These firms were started by entrepreneurs who are labeled as ‘digital entrepreneurs’ and they operate within a digital entrepreneurial ecosystem (Sussan and Acs 2017). A digital entrepreneurial ecosystem is built on the foundation of abundance instead of scarcity of information (Shapiro and Varian 1999), open sourced instead of legacy software (e.g., Linux, R), voluntary user-generated content instead of expensive and inadequate customer feedback solicited by firms (e.g., third party website of Tripadvisor), and a democratic horizontal worldview coupled with a mechanism that enables value co-creation among firms, among consumers, and among firms and consumers surrounding platforms connecting partners (Knickrehm et al. 2016). With these drastic changes, can an EE that was built on supporting traditional industries (e.g., entertainment in the form of brick-and-mortar casinos in Las Vegas) be transformed to support digital entrepreneurs and their firms? Our main research question goes beyond whether an EE is versatile enough to support various industries, a question already partially answered by cluster-related research that the cluster of a particular industry does not support unrelated industry (Delgado et al. 2010), it opens up areas for further enquiry on whether an EE is needed and whether knowledge spillover is needed for the digital economy. In asking about knowledge, we also raise questions about the emergence of and the global nature of digitized knowledge stored in big data (both internal and external to firms) and its relation to ‘place-based’ knowledge spillover mechanisms for entrepreneurs in the digital economy.

To answer these questions, we begin in the following by examining the economic history of Las Vegas, highlighting the entrepreneurs who founded the City, followed by surveying the major entrepreneurial activities related to the digital economy in the past decade. We then use EE vibrancy (Stangler and Bell-Masterson 2015) to examine the city with a static lens. From the analysis of

our data, we provide insights for managers and recommendations for policy makers in EE building in the digital economy.

## **2 Historical and Industry Background of the Las Vegas EE**

Las Vegas (meadows in Spanish), famous for its casinos and gangster characters like ‘Bugsy’, began as a mail post between Salt Lake City and California in 1829. The place was viewed as an oasis in desert land with good natural resources nearby such as crystal salt, timber, lead, silver, and agriculture. Mining was predominant until the 1850s, and it switched to focus more on agriculture. The construction of a railroad in the early 1900s brought about jobs, entrepreneurial activities, and population growth. By late 1920s, the construction of Boulder Dam (now named Hoover Dam) brought more workers to Las Vegas for enjoying its lax rules of gambling and liquor. It is not until 1941 that the first iconic motel-casino El Rancho was opened as an anchor for the famous Strip. El Rancho was built by a Chicago business man Thomas Hull. The second motel ‘Last frontier’ was built by Texan R. E. Griffith, and the third motel Flamingo by Californian Kirk Kerkorian. The development of the Strip is followed by the arrival of the New York gangster Benjamin Siegel “Bugsy” in the 1940s starting a new era of mafia-dominated Las Vegas that would last for four decades (Bernhard et al. 2008). In brief, the early entrepreneurs of Las Vegas were Thomas Hull, R.E. Griffith, and Kirk Kerkorian. They came from out of state and they were in the casino and hotel business.

A boost of the development of the Strip or the gaming and entertainment industry in Las Vegas took place when atomic bomb testing began in early 1950s. The result of this unplanned cold-war activity boosted tourism as the Chamber of Commerce advertised bomb detonation times and best spots for watching mushroom clouds. “Miss Atomic Energy Pageants” were held until early 1960s, hotels offered “atomic cocktails”, “dawn bomb parties”, among others. Amidst the fear of cold war and the preparedness of nuclear war, tourists in the area were fascinated by the mushroom clouds unlocking the secrets of the atom. This went on until the late 1960s when the public battled to end the atomic bomb tests.

Tangential to the atomic bomb tests and the cold war, a dominant Texan entrepreneur Howard Hughes, arrived at the scene of Las Vegas in the 1960s. Hughes, a lifelong pilot and aviation enthusiast, already owned Hughes Aircraft Company from California and had substantial aerospace defense contracts during World War II. He was a maverick movie tycoon and had purchased 73,000 acres of desert land in Northern Nevada in the 1950s. Hughes began his spending spree in the Strip in 1966 by purchasing six hotels in Las Vegas. Hughes structured his purchases in such a way that they are considered by some the beginning of the corporate-owned casinos that evolved to the MBA-led venture-capital funded luxury casino empires in the 1990s by people like Gary Loveman (Harrah’s), Steve

Wynn (Mirage that relied on junk bond king Michael Milken), J. Terrence Lannie (MGM), and Glenn Schaeffer (Mandalay).

The 1990s and early 2000s saw an emergence of new major hotels and resorts on the Strip (e.g., Aladdin, Bally's, Bellagio, Caesars Palace, Excalibur, Luxor, MGM Grand, Sahara, Stratosphere Tower, Tropicana, and off the Strip (e.g., Rio, Gold Coast Hotel, The Orleans Hotel, The Suncoast Hotel). Large scale shopping malls were also built within (e.g., Venetian's Canal Shoppes) and outside of hotels (e.g., FashionShow Mall). Together these serve the increasing tourist visits and population growth in Las Vegas.

The advent of the Internet also impacted Las Vegas. The first prominent e-business that caught the headline was Zappos, an Amazon subsidiary. Zappos, founded in 1999 by a Harvard computer science graduate Tony Hsieh (a serial entrepreneur), is an online retail platform for consumer goods, began in San Francisco and later in 2004 moved to Henderson, a suburb contiguous to Las Vegas. After being purchased by Amazon for 1.2 billion US dollars in 2009, Zappos moved to downtown Las Vegas in 2012. Zappos presence in downtown Las Vegas is important in two dimensions. One, it provides a positive signal that downtown Las Vegas can be a hub for new firms born out of the digital economy. Second, the owner of Zappos Tony Hsieh committed 350 million US dollars for the Las Vegas Downtown Project aiming to revitalize the area or replicate a Silicon Valley-like area by creating a non-exclusive campus for high tech firms (Semuels 2015). Revitalizing downtown has always been an agenda for the City of Las Vegas since the 1980s but the agenda varied across decades. Very briefly, in the 1980s, the revitalization focused on alluring office building. In the 1990s the focus was on attracting residential. By 2008 financial crisis, all revitalization plans were stopped. At the time of this writing, entrepreneurial ecosystems in Las Vegas are much discussed in the press and the verdict on its success is mixed. There were both optimistic and pessimistic reports. In the following, we will first analyze the EE vibrancy of Las Vegas in Sects. 3 and 4, followed by an entrepreneur-centric analysis of who's who in Las Vegas EE in Sect. 5. Section 6 is a discussion of our findings and Sect. 7 is the conclusion.

### **3 Kaufman Foundation Indicators of Entrepreneurial Ecosystem Vibrancy**

The US federal government spent almost \$2 billion per year on entrepreneurial and small business support through technical commercialization, financial assistance, and government contracting assistance (GAO 2013). More important, technology commercialization serves as an impetus to promote entrepreneurship which comprised much of the spending by the federal government to promote entrepreneurial and small business support (Qian and Haynes 2014). In addition, states also enacted their own programs to assist the promotion of entrepreneurship. In 2012, twenty-two states offered early-stage investment tax credits as means of supporting

early-stage development or attracting early-stage investment firms (Austrian and Piazza 2014).

This section uses Stangler and Bell-Masterson's (2015) four indicators: density, fluidity, connectivity, and diversity to gauge the Las Vegas EE vibrancy. While these four indicators were not explicitly defined, density very loosely refers to the number of entrepreneurs you'll run into walking across the street (Feld 2012), fluidity refers to the movement of people and firms, connectivity is about networks of people and resources, and diversity refers to a wide range of specializations and economic mobility. We collected data for Las Vegas-Henderson-Paradise, Nevada (NV) Metropolitan Statistical Area (MSA) from the Bureau of the Census, Business Dynamics of Statistics (BDS), Crunchbase, and others. The sources of data are listed in Table 1.

Before we go into the details of the measurements, we provide a survey of industries that hire the most people in Las Vegas. Table 2 reports the top ten major industries in the area with the most employees, and spectator sports lead the list. Next on the list is restaurants and eating places being the second largest employer in the area.

Now that we know most employees work for the service industry, we report the results of the EE vibrancy in Figs. 1 and 2, and Tables 3 and 4. For the density measurement, we collected the number of new and young companies, their employment level, and the extent to which those companies function in similar sectors of the local economy. In other words, these measurements reflect the intent of the regional economy being able to attract entrepreneurs to its locality and in a generic way means that the greater the density, the more entrepreneurial vibrancy.

In the Las Vegas area, the number of new and young firms per 1000 residents was 0.775 firms in 2010 and remained similar in 2013 except a big drop in 2012. Despite the stable number of new and young firms, the total number of new and young firms decreased 3.28% from 12,549 in 2010 to 12,151 in 2013. More important, the number of people employed by the new and young firms also decreased from 2010 to 2013. Overall, the total number of people employed by new and young firms decreased 7.44% from 124,782 in 2010 to 115,498 in 2013. This is indicative of a decrease of overall entrepreneurial activity in the Las Vegas area.

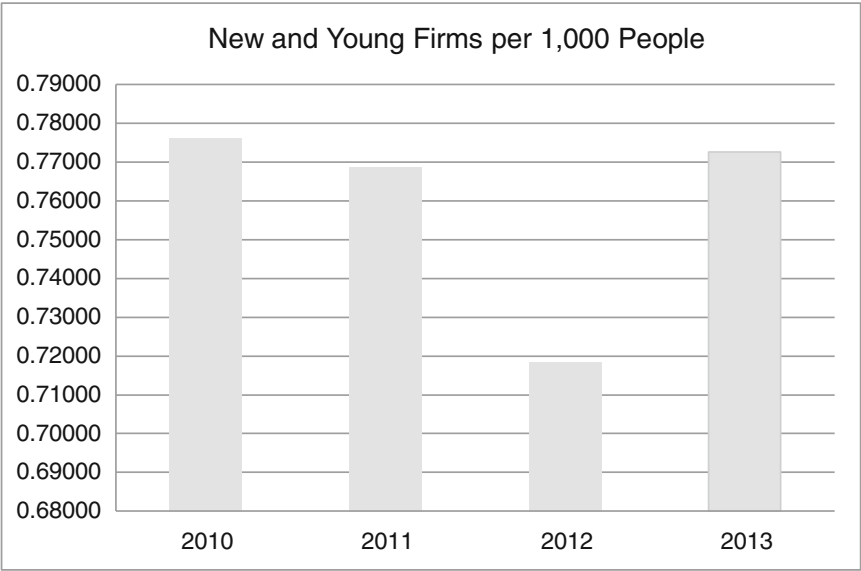
**Table 1** Data sources

| Indicator    | Measure                                          | Source                                                                      |
|--------------|--------------------------------------------------|-----------------------------------------------------------------------------|
| Density      | New and young firms per 1000 people              | Business Dynamics Statistics (BDS), Bureau of the Census                    |
|              | Share of employment in new and young firms       | Business Dynamics Statistics (BDS), Bureau of the Census                    |
| Fluidity     | Labor market reallocation                        | Quarterly Workforce Indicators (QWI), Bureau of the Census                  |
| Connectivity | Quality of network of entrepreneurial activities | CrunchBase                                                                  |
| Diversity    | Multiple economic specializations                | Quarterly Census of Employment and Wages (QCEW), Bureau of Labor Statistics |

**Table 2** Top ten major industries with the most employees in the Las Vegas-Henderson-Paradise, NV MSA

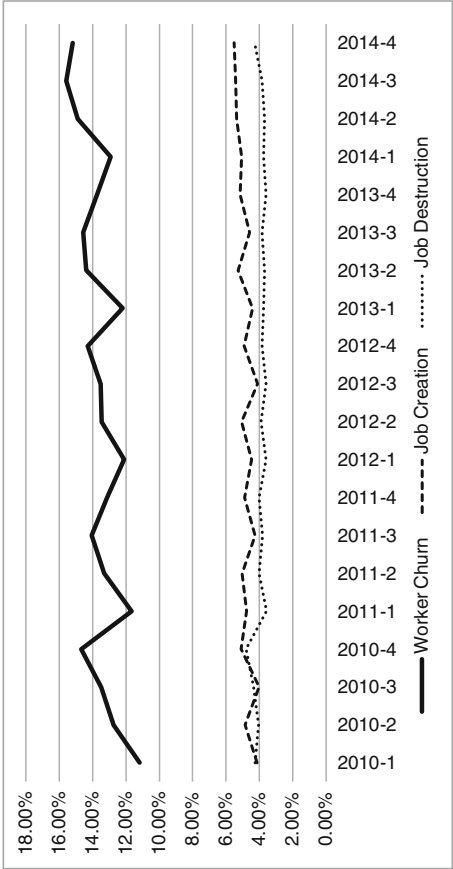
| NAICS Code | Description                                   | Employment | Earnings average |
|------------|-----------------------------------------------|------------|------------------|
| 7211       | Spectator Sports                              | 175,824    | 3275             |
| 7225       | Restaurants and other eating places           | 78,960     | 1900             |
| 5617       | Services to buildings and dwellings           | 17,795     | 2175             |
| 5511       | Management of companies and other enterprises | 17,179     | 8274             |
| 4451       | Grocery stores                                | 15,405     | 2679             |
| 6221       | General medical and surgical hospitals        | 15,191     | 6203             |
| 5613       | Employment services                           | 14,550     | 2051             |
| 4481       | Clothing stores                               | 14,391     | 2130             |
| 2382       | Building equipment contractors                | 13,909     | 5106             |
| 6211       | Offices of physicians                         | 13,221     | 7110             |

Source Authors calculations using the Quarterly Workforce Indicators, Bureau of the Census 2014



**Fig. 1** EE Density for Las Vegas-Henderson-Paradise, NV MSA. Source Business Dynamics Statistics (BDS), Bureau of the Census 2014

The next measurement we look at is fluidity. We collected data from the Quarterly Workforce indicators and calculated quarterly worker churn, job churn, and job destruction from year 2010 to 2014. The results are depicted in Fig. 2. Worker churn represents the dynamic movement of workers across jobs not accounted for by new job creation or job destruction by established firms. In other



**Fig. 2** EE Fluidity—Las Vegas-Henderson-Paradise, NV MSA Labor Market Reallocation. *Source* Authors calculations based on the data from the Quarterly Workforce Indicators (QWI)

**Table 3** EE connectivity—Las Vegas-Henderson-Paradise NV MSA

|                    |                               |
|--------------------|-------------------------------|
| DealMakers         | Program connectivity          |
| Investors 0.70     | Business associations 2.4     |
| Universities 0.005 | Professional associations 1.1 |

Source Crunchbase, calculated by authors

**Table 4** EE diversity—Las Vegas-Henderson-Paradise NV MSA

|                                                           | Employment | Percentage of total (%) | Median earnings |
|-----------------------------------------------------------|------------|-------------------------|-----------------|
| Food preparation and serving related occupations          | 139,410    | 16.01                   | 23,400          |
| Office and administrative support Occupations             | 134,590    | 15.46                   | 32,200          |
| Sales and related occupations                             | 99,080     | 11.38                   | 25,130          |
| Transportation and material moving occupations            | 60,120     | 6.90                    | 29,560          |
| Personal care and service occupations                     | 58,130     | 6.68                    | 20,170          |
| Building and grounds cleaning and maintenance occupations | 56,770     | 6.52                    | 32,290          |
| Total occupations (All)                                   | 870,740    | 100                     | 33,160          |

Source Occupational Employment Statistics, Bureau of Labor Statistics, 2014

words, this is a job-to-job movement among workers within a labor market. For example, when a worker leaves a gaming dealer job for a higher-paying one across town, the move counts as one separation and one hire. Unless her old firm decides not to replace her, someone new will be hired to take her post. This measurement is thus a proxy for information spillovers within the Las Vegas-Henderson-Paradise, NV MSA. Worker churn in the Las Vegas area increased between 2010 and 2014 from an annual high of 11.19% in the first quarter of 2010 to 15.6% in the third quarter of 2014.

In Fig. 2, the unadjusted quarterly estimates showed that gross job relocation in the Las Vegas area is relatively stable with higher levels of job creation than that of job destruction for the majority of the period.

The third measurement is connectivity. As we have loosely defined earlier in the paper, connectivity is about networks of people and resources. For an entrepreneur connectivity matters because it helps them solve problems, find talent, attract funding, and build the relationships that leads to customers, product innovation, and diffusion of new products. In order to measure (1) dealmaker networks, (2) spinoff rates, and (3) connections between programs and resources suggested by Stangler and Bell-Masterson (2015), we turn to Crunchbase database for surrogate measurements for dealmaker networks and program connectivity among resources. The Crunchbase database lists major companies, startups, and investors active in the technology sectors. By estimating regional participation of investors, companies,

and universities via Crunchbase we can examine the connectedness of regional entrepreneurial activity (Dani 2016).

Table 3 shows the connectivity. In 2014, the Las Vegas-Henderson-Paradise, NV MSA economy comprised of 28,340 firms employing 801,495 employees. Correspondingly, there were 240 business associations and 100 professional associations operational in this metropolitan area. This concentration of associations represented 2.4 business associations per 1000 firms and 1.1 professional associations per 1000 firms. We use these two measurements as ‘connections between programs and resources’ similar to prior research by Dani (2016). Normalized by 1000 firms, both of these ratios in Las Vegas are low, contrasting with a report of Washington DC having 10.4 business associations per 1000 forms and 7.3 professional associations per 1000 firms by Dani (2016).

In addition, this MSA has a concentration of investors (0.70) and universities (0.005) active in the Crunchbase database. Universities and investors are surrogate measures for dealmakers (Dani 2016). The investors are the ones who provide startup capital for entrepreneurs. Universities are the entities or places that enable entrepreneurs and researchers to meet, collaborate, exchange ideas, and incubate startups. Las Vegas numbers are relative low in both of these concentrations when compared to Washington DC metro at 1.1 and 0.1 respectively. In other words, Las Vegas does not have many dealmakers in relation to entrepreneurship.

The last measurement is diversity. We collected data about the occupational diversity of workers in the Las Vegas-Henderson-Paradise, NV MSA from the Bureau of Labor Statistics and reported the top six occupations in the area and their market share and median earnings in Table 4. All of the major occupations in this metro area fall under the general category of services.

Table 5 below presents the detailed occupations by location quotient (LQ). LQ is the occupational concentrations of the work force of the Las Vegas-Henderson-Paradise, NV MSA which reveal regional characteristics. More specifically, the top five location quotients (LQ) are within the gaming industry. More specifically, Las Vegas-Henderson-Paradise, NV MSA has high detailed occupational LQs for Gaming Supervisors (LQ = 37.98); Gaming Dealers (LQ = 31.55); Gaming Service Workers, All Other (LQ = 24.60); Gaming Cage Workers (LQ = 20.90) and Gaming Managers (LQ = 16.43).

To provide some perspective on how Las Vegas fairs within the U.S., we collected data at the state level and compare small businesses performance in Nevada to the rest of the country. Our findings revealed that the small business and entrepreneurial development in Nevada represents a smaller part of its economy which is smallest level of development in the United States. More specifically, Nevada state ranks last in the nation in both its level of entrepreneurship and the share of employment in small businesses (firms with fewer than 10 employees). Nevada provides the least amount of private capital for its entrepreneurs and small businesses or finance provided to small businesses. Finally, Nevada falls short on virtually every CFED *Asset and Opportunity Scorecard* measure of small business policy. Table 6 shows the small business characteristics of Nevada in comparison with the United States.

**Table 5** 15 top occupations by location quotient—Las Vegas-Henderson-Paradise NV MSA

|                                                              | LQ    | Employment | Median Earnings |
|--------------------------------------------------------------|-------|------------|-----------------|
| Gaming supervisors                                           | 37.98 | 5900       | 55,620          |
| Gaming dealers                                               | 31.55 | 19,530     | 17,530          |
| Gaming service workers, all other                            | 24.60 | 2570       | 28,680          |
| Gaming cage workers                                          | 20.90 | 2200       | 33,380          |
| Gaming managers                                              | 16.43 | 410        | 87,700          |
| Dancers                                                      | 13.98 | 1010       | Not available   |
| Gaming and sports book writers and runners                   | 11.80 | 920        | 23,320          |
| Gaming change persons and booth cashiers                     | 11.61 | 1470       | 24,310          |
| Baggage porters and bellhops                                 | 10.19 | 2900       | 27,760          |
| Taxi drivers and chauffeurs                                  | 9.76  | 11,210     | 29,470          |
| Gaming surveillance officers and gaming investigators        | 8.75  | 570        | 39,720          |
| Food preparation and serving related workers, all other      | 7.32  | 2150       | 28,300          |
| Costume attendants                                           | 7.19  | 290        | 49,260          |
| Coin, vending, and amusement machine servicers and repairers | 6.09  | 1210       | 46,820          |
| Stonemasons                                                  | 5.37  | 390        | 29,460          |
| Total                                                        |       | 870,740    | 33,160          |

Source Occupational employment statistics, Bureau of Labor Statistics, 2014

**Table 6** Small business indicators for Nevada compared to the United States (in parentheses indicates the ranking for Nevada)

| Small business indicators                      | Nevada     | United States      |
|------------------------------------------------|------------|--------------------|
| Microenterprise ownership rate <sup>d</sup>    | 15.4% (22) | 16.6% <sup>a</sup> |
| Small business ownership rate <sup>e</sup>     | 1.19% (50) | 1.38% <sup>b</sup> |
| Business creation rate <sup>f</sup>            | 9.8 (22)   | 9.3 <sup>c</sup>   |
| Private loans to small businesses <sup>g</sup> | \$939 (50) | \$1318             |

<sup>a</sup>The best state is Florida which has the rate of 21.8%

<sup>b</sup>The best state is Montana with 2.02%

<sup>c</sup>The best state is Montana with 17.1

<sup>d</sup>CFED, *Asset and Opportunity Scorecard*, Total number of firms with 1–4 employees and non-employer firms as a percentage of people in the labor force, 2013 which is available at <http://assetsandopportunity.org/scorecard/>

<sup>e</sup>Ibid. Total number of firms with 5–99 employees as a percentage of people in the labor force, 2013

<sup>f</sup>Ibid. Number of quarterly establishment openings per 1000 workers, 2011

<sup>g</sup>Ibid. The dollar amount of small business loans (defined as private business loans of \$1 million or less) made, per person in the labor force, 2012

## 4 Kaufman Growth Index Report

Another measurement of EE can be found in the Kaufman Growth Index. In their 2016 report, the Las Vegas metro area ranks 32nd out of 40 cities reported. It dropped 3 rankings when compared to 2015. It has a 55.4% rate of startup growth, an increase from their 38% growth rate in 2015. This rate represents funding growth after 5 years measured by growth in employment. The area has a moderate 1.5% share of scaleup, with the market ranges from 0.79 to 2.34. This rate represents growth to 50 employees or more by 10th year as a percentage of all employer firms ten years or younger. Las Vegas had 61.7% high growth company density in 2016, compared to its 83.5% in 2015. The range of this rate was 40.4– 271% for the 40 cities reported in 2016. This rate measures the number of private businesses with at least \$2 million in annual revenue reaching three years of 20% annual revenue growth normalized by total business population. According to this Index, Las Vegas is a slower growth city.

## 5 An Entrepreneurial-Centric Analysis of the Las Vegas EE

As the previous measurements of EE vibrancy and Kaufman Growth Index are at the macro level, we next turn to a more ‘entrepreneur’ centric approach to add depth to the inner workings of the Las Vegas EE for the past decade. In this section, we report entrepreneurs as the focal players and list the activities and funding they received in Table 7. We collected data from Crunchbase and other web-based sources. Before 2009, two major tech entrepreneurs emerged in Las Vegas. The first one is Zappos, an online retailer of shoes started in California in 1999 and relocated to Henderson (suburb of Las Vegas) in 2002, and subsequently sold to Amazon for 1.2 billion in 2009. The second one is Switch SUPERNAP, a data center started in Las Vegas when Rob Roy purchased a facility in Las Vegas (formerly owned by Enron) constructed right over the backbone of fiber optic cables providing services to technology nationwide in 2002.

Most of these companies are funded through various angel investors and crowdfunding except for Downtown Project and Vegas TechFund which are directly funded by Tony Hsieh. Banjo received the highest funding amount at 121 million and its business is in big data analytics. The second one is Orderwithme which received 37 million, a fintech firm that has Zappo as a client. Zirtual, a platform for virtual executive assistant, and Launchkey, a cyber security firm, received 5 and 4 million respectively. Jobhives (platform for hiring), MiToo sports (platform for sports), and Wedgies (data analytics) received more than 2 million dollars each. Alongside these startups and scaleups, Las Vegas also attracted established tech businesses such as Microsoft Licensing, Amazon, Tesla, and

**Table 7** Examples of major startups and scaleups in Las Vegas since 2009

| Company            | Founded | Funding 2017                     | Headquarter               | Industry and remarks                                      |
|--------------------|---------|----------------------------------|---------------------------|-----------------------------------------------------------|
| Alice receptionist | 2012    | Equity crowdfunding, undisclosed |                           | Information technology                                    |
| Banjo              | 2012    | 121 million, 4 rounds            | California                | Big data analysis firm                                    |
| Checkio            | 2012    | 850 k                            | Las Vegas                 | Game for learning Python                                  |
| Crowdhall          | 2012    | 770 k                            | Las Vegas                 | Crowdsource townhall                                      |
| Digital Royalty    | 2009    | 500 k                            | Las Vegas                 | Digital media                                             |
| Discotech          | 2012    | 500 k                            | California                | Hospitality—sell tickets                                  |
| Downtown Project   | 2012    | 350 million                      | Las Vegas                 | Zappo's Tony Hsieh                                        |
| Fandeavor          | 2011    | 525 k                            | Las Vegas                 | Booking platform. Ex-Zappo employees                      |
| Ginx               | 2012    | 700 k, 3 rounds                  | Las Vegas                 | Marketing and analytics—Image based                       |
| JobHive            | 2013    | 2.9 million                      | Las Vegas                 | Hiring automation platform                                |
| LaunchBit          | 2012    | 0.96 million                     | California                | Internet ad—Sold to BuySellAd in 2014                     |
| LaunchKey          | 2012    | 4 million                        | Las Vegas                 | Cybersecurity—eliminate use of passwords                  |
| MiToo Sports       | 2011    | 2.7 million                      | California, Mitoo from UK | Platform for world sports                                 |
| Orderwithme        | 2011    | 37 million                       | Las Vegas                 | Fintech. Streamline retailer backoffice                   |
| Rolltech           | 2011    | 3 million                        | Las Vegas                 | Sports—Mobile app for bowlers                             |
| Tracky             | 2010    | 1.5 million                      | Las Vegas                 | Information Technology—An app that removes clutter        |
| Vegas TechFund     | 2012    | 50 million                       | Las Vegas                 | Zappo's Tony Hsieh, part of Downtown Project              |
| Wedgies            | 2012    | 2.4 million                      | Las Vegas                 | Data and analytics, Online polling collection             |
| Zirtual            | 2011    | 5 million                        | Las Vegas                 | Professional service—Virtual executive assistant platform |

Source Crunchbase, Foundergrid

Apple. Switch Supernap opened a 65,000 square foot Innovation Center that houses startups and scaleups like Banjo, Originate, and Telesphere.

Vegas has recently been touted by the media (e.g., Forbes) as a hot place for startups. In addition to the Downtown project funded by Tony Hsieh, the 24 h nature of the Strip and Freemont Street, the inexpensive real estate prices, the Wild

West culture, lots of sunshine, and the pedestrian friendly downtown are all the reasons listed as an ideal place for small businesses, entrepreneurs, and tech startups (Farr 2013; Schupp 2013). However, critics are concerned about the lack of supporting institutions necessary for making Las Vegas a vibrant EE. Despite the 350 million dollars pumped into downtown to turn it into a tech city, there is not enough tech talents, startup infrastructure, and research universities that naturally feed students and startups (Semuels 2015).

## 6 Discussions

### 6.1 *Continuous Flow of Entrepreneurs*

For the past 100 years, Las Vegas has seen important entrepreneurs ranging from perhaps the most famous Howard Hughes in Hughes Aviation, Kekorian who founded Flamingo to the more recent “Uncrowned King of Las Vegas” Steve Wynn of Mirage, Bellagio, and others (Munk 2005). The core cluster of industry of Las Vegas is still entertainment (i.e., casinos, hotels, resorts). However, the advent of the digital economy has resulted in a new class of entrepreneurs deviated from entertainment business. Tony Hsieh of Zappos and Rob Roy of Switch SUPERNAP are icons of the Las Vegas digital economy. Switch SUPERNAP has emerged to become the most advanced data centers and most powerful technology ecosystems on the planet (Switch Company Website). Given the contrast of the glitzy and ultra-luxury resort business versus the downtown Freemont street mirroring the two generations of entrepreneurs—one of glamorous entertainment business and one of tech business, it will be interesting to see more interactions between these two groups of entrepreneurs developing Las Vegas as a venue for supporting entrepreneurs in the digital economy.

### 6.2 *Homegrown Versus Relocated Entrepreneurs*

Since the beginning of the 1900s, all successful entrepreneurs in Las Vegas came from somewhere other than Nevada. In the first wave of entrepreneurs, Thomas Hull of El Rancho came from Chicago, R.E. Griffith of Last Frontier from Texas, and Kirk Kerkorian of Flamingo from California. Later on Howard Hughes came from Texas, and Steve Wynn came from Connecticut. In the most recent wave of tech entrepreneurs, Tony Hsieh grew up in California, and Rob Roy grew up in Wisconsin. This reflects Las Vegas as a true All-American city (Firat 2001). From an entrepreneurial firm perspective, Las Vegas has both local startups and relocated scaleups. Zappos and Banjo are both relocated scaleups. For the past decade, major tech firms—Apple, Switch, Tesla, Amazon, and Microsoft Licensing—also opened

offices in Las Vegas. The balance between local startups and relocated scaleups is important from the perspective of human capital retention and management. Since not all jobs can attract relocation of talents from out of state, some level of local talents are necessary in order to support entrepreneurial activities and grow a sense of community for existing and future entrepreneurs. Las Vegas may suffer in developing home grown tech entrepreneurs as there is a lack of research active universities to provide the infrastructure that fills the pipeline of student-turn-entrepreneurs (Semuels 2015).

### 6.3 *Cluster and EE Vibrancy*

From a cluster and entrepreneurship perspective (Delgado et al. 2010), the strength of Las Vegas is in gambling, hotels, and entertainment. The positive network externalities that entrepreneurs in Las Vegas will benefit thus surround the industry of gaming, hotels, and entertainment. Exactly how these legacy industries support entrepreneurs in Las Vegas in the digital economy remains a puzzle. Prior literature mentioned that unrelated industry will not benefit from clusters or agglomeration positive network externalities (Delgado et al. 2010). In other words, unrelated industry will not gain support from the existing cluster. Perhaps our case study of Las Vegas adds empirical evidence to support this theory. Academics and think tanks already criticized and projected the failure of Las Vegas as a tech hub because of the lack of startup culture, infrastructure, and human capital (Semuels 2015). The legacy industries of vast casino empires maneuvered by hedge funds and Wall Street (Cohan 2015) seem to be disconnected from the new tech industries. In fact, strong clusters may not foster entrepreneurship at all (Delgado et al. 2010).

On the other hand, alternative theory suggests that diversification or heterogeneity of industries is better for entrepreneurship. Our observation thus far showed that the new tech industry (loosely as Downtown) and legacy entertainment industry (loosely as the Strip) co-exist without much interaction. Does the Strip have negative network externalities to Downtown tech hub? There are industry reports suggesting that the over-emphasis of ‘happiness’ for startup culture in Downtown may end up creating more unnecessary stress and suicides for entrepreneurs (Bowles 2014). The pursuit of happiness has always been the DNA of this All-American neon city (Firat 2001). Tony Hsieh emphasizes this as he is the advocate of the ‘pursuit of happiness’ in his book and evangelizes it as the reason to be a tech entrepreneur in Downtown Las Vegas (Griswold 2014). It is unclear if the partying culture created by legacy industry in Las Vegas have negatively influenced the newly relocated tech entrepreneurs in pushing them too far emotionally by living and working in an isolated desert town without pre-existing social networks and roots (Bowles 2014).

Auerswald and Dani (2017) argued that entrepreneurs and entrepreneurship are at the center of an EE but also that an EE cuts across industries. They continue to suggest that “the best opportunities for policy-making arguably arise neither from

correcting conventionally defined market failures, nor from a singular focus on context-independent “best practices.” Instead the best opportunities for policy-making may arise from understanding, and capitalizing upon, evolutionary transitions within regional entrepreneurial ecosystems and subsidiary industry clusters.” (p. 113). This case seems to suggest the importing of entrepreneurs and the support of the EE by entrepreneurs gives rise to a start-up and scale-up culture (in the pursuit of happiness) that may not benefit from the local cross-industry potential. The perceptions of failing government policy therefore may be based in the view that the Nevada governments (local and state) do not understand and capitalize on the evolutionary transition that leverages the entertainment cluster strength but instead the history of Las Vegas has biased an import mentality rather than the cultivation of strength.

Although we identified Tony Hsieh and Rob Roy as the new iconic tech entrepreneurs that define Las Vegas in the digital age, at the time of this writing, Tony Hsieh’s 50 million dollar Vegas TechFund that was intended to invest in Las Vegas had to invest outside of Las Vegas as they could not find good enough deals locally (Carr 2016).

More interestingly, in the digital economy where most communications with customers and among workers are conducted via bytes, how important does the economic geography-based cluster concept remain? Is it possible that Internet-based relationships with customers and workers transcend that of geographically-bound relationships? Can entrepreneurs find equal support in a *virtual* entrepreneurial ecosystem?

Of particular interest in the Las Vegas case is the role of the large attracted tech firms, e.g. Apple, Switch, Tesla, Amazon, and Microsoft Licensing. Different to the start-up culture, these companies offer unique opportunities to anchor new capability adjacent to the strengths or leveraging the strengths of the Las Vegas economy. For example, Tesla opened its battery plant in Northern Nevada and Switch SUPERNAP followed and announced it would build a data fortress adjacent to Tesla (Akers 2017). Amazon built a fulfillment center in Northern Nevada (Raz 2016). These tech firms add new capabilities of Las Vegas beyond their traditional gaming and entertainment business. These large tech firms benefited from favorable tax rebate from the local government (Raz 2016).

## 6.4 *EE Vibrancy Measurement in the Digital Economy*

The last issue that we need to discuss is about how to measure EE vibrancy in the digital economy. We have used the indicators from Stangler and Bell-Masterson (2015) to measure the EE vibrancy for Las Vegas, but what do the results tell us? In terms of diversity, we found the majority of workers are in the service industry. What does that mean to entrepreneurship ecosystem? It is no secret that Las Vegas is a gambling and tourist town, and food preparation and service labor tops the list. How does this information help policy makers, potential entrepreneurial actors (i.e.,

individuals, universities, corporations) to decide whether to enter the market? For fluidity, we found worker churn rate in Las Vegas three times that of job creation and destruction rates. What does high workers churn rate mean to entrepreneurs? For density of new and young firms, it is below one percent, which is low. Kaufman foundation growth index concluded that entrepreneurial growth rate in Las Vegas is low. Our list of recent tech entrepreneurs suggest that there are some scaleups there. In our three separate sets of measurements, we provided both a macro and a micro lens to the Las Vegas EE. As for the digital economy, digital infrastructure and capabilities in Las Vegas is quite good ranking 12 in Internet access, 7 in online speed, and 20 in household Internet access (Internet Access Rankings 2017). But we know that the digital economy is more than access as entrepreneurs in the digital age require an optimal combination and balance of digital governance, digital citizenship, and digital marketplace to make an EE sustainable (Sussan and Acs 2017). Will Las Vegas tech EE be sustainable?

## 7 Conclusion

We began the chapter with a historical background of Las Vegas and examined its history of entrepreneurs leading to the birth of its legacy industry in gambling, hotel, and entertainment. We found that the legacy industry in the past three decades, although still consist of iconic entrepreneurs, is mainly Wall Street owned and directed. We then presented an analysis of the Las Vegas entrepreneurial vibrancy using data at the macro level to reflect entrepreneurial density, diversity, connectivity, and fluidity. To address our interest in entrepreneurship of Las Vegas in the digital economy, we took an approach at the micro level and followed the development of the major startups and scaleups for the past decade and concluded that there were a few stars but that the overall entrepreneurial activities are moderate.

Measuring EE ecosystem vibrancy in the digital economy proves to be more difficult than expected. The macro data collected using Stangler and Bell-Masterson (2015) framework did not provide any particular insight into the vibrancy of Las Vegas EE except its workforce is mainly engaged in the service industry with a concentration in gaming, that its workers are mobile, the connectivity among entrepreneurs appears to be low, and that the occupation of the workforce consist mainly of food preparation, administrative support, and sales people. The data of Internet access only provided a partial picture of the digital economy. Future research should consider measurements of how digital technologies are used across industries among entrepreneurs in Las Vegas to gauge EE vibrancy in the digital economy. We attempted to list the major tech startups and scaleups but stopped short in analyzing them across industry and their possible impact on the digital economy.

The continuation of legacy industry toward tech industry in the digital economy from a cluster perspective needs more investigation. Theoretically, if online is analogous to face-to-face interactions and knowledge exchange or spillover, we need to rethink whether the advantages of being in a geographically-bound physical

cluster of place-based entrepreneurship remain in the digital economy. Specifically our case showed that Las Vegas was built on the historical importing of the gaming and entertainment industry enabled by lax regulations, a local under-met demand from an isolated community with a growth population around mining and resources, the tourism from atomic testing that each contributed to creating the dynamics to build an export industry—i.e., a local service base attracting high flyers and tourists of all types. Will these same dynamic capabilities continue to attract or support Las Vegas-based digital entrepreneurs? Since digital entrepreneurs with virtual business can more or less operate anywhere that they find the best conditions for a business to flourish, Las Vegas was able to attract entrepreneur firms Zappos, then Switch SUPERNAP, then subsequently larger tech business Microsoft Licensing, Amazon, Tesla, and Apple. These firms are remotely related to gaming and entertainment business. Perhaps the tech entrepreneurs are gradually building a tech culture and tech human capital base that can possibly support these larger tech businesses? Entrepreneurs in the digital economy may need and generate a different kind of network externality than the existing cluster theory has to offer. Future research should investigate what network externalities exist in the online environment without the need to rely on geographically-bound network externalities for entrepreneurs to be supported in the digital economy.

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## References

- Akers M (2017) Switch data center opens near tesla battery plant in Northern Nevada. <https://vegasinc.lasvegassun.com/business/2017/feb/14/switch-data-center-opens-near-tesla-battery-plant/>
- Auerswald PE, Dani L (2017) The adaptive life cycle of entrepreneurial ecosystems: the biotechnology cluster. *Small Bus Econ* 49:97–117
- Austrian Z, Piazza MC (2014) Barriers and opportunities for entrepreneurship in older industrial regions. The road through the rustbelt, pp 215–243
- Bernhard BJ, Green MS, Lucas AF (2008) From Maverick to Mafia to MBA gaming industry leadership in Las Vegas from 1931 through 2007. *Cornell Hospitality Q* 49(2):177–190
- Bowles N (2014) The downtown project suicides: can the pursuit of happiness kill you? building a startup city in the desert is lonely and hard. <https://www.recode.net/2014/10/1/11631452/the-downtown-project-suicides-can-the-pursuit-of-happiness-kill-you>
- Bunker D (2016) Why the Nevada Tech Zone is surging. Retrieved 5 July 2017 from <https://innotechtoday.com/nevadas-tech-zone-surge/>
- Carr PD (2016) 75% of Vegas Tech Fund's investments are now outside of Vegas (and they've dropped Vegas from their name). Retrieved 5 July 2017 from <https://pando.com/2016/07/18/75-vegas-tech-funds-investments-are-now-outside-vegas-and-theyve-dropped-vegas-their-name/>
- Cohan WD (2015) A private equity gamble in Vegas gone wrong. *Fortune* June 5. Retrieved 5 July 2017 from <http://fortune.com/2015/06/05/caesars-losing-las-vegas/>

- Dani L (2016) Washington, D.C. metropolitan statistical area: entrepreneurial ecosystem metrics summary (15 Jan). Center for Regional Analysis, George Mason University. Retrieved from [http://cra.gmu.edu/pdfs/studies\\_reports\\_presentations/Washington\\_Metrics.pdf](http://cra.gmu.edu/pdfs/studies_reports_presentations/Washington_Metrics.pdf)
- Delgado M, Porter ME, Stern S (2010) Clusters and entrepreneurship. *J Econ Geogr* 10(4):495–518
- Farr C (2013) Why investors tech startups are flocking to downtown Vegas. Retrieved 5 July 2017 from <https://venturebeat.com/2013/06/26/part-3-why-investors-tech-startups-are-flocking-to-downtown-vegas/>
- Feld B (2012) Startup communities: building an entrepreneurial ecosystem in your city. Wiley, Hoboken
- Firat AF (2001) The meanings and messages of Las Vegas: the present of our future. *M@n@gement* 4(3):101–120
- GAO (2013) U.S. Government Accountability Office. Hughes changed strip landscape with corporately-owned casinos. Retrieved 25 May 2017 from <https://lasvegassun.com/news/2008/may/15/how-vegas-went-mob-corporate/>
- Griswold A (2014) The dark side of techtopia. Accessed from [http://www.slate.com/blogs/moneybox/2014/10/02/tony\\_hsieh\\_and\\_the\\_downtown\\_project\\_the\\_dark\\_side\\_of\\_a\\_las\\_vegas\\_techtopia.html](http://www.slate.com/blogs/moneybox/2014/10/02/tony_hsieh_and_the_downtown_project_the_dark_side_of_a_las_vegas_techtopia.html)
- Internet Access Rankings, US news. Retrieved 5 July 2017 from <https://www.usnews.com/news/best-states/rankings/infrastructure/internet-access>
- Knickrehm M, Berthon B, Daugherty P (2016) Digital disruption: the growth multiplier. Accenture Strategy, Tech Rep
- Mack E, Mayer H (2016) The evolutionary dynamics of entrepreneurial ecosystems. *Urban Stud* 53(10):2118–2133
- Munk N (2005) Steve Wynn's biggest gamble. Retrieved 5 July 2017 from <http://www.vanityfair.com/news/2005/06/steve-wynn-las-vegas-resort>
- Qian H, Haynes KE (2014) Beyond innovation: the small business innovation research program as entrepreneurship policy. *J Technol Transfer* 39(4):524
- Raz N (2016) Amazon plans new North Las Vegas fulfillment center with estimated 1 k jobs. <https://www.reviewjournal.com/business/amazon-plans-new-north-las-vegas-fulfillment-center-with-estimated-1k-jobs/>
- Schupp S (2013) 5 Reasons you should consider starting up in Vegas. <https://www.forbes.com/sites/yec/2013/03/18/5-reasons-you-should-consider-starting-up-in-vegas/#3e29fb7d7287>
- Samuels A (2015) \$350 million might not be enough to save Las Vegas. Retrieved 5 July 2017 from <https://www.theatlantic.com/business/archive/2015/03/350-million-might-not-be-enough-to-save-las-vegas/386213/>
- Shapiro C, Varian H (1999) Information rules. Harvard Business School Press, Boston, MA
- Stangler D, Bell-Masterson J (2015) Measuring an entrepreneurial ecosystem. Kauffman Foundation Research Series on City, Metro, and Regional Entrepreneurship
- Sussan F, Acs ZJ (2017) The digital entrepreneurial ecosystem. *Small Bus Econ* 49(1):55–73
- Switch Company Website. Retrieved 5 July 2017 from <https://www.switch.com/data-centers/>

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# Measuring Entrepreneurial Ecosystems

Erik Stam

**Abstract** How can entrepreneurial ecosystems and productive entrepreneurship be traced empirically and how is entrepreneurship related to entrepreneurial ecosystems? The analyses in this chapter show the value of taking a systems view on the context of entrepreneurship. We measure entrepreneurial ecosystem elements and use these to compose an entrepreneurial ecosystem index. Next, we measure the output of entrepreneurial ecosystems with different indicators of high-growth firms. We use the 12 provinces of the Netherlands as a test case for measuring the entrepreneurial ecosystem elements, composing an entrepreneurial ecosystem index and relate this to entrepreneurial outputs. The prevalence of high-growth firms relates to the overall value of the entrepreneurial ecosystem index, but not to individual elements of the entrepreneurial ecosystem. The model fit increases once we introduce a multiplicative index and a non-linear model. By measuring entrepreneurial ecosystems and their outputs in this way we move from the ecosystem metaphor to a complex system model of the entrepreneurial economy.

**Keywords** Entrepreneurial ecosystems • Complex systems • Regional development • Entrepreneurship • High-growth firms

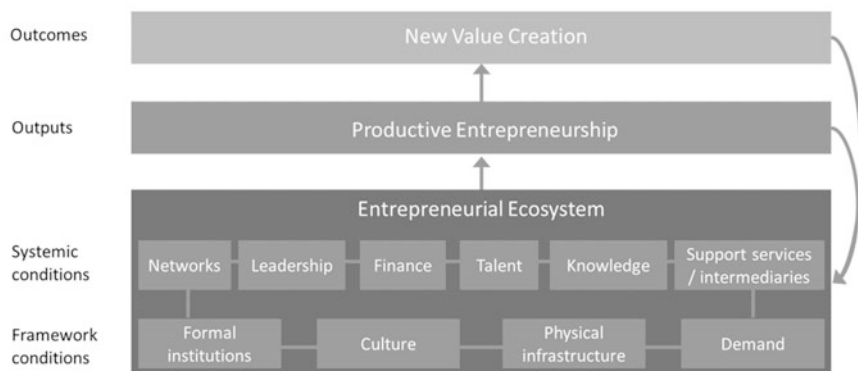
## 1 Introduction

The entrepreneurial ecosystem approach has gained prominence amongst entrepreneurs, entrepreneurship supporters (Feld 2012), and in public policy (Isenberg 2010). Entrepreneurial ecosystems have revealed to be a mobilizing metaphor in the public and private sector, also bridging the boundary between the two. The popularity of the concept is however not a sign of its robustness: the approach is highly under theorized, and not yet adequately measured (Stam 2015). This chapter offers measures and methods for analysing entrepreneurial ecosystems as a system,

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**Fig. 1** Key elements, outputs and outcomes of the entrepreneurial ecosystem (based on: Stam 2015)

moving beyond the metaphorical use of the entrepreneurial ecosystem concept. We use complex systems—systems composed of many components which may interact with each other—reasoning to improve insight into how entrepreneurial ecosystems function as a whole system. Complexity economics (Arthur 2013) provides the conceptual basis for analysing the relation of entrepreneurial ecosystems with outputs (entrepreneurship) and outcomes (value creation and structural change).

Several studies (Stam 2015; Spigel 2017; Stam and Spigel 2018) have shown that the entrepreneurial ecosystem approach can be used to synthesize prior academic research on the geography of entrepreneurship (see e.g. Sternberg 2009; Stam 2010) and entrepreneurship and regional development (see e.g. Fritsch 2013). The synthesizing model (see Fig. 1) includes insights from the academic literature (i.e. the aspects that have been deemed important elements of entrepreneurial ecosystems), but most importantly it provides more causal depth with four ontological layers (framework conditions, systemic conditions, outputs, and outcomes), including the upward and downward causation, and intra-layer causal relations (Stam 2015). Upward causation reveals how the fundamental causes of new value creation are mediated by intermediate causes, while downward causation shows how outcomes and outputs of the system over time also feed back into the system conditions. Intra-layer causal relations refer to the interaction of the different elements within the ecosystem, and how the different outputs and outcomes of the ecosystem might interact (Stam 2015). The model is distinctive of existing measurements of entrepreneurial (eco) systems that do not separate inputs and entrepreneurial outputs of the system (e.g. Acs et al. 2014; Bell-Masterson and Stangler 2015).

The entrepreneurial ecosystem approach can go beyond a metaphorical approach by constructing a complex systems approach to entrepreneurship and structural economic change. The economy is a constantly evolving system in which economic agents at the micro level experiment and interact with each other. Many of these

experiments fail, but some succeed as innovations and create wealth for society (Beinhocker 2006). Economic development does not emerge automatically: entrepreneurs are needed to create new value (Fayolle 2007; Schumpeter 1934). This new value creation is an emerging property of a complex system of economic agents and their interactions: the entrepreneurial ecosystem. Entrepreneurs might structurally change the economy and society, as exemplified with new sets of technologies, institutions and organizational arrangements (Arthur 2013). An entrepreneurial ecosystem can be defined as a set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship within a particular territory (Stam 2015; Stam and Spigel 2018). Entrepreneurship is both the result of and the mediator of evolution (Day 1987): entrepreneurial behaviour as an output is enabled by the system, while the new value created, and potential structural change as an outcome of the system is mediated by entrepreneurship. This outcome is an emergent property of the system, and redefines the nature of the system via feedback effects. Such feedback effects mean that the system and its outputs should not be interpreted as a one way relation, as the current state of the system might be affected by previous outcomes. This comes close to the statistics problem of simultaneity, which “arises when one or more of the explanatory variables is jointly determined with the dependent variable [...]” (Wooldridge 2013, p 530). However, in dynamic systems analysis this is not a problem to be evaded, but an inherent characteristic of system dynamics. In the next section, we discuss how we can empirically trace the elements of entrepreneurial ecosystems as a first step towards analysing the entrepreneurial economy (Thurik et al. 2013) as a complex system.

## 2 Elements of Entrepreneurial Ecosystems

Entrepreneurial ecosystems are constituted by framework conditions and systemic conditions (Stam 2015). Both are summarized in Fig. 1. The framework conditions include the social (formal institutions and culture) and physical conditions enabling or constraining human interaction. In addition, access to a more or less exogenous demand for new goods and services is also of great importance. This access to buyers of goods and services, however, is likely to be more related to the relative position of the ecosystem than its internal conditions. These conditions might be regarded as the fundamental causes of value creation in the entrepreneurial ecosystem (cf. Acemoglu et al. 2005). However, to fully understand how these fundamental causes lead to this outcome, we first need to gain insight into how systemic conditions lead to entrepreneurial activity (cf. Cooke 2001).

Systemic conditions are the heart of the ecosystem: networks of entrepreneurs, leadership, finance, talent, knowledge, and support services. The presence of these elements and the interaction between them are said to be of crucial relevance for the

success of the ecosystem. Networks of entrepreneurs provide information flows, enabling an effective distribution of knowledge, labour and capital. Leadership provides direction and role models for the entrepreneurial ecosystem. This leadership has been said to be critical in building and maintaining a healthy ecosystem: this involves a set of ‘visible’ entrepreneurial leaders who are committed to the region (Feld 2012). Access to finance—preferably provided by investors with entrepreneurial knowledge—is crucial for investments in uncertain entrepreneurial projects with a long-term horizon (see e.g. Kerr and Nanda 2009). But perhaps the most important element of an effective entrepreneurial ecosystem is the presence of a diverse and skilled group of workers (‘talent’: see e.g. Acs and Armington 2004; Lee et al. 2004; Qian et al. 2013). An important source of opportunities for entrepreneurship can be found in knowledge, from both public and private organizations (see e.g. Audretsch and Lehmann 2005). Finally, the supply of support services by a variety of intermediaries can substantially lower entry barriers for new entrepreneurial projects, reduce the time to market of innovations (see e.g. Zhang and Li 2010), and enhance the diffusion of innovations, as input to next generations of innovation (Howells 2006). There is (quantitative) empirical evidence for the effects of all these elements<sup>1</sup> on entrepreneurship and economic growth, most often on the regional level. We thus build on previous studies for selecting empirical indicators of these elements. The challenge is to find empirical indicators that are comparable over time and space.

## 2.1 *Formal Institutions*

Formal and informal institutions (culture) reflect the rules of the game in society (North 1990). For entrepreneurship, the quality and efficiency of institutions matter: the level of perceived corruption and the general regulatory framework within countries. We use data from the Quality of Governance 2012 survey. It consists of data acquired for a large, European Commission-funded project on measuring quality of governance within the EU (Charron et al. 2012). The survey is the largest one ever undertaken to measure quality of governance at the sub-national level so far. It includes approximately 34,000 EU citizens for a total of 172 regions, either at the NUTS1 or NUTS2 level, within the EU Member States. Survey questions are focused on four aspects related to three public services (education, healthcare and law enforcement): corruption, rule of law, government effectiveness, and voice and accountability. Four standardized indicators are provided with and used in the “formal institutions” element of the entrepreneurial ecosystem (for additional details refer to Charron et al. 2012).

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<sup>1</sup>Except on the role of leadership (see Beer and Clower 2014 for a recent discussion).

## 2.2 *Culture*

Entrepreneurship culture (as an informal institution) reflects the degree to which entrepreneurship is valued in society. We measure entrepreneurship culture indirectly with the prevalence of new firms, which indicates how ‘common’ starting up a business is in a particular region.

Entrepreneurship culture could also be measured with the degree to which self-employment is seen as a viable career choice and the degree to which successful entrepreneurs are valued (both derived from the Global Entrepreneurship Monitor). However, this measure is not readily available for regions within the Netherlands.

## 2.3 *Physical Infrastructure*

Physical infrastructure is a composite measure including indicators of motorway and railway potential accessibility and the number of passenger flights (see Annoni and Dijkstra 2013). Motorway accessibility includes the population living in surrounding regions weighted by travel time along motorways, while railway accessibility includes the population living in surrounding regions weighted by travel time along railways. Motorway and railway potential accessibility indicators take into account ferry networks allowing for correcting islands penalization. Potential accessibility is a construct of two functions, the activity function representing the activities or opportunities to be reached and the impedance function representing the effort, time, distance or cost needed to reach them (Spiekermann et al. 2002). For potential accessibility the two functions are combined multiplicatively, i.e. they are weights to each other and both are necessary elements of accessibility. The interpretation is that the greater the number of attractive destinations in areas  $j$  and the more accessible areas  $j$  are from area  $i$ , the greater the accessibility of area  $i$ . The accessibility model used is based on the work of Spiekermann and Wegener (1996) and uses centroids of NUTS 2 regions as origins and destinations. The accessibility model calculates the minimum paths for the road network, i.e. minimum travel times between the centroids of the NUTS 2 regions. For each region the value of the potential accessibility indicator is calculated by summing up the population in all other regions weighted by the travel time to go there. For access to the region to itself, the time to the centroid of the region is used, while for access to other regions: (i) travel time over the network between the two centroids plus the (ii) access from the destination centroid to the destination region are used. The potential accessibility indicators use population and give the highest weight to the population that can be reached within four hours (Annoni and Dijkstra 2013).

The indicator on passenger flights is from Eurostat/EuroGeographics/National Statistical Institutes and corresponds to the daily number of passenger flights accessible within a 90 min’ drive from the region center.

## **2.4 Demand**

Demand is measured as a composite consisting of disposable income per capital and two measures of potential market demand. Disposable income is included as income per capita. The two indicators on potential market demand provide an estimate of the GDP and population available within a pre-defined neighbourhood. They are expressed respectively in purchasing power standards and population size (EU average set to 100). See Annoni and Kozovska (2010) for details on the computation of potential market demand indicators.

## **2.5 Networks**

Networks indicate the connectedness of businesses for new value creation, which is measured as the percentage of businesses (with at least 10 employees) in a region that collaborate for innovation, based on data of the Community Innovation Survey 2010 (CIS; see Arundel and Smith 2013).

## **2.6 Leadership**

Leadership provides guidance for and direction of collective action. Leadership is measured with the prevalence of innovation project leaders. We have constructed a database with information on all the innovation projects in the Netherlands that received (Dutch or European) public subsidies in the period 2010–2013 (see Stam et al. 2016). We selected projects with at least two participating organizations (2231 projects). The geographical origin of these projects is established by taking the province of the main applicant or principal firm. This allowed us to measure the prevalence of innovation project leaders per 1000 businesses in each region (Table 1).

## **2.7 Talent**

Talent can be indicated by the prevalence of individuals with high levels of human capital. This is measured with the share of the population aged 15–65 years with a higher education degree.

Talent could also be measured with the share of the labour force with at least secondary education, but we have chosen for the more general, population based indicator.

**Table 1** Empirical measures of the Entrepreneurial Ecosystem Elements

| Elements                 | Description                                                               | Empirical indicators                                                                                                                                                                                                                                                                                                                                                                                | Data sources                                 |
|--------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Formal institutions      | The rules of the game in society, in particular the quality of government | Four components: corruption, rule of law, government effectiveness and voice and accountability                                                                                                                                                                                                                                                                                                     | Quality of Government Survey 2012 (RCI 2013) |
| Entrepreneurship culture | The degree to which entrepreneurship is valued in a region                | New firms registered per 1000 inhabitants                                                                                                                                                                                                                                                                                                                                                           | CBS (Netherlands Census Bureau) 2012         |
| Physical infrastructure  | Physical infrastructure and the position of a region                      | Three components: accessibility via road, accessibility via railroad, accessibility via airports (number of passenger flights within 90 minutes drive)                                                                                                                                                                                                                                              | RCI 2013                                     |
| Demand                   | Potential market demand                                                   | Three components: purchasing power per capita, regional product, total human population                                                                                                                                                                                                                                                                                                             | RCI 2013                                     |
| Networks                 | The connectedness of businesses for new value creation                    | Percentage of firms in the business population that collaborate for innovation                                                                                                                                                                                                                                                                                                                      | Community Innovation Survey 2010             |
| Leadership               | Leadership that provides guidance for and direction of collective action  | Leadership is measured with the prevalence of innovation project leaders per 1000 businesses, derived from a database with information on all the innovation projects in the Netherlands that received (Dutch or European) public subsidies in the period 2010–2013. The geographical origin of these project leaders is established by taking the province of the main applicant or principal firm | Birch Consultants (see Stam et al. 2016)     |
| Talent                   | The prevalence of individuals with high levels of human capital           | Percentage of higher-educated in the adult population                                                                                                                                                                                                                                                                                                                                               | CBS (Netherlands Census Bureau) 2013         |
| Finance                  | The supply and accessibility of finance for new and small firms           | Percentage of SMEs that have applied for bank loans and also received this                                                                                                                                                                                                                                                                                                                          | EIM 2009                                     |
| New knowledge            | Investments in new knowledge                                              | Percentage of gross domestic product invested in R&D (by public and private organizations)                                                                                                                                                                                                                                                                                                          | CBS (Netherlands Census Bureau) 2009         |
| Intermediate services    | The supply and accessibility of intermediate business services            | Percentage of business service firms in the business population                                                                                                                                                                                                                                                                                                                                     | CBS (Netherlands Census Bureau) 2013         |

## **2.8 Finance**

The supply and accessibility of finance for new and small firms is an important condition for their growth and survival. We use the percentage of SMEs that applied for bank finance and received it as an indicator for the finance element. This measure is based on a study amongst 3027 SMEs in the Netherlands on their need for bank finance and the success rate in acquiring bank finance (Snoei and Ichou 2010). High-growth SMEs in the Netherlands do not more often apply for bank finance than other SMEs, but are more successful in getting bank finance, and use more credit facilities at their bank (Braaksma et al. 2014).

Finance can be traced in many other ways: for example with the ease of access to loans (World Economic Forum), the prevalence of informal investors (Global Entrepreneurship Monitor), venture capital, and crowd funding. Data for these measures is available at the national, but not at the regional level.

## **2.9 Knowledge**

Investments in new knowledge are an important source of entrepreneurial opportunities, and if they lead to (better) solutions, they are also a source of prosperity. New knowledge is created in many ways, but probably the best measured activity is investments in (public and private) research and development. Our indicator for the knowledge element is the percentage of gross domestic product invested in R&D (by public and private organizations).

## **2.10 Intermediate Services**

The supply and accessibility of intermediate business services can substantially lower the barriers and increase the speed of new value creation. Our indicator for intermediate services is the percentage of business service firms in the business population.

# **3 Entrepreneurship Outputs**

A ‘healthy’ entrepreneurial ecosystem is said to produce entrepreneurship as an output and ultimately aggregate value as outcome. There are no perfect measures of either entrepreneurship nor aggregate value creation. To capture both output and outcome we use the concept of productive entrepreneurship (see Stam 2015;

Stam and Spigel 2018). Productive entrepreneurship refers to “any entrepreneurial activity that contributes directly or indirectly to net output of the economy or to the capacity to produce additional output” (Baumol 1993, p 30). We interpret this as entrepreneurial activity that creates aggregate welfare.

We have proxied productive entrepreneurship with the prevalence of high-growth firms (Henrekson and Johansson 2010; OECD 2011; Stam and Bosma 2015). These high-growth firms are rare, but not so rare as “unicorns” (start-ups valued over \$1 billion). Taking “unicorns” as entrepreneurial output, would leave many regions with zero output. We could also start at the other side of the ‘entrepreneurship funnel’, and count the share of the population that has the intention to start a business, or has just started a business. But we regard this to be an indicator of entrepreneurial culture in a region, not as entrepreneurial output. However, one might take a more process view of entrepreneurial outputs, and differentiate the entrepreneurial ecosystem contexts per phase of the entrepreneurial process (see Stam and Bosma 2015). This is probably most relevant for the finance element, with nascent entrepreneurs, start-ups, moderately growing, high-growth and unicorn firms having substantially different finance needs.

We use three measures of high-growth firms: ambitious entrepreneurs, high-growth businesses, and gazelles. The academic literature also proposes other measures (see: Stam 2007; Bos and Stam 2013; Parker et al. 2010; Daunfeldt et al. 2014), but these are not readily available for regions within the Netherlands in recent years.

### ***3.1 Ambitious Entrepreneurs***

To create a scale-up, entrepreneurs need to have the ambition to create a substantial new organization (Stam et al. 2011, 2012). The share of the adult population in a region that has the ambition to grow a new business to a size of 20 employees within 5 years, based on Global Entrepreneurship Monitor data, so-called ambitious entrepreneurs (see Stam et al. 2011, 2012). This is a measure of potential productive entrepreneurship, not of realized productive entrepreneurship. Having the ambition to grow a business is close to a necessary condition for subsequent growth, but it is far from a sufficient condition: only about 1 out of 5 entrepreneurs with a growth ambition realizes this within a two-year period (Stam et al. 2012, p 97).

This measure is better available on a global scale than most other proxies of productive entrepreneurship. We analysed the prevalence of individuals in the adult population that are involved in setting up a business, or own a young business (less than 42 months old), who expect this new business to have at least 20 employees within 5 years (see Bosma et al. 2009; Stam et al. 2009, 2011; Stam and Van Stel 2011). Given the low frequency of this kind of activity we took a longer time frame (2006–2014) to compute the annual average per province over this period.

### 3.2 *High-Growth Businesses*

There is an emerging consensus on seeing high-growth businesses as business that grow substantially consistently: businesses that have grown with at least 20% per year in turnover or employment over a three-year period (OECD 2011). Statistics Netherlands (CBS) annually composes a dataset of high-growth businesses in the Netherlands. The regionalized version of this dataset consists of businesses that have grown with at least 20% per year in turnover or employment over a three-year period, with at least 10 employees at the start of this three-year period. The number of high-growth businesses has declined from 20,590 businesses in 2010 to 17,020 in 2015. As a share of the total business population, it has gone down from 16.7% in 2010 to 12% in 2015. There is some regional variation, ranging from 13.7% in Zuid-Holland and Limburg to 9.7% in Friesland (in 2014).

### 3.3 *Gazelles*

The Dutch Financial Times (Financieele Dagblad), in collaboration with the Chambers of Commerce, has developed a somewhat more selective measure of high-growth businesses in the Netherlands: the number of independent firms with a profitable growth in turnover of at least 20% per year over three years (i.e. at least 72.8% over the full three-year period). The selection logic for the 2013 sample is as follows (FD Gazellen 2013):

1. There are about 2 million registered firms in the Netherlands
2. 825,000 of these firms are obliged to publish their annual financial details
3. 11,400 of these firms have published annual financial reports in time
4. Only 1750 of these firms had an average turnover growth of at least 20% over the last three years
5. 784 of these also fulfilled the following requirements: profitable, financial position, payment behaviour
6. After a quality check, 395 gazelles remained (in 2014: 332).

Table 2 shows the shares of the three measures of entrepreneurship in the Dutch provinces, and the rankings of the provinces based on their performance on these measures.

There is some convergence in the province rankings: the Northern peripheral provinces of Groningen, Friesland and Drenthe never rank higher than 6, and the southern provinces of Noord-Brabant and Limburg never rank lower than 6, just like the urbanized province Zuid-Holland. For the other provinces, there is quite some divergence over the rankings of the different entrepreneurship measures.

For our analyses, we take the most selective measure of productive entrepreneurship, the share of gazelles. In the next section, we will analyse the effects of the individual components on the shares of gazelles.

**Table 2** Shares and rankings of the three measures of entrepreneurship in Dutch provinces

| Province      | Share ambitious entrepreneurship, 2006–2014 (%) |    | Share of high-growth businesses, 2014 (%) |    | Share gazelles, 2014 (%) |    |
|---------------|-------------------------------------------------|----|-------------------------------------------|----|--------------------------|----|
| Groningen     | 0.399                                           | 9  | 11.2                                      | 11 | 0.020                    | 6  |
| Friesland     | 0.337                                           | 10 | 9.7                                       | 12 | 0.018                    | 9  |
| Drenthe       | 0.465                                           | 8  | 13.2                                      | 6  | 0.003                    | 12 |
| Overijssel    | 0.312                                           | 12 | 13.4                                      | 4  | 0.025                    | 4  |
| Gelderland    | 0.482                                           | 7  | 12.4                                      | 9  | 0.014                    | 10 |
| Flevoland     | 0.559                                           | 5  | 13.5                                      | 3  | 0.010                    | 11 |
| Utrecht       | 0.494                                           | 6  | 12.4                                      | 9  | 0.045                    | 1  |
| Noord-Holland | 0.706                                           | 2  | 12.8                                      | 8  | 0.029                    | 2  |
| Zuid-Holland  | 0.847                                           | 1  | 13.7                                      | 1  | 0.024                    | 5  |
| Zeeland       | 0.331                                           | 11 | 13.0                                      | 7  | 0.020                    | 6  |
| Noord-Brabant | 0.561                                           | 4  | 13.3                                      | 5  | 0.027                    | 3  |
| Limburg       | 0.631                                           | 3  | 13.7                                      | 1  | 0.020                    | 6  |

## 4 Entrepreneurial Ecosystem Elements and Entrepreneurship Outputs

There is substantial variation in the values of the different entrepreneurial ecosystem elements, even within a small country like the Netherlands (see Table 3). Very often the highest ranked region has a value that is more than double the value of the lowest ranked region. There is also a 15-fold difference in the rate of gazelles between regions within the Netherlands. This regional heterogeneity in the prevalence of gazelles is much more substantial than the heterogeneity in the prevalence of start-ups, as captured with the entrepreneurship culture element (cf. Stam 2005).

Several entrepreneurial ecosystem elements are highly correlated in this Dutch dataset (see Table 3), as might be expected. The infrastructure index and demand index are strongly positively correlated. Education is highly positively correlated with new firm formation and the prevalence of intermediate services. There seems to be a cluster of knowledge economy variables, including Research and Development, education, and (knowledge-intensive) intermediate services. Remarkably, the quality of government index is strongly negatively correlated with the infrastructure and demand indices, pointing at the peculiarities of the value of this index in Dutch regions (especially the relatively low quality of government in Noord-Brabant and Limburg).

These high correlations between predictor variables pose the statistics problem of ‘multicollinearity’. One predictor variable (e.g. infrastructure) can be linearly predicted from the others (e.g. market access) with a substantial degree of accuracy. In a situation of multicollinearity, the coefficient estimates of the multiple regression may change erratically in response to small changes in the model or the data.

Table 3 Correlation table entrepreneurial ecosystem elements and output

|           | Min     | Max     | Mean     | SD      | Gazelles | RCL_Ins  | NewFiFo  | RCL_Infra |
|-----------|---------|---------|----------|---------|----------|----------|----------|-----------|
| Gazelles  | 0.0030  | 0.00450 | 0.00213  | 0.00105 | 1        |          |          |           |
| RCL_Ins   | 1.12    | 1.59    | 1.3333   | 0.01689 | -0.355   | 1        |          |           |
| NewFiFo   | 9.69    | 16.07   | 11.8555  | 1.9331  | 0.0622*  | -0.260   | 1        |           |
| RCL_Infra | -0.16   | 1.99    | 1.1092   | 0.07307 | 0.0468   | -0.922** | 0.0386   | 1         |
| RCL_Ma    | -0.24   | 1.45    | 0.07383  | 0.05837 | 0.0503   | -0.846** | .0488    | 0.0948**  |
| FirmCol   | 0.00990 | 0.01430 | 0.01160  | 0.00138 | -0.547   | 0.0276   | -0.291   | -0.456    |
| ProjLd    | 0.033   | 2.45    | 1.3342   | 0.07465 | 0.0449   | -0.384   | 0.0473   | 0.0476    |
| SucFin    | 37      | 95      | 56.50    | 16.065  | 0.0130   | 0.0271   | -0.039   | -0.375    |
| HiEdp     | 0.02000 | 0.03700 | 0.02674  | 0.00477 | 0.0756** | -0.135   | 0.0845** | 0.0329    |
| RD        | 0.040   | 2.40    | 1.5083   | 0.06142 | 0.0501   | -0.615*  | 0.0466   | 0.0664*   |
| ImServ    | 0.0131  | 0.0269  | 0.01858  | 0.00380 | 0.0713** | -0.241   | 0.0871** | 0.0472    |
|           | RCL_Ma  | FirmCol | ProjLd   | SucFin  | HiEdp    | RD       |          | ImServ    |
| Gazelles  |         |         |          |         |          |          |          |           |
| RCL_Ins   |         |         |          |         |          |          |          |           |
| NewFiFo   |         |         |          |         |          |          |          |           |
| RCL_Infra |         |         |          |         |          |          |          |           |
| RCL_Ma    | 1       |         |          |         |          |          |          |           |
| FirmCol   | -0.458  | 1       |          |         |          |          |          |           |
| ProjLd    | 0.0433  | -0.179  | 1        |         |          |          |          |           |
| SucFin    | -0.407  | 0.0146  | 0.0204   | 1       |          |          |          |           |
| HiEdp     | 0.0398  | -0.267  | 0.0658*  | 0.078   | 1        |          |          |           |
| RD        | 0.0599* | -0.068  | 0.0821** | 0.177   | 0.613*   | 1        |          |           |
| ImServ    | 0.0565  | -0.329  | 0.0552   | -0.156  | 0.948**  | 0.601*   | 1        |           |

\*\*Correlation is significant at the 0.01 level (2-tailed)

\*Correlation is significant at the 0.05 level (2-tailed)

Multicollinearity does not reduce the predictive power or reliability of the overall model, it only affects calculations regarding individual predictors. A multiple regression model with correlated predictors can indicate how well the entire bundle of predictors forecasts the outcome variable (i.e. a high  $R^2$ ), but it may not give valid results about any individual predictor, or about which predictors are redundant with respect to others. This multicollinearity is also a reason why an index value might better represent the ‘quality’ of the entrepreneurial ecosystem than a set of independent variables. Despite these collinearity problems, we are interested in how and to what extent the decomposable parts of the entrepreneurial ecosystem, statistically explain differences in entrepreneurship output measures. We first performed a correlation analysis to see whether there are bivariate relations between the individual elements on the one hand and the shares of gazelles on the other hand. The correlation table (Table 3) shows that only three variables are (positively) statistically related to the share of gazelles, namely entrepreneurship culture, talent and intermediate services.

The ‘standard’ methodological procedure in social science for tracing the effects of individual independent variables, controlling for the effects of the other independent variables, is a multivariate regression model. We executed a multivariate linear regression model with ten independent variables reflecting the ten elements of the entrepreneurial ecosystem, with the share of gazelles per province as dependent variable (see Table 4). Despite of the high  $R^2$  (0.940) of the model, none of the independent variables has a statistically significant effect on the dependent variable, and several even have negative coefficients. Even if we reduce the model from 10 to the 5 variables with positive coefficients ( $R^2$  0.703), there is only one independent variable (higher education) with a weak ( $p$  0.096) statistically significant effect. The lack of statistical significant relations of ‘predictor’ variables with entrepreneurial output does not mean that the elements are unimportant. They may all be important, but are perhaps already at a value that is beyond a necessary threshold value (such as institutions and infrastructure, which are all at the top levels within Europe; see Annoni and Dijkstra 2013). The diagnostics question is whether a weakest link

**Table 4** Multivariate linear regression of ten entrepreneurial ecosystem elements on gazelles

|            | B       | SE    | Sign  |
|------------|---------|-------|-------|
| (Constant) | 0.032   | 0.097 | 0.796 |
| RCI_InsLN  | 0.0007  | 0.029 | 0.845 |
| NewFiFo    | 0.0003  | 0.010 | 0.826 |
| RCI_InfLN  | -0.0007 | 0.007 | 0.481 |
| RCI_MaLN   | 0.016   | 0.020 | 0.559 |
| FirmCol    | -0.654  | 0.514 | 0.424 |
| ProjLd     | -0.024  | 0.024 | 0.500 |
| SucFin     | -0.001  | 0.001 | 0.689 |
| HiEdp      | 0.973   | 0.919 | 0.482 |
| RD         | 0.041   | 0.055 | 0.590 |
| ImServ     | -1.431  | 1.953 | 0.597 |

$R^2$ : 0.940

analysis (cf. Acs et al. 2014; Szerb and Acs 2011) is appropriate in this context, because a relatively low value of an element that is beyond the necessary minimum level (e.g. formal institutions) is not a substantial constraint. And on the other hand, the Netherlands performs rather bad on accessibility of SME loans (see Gomez and Stam 2017), which means that a relatively high score on the finance element might still be a substantial constraint in the ecosystem.

Decomposing the entrepreneurial ecosystem into a set of elements and then regressing these on the output of the entrepreneurial ecosystem, does not seem to be adequate both for substantive and statistical reasons. The substantive reason is that the entrepreneurial ecosystem should be treated as one system, not as set of independent elements. In an ecosystem there are no direct, one-to-one relationships. The statistical reason is that the individual elements do not reveal to be statistically significantly related to the prevalence of gazelles, despite the large explained variance of the model.

In the next section we will take a systems analytical strategy, and analyse how the entrepreneurial ecosystem index is related to the share of gazelles, as the entrepreneurial ecosystem output measure.

## 5 Entrepreneurial Ecosystem Index

For mapping the ‘health’ or ‘quality’ of the entrepreneurial ecosystem we have constructed an entrepreneurial ecosystem index. The index is created to compare different units (regions, countries) and a rank in terms of multiple features (elements). The unit may be regions or countries, depending on the (policy) audience to which it is targeted and/or which spatial unit of analysis most adequately covers the relevant mechanisms in the context of entrepreneurship. Since one unit is stronger in one particular feature and the other in another feature, it is necessary to find a universal way to compare and summarize them in one index. This results in the following transformation (Acs et al. 2011):

$$P = [y_{ij}] \rightarrow I = [I_i] \quad (1)$$

$P$ : is a matrix of the dataset containing  $n \times k$  elements

$n$ : is the number of units (country, region, etc.)

$k$ : is the number of variables

$y_{ij}$ : is the observed value of unit  $i$  with respect to feature  $j$

$I_i$ : is the index associated to the  $i$ th unit ( $i = 1, 2, \dots, n$ )

On the basis of existing geography of entrepreneurship studies (see Stam 2015; Stam and Spigel 2018), an entrepreneurial ecosystem index is proposed which includes ten elements. The elements included in the entrepreneurial ecosystem index are listed in Table 1. The index compresses a large amount of data: the Dutch entrepreneurial ecosystem index, with 12 regions (units), is based on ten thousands of datapoints (for example the value of the finance element is based on a survey of

3027 SMEs, and the value of the leadership element is based on 2231 innovation projects). To develop an entrepreneurial ecosystem index based on a set of elements (features) we have used insights from other related indices, such as the World Economic Forum Global Competitiveness index (Porter et al. 2004), the European Commission Regional Competitiveness Index (RCI) (Dijkstra et al. 2011), and the Global Entrepreneurship and Development Index (GEDI) (Szerb and Acs 2011).

The ten elements of the entrepreneurial ecosystem can be quantified (see Sect. 2), and be given a comparable value. This is done by normalizing the average value of each element to 1 and then let all deviations be relative to one: with elements in regions performing less than the average having a value below 1, and elements in regions performing better than the average having a value above 1. The advantage is that this allows us to compose an index value, and compare the ‘strength’ or ‘health’ of different entrepreneurial ecosystems. This index value is computed in an additive way (see Formula 2).

$$I_i = \sum [E_{ki}] \quad (2)$$

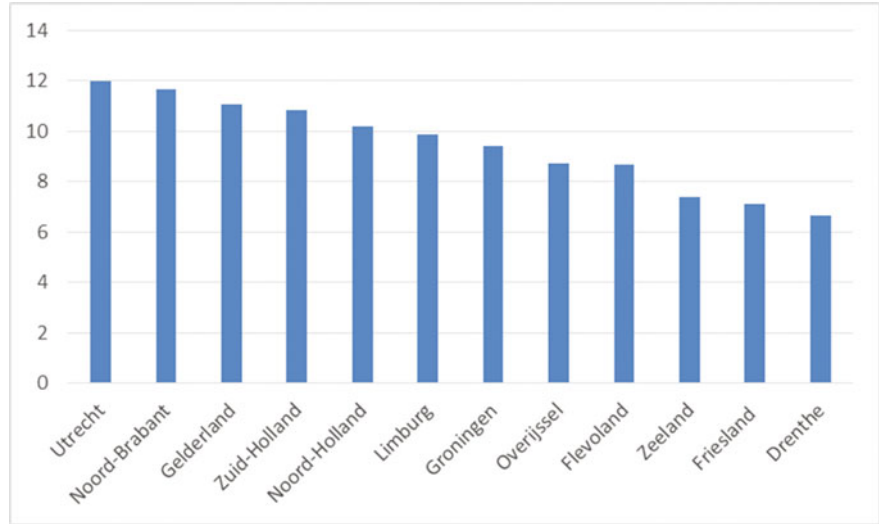
$I_i$ : is the index value of unit  $i$ , computed as the sum of the normalized values of each element  $E_{ki}$

$E_{ki}$ : is the normalized value of element  $k$ , of unit  $i$

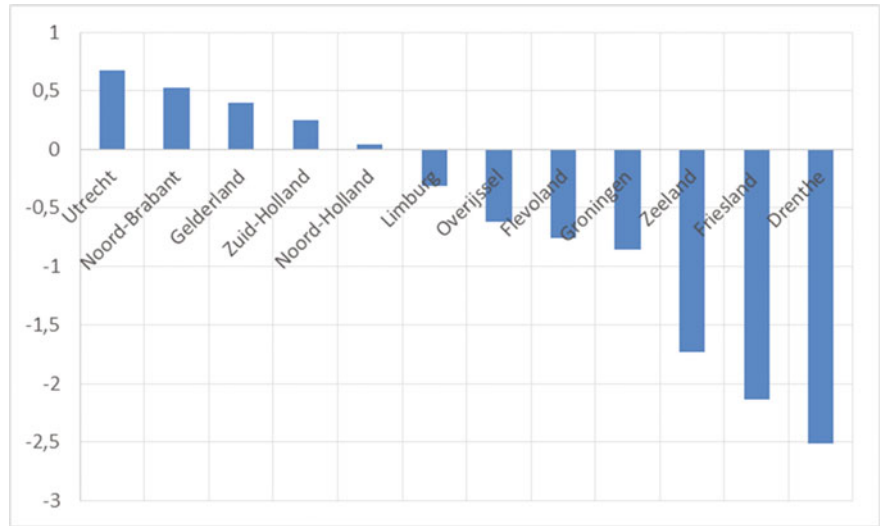
The elements of the index all get the same weight. In a later research phase other weighting techniques than the equal weighting methodology may be applied, based on either the opinion of experts or based on statistical properties of the data. The elements are here summed into one index value, which moves around 10, with regions performing on the average for all elements scoring an index value of 10, while regions performing above the average for all elements scoring an index value higher than 10. This is shown in Fig. 2 for provinces in the Netherlands, revealing variation from 6.66 (Drenthe) to 11.97 (Utrecht).

The disadvantage of this index construction is that elements with above average value (ranging from 1 to infinity) can have a stronger effect on the index than elements with below average value (ranging from 0 to 1). To solve this we take the natural logarithm of the elements, so that these symmetrically oscillate around 0, with negative values for regions below average, and positive values for regions above average. This also means that the total index value oscillates around 0 and not around 10 (see Fig. 3). The index values now vary between  $-2.52$  (Drenthe) and  $0.67$  (Utrecht). The rank order of provinces remains largely the same, with Groningen moving from the 7th to the 8th place.

The essence of ecosystems is the interaction among its elements. This interaction is not adequately covered when an index is constructed as a sum of its elements. If we take the interactive nature of the system seriously, and the resulting non-linear relations, the index should be constructed differently. For this we compute an index that is not additive ( $E_1 + E_2 + \dots E_n$ ) but multiplicative ( $E_1 * E_2 * \dots E_n$ ). This leads to index values with much larger variation, as the effect of deviations of the average is now much more substantial. The index values now vary between 0.003

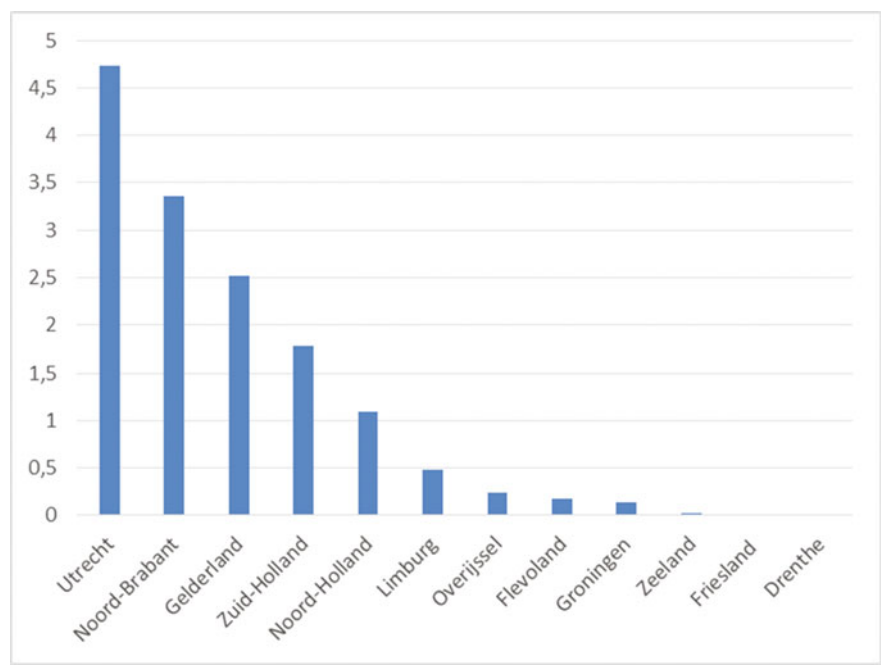


**Fig. 2** Entrepreneurial ecosystem index Netherlands provinces (additive)



**Fig. 3** Entrepreneurial ecosystem index Netherlands provinces (natural logarithm values)

(Drenthe) and 4.727 (Utrecht) (see Fig. 4). This leads to substantially more variation in the index value, which might not be valid in a small country such as the Netherlands with relatively small regional differences in entrepreneurial activity (Stam 2005).



**Fig. 4** Entrepreneurial ecosystem index Netherlands provinces (multiplicative)

**6 Entrepreneurial Ecosystem Index and Its Output**

To what extent is entrepreneurship related to the ‘health’ of entrepreneurial ecosystems? The three selected entrepreneurship measures do positively correlate with the entrepreneurial ecosystem values, but only the gazelles measure is strongly and statistically significantly correlated to the entrepreneurial ecosystem index (Table 5).

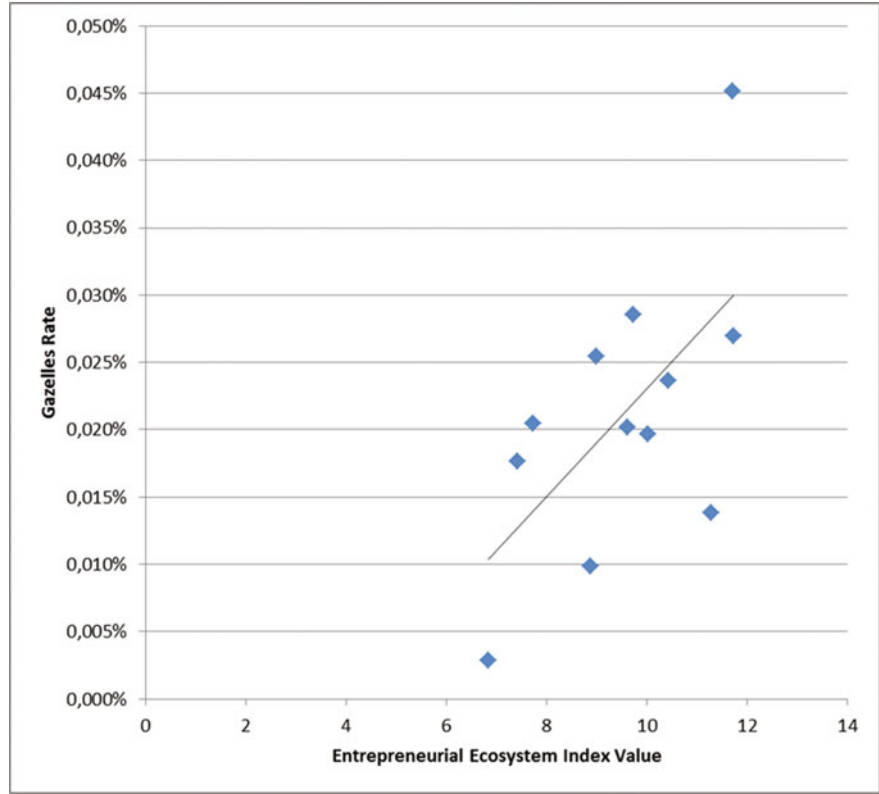
As a robustness check and perhaps demarcation of the boundary conditions of an entrepreneurial ecosystem approach, we also used three other proxies of productive entrepreneurship, namely the internationalization of firms (based on Statistics Netherlands) and innovators (based on Community Innovation Survey 2010 data). Internationalization is a measure of success beyond national context—an indicator of the ‘competitiveness’ of regional firms. Innovators is measured as being an innovator or not (having introduced a new good or service) and the share of innovative sales. All measures are aggregated at the regional level, indicating the average prevalence of these proxies in the regions. There is hardly any correlation with the strength of the entrepreneurial ecosystem and international traders (0.082), somewhat stronger but not statistically significant with turnover share innovation

**Table 5** Relation of the entrepreneurial ecosystem index (EEI) values with entrepreneurship measures

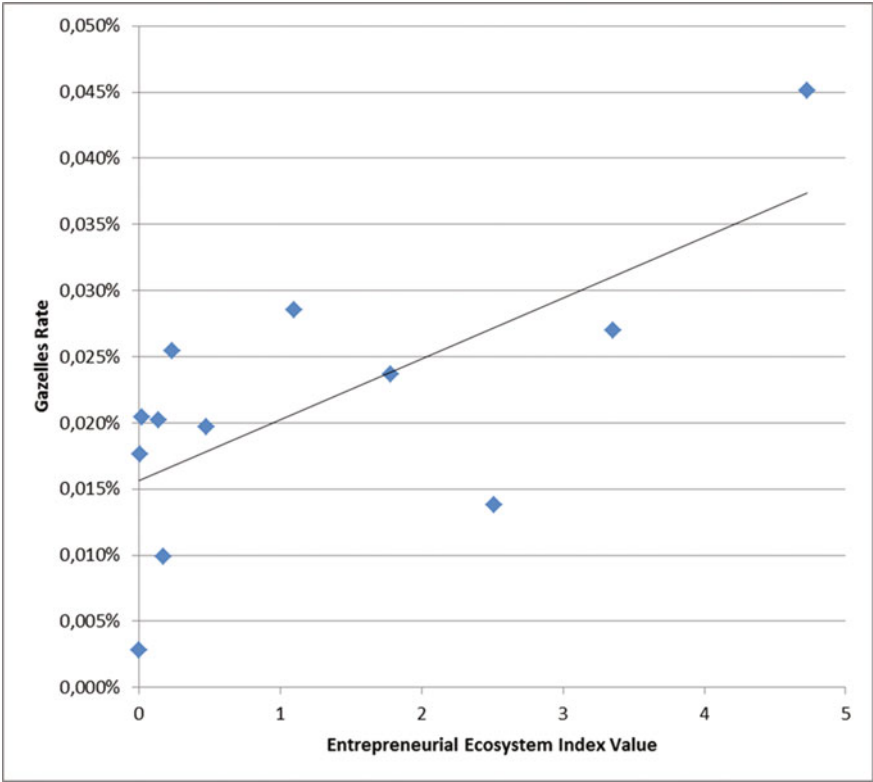
|                                                  | Share ambitious entrepreneurship, 2006–2014 | Share of high-growth businesses, 2014 | Share gazelles, 2014 |
|--------------------------------------------------|---------------------------------------------|---------------------------------------|----------------------|
| Correlation with entrepreneurial ecosystem index | 0.448                                       | 0.234                                 | 0.629                |

(0.323) and even a negative (but not significant) correlation with the share of business innovators (−0.417).

To what extent is the prevalence of gazelles a function of the ‘quality’ of the entrepreneurial ecosystem? To trace this, we have constructed several statistical models based on different computations of the entrepreneurial ecosystem index



**Fig. 5** Relation entrepreneurial ecosystem (additive) and gazelles



**Fig. 6** Relation entrepreneurial ecosystem (multiplicative) and gazelles

values. First, we use the additive index to create a linear model, which has a  $R^2$  of 0.39 (see Fig. 5). Second, we use the multiplicative index to create a linear model, which has an improved  $R^2$  of 0.47 (Fig. 6). Third, we use the multiplicative index and create a non-linear (polynomial) model, which further improves the  $R^2$  to 0.52 (Fig. 7).

Figures 5, 6 and 7 show that taking non-linearities into account increases the ‘explanatory’ power of the model. However, there might be a too limited number of data points available for creating systematic patterns, and for example tracing threshold values. Visual inspection of Fig. 7 shows that even though there might be a threshold index value of around 1.5–2.0 beyond which the rate of gazelles substantially increases, the individual data points do not suggest such a threshold.

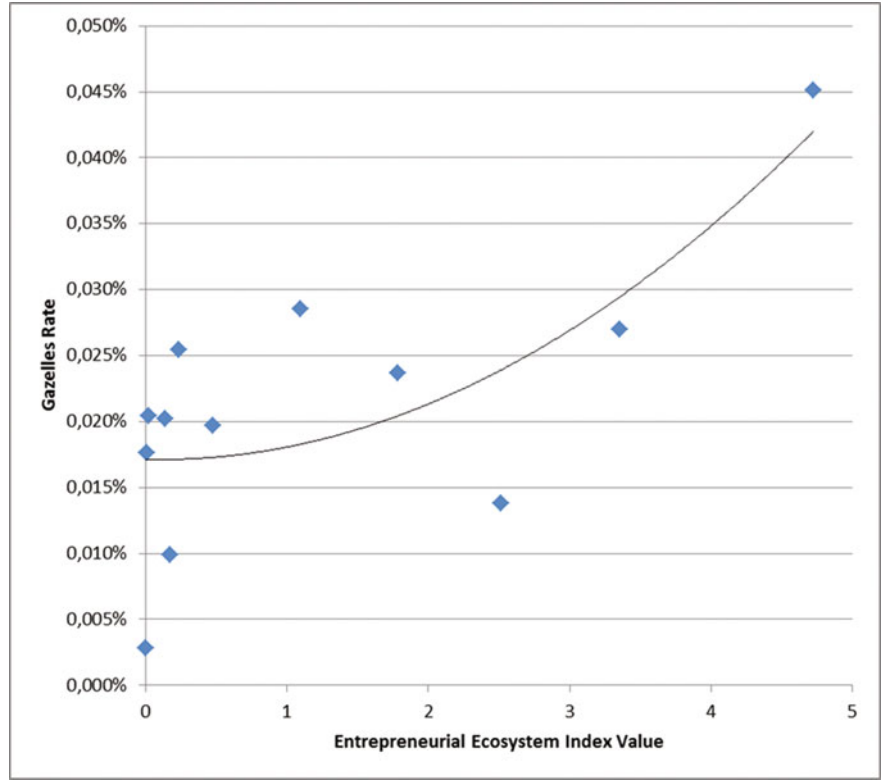


Fig. 7 Relation entrepreneurial ecosystem (multiplicative) and gazelles (polynomial function)

7 Discussion

The aim of the entrepreneurial ecosystem approach is not to predict, but to better understand how (entrepreneurial) economies function and in particular how they ‘produce’ entrepreneurship as an emerging property of the system. In this chapter we have explored how entrepreneurial ecosystems and productive entrepreneurship can be traced empirically and how entrepreneurship is related to entrepreneurial ecosystems. We have been looking for tendencies at one period in time. The analyses show that taking a systems view on the context of entrepreneurship necessitates measuring the quality of the context as a system. In this sense, we moved from the ecosystem metaphor to a complex system model of the entrepreneurial economy, at least from an epistemological point of view (Martin and Sunley 2007). Our analysis is based on a relatively small set of regions in one period in time. To arrive at more robust findings, this analysis should be repeated in multiple periods. This not only delivers more data points of entrepreneurial ecosystem index values and entrepreneurial outputs, but would also allow for feedback effects of the

entrepreneurial output on the entrepreneurial ecosystem. The analyses should also be repeated in other contexts, potentially revealing different relations between the entrepreneurial ecosystem and its output.

Several other issues for further research remain. We discuss three of them: the boundary and openness of the system, functional decomposability of the system, and structural change as an emerging property of the system.

## ***7.1 Boundary and Openness of the System***

We have been quite pragmatic in selecting the ‘least worst’, or most readily available measures of elements of entrepreneurial ecosystems in Dutch provinces. However, it may be debated whether the provincial border provides the most adequate boundary of entrepreneurial ecosystems. The boundaries are almost always arbitrary, most likely somewhere in between the municipality and the national level. Is the province the best unit of analysis, or should entrepreneurial ecosystems perhaps be analysed in a more nested or polycentric (Ostrom 2010) way, further problematizing the territorial view ‘borrowed’ from the ecological analogy? If we take the openness of the system serious this also opens ‘explanatory power’ of events and elements outside the current regional boundary, affecting the prevalence of entrepreneurship beyond regional boundaries.

## ***7.2 Functional Decomposability***

The established empirical literature on the geography of entrepreneurship and economic development has revealed several factors to be of relevance in explaining the spatial heterogeneity in entrepreneurship. This suggests that there is a limited set of factors, or elements that affects the prevalence of entrepreneurship in a region. However, because of its inherent connectivity, nonlinearity and openness, a complex system affords limited functional decomposability (Martin and Sunley 2007), which suggests that the overall functioning of the entrepreneurial ecosystem cannot be deduced from knowledge of the function of its elements. We have seen this in the non-significance of the individual elements in explaining the rate of gazelles. By constructing an index value, also by multiplying the composing elements, we have done more justice to the systemic nature of the ecosystem than can be done with traditional multivariate regressions. However, the connectivity, nonlinearity and openness can be taken into account in a more adequate fashion in future studies.

### 7.3 *Structural Change*

Complex systems may be best applicable for gaining understanding in structural change of the system, e.g. radical changes in technology, institutions and organizational arrangements (Arthur 2013). Entrepreneurship is said to be a driving force of such structural change (Schumpeter 1934). Prior research has shown that ambitious entrepreneurship has stronger effects on economic growth than other types of entrepreneurship (Stam et al. 2011; Wong et al. 2005), and that young firms are a driver of job creation (Haltiwanger et al. 2013; Criscuolo et al. 2014), and that young high-growth firms accelerate the reallocation of jobs from old to new industries (Bos and Stam 2013). However, it may be the case that our—already very refined—measure of entrepreneurship does not sufficiently cover structural change, and is not an adequate indicator of emerging properties of the system.

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## References

- Acemoglu D, Johnson S, Robinson JA (2005) Institutions as a fundamental cause of long-run growth. In: Aghion P, Durlauf S (eds) *Handbook of economic growth*. Elsevier, Amsterdam
- Acs ZJ, Armington C (2004) The impact of geographic differences in human capital on service firm formation rates. *J Urb Econ* 56(2):244–278
- Acs ZJ, Rappai G, Szerb L (2011) Index-building in a system of interdependent variables: The penalty for bottleneck. George Mason University School of Public Policy Research Paper No. 2011–2024
- Acs ZJ, Autio E, Szerb L (2014) National systems of entrepreneurship: measurement issues and policy implications. *Res Policy* 43(3):476–494
- Annoni P, Dijkstra L (2013) EU Regional Competitiveness Index RCI 2013. European Commission, Brussels
- Annoni P, Kozovska K (2010) EU Regional Competitiveness Index 2010. European Commission, Joint Research Centre, Brussels
- Arthur WB (2013) *Complexity economics: a different framework for economic thought*. Santa Fe Institute Working Paper, New Mexico
- Arundel A, Smith K (2013) History of the community innovation survey. *Handbook of innovation indicators and measurement*, pp 60–87
- Audretsch DB, Lehmann EE (2005) Does the knowledge spillover theory of entrepreneurship hold for regions? *Res Policy* 34(8):1191–1202
- Baumol WJ (1993) *Entrepreneurship, management and the structure of payoffs*. MIT Press, London
- Beer A, Clower T (2014) Mobilizing leadership in cities and regions. *Reg Stud Reg Sci* 1(1):5–20
- Beinhocker ED (2006) *The origin of wealth: evolution, complexity, and the radical remaking of economics*. Harvard Business Press, Boston
- Bell-Masterson J, Stangler D (2015) *Measuring an entrepreneurial ecosystem*. Kauffman Foundation, Kansas City
- Bos JW, Stam E (2013) Gazelles and industry growth: a study of young high-growth firms in The Netherlands. *Ind Corp Change* 23(1):145–169

- Bosma N, Schutjens V, Stam E (2009) Entrepreneurship in European regions: implications for public policy. In: Leitao J, Baptista R (eds) *Public policies for fostering entrepreneurship: A European perspective*. Springer, New York, pp 59–89
- Braaksma R, Verhoeven W, Smit L, Span T (2014) *Financieringsmonitor 2014-1. Onderzoek naar de financiering van het Nederlandse bedrijfsleven*. Zoetermeer, Panteia
- Charron N, Lapuente V, Dijkstra L (2012) *Regional governance matters: a study on regional variation in quality of government within the EU*. DG Regional Policy Working papers WP01/2012
- Cooke P (2001) Regional innovation systems, clusters, and the knowledge economy. *Ind Corp Change* 10:945–974
- Criscuolo C, Gal PN, Menon C (2014) *The dynamics of employment growth*. OECD, Paris
- Daunfeldt SO, Elert N, Johansson D (2014) The economic contribution of high-growth firms: do policy implications depend on the choice of growth indicator? *J Ind Compet. Trade* 14(3):337–365
- Day R (1987) The general theory of disequilibrium economics and of economic evolution. In: D. Batten, J. Casti and B. Johansson (eds), *Lecture notes in economics and mathematical systems: economic evolution and structural adjustment*, Springer, pp 46–63
- Dijkstra L, Annoni P, Kozovska K (2011) A new regional competitiveness index: theory, methods and findings. European Union, Brussels
- Fayolle A (2007) *Entrepreneurship and new value creation: the dynamic of the entrepreneurial process*. Cambridge University Press, Cambridge
- Feld B (2012) *Startup communities: building an entrepreneurial ecosystem in your city*. Wiley, New York
- FD Gazellen (2013) *Financieele Dagblad*. Amsterdam, FD Media Group
- Fritsch M (2013) New business formation and regional development—a survey and assessment of the evidence. *Found Trends Entrep* 9:249–364
- Gomez T, Stam E (2017) *Ondernemen in Nederland: Bedrijvendynamiek en Financiering*. Utrecht University School of Economics, Utrecht
- Haltiwanger J, Jarmin RS, Miranda J (2013) Who creates jobs? small versus large versus young. *Rev Econ Stat* 95(2):347–361
- Henrekson M, Johansson D (2010) Gazelles as job creators: a survey and interpretation of the evidence. *Small Bus Econ* 35(2):227–244
- Howells J (2006) Intermediation and the role of intermediaries in innovation. *Res Policy* 35(5):715–728
- Isenberg DJ (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):41–50
- Kerr WR, Nanda R (2009) Democratizing entry: banking deregulations, financing constraints, and entrepreneurship. *J Financ Econ* 94(1):124–149
- Lee SY, Florida R, Acs ZJ (2004) Creativity and entrepreneurship: a regional analysis of new firm formation. *Reg Stud* 38(8):879–891
- Martin R, Sunley P (2007) Complexity thinking and evolutionary economic geography. *J Econ Geogr* 7(5):573–601
- North DC (1990) *Institutions, institutional change and economic performance*. Cambridge University Press, New York
- OECD (2011) *Entrepreneurship at a glance 2011*. OECD Publishing. <https://doi.org/10.1787/9789264097711-en>
- Ostrom E (2010) Beyond markets and states: polycentric governance of complex economic systems. *Am Econ Rev* 100:1–33
- Parker SC, Storey DJ, Van Witteloostuijn A (2010) What happens to gazelles? The importance of dynamic management strategy. *Small Bus Econ* 35(2):203–226
- Porter ME, Schwab K, Sala-i-Martin X, Lopez-Claros A (eds) (2004) *The global competitiveness report 2004–2005*. Palgrave Macmillan, World Economic Forum, New York
- Qian H, Acs ZJ, Stough RR (2013) Regional systems of entrepreneurship: the nexus of human capital, knowledge and new firm formation. *J Econ Geogr* 13(4):559–587

- Schumpeter JA (1934) *The theory of economic development*. Harvard University Press, Cambridge MA
- Snoei J, Ichou A (2010) *Financiering in het MKB. Onderzoek naar de financieringsbehoefte per provincie*. EIM, Zoetermeer
- Spiekermann K, Wegener M (1996) Trans-European networks and unequal accessibility in Europe. *Eur J Reg Dev* 4:35–42
- Spiekermann K, Wegener M, Copus A (2002) Review of peripherality indices and identification of 'baseline indicator: deliverable 1 of AsPIRE—aspacial peripherality, innovation, and the rural economy. S&W, IRPUD, SAC, Dortmund/Aberdeen
- Spigel B (2017) The relational organization of entrepreneurial ecosystems. *Entrep Theory Pract* 41:49–72
- Stam E (2005) The geography of gazelles in the Netherlands. *Tijdschr voor econ en soc geografie* 96(1):121–127
- Stam E (2007) Why Butterflies don't leave. *Locational behavior of entrepreneurial firms. Economic Geography* 83(1):27–50
- Stam E (2010) Entrepreneurship, evolution and geography. In: *The handbook of evolutionary economic geography*, pp. 307–348
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plan Stud* 23(9):1759–1769
- Stam E, Bosma NS (2015) Local policies for high-growth firms. In: Audretsch D, Link A, Walshok A (eds) *The oxford handbook of local competitiveness*. Oxford University Press, Oxford, pp 286–305
- Stam E, Spigel B (2018) Entrepreneurial ecosystems. In: Blackburn R, De Clercq D, Heinonen J (eds) *The SAGE handbook of small business and entrepreneurship*. SAGE, London, pp 407–422
- Stam E, Van Stel A (2011) Types of entrepreneurship and economic growth. In: Goedhuys M, Naudé W, Szirmai E (eds) *Innovation, entrepreneurship and economic development*. Oxford University Press, Oxford, pp 78–95
- Stam E, Suddle K, Hessels J, Van Stel A (2009) High-growth entrepreneurs, public policies and economic growth. In: Leitao J, Baptista R (eds) *Public policies for fostering entrepreneurship: a european perspective*. Springer, New York, pp 91–110
- Stam E, Hartog C, Van Stel A, Thurik R (2011) Ambitious entrepreneurship and macro-economic growth. In: Minniti M (ed) *The dynamics of entrepreneurship. Evidence from the global entrepreneurship monitor data*. Oxford University Press, Oxford, pp 231–249
- Stam E, Bosma N, Van Witteloostuijn A, De Jong J, Bogaert S, Edwards N, Jaspers F (2012) *Ambitious entrepreneurship. A review of the academic literature and new directions for public policy*. Advisory Council for Science and Technology Policy (AWT), The Hague
- Stam E, Romme AGL, Roso M, Van den Toren JP, Van der Starre BT (2016) *Knowledge triangles in the Netherlands. An entrepreneurial ecosystem approach*, Report for the OECD Technology and Innovation Policy Working Group
- Sternberg R (2009) Regional dimensions of entrepreneurship. *Found Trends® Entrep* 5(4): 211–340
- Szerb L, Acs ZJ (2011) The global entrepreneurship and development index methodology
- Thurik R, Stam E, Audretsch D (2013) The rise of the entrepreneurial economy and the future of dynamic capitalism. *Technovation* 33(8–9):302–310
- Wong PK, Ho YP, Autio E (2005) Entrepreneurship, innovation and economic growth: evidence from GEM data. *Small Bus Econ* 24(3):335–350
- Wooldridge JM (2013) *Introductory econometrics—a modern approach*. South-Western Cengage Learning, 5th edn
- Zhang Y, Li H (2010) Innovation search of new ventures in a technology cluster: the role of ties with service intermediaries. *Strateg Manag J* 31(1):88–109

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